Arithmetic of higher coherent cohomology of Shimura varieties

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Abstract

In this thesis, we study various arithmetic properties of the higher (namely, H^i with $i > 0$) coherent cohomology of Shimura varieties and their applications on automorphic forms. Coherent cohomology, being only loosely connected to the theory of motives, carries interesting information that cannot be seen in other cohomology theories. On the other hand, its indirect definition makes a coherent cohomology class a highly interesting object which requires careful interpretation. The current thesis hopes to suggest that understanding various aspects of the mystery of coherent cohomology often yields fruitful and interesting study of arithmetic of automorphic forms.

This thesis is naturally divided into three parts. In the first part, we interpret *harmonic Maass forms*as classes in the *local cohomology* of modular curves. This reinterpretation suggests natural generalizations of harmonic Maass forms to other Shimura varieties, which are engineered to avoid the Koecher's principle. In the second part, we suggest a conjecture on the relationship between coherent cohomology of Shimura varieties of different cohomological degrees. The conjecture is motivated by a philosophy of Venkatesh, which suggests that each such part of cohomology is a graded module over the exterior algebra of the motivic cohomology of the corresponding adjoint motive. The case of low-weight automorphic forms on Shimura varieties is in some sense a degenerate case where more structures get involved, such as archimedean L-packets. In the final part, we consider the theme of p-adic variation of higher coherent cohomology, which is recently developed under the name of *higher Hida theory.*We carry out this strategy for the base-change L-function of $\mathrm{U}(2,1)$, following the approach of Loeffler–Pilloni–Skinner–Zerbes for the spin L-function of GSp_4. As such strategy relies heavily on the cohomological period integrals, we suggest a more streamlined approach based on the *generalized Whittaker models.*

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Chapter 1

Introduction

A major tool in the study of automorphic forms is the cohomology of locally symmetric spaces. Certain kinds of locally symmetric spaces can be endowed with the structure of complex algebraic varieties, and they, in fact, have models defined over the field of algebraic numbers $\overline{\mathbb{Q}}$. This fact, that some locally symmetric spaces are algebraic varieties defined over $\overline{\mathbb{Q}}$, is arguably the most important fact in the context of studying arithmetic properties of automorphic forms. For example, one proves that the space of modular forms has a basis consisted of Hecke eigenforms with eigenvalues in $\overline{\mathbb{Q}}$ by regarding modular forms as sections of a line bundle over a modular curve. In general, whenever an automorphic form is holomorphic, it can be realized as a section of a vector bundle over algebraic locally symmetric spaces, called Shimura varieties.

Even though one abstractly associates other types of automorphic forms to the *higher* coherent cohomology of Shimura varieties, namely H^i for $i > 0$, a class in the higher coherent cohomology does not represent a clean geometric object, unlike that a class in H^0 represents a holomorphic section of a holomorphic vector bundle. Indeed, one can try to associate a Čech cocycle to a coherent cohomology class as a representative, but this association is often hard to directly manipulate with. For example, an honest (e.g. without using Serre duality, automorphic forms, Hodge decomposition,

etc.) computation of higher Čech cohomology of arithmetically interesting varieties, such as Hilbert modular varieties, is an extremely challenging problem.

On the other hand, a higher coherent cohomology class possesses a homological-algebraic information. For example, the punctured affine plane $\mathbb{A}^2 - \{0\}$ has the same H^0 as the affine plane $\mathbb{A}^2$ by the Hartogs' lemma, which says that a holomorphic function automatically extends over a codimension ≥ 2 closed subset. Thus, $\mathbb{A}^2 - \{0\}$ cannot be affine, and the "obstruction to extension" is rather captured by its H^1, namely $H^1(\mathbb{A}^2) = 0$ whereas $H^1(\mathbb{A}^2 - \{0\}) \neq 0$. This obstruction can also be expressed in terms of a tangible non-holomorphic $(0,1)$-form, $\frac{-z_2 d\bar{z}_1 + z_1 d\bar{z}_2}{|z_1|^2+|z_2|^2}$, where z_1, z_2 are the affine coordinates of $\mathbb{A}^2$. The theme of a higher coherent cohomology class being an obstruction continues in the context of Shimura varieties; for example, for a compact modular curve X and its weight-k line bundle ω^k, $H^1(X, \omega^k)$ is the space of obstructions to finding a meromorphic section of ω^k which is holomorphic over the open modular curve and has given singularities at cusps ([Bor, Thm. 3.1]).

In this thesis, we will study the higher coherent cohomology of Shimura varieties using its various interpretations (e.g. Čech cohomology, Zariski cohomology, Dolbeault cohomology, Lie algebra cohomology). The interaction between various incarnations of coherent cohomology will yield fruitful results on the arithmetic of automorphic forms. Along the way, we face an abundance of new kinds of objects and structures natural from one interpretation but not known in the other incarnations. Indeed, the way that the coherent cohomology gets related to motives is via the E_1-degeneration of Hodge-to-de Rham spectral sequence, and this inherently exploits the Hodge theory, that Shimura varieties have structures of Kähler manifolds. Even more mysteriously, these graded pieces of de Rham cohomology each have their own $\overline{\mathbb{Q}}$-rational structure, whose relation with the rational structures of other cohomologies is unclear.

This thesis is consisted of three parts, where each part is about a separate topic

that is independent of one another. Each part is about an object or a phenomenon in the context of coherent cohomology of Shimura varieties and its consequences in terms of automorphic forms: in Part I, we study the *local cohomology* of Shimura varieties, which is translated into *harmonic Maass forms* and *q-expansions*; in Part II, we study the appearance of an *archimedean L-packet* in the cohomology of Shimura varieties and its arithmetic; in Part III, we study the *p-adic variation of higher coherent cohomology*, and its application to the construction of a *p-adic L-function*.

1.1 Part I

In this part, we consider the local cohomology of Shimura varieties. Local cohomology is the algebraic analogue of relative singular cohomology, and it was developed by Grothendieck to develop an appropriate duality theory (called *Grothendieck duality* or *coherent duality*) generalizing the Serre duality. It satisfies most properties that are expected for the analogue of relative singular cohomology:

- they fit into a long exact sequence with other coherent cohomology groups;
- they satisfy the excision property;
- they have a Mayer–Vietoris sequence, etc.

It was recently observed that the *overconvergent modular forms* should be considered in the context of local cohomology of Shimura varieties in [BoPi] and [Pa]. They consider the local cohomology with support in a stratum in the Bruhat–Tits stratification, which is a refinement of the more classical Ekedahl–Oort stratification. In the case of modular curves, this is just the stratification of ordinary locus and supersingular points.

The Part I take the local cohomology with a slightly different support, namely the support in the *cuspidal divisor*. We assert that the local cohomology with the

support in the cuspidal divisor is a natural place to consider harmonic Maass forms.

Theorem 1. *Let $H_k^+(\Gamma)$ be the space of harmonic Maass forms of weight k and level Γ whose shadows are cuspidal. Let $M_k(\Gamma)$ be the space of modular forms of weight k and level Γ. Let $X(\Gamma)$ be the compact modular curve of level Γ, $D \subset X(\Gamma)$ be the cuspidal divisor, and let ω be the (canonical extension of the) Hodge bundle. Then, there is a natural isomorphism*

$$\frac{H_k^+}{M_k} \xrightarrow{\sim} H_D^1(X(\Gamma), \omega^k),$$

$$f \mapsto (\overline{\partial} f, f),$$

where the target denotes the local cohomology*.*

This harmonizes well with the analogy of Darmon and coauthors between harmonic Maass forms, the archimedean generalization of modular forms, and overconvergent modular forms, the p-adic generalization of modular forms; namely, the supersingular points are the p-adic analogues of cusps.

One immediate consequence of our interpretation is that we can give a robust notion of arithmeticity in the theory of harmonic Maass forms. For example, it would make sense to talk about the harmonic Maass forms over $\mathbb{Z}_p$ or $\mathbb{F}_p$. Another important consequence is that we can generalize the notion of harmonic Maass forms to other Shimura varieties. For example, the Hilbert-modular analogue would be the following.

Definition 1. *Let $\Gamma \subset \mathrm{SL}_2(\mathcal{O}_F)$ be a congruence subgroup of a totally real field F of degree $d > 1$, and let $Y(\Gamma)$ be the Hilbert modular variety for F with level Γ. Let ω be the Hodge line bundle over $Y(\Gamma)$. For $k \in \mathbb{Z}$, we define the space of weakly holomorphic Hilbert modular forms $M_k^!(\Gamma)$ as*

$$M_k^!(\Gamma) := H^{d-1}(Y(\Gamma), \omega^k).$$

Let $X(\Gamma)^{\mathrm{tor}}$ be a toroidal compactification of $Y(\Gamma)$ with respect to a sufficiently fine rational polyhedral cone decomposition, and let $D \subset X(\Gamma)^{\mathrm{tor}}$ be the cuspidal divisor. The space of principal parts of harmonic Hilbert Maass forms *is defined to be*

$$\frac{H_k^+(\Gamma)}{M_k(\Gamma)} := H_D^d(X(\Gamma)^{\mathrm{tor}}, \omega^k),$$

where ω on $X(\Gamma)^{\mathrm{tor}}$ is, by abuse of notation, the canonical extension of ω.

Indeed, such generalization was not known, mostly due to the *Koecher's principle* that exists in a higher-dimensional Shimura variety: the Koecher's principle says that moderate growth condition is automatically imposed in such varieties, and for the definition of harmonic Maass forms, it is crucial to allow the function to blow up at infinity. Even though a function is no longer allowed to blow up at infinity by the Koecher's principle, it turns out that Dolbeault $(0, i)$-forms can still blow up at infinity for $i > 0$, which was first realized by Lan (see [Lan2]). We insist that the discrepancy at this cohomological degree encodes the higher-dimensional analogue of harmonic Maass forms. This is in accordance with another principle that a harmonic Maass form encodes "Ext^1 of modular forms."

From this, we finish with a few applications. We first relate our Hilbert-modular harmonic Maass forms with [Br2], which is idiosyncratic in the sense that it defines what the Borcherds lift of a Hilbert-modular harmonic Maass form is *without defining what a Hilbert-modular harmonic Maass form is.* Instead, Bruinier avoids the Koecher's principle by instead working with Whittaker functions, which do not have to satisfy the Koecher's principle; indeed, the broadly used Whittaker functions in the theory of automorphic forms are classically defined as the good solution (i.e. that of moderate growth towards infinity) of the Whittaker differential equation, which is a differential equation with irregular singularity at infinity, but there is also the bad solution (i.e. that blows up exponentially at infinity) because of the irregular

singularity, which can be used as well. We relate the Whittaker functions studied in [Br2] with our Hilbert-modular generalization of harmonic Maass forms. We hope to convey a sense that the reason why automorphic representations are semisimple is because one imposes the moderate growth condition, and one can expect to see nonsplit extensions between automorphic representations when one relaxes the moderate growth condition to a more relaxed growth condition (e.g. linear exponential growth). We also investigate what kind of arithmetic information one could get from the local cohomology interpretation.

1.2 Part II

By the relationship between the cohomology of locally symmetric spaces and automorphic forms (see §2.2), the arithmetic of automorphic forms is most well studied via the cohomology of locally symmetric spaces or Shimura varieties. On the other hand, whether some automorphic form appears in H^i or not, or the multiplicity of its appearance, can be computed solely in terms of automorphic data. As each H^i seems to convey a different arithmetic information, it is a natural question to ask the relationship between the various arithmetic structures induced by various H^i at which the automorphic form appears. This is the context of the philosophy of Venkatesh, that the appearance of an automorphic form in multiple cohomological degrees can be explained by the action of an appropriate *motivic cohomology group*. Furthermore, this degree-shifting action is in fact a part of the *derived Hecke algebra*, an enhancement of Hecke algebra, and this can be also seen in the corresponding Galois side.

There are several reasons why a single automorphic representation can appear in multiple cohomological degrees. On one hand, if the automorphic representation is non-tempered, then it can appear in multiple degrees of cohomology, and these

appearances are related by the action of the Lefschetz SL_2. On the other hand, if the group in concern, denoted G, has no compact Cartan subgroup, then the Lie algebra of the split part of the fundamental Cartan (i.e. maximally compact Cartan) naturally acts as a degree-shifting action.

Even if neither of the above is true, namely if $\operatorname{rank} G = \operatorname{rank} K$, a tempered cuspidal automorphic representation can appear in multiple degrees of cohomology, if we consider the *coherent cohomology* of Shimura varieties. This corresponds to the Hodge decomposition of the de Rham cohomology, and especially for automorphic forms of irregular weight (namely, the infinitesimal character of the infinity type is singular, i.e. orthogonal to some root), there are multiple parts of the Hodge decomposition that arise as the cohomology of a single coefficient bundle with different cohomological degrees. The most evident example is the case of modular forms of weight one: a cusp form of weight one can be seen as either an element of $H^0(X, \omega)$ or $H^1(X, \omega)$, where X is a compact modular curve and ω is the (canonical extension of the) weight one line bundle. Ultimately this is related to the fact that there are automorphic representations with the same finite part but with different infinity types, or more concisely the existence of *archimedean L-packet*.

For example, the motive M_f of a weight k modular eigenform f, which is of weight $k-1$, has a Hodge decomposition

$$(M_f)_B = H^{0,k-1} \oplus H^{k-1,0},$$

where f can be naturally seen as an element of $H^{k-1,0}$. On the other hand, $H^{k-1,0}$ has $\overline{f}$ as a natural element, and it represents another automorphic representation having the same finite part but having a different, complex conjugate, infinity type. If $k = 1$, both the $(0, k-1)$-part and the $(k-1, 0)$-part become the $(0, 0)$-part with rank 2, and it is even more difficult to distinguish f out of the rank 2 vector space.

We suggest a conjecture regarding the arithmetic structures of coherent cohomology of Shimura varieties when such coincidence of Hodge decomposition occurs. The situation is in stark contrast with the case of regular weight, where the infinity type of an automorphic reprsentation can be determined by the finite part and the cohomological degree. The conjecture is roughly of the following form:

Conjecture 1. *Let λ be a nondegenerate singular analytically integral character, and let Π satisfy Assumption 10.1.5, with $\Pi_\infty = A_C(\lambda)$ (using the notation of [VZ]) for some Weyl chamber C containing λ (necessarily in one of the walls of C). Let $\mathcal{M} = H^1_M((\operatorname{Ad}\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$ and $\mathcal{H}^i = H^i(X_G(\Gamma)_{\overline{\mathbb{Q}}}, [V])[\Pi_f]$, where both are regarded as $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian bilinear forms, induced from a fixed admissible bilinear form on $\mathfrak{g}_{\mathbb{C}}$ (see §10.2). Then, there is an isometry between graded $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian metrics,*

$$\wedge^* \mathcal{M}^* \otimes \mathcal{H}^{i_{\min}} \cong \bigoplus_{i=i_{\min}}^{i_{\max}} \mathcal{H}^i,$$

where V is the automorphic vector bundle coming from Levi (see Notation) such that Π_f appears in multiple degrees of its cohomology, $i_{\min} = \min\{i \mid H^i(X_G(\Gamma), [V])[\Pi_f] \neq 0\}$, and $i_{\max}$ is defined analogously.

We prove that the conjecture is compatible with the standard conjectures on automorphic periods, such as the refined Gan–Gross–Prasad conjectures, the Lapid–Mao conjecture, and the Beilinson conjecture.

Theorem 2. *Under certain mild conditions and several standard conjectures on periods (see below), for $G = \mathrm{Sp}_4$ and $\mathrm{SU}(2,1)$, Conjecture 1 is true. In other words, the action of adjoint motivic cohomology group on the coherent cohomology groups of Shimura variety respects $\overline{\mathbb{Q}}^\times$-structure.*

The outstanding question that stems out of this investigation is a quest for a geometric correspondence that distinguishes different infinity types. Indeed, in the

case of modular forms, the operation of *complex conjugation* could distinguish the (holomorphic) modular form from its (anti-holomorphic) complex conjugate modular form. There should be such operations for more general infinity types, which we call *generalized complex conjugations*. By our conjecture, it is expected that the generalized complex conjugations are not algebraic, but only analytic. We expect to construct such operations using complex analytic geometry that goes beyond the setup of Shimura varieties, e.g. using flag domains, which will be a subject of the author's future research.

1.3 Part III

The classical *Hida theory* concerns the p-adic interpolation of ordinary modular cusp forms and their associated Galois representations. In more concrete terms, the Hida theory interpolates a part of $H^0(X, \omega^k(-D))$ (using the notation as above) into the space of $\Lambda = \mathbb{Z}_p[[X]]$-adic modular forms, and a part of $H^1_{\text{ét}}(X_{\overline{\mathbb{Q}}})$ into the Galois representation $G_{\mathbb{Q}} \to \mathrm{GL}_2(\Lambda)$. There are many generalizations of Hida theory, either interpolating other cohomology groups such as de Rham cohomology or (φ, Γ)-modules (see [Cai1], [Cai2]) or considering higher dimensional Shimura varieties. However, the interpolation has been limited to either Galois representations or the space of holomorphic automorphic forms (i.e. H^0) for Shimura varieties other than modular curves. The first difficulty for H^i with $i > 0$ is that we do not have an analogue of q-expansion principle which enabled us to identify holomorphic automorphic forms in terms of numerical data.

Recently, Pilloni and his coauthors have developed the *higher Hida theory*, the higher coherent analogue of Hida theory, using the p-adic geometry of Shimura varieties. This already has various arithmetic applications:

- one can p-adically interpolate the space of holomorphic automorphic forms of

irregular weight, which has applications to potential modularity of abelian surfaces (see [BCGP]);

- one can p-adically interpolate each part of the *cohomological period integral*. which is a period integral that can be interpreted as a cup product pairing in coherent cohomology, which has applications to the "de Rham" construciton of p-adic L-functions (see [LPSZ]).

We carry out the second application using the cohomological period integral that represents the base-change L-function for $\mathrm{U}(2,1)$.

Theorem 3. *Suppose* Π *is a generic, stable, cuspidal automorphic representation of* $\mathrm{GU}(2,1)$ *of infinitesimal character* $(k_1, k_2, k_3; c)$ *(see §16.5). If* $k_1 - k_2 \geq 2$ *and* $k_2 + k_3 \geq 2$*, there is an element* $\mathcal{L}(\Pi, \mathbf{j}) \in \Lambda_{\overline{\mathbb{Q}}_p}(\mathbb{Z}_p^\times)$*, the Iwasawa algebra, such that, for any Dirichlet character* χ *of* p*-power conductor and any critical value* a *of* $L(\Pi, \chi, s)$*,*

$$\mathcal{L}(\Pi, a + \chi) = Z_\infty(\Pi, \chi, a) Z_p(\Pi, \chi, a) L(BC(\Pi) \otimes \chi^{-1}, a),$$

where Z_∞ *and* Z_p *are certain local factors at* ∞ *and* p*, respectively.*

The underlying integral representation of L-function was originally due to Gelbart–Piatetski-Shapiro (see [GPS]). Although this is a degenerate case of the period integral of Gan–Gross–Prasad type, we also systematically develop the study of the integral representation of L-functions using the *generalized Whittaker models*.

To construct the p-adic L-function, the key ingredient is the higher Hida theory for $\mathrm{U}(2,1)$. Fortunately, this is developed in the thesis of M. T. Nguyen [Ng], which we heavily use. In some sense, the higher Hida theory for $\mathrm{U}(2,1)$ is actually easier to develop than that of GSp_4. In very rough terms, higher Hida theory is about a careful study of arithmetic of integral (i.e. as opposed to rational) Hecke operators acting on the coherent cohomology of integral models of Shimura varieties. This is in

particular not subsumed by the higher Coleman theory (as opposed to what a naive layman might think), partly because the higher Coleman theory loses track of integral structures, and also because it does not try too hard to specify which slope is the least p-divisible slope ("ordinary part") that could appear in cohomology. In the case of [Pi] on the higher Hida theory for GSp_4, *op. cit.* actually *uses the higher Coleman theory* to finish the proof of the integral control theorem of the higher Hida theory. This is partly because the arithmetic of p-integral p-Hecke operator is still mysterious; as the correspondence spaces for the p-Hecke operators do not admit any known integral models (as it is the GSp_4-Shimura variety with level *deeper than parahoric level*), *op. cit.* comes up with a modified version of the Hecke operator that is still good for the purpose of detecting the ordinary part of the cohomology. This modified Hecke operator is the composition of two different Hecke corresopndences, each of which is supported by a parahoric level integral model of GSp_4-Shimura variety, so its arithmetic is hard to approach explicitly. On the other hand, for $\mathrm{U}(2,1)$, the p-Hecke correspondence is supported by a single parahoric level Shimura variety, so [Ng] could actually deduce a strong enough integral control theorem without appealing to higher Coleman theory. In this way, many parts on the interpolation of H^1 in [LPSZ] are not necessary in our case.

The integral representation we use is that of Gelbart–Piatetski-Shapiro, [GPS], whose archimedean local computation is completed in [KO]. It is a basic application of the notion of *Fourier expansion of an automorphic form along a unipotent subgroup.* A naive Fourier expansion of an automorphic form f along a unipotent subgroup N would be a decomposition of form

$$f(ng) = \sum_{\rho \in \widehat{N}} f_\rho(ng),$$

where $\widehat{N}$ is the unitary dual of N; namely, one sees $n \mapsto f(ng)$ as a representation

of N and decompose this representation as a sum of irreducible N-representations. This (or subset of it) is how the notion of Fourier expansion is used in most of the literature:

- The q-expansion of a modular form, or more generally the *Fourier–Jacobi expansion* of a holomorphic automorphic form f, is a Fourier expansion with respect to the unipotent radical N of a certain maximal parabolic subgroup (Fourier–Jacobi type). Here, the Fourier–Jacobi coefficients are certain analogues of *Jacobi forms*, and the expansion is indexed by the characters of the center of $N(\mathbb{R})$. The reason why we only need to consider the characters of $Z(N(\mathbb{R}))$ is because the infinity type of f, a *holomorphic discrete series representation*, has the smallest possible Gelfand–Kirillov dimension, so any other less degenerate Whittaker model is zero. In this case, the characters of $Z(N(\mathbb{R}))$ correspond to the Weil representations of $N(\mathbb{R})$ having the corresponding central character via the Stone–von Neumann theorem.

- In other contexts, certain parts of the expansion (not the whole expansion) are used. The most used Fourier coefficients are *Whittaker coefficients*, which correspond to the coefficients of the characters of N for the Fourier expansion with repsect to the unipotent radical N of a Borel subgroup.

In most cases, the Fourier coefficients are considered only if they are supported by a *multiplicity one model*. For example, the Whittaker model, if exists (namely, if it is *generic*), is of multiplicity one, both globally and locally. More interestingly, if a representation is not generic, then still there are many variants of Whittaker models that could be considered and are not zero. These are, using the term of [MW], called *degenerate Whittaker models*. For each choice of unipotent subgroup N and a unitary irreducible representation ρ of N, there is an associated degenerate Whittaker model. The behavior of a degenerate Whittaker model is similar for those with the same N

and the same Gelfand–Kirillov dimension of ρ. For example, in [MW], it is shown that the degenerate Whittaker model for a representation is of finite multiplicity for the choices of (N, ρ) that correspond to the *wave-front set* of the representation. On the other hand, any degenerate Whittaker model more degenerate than this will have infinite dimension. For example, the Fourier–Jacobi expansion exploits the fact that, for the holomorphic discrete series, the (N, ρ) corresponding to the Fourier–Jacobi coefficients is the "least degenerate" nonzero degenerate Whittaker model so that it enjoys multiplicity one property.

Along the way of developing Part III, we initiate a program to study the degenerate Whittaker models of infinite dimension more systematically. In the case of $\mathrm{U}(2,1)$, the Fourier expansion along the unipotent radical N of the Borel subgroup B is

$$f = \underbrace{\sum_{\chi \text{ character of } N} f_\chi}_{\text{Whittaker coefficients}} + \underbrace{\sum_{\psi \text{ character of } Z(N)} f_\psi}_{\text{Fourier–Jacobi coefficients}} .$$

We are now interested in an integral of the form

$$\int_{[\mathrm{U}(1,1)]} f(h)g(h)dh,$$

where $g(h)$ is another automorphic form on $\mathrm{U}(1,1)$, and $\mathrm{U}(1,1) \subset \mathrm{U}(2,1)$ is embedded naturally. The integral representation of [GPS] considered the case when g is Eisenstein. The integral then unfolds to ignore the contribution of Fourier–Jacobi coefficients and the period of "constant terms" of g (one can think of the constant term as the least degenerate Whittaker coefficient) are paired with the Whittaker coefficients of f, yielding a nice Eulerian integral. A sample theorem is as follows:

Theorem 4. *Let Π, σ be tempered cuspidal automorphic representations of $\mathrm{GU}(2,1)(\mathbb{A}_\mathbb{Q})$ and $\mathrm{GU}(1,1)(\mathbb{A}_\mathbb{Q})$, respectively, and let μ be a character of $\mathrm{GU}(1)(\mathbb{A}_\mathbb{Q})$. Suppose that $\chi_\Pi \chi_\sigma \mu = 1$, where χ_Π and χ_σ are the central characters of Π and σ, respectively. For*

$f \in \Pi$ and $g \in \sigma$, let

$$Z(f,g,\mu) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)g(h)\mu(h)dh,$$

be the corresponding period integral of Gan–Gross–Prasad type. Then, $Z(f,g,\mu)$ is the Euler product of local zeta integrals,

$$Z(f,g,\mu) = \prod_v Z_v(f,g,\mu), \qquad Z_v(f,g,\mu) = \int_{Z_H(\mathbb{Q}_v)N_{B_H}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_{f,v}^{\psi_v,\mu_v^{-1}}(h)W_{g,v}^{\psi_v}(h)dh,$$

where $W_{f,v}^{\psi,\mu_v^{-1}}(h)$ and $W_{g,v}^{\psi}$ belong the generalized Whittaker model for Π_v with respect to $\mu_v^{-1} \otimes \omega_{\psi_v}$ and the Whittaker model for σ_v with respect to ψ_v, respectively (see §19 for explanation of the terms).

The Ichino–Ikeda formula expresses the norm square of the integral as an L-value, but the Eulerianity of the integral itself is, to our knowledge, a new result.

This integral fits into a general philosophy of Gan–Gross–Prasad conjectures, so the same integral should have a connection to L-functions even when g is a cuspidal holomorphic automorphic form of $\mathrm{U}(1,1)$. On the other hand, the same unfolding pairs the Whittaker coefficients of g (which are the next least degenerate Whittaker coefficients of g) with the Fourier–Jacobi coefficients of f. If the Fourier–Jacobi coefficients are the least degenerate nonzero degenerate Whittaker model for f, which is the case when f is holomorphic, this would again yield an Eulerian integral. On the other hand, if f is generic, then the Whittaker coefficients are nonzero, and the Fourier–Jacobi coefficients are supported by degenerate Whittaker models of *infinite dimension.*

We suggest that this infinite dimensionality of degenerate Whittaker models can be systematically resolved by using the *generalized Whittaker models* of Yamashita and others. In fact, the generalized Whittaker model for $\mathrm{U}(2,1)$ was already considered

long before by Gelbart and Rogawski in [GR]. The key is to not just consider the action of N but to also include the action of some part of the Levi; in the case of $\mathrm{U}(2,1)$, concretely one needs to consider $R = S \ltimes N$ where $S \subset M$ is the stabilizer of the center of N, so $S = \mathrm{U}(1) \times \mathrm{U}(1) \subset \mathrm{U}(1,1) \times \mathrm{U}(1)$. It is a theorem of Gelbart–Rogawski that $L^2([R])$ has multiplicity one property (as an R-representation). This fits in a more general framework of generalized Whittaker models where one expects similar finite multiplicity / multiplicity one property. In particular, the Fourier–Jacobi coefficient f_ψ decomposes further into

$$f_\psi = \sum_{\nu \text{ character of } S} f_{\psi,\nu},$$

where $f_{\psi,\nu}$ is now supported by a multiplicity one model. In the unfolding of the GGP integral against a cuspidal g, it is this "refined Fourier–Jacobi coefficient" that gets paired with a Whittaker coefficient of g. This new viewpoint of exploiting degenerate Whittaker models has potentially many applications in the theory of automorphic forms, which is a sentiment we try to convey in Part III.

Chapter 2

Preliminaries

We collect some preliminary materials that do not seem to fit into the middle of the thesis. The materials of this chapter are not original, and the exposition of the materials is borrowed from many other sources, including [Su], [Mi], [Kn]. Note that the Parts of this thesis will have their own specialized preliminaries.

2.1 Shimura varieties

Let G be a connected reductive algebraic group over $\mathbb{Q}$. To not get into trouble with the issue of quotient-by-$\mathbb{R}^\times$, we assume that the maximal $\mathbb{R}$-split torus of Z_G, the center of G, is the $\mathbb{R}$-base change of the maximal $\mathbb{Q}$-split torus of Z_G.

Definition 2.1.1. A Shimura datum is a pair (G, X) where X is a $G(\mathbb{R})$-conjugacy class of homomorphisms $h : \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\, \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{R}}$ such that the following conditions hold.

1. For all $h \in X$, it induces a Hodge structure on $\mathfrak{g} = \mathrm{Lie}\, G_{\mathbb{R}}$, with the Hodge decompositoin being
$$\mathfrak{g} = \mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{0,0} \oplus \mathfrak{g}^{1,-1}.$$

2. For all $h \in X$, $\mathrm{Ad}(h(i))$ is a Cartan involution on $G_{\mathbb{R}}^{\mathrm{ad}}$.

3. There is no $\mathbb{Q}$-factor of G^{ad} whose projection of h is trivial.

The set of conjugacy classes X is topologized using $X \cong G(\mathbb{R})/K_\infty$, for the stabilizer K_∞ of $h \in X$, and it is known that, under the conditions of a Shimura datum, X is a finite union of Hermitian symmetric domains. More precisely, one can identify $\operatorname{Lie} K_\infty = \mathfrak{g}^{0,0} =: \mathfrak{k}$, and there is a parabolic subgroup $P \subset G(\mathbb{C})$ whose Lie algebra is identified with $\mathfrak{p} = \mathfrak{g}^{0,0} \oplus \mathfrak{g}^{1,-1}$. Then, $X = G(\mathbb{R})/K_\infty$ is an open $G(\mathbb{R})$-orbit inside the flag variety $\check{X} = G(\mathbb{C})/P$, which we call the *compact dual* of X. The natural embedding $\beta : X \hookrightarrow \check{X}$ is called the *Borel embedding*.

Definition 2.1.2. Given a Shimura datum (G, X), for a compact open subgroup $\Gamma \subset G(\mathbb{A}_f)$, let

$$\mathrm{Sh}_\Gamma(G, X) = G(\mathbb{A})\backslash X \times G(\mathbb{A}_f)/\Gamma.$$

To avoid technicalities, we assume that Γ is neat (cf. [Har2, §1.1]): for each $\gamma \in \Gamma$, there is a prime p and a faithful finite-dimensional $\mathbb{Q}_p$-rational representation $\rho : G(\mathbb{Q}_p) \hookrightarrow \mathrm{GL}_n(\mathbb{Q}_p)$ such that the subgroup of $\overline{\mathbb{Q}}_p^\times$ generated by the eigenvalues of $\rho(\gamma_p)^{\mathrm{ss}}$ contains no roots of unity. If Γ is neat, then $\mathrm{Sh}_\Gamma(G, X)$ naturally has a structure of a complex manifold, which we call a Shimura manifold.

Theorem 2.1.3 (Shimura)**.** *The Shimura manifold* $\mathrm{Sh}_\Gamma(G, X)$ *is the analytification of a complex algebraic quasiprojective variety. Furthermore, it has a canonical model* Y_Γ *defined over the* reflex field*,* $E(G, X)$*, which is defined as the field of definition of the conjugacy class of the Hodge cocharacter* $h|_{\mathbb{C}^\times} : \mathbb{C}^\times \to G_{\mathbb{C}}$*. We call* Y_Γ *a* Shimura variety*.*

Remark 2.1.4. The reflex field is most notably a number field, and the definition of canonical model utilizes how the Galois group acts on the CM points. A less known fact is that the reflex field is defined as a subfield of $\mathbb{C}$. This implies that the machinery of Shimura varieties also spits out a *preferred* archimedean place of the reflex field.

There are naturally defined vector bundles, called *automorphic vector bundles*, over the Shimura variety Y_Γ defined as follows. Given an algebraic representation V of P, one defines a holomorphic vector bundle $[V]$ on $\check{X} = G(\mathbb{C})/P(\mathbb{C})$ as follows:

$$\mathcal{E}_{[V]} := G(\mathbb{C}) \times^{P(\mathbb{C})} V \to \check{X}.$$

By [Har1], it is known that the restriction $[V]|_X$ defines a complex algebraic vector bundle over $\mathrm{Sh}_\Gamma(G, X)$ that also descends to an algebraic vector bundle over Y_Γ.

To prove that $\mathrm{Sh}_\Gamma(G, X)$ is a complex algebraic quasiprojective variety, one needs a compactification of $\mathrm{Sh}_\Gamma(G, X)$ into a compact complex manifold. This is given by the *minimal compactification*, which is often decorated with a superscript $^{\min}$. It is minimal in the sense that, for any compactification $\mathrm{Sh}_\Gamma(G, X) \hookrightarrow Z$ with $Z - \mathrm{Sh}_\Gamma(G, X)$ a normal crossings divisor, there is a unique map $\mathrm{Sh}_\Gamma(G, X)^{\min} \to Z$ that is compatible with the inclusion of $\mathrm{Sh}_\Gamma(G, X)$. On the other hand, $\mathrm{Sh}_\Gamma(G, X)$ is in general a highly singular algebraic variety.

At the other extreme, there is a smooth compactification of $\mathrm{Sh}_\Gamma(G, X)$ whose boundary is a normal crossings divisor, called a *toroidal compactification*. We decorate it with a superscript ^{tor}. Its construction is highly dependent on choice; on the other hand, such choice does not matter very much, especially in terms of its cohomology, that the isomorphism class of its cohomology as a Hecke module is independent of choice. Moreover, a toroidal compactification also has a canonical model over the reflex field $E(G, X)$. There are various ways of extending an automorphic vector bundle $[V]$ to a toroidal compactification, but there are two prominent ways, the *canonical extension* $[V]^{\mathrm{can}}$ and the *subcanonical extension* $[V]^{\mathrm{sub}}$. The canonical extension $[V]^{\mathrm{can}}$ is complex-analytically defined as the sheaf of sections of at most logarithmic growth at infinity, which is precisely the canonical extension in the sense of Deligne in [De]. On the other hand, $[V]^{\mathrm{sub}} = [V]^{\mathrm{can}}(-D)$, where D is the re-

duced boundary divisor. These vector bundles also have canonical models over the toroidal compactifciation X_Γ^{tor}. For more refined properties regarding the two types of compactifications of Shimura varieties, see §5.1.

2.2 Coherent cohomology of Shimura varieties

We first clarify what we mean by the term *coherent cohomology*. Depending on the context, it can mean two different things. In the complex analytic setting, this means that we are taking the cohomology of a coherent sheaf over a complex manifold with respect to the complex analytic topology. In the algebraic setting, this means that we are taking the Zariski cohomology of a coherent sheaf over an algebraic variety. If the underlying manifold is a compact complex manifold, then by Serre's GAGA, the two coherent cohomologies coincide. In general, however, the two need not coincide; the algebraic coherent cohomology $H^0(\mathbb{A}^1_{\mathbb{C}}, \mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}}) \cong \mathbb{C}[t]$ is the space of polynomials with complex coefficients, while the analytic coherent cohomology $H^0((\mathbb{A}^1_{\mathbb{C}})^{\mathrm{an}}, (\mathcal{O}_{\mathbb{A}^1_{\mathbb{C}}})^{\mathrm{an}})$ is the space of holomorphic functions on $\mathbb{C}$, which is much larger than the space of polynomials.

The analytic coherent cohomology can be computed using the Hodge theory and the $\overline{\partial}$-Poincaré lemma. This is called the *Dolbeault cohomology*.

Theorem 2.2.1. *Let X be a complex manifold and E be a holomorphic vector bundle over X. Let $\mathcal{A}^{0,i}(X,E)$ be the sheaf of smooth (=real analytic) $(0,i)$-forms with coefficients in E. Then, the* Dolbeault complex

$$0 \to E \to \mathcal{A}^{0,0}(X,E) \xrightarrow{\overline{\partial}} \mathcal{A}^{0,1}(X,E) \xrightarrow{\overline{\partial}} \mathcal{A}^{0,2}(X,E) \xrightarrow{\overline{\partial}} \cdots,$$

is a fine resolution of E. Therefore, the (analytic) coherent cohomology $H^i(X,E)$ can

be computed using the complex

$$\cdots \xrightarrow{\overline{\partial}} \Gamma(X, \mathcal{A}^{0,i-1}(X,E)) \xrightarrow{\overline{\partial}} \Gamma(X, \mathcal{A}^{0,i}(X,E)) \xrightarrow{\overline{\partial}} \Gamma(X, \mathcal{A}^{0,i+1}(X,E)) \xrightarrow{\overline{\partial}} \cdots .$$

If we apply this construction to the computation of coherent cohomology of an automorphic vector bundle over a Shimura variety, we have a simple description in terms of automorphic forms (and their relatives). The idea is that the sheaf of holomorphic differential i-forms over $\mathrm{Sh}_\Gamma(G,X)$ is an automorphic vector bundle corresponding to the representation $\wedge^p(\mathfrak{g}/\mathfrak{p})^*$.

For simplicity, we further assume that the maximal $\mathbb{Q}$-split torus of Z_G is trivial. Then one can naturally identify

$$\Gamma(\mathrm{Sh}_\Gamma(G,X), \mathcal{A}^{0,i}(\mathrm{Sh}_\Gamma(G,X),[V])) \cong \mathrm{Hom}_{K_\infty}(\wedge^i(\mathfrak{p}/\mathfrak{k}), C^\infty_{K_\infty\text{-finite}}(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma)\otimes V).$$

The right hand side is the i-th term of the Chevalley–Eilenberg complex for computing the relative Lie algebra cohomology with respect to $(\mathfrak{p}, K_\infty)$ of $C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma)\otimes V$, so we see that

$$H^i(\mathrm{Sh}_\Gamma(G,X),[V]) \cong H^i(\mathfrak{p}, K_\infty; C^\infty_{K_\infty\text{-finite}}(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma)\otimes V).$$

The downside of this computation is that this computes analytic coherent cohomology, and, as we have seen above, the analytic coherent cohomology of a noncompact complex manifold is extremely huge so that it has almost nothing to do with algebra. To compute the algebraic coherent cohomology, we would have to deal with the coherent cohomology of compactifications of Shimura varieties.

We draw a motivation from the analogue of this computation in the setting of singular cohomology of locally symmetric spaces.

Theorem 2.2.2 (Matsushima's formula)**.** *For a semisimple reductive $\mathbb{Q}$-group G and*

a neat open compact subgroup $\Gamma \subset G(\mathbb{A}_f)$*, let*

$$S_\Gamma := G(\mathbb{Q})\backslash G(\mathbb{A})/K_\infty\Gamma,$$

be the associated Riemannian locally symmetric space, where K_∞ *is a maximal compact subgroup of* $G(\mathbb{R})$*. For an algebraic representation* V *of* G*, let* $[V]$ *be the local system over* S_Γ *defined by*

$$G(\mathbb{Q})\backslash(V \times G(\mathbb{A}))/K_\infty\Gamma \to S_\Gamma.$$

Then,

$$H^i(S_\Gamma, [V]) \cong H^i(\mathfrak{g}, K_\infty; C^\infty_{K_\infty\text{-finite}}(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma) \otimes V).$$

Its connection to automorphic forms comes from the following regularization theorem.

Theorem 2.2.3 (Franke, [Fra])**.** *With the notation as above, we have*

$$H^i(S_\Gamma, [V]) \cong H^i(\mathfrak{g}, K_\infty; \mathcal{A}([G])^\Gamma \otimes V),$$

where $\mathcal{A}([G])$ *is the space of automorphic forms on* $[G] = G(\mathbb{Q})\backslash G(\mathbb{A})$*, namely the space of* K_∞*-finite* $Z(\mathfrak{g})$*-finite function of moderate growth.*

There is a similar regularization theorem for the coherent cohomology of Shimura varieties.

Theorem 2.2.4 (Su, [Su])**.** *Suppose that* G *is a connected reductive group over* $\mathbb{Q}$ *where* Z_G *has no nontrivial* $\mathbb{R}$*-split torus and* $X \cong G/K_\infty$ *is a Hermitian symmetric domain. For an algebraic representation* V *of* P*, we have*

$$H^i(\mathrm{Sh}_\Gamma(G, X)^{\mathrm{tor}}, [V]^{\mathrm{can}}) \cong H^i(\mathfrak{p}, K_\infty; \mathcal{A}([G])^\Gamma \otimes V).$$

The proof of these two theorems are extremely delicate. For the computation of the algebraic coherent cohomology of $\mathrm{Sh}_\Gamma(G, X)$, see Part I.

2.3 Discrete series

Discrete series representations are very special types of irreducible admissible representations of reductive Lie groups which are regarded as building blocks of all irreducible admissible representations. Discrete series representations are often easier to deal with, as we know much better about them.

Definition 2.3.1. Let G be a Lie group in Harish-Chandra's class (see [Kn]). An irreducible unitary representation π of G is a *discrete series representation* if π if π is equivalent with a direct summand of the right-regular representation $L^2(G)$.

Theorem 2.3.2 (Harish-Chandra)**.** *A Lie group G possesses a discrete series representation if and only if* $\operatorname{rank} G = \operatorname{rank} K$*, where K is a maximal compact subgroup of G. Equivalently, there is a discrete series representation for G if and only if G has a compact Cartan subgroup T.*

For each nonsingular weight $\lambda \in (i\mathfrak{t})'$, if $\lambda + \rho_G$ is analytically integral, then there exists a discrete series representation π_λ of G whose infinitesimal character is given by χ_λ, the character of $Z(\mathfrak{g})$ corresponding to λ via the Harish-Chandra isomorphism. Here, ρ_G is the half-sum of positive roots. Furthermore, any discrete series representation is equivalent to π_λ for some λ as above.

In the study of admissible representations of G, there are more algebraic approaches. Namely, for an admissible representation (π, V) of G, the space of smooth K-finite vectors of V forms a $(\mathfrak{g}, K)$-module. Two admissible representations are *infinitesimally equivalent* if the corresponding $(\mathfrak{g}, K)$-modules are equivalent. A *K-type* of a $(\mathfrak{g}, K)$-module is an irreducible sub-K-representation of it, and a *minimal K-type*

of a $(\mathfrak{g}, K)$-module is a K-type whose highest weight is minimal among the highest weights of the K-types of the $(\mathfrak{g}, K)$-module.

Oftentimes the information on minimal K-types is strong enough to characterize an admissible representation. This is the case for discrete series.

Theorem 2.3.3 (Harish-Chandra, Vogan)**.** *For a discrete series representation π_λ, there is a unique minimal K-type, whose highest weight is $\Lambda = \lambda + \rho_G - 2\rho_K$, where ρ_K is the half-sum of positive compact roots. Furthermore, any K-type of π_λ has the highest weight of form*

$$\Lambda + \sum_{\alpha \in \Delta^+} n_\alpha \alpha, \qquad n_\alpha \geq 0,$$

where Δ^+ is the set of positive simple roots with respect to λ, i.e. simple roots α such that $\langle \lambda, \alpha \rangle > 0$.

We call λ the *Harish-Chandra parameter* of π_λ, and Λ the *Blattner parameter* of π_λ.

There is an irreducible admissible representation of G that nearly misses being a discrete series representation.

Theorem 2.3.4. *For a weight $\lambda \in (i\mathfrak{t})'$ with $\lambda + \rho_G$ analytically integral and a positive system Δ^+ that makes λ dominant, there is a unique irreducible admissible $(\mathfrak{g}, K)$-module, up to infinitesimal equivalence, $\pi(\lambda, \Delta^+)$, which satisfies the following.*

1. *It is tempered.*

2. *It has a unique minimal K-type, and its highest weight is given by the formula*

$$\Lambda = \lambda + \rho_G - 2\rho_K.$$

3. Any of its K-type has the highest weight of form

$$\Lambda + \sum_{\alpha \in \Delta^+} n_\alpha \alpha, \qquad n_\alpha \geq 0.$$

Note that there is a unique choice of Δ^+ if λ is nonsingular, and $\pi(\lambda, \Delta^+)$ is just the discrete series representation π_λ. On the other hand, if λ is singular, we call $\pi(\lambda, \Delta^+)$ a *limit of discrete series representation.*

Depending on λ, there are two types of limit of discrete series representations. If λ is orthogonal to a compact root, we call $\pi(\lambda, \Delta^+)$ a *degenerate limit of discrete series.* If not, we call it a *nondegenerate limit of discrete series.* These are named as such because $\pi(\lambda, \Delta^+)$ has global character 0 if and only if it is a degenerate limit of discrete series. Furthermore, a discrete series or a nondegenerate limit of discrete series representation has a nontrivial $(\mathfrak{g}, K)$- or $(\mathfrak{p}, K)$-cohomology, whereas a degenerate limit of discrete series representation always has a vanishing $(\mathfrak{g}, K)$- and $(\mathfrak{p}, K)$-cohomology (see [So]). Thus, there is so far no cohomological way to detect degenerate limits of discrete series; however, see [Car] and Part II.

Part I

Local cohomology and harmonic Maass forms

Chapter 3

Introduction

The theory of harmonic Maass forms was initiated by Ramanujan and its theoretical foundation was set by Zwegers in his thesis. Although harmonic Maass forms have been shown to be closely related to arithmetic, they have not been connected to the realm of arithmetic geometry or automorphic forms. Indeed, it was spectulated that harmonic Maass forms are related to a still-speculative concept of "extensions of automorphic representations" (e.g. [BK]), which does not exist in the current theory of automorphic representations.

In this part, we recast the theory of harmonic Maass forms in terms of complex/arithmetic geometry of modular curves. In particular, we give a purely algebro-geometric interpretation of the *principal parts* of harmonic Maass forms using the algebraic local cohomology of modular curves at cusps[1]. Such interpretation would make sense over any base, for example $\mathbb{Z}$ or $\mathbb{F}_q$. On the other hand, harmonic Maass forms are expressed using the complex geometry of modular curves, by using Hodge theory and the relative Dolbeault cohomology of modular curves. We believe that this re-interpretation is new, and this would help us to set up a correct picture of the theory, amenable to further development.

[1]This is in accordance with the usual analogy between harmonic Maass forms and overconvergent modular forms. Indeed, overconvergent modular forms are recently also realized to be related with local cohomology at a cusp (see [BoPi], [Pa]).

As an example, we develop a generalization of harmonic Maass forms in the context of Hilbert modular forms. Such generalization, or even a generalization of weakly holomorphic modular forms (namely, modular holomorphic functions on the upper half plane of at most linear exponential growth), has not been realized, due to the *Koecher's principle*. Namely, any singularity of a holomorphic function of codimension ≥ 2 can always be filled, and Hilbert modular varieties can be compactified so that the boundary is a finite collection of points, which is of codimension ≥ 2 as soon as one escapes the case of modular curves.

Under the coherent-cohomological interpretation of harmonic Maass forms developed in this paper, we realize that one can avoid Koecher's principle using higher coherent cohomology of Hilbert modular varieties. Namely, it is realized in [Lan2, §9] that the analogue of Koecher's principle for coherent cohomology of Shimura varieties *must fail* in a certain cohomological degree, which is usually not H^0. For example, for a Hilbert modular variety associated to a totally real field F of degree d, the failure of Koecher's principle occurs in H^{d-1} (and H^d, although the statement is trivial in that case). We define (principal parts of) *weakly holomorphic Hilbert modular forms* and *harmonic Hilbert Maass forms* using the coherent cohomology of an open Hilbert modular variety and its toroidal compactification.

To have a hold on them, we compute the space of principal parts of harmonic Hilbert Maass forms using the analogue of Fourier expansions for higher coherent cohomology classes. The computation exploits the geometry of toroidal compactification. In contrast to the q-expansion of holomorphic Hilbert modular forms, we see the following exotic features.

1. This "higher" q-expansion is a U_F-coinvariance class of Laurent polynomials with totally negative exponents, where $U_F = \mathcal{O}_F^\times$. This is in contrast with the fact that the q-expansion of holomorphic Hilbert modular forms has only totally positive exponents.

2. The higher q-expansion of those coming from cuspidal holomorphic Hilbert modular forms is zero. Thus, the higher q-expansion map exactly detects the principal parts of weakly holomorphic Hilbert modular forms.

In [Br2], an attempt to overcome Koecher's principle for Hilbert modular forms was made. As harmonic Maass forms are spanned by non-holomorphic Poincaré series, built out of "bad" Whittaker function, *op. cit.* uses bad Whittaker functions, named *harmonic Whittaker forms*, in place of harmonic Hilbert Maass forms. In particular, the Borcherds lift of harmonic Whittaker forms is constructed, and its relation to Kudla–Millson lift is studied as in [BF].

We will also observe that the standard harmonic Whittaker forms are harmonic Hilbert Maass forms with q-expansion q^ν for some $\nu < 0$. In this way, the Borcherds lift of [Br2] can be regarded as lifting harmonic Hilbert Maass forms in our sense.

3.1 Related problems

As the primary objective of this manuscript is to suggest a correct viewpoint, various directions can stem out of it.

1. Clearly, one can try to generalize the concept of harmonic Maass forms to other higher-dimensional Shimura varieties. The analogues of Whittaker functions of exponential growth for Sp_{2n} is developed in [BFK], which can be useful in developing the theory for Siegel modular varieties.

2. One can explicitly compute the Dolbeault representative of harmonic Hilbert Maass forms. This might suggest how to directly connect Whittaker functions with the harmonic Hilbert Maass forms, or to directly construct the Borcherds lift of harmonic Hilbert Maass forms using their Dolbeault representatives.

3. It would be interesting to investigate the principal parts of harmonic Maass

forms over more general base ring, say $\mathbb{Z}$ or $\mathbb{F}_q$. Such consideration can make the analogy between harmonic Maass forms and overconvergent modular forms more rigorous.

4. It is still unclear how the full harmonic Maass forms are related to arithmetic geometry. For weakly holomorphic modular forms, one can indeed just use the cohomology of open modular curve. On the other hand, the local cohomology long exact sequence starts from the cohomology of closed modular curve, and it is not clear how to detach this term from others.

3.2 Notation

The upper half plane $\mathbb{H}$ is defined as $\{\tau = u + iv \in \mathbb{C} \mid v > 0\}$. Every algebraic variety is defined over $\mathbb{C}$ unless otherwise noted. Any complex analytic manifold will have an in its superscript, to distinguish it from its algebraization. For example, $H^0(\mathbb{A}^1, \mathcal{O}) = \mathbb{C}[z]$ is the algebraic coherent cohomology, whereas $H^0(\mathbb{A}^{1,\mathrm{an}}, \mathcal{O})$ is the space of all holomorphic functions on $\mathbb{C}$. Given an automorphic vector bundle $\mathcal{E}$ over an open Shimura variety M, $\mathcal{E}^{\mathrm{can}}$, $\mathcal{E}^{\mathrm{sub}}$ are the canonical and subcanonical extensions of $\mathcal{E}$ over a toroidal compactification M^{tor} (where we do not specify the choice of a datum if possible), respectively, and $\mathcal{E}$ will also denote its natural extension to the minimal compactification M^{min}. The boundary divisors $M^{\mathrm{tor}} - M$ and $M^{\mathrm{min}} - M$ are all denoted D if there is no confusion. Over M^{tor}, we also tend to use the same letter $\mathcal{E}$ for its canonical extension and $\mathcal{E}(-D)$ for its subcanonical extension.

Chapter 4

Cohomological interpretation of harmonic Maass forms

We first reinterpret the classical theory of harmonic Maass forms in terms of coherent cohomology and Dolbeault cohomology.

4.1 Harmonic Maass forms

Definition 4.1.1. Let $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup. A *harmonic Maass form* of level Γ and weight k is a real analytic function $f : \mathbb{H} \to \mathbb{C}$ such that

1. for all $\gamma \in \Gamma$, $f(\gamma\tau) = j(\gamma, \tau)^k f(\tau)$, where $j(\gamma, \tau)$ is the usual factor of automorphy,

2. f is at most of exponential growth towards $+i\infty$,

3. and $\Delta_k f = 0$, where Δ_k is the hyperbolic Laplacian of weight k,

$$\Delta_k = -v^2 \left(\frac{\partial^2}{\partial u^2} + \frac{\partial^2}{\partial v^2} \right) + ikv \left(\frac{\partial}{\partial u} + i \frac{\partial}{\partial v} \right).$$

The space of such forms is denoted $H_k(\Gamma)$.

Since $\Delta_k = -R_{k-2}L_k$, where R_k and L_k are the raising and lowering operators, respectively, it follows that, for a harmonic Maass form f, $L_k f$ is closely related to a (weakly) anti-holomorphic modular form.

Definition 4.1.2. For $f \in H_k(\Gamma)$, let

$$\xi_k(f)(\tau) := v^{k-2}\overline{L_k f(\tau)}.$$

This is an antilinear map $\xi_k : H_k \to M^!_{2-k}$. We let H_k^+ be $\xi_k^{-1}(S_{2-k})$.

Under this definition, there is a short exact sequence

$$0 \to M_k^! \to H_k^+ \xrightarrow{\xi_k} S_{2-k} \to 0.$$

In terms of coherent cohomology of open and closed modular curves $Y(\Gamma)$ and $X(\Gamma)$, we can express two terms of the three in above:

$$0 \to H^0(Y(\Gamma), \omega^k) \to H_k^+ \to H^1(X(\Gamma), \omega^k) \to 0.$$

Here, the cohomology groups are coherent cohomology groups of complex algebraic varieties. A variant of this would be:

$$0 \to H^0(X(\Gamma), \omega^k) \to H^0(Y(\Gamma), \omega^k) \to \frac{H_k^+}{M_k} \to H^1(X(\Gamma), \omega^k) \to 0.$$

There is a very natural cohomology space that should lie in the place of $\frac{H_k^+}{M_k}$, the *local cohomology*. We can think of a local cohomology exact sequence in algebraic geometry:

$$0 \to H^0(X(\Gamma), \omega^k) \to H^0(Y(\Gamma), \omega^k) \to H^1_D(X(\Gamma), \omega^k) \to H^1(X(\Gamma), \omega^k) \to 0,$$

where $D = X(\Gamma) - Y(\Gamma)$; the first term is exact trivially, and the last term is exact as $Y(\Gamma)$ is affine.

Therefore, we can suspect that $H^1_D(X(\Gamma), \omega^k) \cong \frac{H_k^+}{M_k}$ in a natural way.

4.2 Local cohomology as relative Dolbeault cohomology

As the definition of harmonic Maass forms is analytic in its nature, we would like a way to compute $H^1_D(X(\Gamma), \omega^k)$ using the analytic geometry of $X(\Gamma)$. This is indeed made possible via the notion of *relative Dolbeault cohomology.*

Definition 4.2.1 ([Suw, Definition 2.14])**.** Let $i : Y \hookrightarrow X$ be an open embedding of topological spaces, and let $\mathcal{F}^\bullet$ be a cohomologically graded complex of sheaves on X. Then, the *relative sheaf cohomology* $H_i^*(\mathcal{F}^\bullet)$ is the cohomology of the complex $\widetilde{\mathcal{F}}^\bullet$, where

$$\widetilde{\mathcal{F}}^q = \mathcal{F}^q(X) \oplus \mathcal{F}^{q-1}(Y), \quad d(a, b) = (da, i^*a - db).$$

If X, Y are complex manifolds and $\mathcal{F}^\bullet$ is the Dolbeault complex of a coherent sheaf $\mathcal{F}$ on X, then $H^*_{Y-X}(X, \mathcal{F}) := H_i^*(\mathcal{F}^\bullet)$ is also called the *relative Dolbeault cohomology.*

Proposition 4.2.2 ([Suw, Theorem 2.23])**.** *There is a canonical isomorphism between the relative Dolbeault cohomology and the analytic local cohomology, respecting the long exact sequences.*

This is not difficult, as $\widetilde{\mathcal{F}}^\bullet$ is just constructed as the *mapping cone* of two Dolbeault complexes. We cannot just naively use this version of relative Dolbeault cohomology as it computes *analytic* local cohomology. We therefore need to recast algebraic coherent cohomology in terms of analytic coherent cohomology. The key is that, if we denote $i : Y(\Gamma) \hookrightarrow X(\Gamma)$ be the natural open embedding, then $\mathrm{R}^a i_* \omega^k = 0$ for

$a > 0$, and $i_*\omega^k = \varinjlim_{n\to\infty} \omega^k(nD)$. Thus, by Leray spectral sequence and GAGA over $X(\Gamma)$,

$$H^i(Y(\Gamma), \omega^k) = \varinjlim_{n\to\infty} H^i(X(\Gamma), \omega^k(nD)) = \varinjlim_{n\to\infty} H^i(X(\Gamma), \omega^k(nD))^{\mathrm{an}}.$$

Thus, by the mapping cone construction as above, $H^1_D(X(\Gamma), \omega^k)$ is H^1 of the complex

$$0 \to \mathcal{A}^{0,0}(X(\Gamma), \omega^k) \to \mathcal{A}^{0,1}(X(\Gamma), \omega^k) \oplus \varinjlim_{n\geq 0} \mathcal{A}^{0,0}(X(\Gamma), \omega^k(nD)) \to \varinjlim_{n\geq 0} \mathcal{A}^{0,1}(X(\Gamma), \omega^k(nD)) \to 0.$$

Since $\varinjlim_{n\geq 0} \mathcal{A}^{0,0}(X(\Gamma), \omega^k(nD))$ is precisely consisted of real analytic functions $\mathbb{H} \to \mathbb{C}$ with modular property and polar growth property, we can denote

$$\mathcal{A}^{0,0}(Y(\Gamma), \omega^k)_{\mathrm{polar}} := \varinjlim_{n\geq 0} \mathcal{A}^{0,0}(X(\Gamma), \omega^k(nD)),$$

namely the space of those of at most polar growth at boundary. Then,

$$H^1_D(X(\Gamma), \omega^k) = \frac{\{(\omega, g) \in \mathcal{A}^{0,1}(X(\Gamma), \omega^k) \oplus \mathcal{A}^{0,0}(Y(\Gamma), \omega^k)_{\mathrm{polar}} : \omega|_{Y(\Gamma)} = \overline{\partial} g\}}{\{(\overline{\partial} f, f|_{Y(\Gamma)}) : f \in \mathcal{A}^{0,0}(X(\Gamma), \omega^k)\}}.$$

Proposition 4.2.3. *There is a natural isomorphism*

$$j : \frac{H_k^+}{M_k} \xrightarrow{\sim} H^1_D(X(\Gamma), \omega^k),$$

given by

$$j(f) = (\overline{\partial} f, f),$$

for $f \in H_k^+$.

Proof. Certainly the above map defines a map $H_k^+ \to H^1_D(X(\Gamma), \omega^k)$, because $\overline{\partial} f = \overline{\xi_k(f)} d\overline{z}$, and this, a priori only defined over $Y(\Gamma)$, extends to $X(\Gamma)$ because $\xi_k(f)$ can be extended. The kernel is precisely consisted of holomorphic f's, namely weakly

holomorphic f's. This constructs an injection,

$$j : \frac{H_k^+}{M_k} \to H_D^1(X(\Gamma), \omega^k).$$

Both sit in a diagram of short exact sequences where the first and the third terms are identical, so j is automatically a surjection by the five-lemma. $\square$

Thus, the space of principal parts of harmonic Maass forms fit perfectly into algebraic geometry of modular curves. In particular, it can be given a natural $\mathbb{Q}$-structure.

Remark 4.2.4. It is easy to see that the natural $\mathbb{Q}$-structure is consisted of harmonic Maass forms whose holomorphic parts (a *mock modular form*) have $\mathbb{Q}$-rational principal parts.

The space of harmonic Maass forms itself, H_k^+, can be also seen as follows.

Proposition 4.2.5. *Let $(\mathcal{A}^{0,*}, \overline{\partial})$ denote the Dolbeault complex of a complex manifold, and let $\mathscr{H}^{p,q}$ be the space of (p,q)-harmonic differential forms. Then H_k^+ is naturally identified with*

$$H_k^+ = \overline{\partial}^{-1}\left(\mathscr{H}^{0,1}(X(\Gamma), \omega^k)\right) \subset \mathcal{A}^{0,0}(Y(\Gamma), \omega^k)_{\text{polar}},$$

and the exact sequence $0 \to M_k^! \to H_k^+ \to S_{2-k} \to 0$ can be naturally identified with

$$0 \to H^0(Y(\Gamma), \omega^k) \to H_k^+ \to H^1(X(\Gamma), \omega^k) \to 0.$$

Proof. Because of Hodge theory, the quotient map $q : \mathcal{A}^{0,1}(X(\Gamma), \omega^k) \to H^1(X(\Gamma), \omega^k)$ from Dolbeault cohomology gives rise to an isomorphism

$$q|_{\mathscr{H}^{0,1}(X(\Gamma),\omega^k)} : \mathscr{H}^{0,1}(X(\Gamma), \omega^k) \xrightarrow{\sim} H^1(X(\Gamma), \omega^k).$$

Thus, $H^1_D(X(\Gamma),\omega^k) = \frac{H_k^+}{M_k}$ can be rewritten as

$$H^1_D(X(\Gamma),\omega^k) = \left\{(\omega, g) \in \mathscr{H}^{0,1}(X(\Gamma),\omega^k) \oplus \frac{\mathcal{A}^{0,0}(Y(\Gamma),\omega^k)_{\text{polar}}}{\omega^k(Y(\Gamma))} : \omega|_{Y(\Gamma)} = \overline{\partial} g\right\},$$

and by the same reason, H_k^+ can be rewritten as

$$H_k^+ = \{(\omega, g) \in \mathscr{H}^{0,1}(X(\Gamma),\omega^k) \oplus \mathcal{A}^{0,0}(Y(\Gamma),\omega^k)_{\text{polar}} : \omega|_{Y(\Gamma)} = \overline{\partial} g\}.$$

Since ω is uniquely determined by g (it is the extension of $\overline{\partial} g$, and it is unique if it exists), one can remove ω and simply identify with the preimage of $\overline{\partial}$. $\square$

Thus, in some sense, weakly holomorphic modular forms and harmonic Maass forms exist *precisely because of the discrepancy between the coherent cohomology of open and closed modular curves.*

It is also notable that the space of principal parts of harmonic Maass forms is something arising from local picture around cusps. Namely, harmonic Maass forms are often regarded analogous to overconvergent modular forms, and the space of overconvergent modular forms is also the "stalk at ∞" (see [Pa, §5.2.4]).

Chapter 5

Beyond Koecher's principle: weakly holomorphic Hilbert modular forms

5.1 Higher Koecher's princple, d'après Lan

Koecher's principle prevents naively generalizing the definition of weakly holomorphic modular forms and harmonic Maass forms to higher rank cases. On the other hand, the slogan can be extended to other Shimura varieties, thanks to the following work of Lan on *higher Koecher's principle*:

Theorem 5.1.1 ([Lan2, Theorem 2.5])**.** *Let M be a Shimura variety, M^{min} be its minimal compactification, and M^{tor} be a toroidal compactification of M, which sit in the diagram*

$$\begin{tikzcd} & M^{\mathrm{tor}} \arrow[d, "\pi"] \\ M \arrow[ur, hook, "j^{\mathrm{tor}}"] \arrow[r, hook, "j^{\mathrm{min}}"'] & M^{\mathrm{min}} \end{tikzcd}$$

Let $\mathcal{E}$ be an automorphic vector bundle, and let $\mathcal{E}^{\mathrm{can}}$ be its canonical extension over

M^{tor}. *Let* $c_M = \mathrm{codim}(M^{\min} - M, M^{\min})$.

1. *The natural map* $\mathrm{R}^i\pi_*(\mathcal{E}^{\mathrm{can}}) \to \mathrm{R}^i j_*^{\min}\mathcal{E}$ *induced by* j^{tor} *is an isomorphism for* $i < c_M - 1$ *and is an injection for* $i = c_M - 1$.

2. *The natural restriction map* $H^i(M^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \to H^i(M, \mathcal{E})$ *is an isomorphism for* $i < c_M - 1$, *and is an injection but not an isomorphism for* $i = c_M - 1$.

We record the following facts used in the course of proof of Theorem 5.1.1.

Proposition 5.1.2. *Retaining the notations of Theorem 5.1.1, we have the following.*

1. *([Lan2, (3.1)])* $j_*^{\mathrm{tor}}\mathcal{E} = \varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)$ *and* $\mathrm{R}^a j_*^{\mathrm{tor}}\mathcal{E} = 0$ *for* $a > 0$, *where* $D = M^{\mathrm{tor}} - M$.

2. *([Lan2, (3.2)])* $\mathrm{R}^a j_*^{\min}\mathcal{E} \cong \varinjlim_{n\geq 0} \mathrm{R}^a\pi_*(\mathcal{E}^{\mathrm{can}}(nD))$ *for* $a \geq 0$.

3. *([Lan2, Theorem 3.9])* $\mathrm{R}^a\pi_*\mathcal{E}^{\mathrm{sub}}(-nD) = 0$ *for any* $n \geq 0$.

Proof. The only difference between this statement and that in [Lan2] is that D' is used in place of D in *op. cit.*, but one can obtain this statement as there is $\ell \in \mathbb{N}$ such that $D \leq D' \leq \ell D$. □

Let us restrict to the case of Hilbert modular forms. Let F be a totally real field of degree $d > 1$, and let $\Gamma \leq \mathrm{SL}_2(\mathcal{O}_F)$ be a congruence subgroup. Then, the Hilbert modular variety $M = Y(\Gamma)$ has $c_M = d-1$. Thus, $H^i(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})$ and $H^i(Y(\Gamma), \mathcal{E})$ can only differ in degrees $i = d-1$ and d:

$$H^i(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \xrightarrow{\sim} H^i(Y(\Gamma), \mathcal{E}), \quad i \leq d-2,$$

$$H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \subsetneq H^i(Y(\Gamma), \mathcal{E}),$$

$$H^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \twoheadrightarrow H^d(Y(\Gamma), \mathcal{E}) = 0.$$

Using the facts we have stated so far, we obtain the following

Proposition 5.1.3.

1. *We have* $H^i_D(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) = 0$ *for* $0 \leq d-1$, *and an exact sequence*

$$0 \to H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \to H^{d-1}(Y(\Gamma), \mathcal{E}) \to H^d_D(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \to H^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \to 0.$$

2. *We have*

$$H^d_D(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) = H^{d-1}\left(X(\Gamma)^{\mathrm{tor}}, \frac{\varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right)$$

$$= \frac{H^0(X(\Gamma)^{\mathrm{min}}, \mathrm{R}^{d-1} j_*^{\mathrm{min}} \mathcal{E})}{H^0(X(\Gamma)^{\mathrm{min}}, \mathrm{R}^{d-1}\pi_* \mathcal{E}^{\mathrm{can}})} = \frac{H^d_D(X(\Gamma)^{\mathrm{min}}, \mathcal{E})}{H^d_\pi(X(\Gamma)^{\mathrm{min}}, \mathcal{E})}.$$

Proof. The first part is a part of the local cohomology exact sequence for $Y(\Gamma) \hookrightarrow X(\Gamma)^{\mathrm{tor}}$. The sequence breaks into parts because of higher Koecher's principle, and the exact sequence in (1) is the end piece of the long exact sequence. By Proposition 5.1.2(1), the local cohomology exact sequence is also the same as long exact sequence of

$$0 \to \mathcal{E}^{\mathrm{can}} \to j_*^{\mathrm{tor}} \mathcal{E} \to \frac{j_*^{\mathrm{tor}} \mathcal{E}}{\mathcal{E}^{\mathrm{can}}} \to 0,$$

because of Leray spectral sequence. This implies

$$H^d_D(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) = H^{d-1}\left(X(\Gamma)^{\mathrm{tor}}, \frac{\varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right).$$

By Leray spectral sequence for $\pi : X(\Gamma)^{\mathrm{tor}} \to X(\Gamma)^{\mathrm{min}}$, we have

$$H^a\left(X(\Gamma)^{\mathrm{min}}, \mathrm{R}^b\pi_* \frac{\varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right) \Rightarrow H^{a+b}\left(X(\Gamma)^{\mathrm{tor}}, \frac{\varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right).$$

We have a long exact sequence

$$\cdots \to \mathrm{R}^i\pi_*\mathcal{E}^{\mathrm{can}} \to \mathrm{R}^i\pi_*(\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)) \to \mathrm{R}^i\pi_*\frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}} \to \cdots .$$

Since $\mathrm{R}^i\pi_*\frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}$ is supported on $X(\Gamma)^{\min} - Y(\Gamma)$,

$$H^a\left(X(\Gamma)^{\min}, \mathrm{R}^b\pi_*\frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right) = 0,$$

for $a > \dim(X(\Gamma)^{\min} - Y(\Gamma)) = 0$. Thus,

$$H^{d-1}\left(X(\Gamma)^{\mathrm{tor}}, \frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right) = H^0\left(X(\Gamma)^{\min}, \mathrm{R}^{d-1}\pi_*\frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right).$$

Also, since $\mathrm{R}^i\pi_*(\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)) = \mathrm{R}^i j_*^{\min}\mathcal{E}$ by Proposition 5.1.2(2), by the higher Koecher's principle the long exact sequence of higher direct image sheaves breaks into pieces, and the end piece is

$$0 \to \mathrm{R}^{d-1}\pi_*\mathcal{E}^{\mathrm{can}} \to \mathrm{R}^{d-1}j_*^{\min}\mathcal{E} \to \mathrm{R}^{d-1}\pi_*\frac{\varinjlim_{n\geq 0}\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}} \to 0,$$

from which we get the rest of (2). □

5.2 Hilbert-modular analogues of weakly holomorphic and harmonic Maass forms

The above situation was entirely analogous to that of harmonic Maass forms. Thus, we are led to the following

Definition 5.2.1. The space of *weakly holomorphic Hilbert modular forms* $M_k^!(\Gamma)$ is

defined to be

$$M_k^!(\Gamma) := H^{d-1}(Y(\Gamma), \omega^k).$$

The space of *principal parts* of weakly holomorphic Hilbert modular forms is defined to be

$$\frac{M_k^!(\Gamma)}{M_k(\Gamma)} := \frac{H^{d-1}(Y(\Gamma), \omega^k)}{H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k)}.$$

Finally, the space of *principal parts of harmonic Hilbert Maass forms* is defined to be

$$\frac{H_k^+(\Gamma)}{M_k(\Gamma)} := H_D^d(X(\Gamma)^{\mathrm{tor}}, \omega^k),$$

so that they fit into the exact sequence

$$0 \to \frac{M_k^!(\Gamma)}{M_k(\Gamma)} \to \frac{H_k^+(\Gamma)}{M_k(\Gamma)} \to H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k) \to 0.$$

Remark 5.2.2. The final part $H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k)$ can be regarded as S_{2-k} by Serre duality, so the situation is entirely analogous to the case of harmonic Maass forms.

We now compute this using relative Dolbeault cohomology. As in the case of harmonic Maass forms, the algebraic local cohomology is not the same as the analytic local cohomology, and we would instead have to use

$$H_D^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) = H^{d-1}\left(X(\Gamma)^{\mathrm{tor}}, \frac{\varinjlim_{n\geq 0} \mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}\right),$$

and use the mapping cone construction over $X(\Gamma)^{\mathrm{tor}}$ where now GAGA can be used. Namely, $H_D^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})$ would be the H^d of the complex

$$0 \to \mathcal{A}^{0,0}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \to \mathcal{A}^{0,1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \oplus \varinjlim_{n\geq 0} \mathcal{A}^{0,0}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD)) \to$$

$$\cdots \to \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \oplus \varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD)) \to \varinjlim_{n} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD)) \to 0.$$

Namely,

$$H_D^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) =$$

$$\frac{\{(\omega_1, \omega_2) \in \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \oplus \varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD)) : \omega_1 = \overline{\partial}\omega_2\}}{\{(\overline{\partial}\nu_1, \nu_1 - \overline{\partial}\nu_2) : (\nu_1, \nu_2) \in \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \oplus \varinjlim_{n\geq 0} \mathcal{A}^{0,d-2}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD))\}}.$$

To obtain the whole space of harmonic Maass forms, we use the Hodge theory of $X(\Gamma)^{\mathrm{tor}}$. Namely, as $X(\Gamma)^{\mathrm{tor}}$ is a compact Kähler manifold, the quotient map for Dolbeault cohomology

$$\mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \twoheadrightarrow H^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}),$$

restricts to an isomorphism

$$\mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \xrightarrow{\sim} H^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}),$$

by Hodge theory, where $\mathscr{H}$ is the space of harmonic forms. This means that any form in $\mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})$ can be modified into $\mathscr{H}^{0,d}$ via adding an appropriate $\overline{\partial}$ of a $(0, d-1)$-form. Thus, $H_D^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})$ has the following description,

$$H_D^d(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) =$$

$$\left\{(\omega_1, \omega_2) \in \mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \oplus \frac{\varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD))}{\ker(\mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \xrightarrow{\overline{\partial}} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})) + \overline{\partial}\left(\varinjlim_{n\geq 0} \mathcal{A}^{0,d-2}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD))\right)} : \omega_1 = \overline{\partial}\omega_2\right\} =$$

$$\left\{\omega \in \frac{\varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD))}{\ker(\mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}) \xrightarrow{\overline{\partial}} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})) + \overline{\partial}\left(\varinjlim_{n\geq 0} \mathcal{A}^{0,d-2}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}}(nD))\right)} : \overline{\partial}\omega \in \mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}^{\mathrm{can}})\right\}.$$

Removing the part in the denominator on cohomology of $\mathcal{E}^{\mathrm{can}}$, we arrive at the following definition.

Definition 5.2.3. Let the space of *harmonic Hilbert Maass forms* $H_k^+(\Gamma)$ be defined

as

$$H_k^+(\Gamma) = \frac{\ker \overline{\partial}^{-1}\left(\mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k)\right)}{\operatorname{im}\overline{\partial}} \subset \frac{\varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD))}{\operatorname{im}\overline{\partial}}.$$

Proposition 5.2.4. *The space of harmonic Hilbert Maass forms $H_k^+(\Gamma)$ sits in a short exact sequence*

$$0 \to M_k^!(\Gamma) \to H_k^+(\Gamma) \to H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k) \to 0.$$

Proof. This is proven similarly as in the case of harmonic Maass forms. Namely, consider the diagram of Dolbeault complexes for $\omega^k \subset \omega^k(\infty D)$ over $X(\Gamma)^{\mathrm{tor}}$:

$$\begin{tikzcd}
\mathcal{A}^{0,0}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[r, "\overline{\partial}"] \arrow[d, hook] & \cdots \arrow[r, "\overline{\partial}"] & \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[r, "\overline{\partial}"] \arrow[d, hook] & \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[d, hook] \\
\varinjlim_{n\geq 0} \mathcal{A}^{0,0}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) \arrow[r, "\overline{\partial}"] & \cdots \arrow[r, "\overline{\partial}"] & \varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) \arrow[r, "\overline{\partial}"] & \varinjlim_{n\geq 0} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)).
\end{tikzcd}$$

Since $\varinjlim_{n\geq 0} H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) = H^d(Y(\Gamma), \omega^k) = 0$, we have a diagram of exact sequences

$$\begin{tikzcd}
0 \arrow[r] & \ker\overline{\partial} \arrow[r] \arrow[d] & \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[r, "\overline{\partial}"] \arrow[d, hook] & \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[r] \arrow[d, hook] & H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker\overline{\partial} \arrow[r] & \varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) \arrow[r, "\overline{\partial}"] & \varinjlim_{n\geq 0} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) \arrow[r] & 0. &
\end{tikzcd}$$

Quotienting out by $\operatorname{im}\overline{\partial}$, we have

$$0 \to H^{d-1}(Y(\Gamma), \omega^k) \to \frac{\varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD))}{\operatorname{im}\overline{\partial}} \to \varinjlim_{n\geq 0} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD)) \to 0.$$

By taking the preimage of the image of $\mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k)$ in $\varinjlim_{n\geq 0} \mathcal{A}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD))$, we get the desired result. $\square$

We now explain elementarily what a harmonic Hilbert Maass form is. Note that the Dolbeault complex for a Shimura variety is identified with the Chevalley–Eilenberg complex for $(\mathfrak{p}, K)$-cohomology of certain functions on adelic quotient of G (cf. [Su]). The harmonic forms $\mathscr{H}^{0,d}(X(\Gamma)^{\mathrm{tor}}, \omega^k)$ are, by Serre duality, all of form $f \wedge_{i=1}^d d\overline{z}_i$, for a Hilbert modular cusp form f of parallel weight $2-k$, antiholomorphic in all variables. Under the same choice of $d\overline{z}_1, \cdots, d\overline{z}_d$, an element of $\mathcal{A}^{0,d-1}(Y(\Gamma), \omega^k)$ is of form

$$\sum_{i=1}^{d} f_i d\overline{z}_1 \wedge \cdots \wedge \widehat{d\overline{z}_i} \wedge \cdots \wedge d\overline{z}_d,$$

where hat means the entry is missing, and f_i is a smooth function over $Y(\Gamma)$ that transforms like a modular form of weight $(2-k, \cdots, k, \cdots, 2-k)$ (k at the i-th entry, $2-k$ at the other entries). Similarly, an element of $\varinjlim_{n\geq 0} \mathcal{A}^{0,d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k(nD))$ is of the same form, where f_i now has at most linear exponential growth towards cusps (when regarded as a function over $\mathbb{H}^d$).

Chapter 6

Fourier expansions of higher coherent cohomology classes

We develop the theory of Fourier expansion for higher coherent cohomology classes of Hilbert modular variety, by computing its local cohomology. This will enable us to relate our definition of harmonic Hilbert Maass forms and weakly holomorphic Hilbert modular forms with the work of [Br2]. Our method of computation of local cohomology using explicit structure of the boundary has its origin in [Fre]. The scope of the theory developed here is only for Hilbert modular varieties, but a similar strategy would work for more general Shimura varieties.

6.1 Analytic q-expansions

In this section, we work with the analytic category. Analytically, the Fourier expansion would be computed after restricting a Dolbeault class to a neighborhood of a cusp. For simplicity, let us assume that $\Gamma = \mathrm{SL}_2(\mathcal{O}_F)$ and $h_F = 1$, so that there is

only one cusp, ∞. The isotropy group of ∞ in Γ is

$$C_\infty := \left\{ \begin{pmatrix} \varepsilon & \mu \\ 0 & \varepsilon^{-1} \end{pmatrix} : \varepsilon \in U_F, \mu \in \mathcal{O}_F \right\} / \{\pm 1\},$$

where U_F is the group of units. More neatly, one can express this as

$$\left\{ \begin{pmatrix} \varepsilon & \mu \\ 0 & 1 \end{pmatrix} : \varepsilon \in U_F^2, \mu \in \mathcal{O}_F \right\}.$$

This contains a subgroup $\Gamma_\infty = \{\left(\begin{smallmatrix} 1 & \mu \\ 0 & 1 \end{smallmatrix}\right) : \mu \in \mathcal{O}_F\}$, and $C_\infty/\Gamma_\infty \cong U_F^2$, which is by Dirichlet's unit theorem isomorphic to $\mathbb{Z}^{d-1}$. A base of open neighborhoods of ∞ in $X(\Gamma)^{\min}$ is given by $\{C_\infty \backslash V_R \cup \{\infty\} : R \gg 0\}$, where

$$V_R = \{(z_1, \cdots, z_d) \in \mathbb{H}^d : \prod_{i=1}^{d} \mathrm{Im}(z_i) > R\}.$$

The q-expansion of a classical Hilbert modular form is obtained by taking the Fourier expansion of the restriction of the Hilbert modular form to $C_\infty \backslash V_R$ for a large enough R, or more precisely its pullback to $C_\infty \backslash V_R$. To deal with higher cohomology of $C_\infty \backslash V_R$, we first prove

Lemma 6.1.1 ([Fre, Hilfssatz 5.1])**.** *For $R \geq 0$, $C_\infty \backslash V_R \cup \{\infty\}$ is Stein.*

Proof. Note that as Γ_∞ acts via translation by real vectors, $\mathrm{Im}(z_i) : V_R \to \mathbb{R}^+$ descends to a function $\Gamma_\infty \backslash V_R \to \mathbb{R}^+$. Furthermore, $U_F^2 = C_\infty/\Gamma_\infty$ acts so that $\prod_{i=1}^d \mathrm{Im}(z_i)$ is unchanged. It is clear that

$$\frac{1}{\prod_{i=1}^d \mathrm{Im}(z_i)} : \Gamma_\infty \backslash V_R \cup \{\infty\} \to \mathbb{R}^+,$$

is a smooth, strictly plurisubharmonic, exhaustive function on $\Gamma_\infty \backslash V_R \cup \{\infty\}$, where the function is defined to be zero at $\{\infty\}$. Thus, by Oka's theorem, it is Stein. $\square$

Now the local cohomology $H^i_{\{\infty\}}(C_\infty \backslash V_R \cup \{\infty\}, \mathcal{F})$ for any coherent sheaf $\mathcal{F}$ fits into the local cohomology long exact sequence

$$\cdots \to H^i_{\{\infty\}}(C_\infty \backslash V_R \cup \{\infty\}, \mathcal{F}) \to H^i(C_\infty \backslash V_R \cup \{\infty\}, \mathcal{F}) \to H^i(C_\infty \backslash V_R, \mathcal{F}) \to \cdots,$$

and by Lemma 6.1.1, we thus know that

$$H^i_{\{\infty\}}(C_\infty \backslash V_R \cup \{\infty\}, \mathcal{F}) \cong H^{i-1}(C_\infty \backslash V_R, \mathcal{F}),$$

for any $i > 1$. Now the excision says that the left hand side is the same as $H^i_{\{\infty\}}(X(\Gamma)^{\min}, \mathcal{F})$, if $\mathcal{F}$ came from a coherent sheaf over $X(\Gamma)^{\min}$ denoted by the same letter $\mathcal{F}$.

Thus, as in [Fre, §5], the Grothendieck spectral sequence degenerates and yields

$$H^{i-1}(C_\infty \backslash V_R, \mathcal{F}) = H^{i-1}(U_F^2, H^0(\Gamma_\infty \backslash V_R, \mathcal{F})),$$

for any coherent sheaf $\mathcal{F}$ on $C_\infty \backslash V_R$ (here, the same letter $\mathcal{F}$ is used for its pullback to $\Gamma_\infty \backslash V_R$). In practice, $\mathcal{F}$ would all come from automorphic vector bundles of the Hilbert modular varieties. There is a canoncial section of $\mathcal{F}$ over $\Gamma_\infty \backslash V_R$ coming from the canonical differential of relative sheaf of differentials on the universal abelian scheme, which enables us to regard a holomorphic section of $\mathcal{F}$ as a holomorphic function. For such functions, we can take Fourier expansion with respect to Γ_∞-invariance,

$$H^0(\Gamma_\infty \backslash V_R, \mathcal{F}) \to \mathbb{C}\langle q^\nu \rangle_{\nu \in \mathcal{O}_F^\vee},$$

where q^ν stands for $\exp(2\pi i \operatorname{Tr} \nu z)$, $\langle - \rangle$ means the formal Laurent series is the Fourier expansion of a germ of a meromorphic function, and $\mathcal{O}_F^\vee = \{\lambda \in F : \operatorname{Tr} \lambda\mu \in \mathbb{Z}\ \forall \mu \in \mathcal{O}_F\}$. Thus, we have a q-expansion map for higher coherent cohomology of Hilbert

modular varieties,

$$q\exp^i : H^i(Y(\Gamma),\omega^k) \to H^{i+1}_{\{\infty\}}(X(\Gamma)^{\min},\omega^k) \xleftarrow{\sim} H^i(C_\infty\backslash V_R,\omega^k) \xrightarrow{\sim} H^i(\mathbb{Z}^{d-1}, H^0(\Gamma_\infty\backslash V_R,\omega^k)) \to H^i(\mathbb{Z}^{d-1},\mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\vee}).$$

We note that the action of $U_F^2 = C_\infty/\Gamma_\infty \cong \mathbb{Z}^{d-1}$ on $\mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\vee}$ is, for $\mu \in U_F^2$, just $\mu\cdot q^\nu = q^{\mu\nu}$. In this optic, we can recast the (failure of) Koecher's principle as follows.

Proposition 6.1.2. *For $i = d-1$, the image of $q\exp^{d-1}$, which a priori lies in the space of U_F^2-coinvariants $(\mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\vee})_{U_F^2}$, lies in $(\mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\vee,\nu\le 0})_{U_F^2}$, where $\nu \le 0$ means ν is either 0 or totally negative. Furthermore,*

$$\mathrm{im}(H^{d-1}(X^{\min},\omega^k) \to H^{d-1}(Y,\omega^k)) = \ker(q\exp^{d-1}).$$

Proof. The second statement is precisely the consquence of local cohomology long exact sequence. We now prove the first statement. it is sufficient to prove that any $\sum_{\nu\in\mathcal{O}_F^\vee,\nu\not\le 0} a_\nu q^\nu \in \mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\varepsilon}$ is in $(U_F^2-1)\mathbb{C}\langle q^\nu\rangle_{\nu\in\mathcal{O}_F^\vee}$. Let $\sigma_1,\cdots,\sigma_d$ be d real embeddings of F, and for $I \subset \{1,\cdots,d\}$ and $x \in F$, we define the expression $x \sim (-1)^I$ to mean that $\sigma_i(x) < 0$ if $i \in I$ and $\sigma_i(x) > 0$ if $i \notin I$. Then, it is sufficient to prove the same for the Laurent series of form $g = \sum_{\nu\in\mathcal{O}_F^\vee,\nu\sim(-1)^I} a_\nu q^\nu$ for $I \neq \{1,\cdots,d\}$.

We can find a unit $\mu_I \in U_F^2$ such that $\sigma_i(\mu_I) < 1$ for $i \in I$, as $I \neq \{1,\cdots,d\}$. Then, consider

$$f(z) := \sum_{\nu\in\mathcal{O}_F^\vee,\nu\sim(-1)^I} a_\nu \left(\sum_{n=1}^\infty q^{\mu_I^n\nu}\right).$$

The infinite sum is uniformly convergent, and defines a germ of a meromorphic function with d variables. Furthermore, by uniform convergence,

$$\mu_I^{-1}\cdot f = \sum_{\nu\in\mathcal{O}_F^\vee,\nu\sim(-1)^I} a_\nu \left(\sum_{n=0}^\infty q^{\mu_I^n\nu}\right),$$

so $(\mu_I^{-1} - 1)f = g$, as desired. $\square$

We would like to say that the q-expansion map precisely detects the failure of Koecher's principle. Proposition 6.1.2 is very close to what we would like, except that the computation relates $H^{d-1}(X(\Gamma)^{\min}, \omega^k)$ instead of $H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k)$. These two spaces are indeed different, but they are not too different: the difference is precisely due to the Eisenstein series (which is one-dimensional in our case, as we assumed that there is only one cusp for simplicity).

Proposition 6.1.3. *The cokernel of the natural map,*

$$H^{d-1}(X(\Gamma)^{\min}, \omega^k) \to H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k),$$

is naturally identified with the dual of the space of Eisenstein series of weight $2-k$.

Proof. Note that there is a long exact sequence of relative sheaf cohomology

$$\cdots \to H^{d-1}(X^{\min}, \omega^k) \to H^{d-1}(X^{\mathrm{tor}}, \omega^k) \to H^d_\pi(X^{\min}, \omega^k) \to H^d(X^{\min}, \omega^k) \to H^d(X^{\mathrm{tor}}, \omega^k) \to 0.$$

Thus,

$$\begin{aligned}
\operatorname{coker}(H^{d-1}(X^{\min}, \omega^k) \to H^{d-1}(X^{\mathrm{tor}}, \omega^k)) &= \frac{H^{d-1}(X^{\mathrm{tor}}, \omega^k)}{\operatorname{im}(H^{d-1}(X^{\min}, \omega^k) \to H^{d-1}(X^{\mathrm{tor}}, \omega^k))} \\
&= \frac{H^{d-1}(X^{\mathrm{tor}}, \omega^k)}{\ker(H^{d-1}(X^{\mathrm{tor}}, \omega^k) \to H^d_\pi(X^{\min}, \omega^k))} \\
&= \operatorname{im}(H^{d-1}(X^{\mathrm{tor}}, \omega^k) \to H^d_\pi(X^{\min}, \omega^k)) \\
&= \ker(H^d(X^{\min}, \omega^k) \to H^d(X^{\mathrm{tor}}, \omega^k)) \\
&= \operatorname{coker}(H^d(X^{\mathrm{tor}}, \omega^k)^* \to H^d(X^{\min}, \omega^k)^*)^*,
\end{aligned}$$

where $(-)^*$ is the $\mathbb{C}$-linear dual. By Kodaira–Spencer isomorphism and (classical)

Koecher's principle,

$$\begin{aligned}\operatorname{coker}(H^d(X^{\mathrm{tor}},\omega^k)^* \to H^d(X^{\min},\omega^k)^*) &= \operatorname{coker}(H^0(X^{\mathrm{tor}},\omega^{2-k,\mathrm{sub}}) \to H^0(X^{\min},\omega^{2-k})) \\ &= \operatorname{coker}(H^0(X^{\mathrm{tor}},\omega^{2-k,\mathrm{sub}}) \to H^0(X^{\mathrm{tor}},\omega^{2-k})),\end{aligned}$$

as desired. □

6.2 Algebraic q-expansions, using the geometry of toroidal compactifications

We now compute the local cohomology group using the geometry of compactification, as in [Lan2]. This will help us identify the image of the q-expansion map.

By Proposition 5.1.3(2), we would like to compute

$$\varinjlim_{n\geq 0} H^0\left(X(\Gamma)^{\min}, \frac{\mathrm{R}^{d-1}\pi_*(\mathcal{E}^{\mathrm{can}}(nD))}{\mathrm{R}^{d-1}\pi_*\mathcal{E}^{\mathrm{can}}}\right) = \varinjlim_{n\geq 0} H^0(X(\Gamma)^{\min}, \mathrm{R}^{d-1}\pi_*\mathcal{Q}_n),$$

where $\mathcal{Q}_n = \frac{\mathcal{E}^{\mathrm{can}}(nD)}{\mathcal{E}^{\mathrm{can}}}$. Since $\mathrm{R}^{d-1}\pi_*\mathcal{Q}_n$ is supported at $\{\infty\}$, we are to compute

$$\varinjlim_{n\geq 0} H^{d-1}((X(\Gamma)^{\mathrm{tor}})^\wedge_\infty, (\mathcal{Q}_n)^\wedge_\infty),$$

where $(X(\Gamma)^{\mathrm{tor}})^\wedge_\infty$ is the pullback of $X(\Gamma)^{\mathrm{tor}}$ under the strict localization at ∞ (regarded as a geometric point), $(X(\Gamma)^{\min})^\wedge_\infty \to X(\Gamma)^{\min}$. This is because $(X(\Gamma)^{\min})^\wedge_\infty$ is affine. Since $\mathcal{Q}_n$ is supported at $D^{\mathrm{tor}} = X(\Gamma)^{\mathrm{tor}} - Y(\Gamma)$, we are to compute

$$\varinjlim_{n\geq 0} H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{Q}_n).$$

We use Serre duality, noting that by Kodaira–Spencer, $\omega^{2,\mathrm{sub}}$ is the dualizing sheaf of

$X(\Gamma)^{\mathrm{tor}}$:

$$H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \mathcal{Q}_n) = H^0(X(\Gamma)^{\mathrm{tor}}, \mathcal{E}xt^1(\mathcal{Q}_n, \omega^{2,\mathrm{sub}}))^* = \left(H^0\left(X(\Gamma)^{\mathrm{tor}}, \frac{\mathcal{H}om(\mathcal{E}^{\mathrm{can}}, \omega^{2,\mathrm{sub}})}{\mathcal{H}om(\mathcal{E}^{\mathrm{can}}, \omega^{2,\mathrm{sub}})(-nD)}\right)\right)^*.$$

In particular, if $\mathcal{E} = \omega^k$, we have

$$H^d_D(X(\Gamma)^{\mathrm{tor}}, \omega^k) = \varinjlim_{n \geq 0} \left(H^0\left(X(\Gamma)^{\mathrm{tor}}, \frac{\omega^{2-k,\mathrm{sub}}}{\omega^{2-k,\mathrm{sub}}(-nD)}\right)\right)^*.$$

Thus, we would like to compute what $H^0\left(X(\Gamma)^{\mathrm{tor}}, \frac{\omega^{2-k,\mathrm{sub}}}{\omega^{2-k,\mathrm{sub}}(-nD)}\right)$ is.

Proposition 6.2.1. *The two Serre duality pairings are compatible with each other, namely*

$$\begin{tikzcd}
H^d(X(\Gamma)^{\mathrm{tor}}, \omega^k) & \times & H^0(X(\Gamma)^{\mathrm{tor}}, \omega^{2-k,\mathrm{sub}}) \arrow[r] \arrow[d] & \mathbb{C} \\
H^d_D(X(\Gamma)^{\mathrm{tor}}, \omega^k) \arrow[u] & \times & H^0\left(X(\Gamma)^{\mathrm{tor}}, \frac{\omega^{2-k,\mathrm{sub}}}{\omega^{2-k,\mathrm{sub}}(-nD)}\right) \arrow[r] & \mathbb{C}
\end{tikzcd}$$

commutes.

Proof. By the setup of coherent duality, namely that the coherent duality is a consequence of the construction of the trace 2-functor as in [Hart, Corollary VII.3.4], the Serre duality is compatible with long exact sequences. □

We now invoke more specific geometry of compactifications, as recalled in [Lan2, §4].

1. There is a split torus $\Xi = \mathrm{Spec}\left(\bigoplus_{\ell \in \mathbb{S}} \Psi(\ell)\right)$ over ∞, with character group $\mathbb{S}$.

2. Each cone $\tau \subset \mathbb{S}^\vee_{\mathbb{R}} = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{S}, \mathbb{R})$ defines a toroidal embedding $\Xi \hookrightarrow \Xi(\tau) = \mathrm{Spec}\left(\bigoplus_{\ell \in \tau^\vee} \Psi(\ell)\right)$, where $\tau^\vee = \{\ell \in \mathbb{S} : \langle \ell, y \rangle \geq 0 \ \forall y \in \tau\}$.

3. Let $\mathfrak{U}_\tau$ be the formal completion of $\Sigma(\tau)$ along $\Sigma(\tau) - \Sigma$. These, for $\tau \in \Sigma^+$ for some rational polyhedral cone decomposition Σ^+, glue to form a formal scheme

$\mathfrak{X}$ over ∞. Let $\widetilde{\mathfrak{N}}$ be the nerve of the covering induced by the closures of the cones in Σ^+.

4. For each $\gamma \in U_F^2$, there is a canonical isomorphism $\gamma : \mathfrak{U}_\sigma \xrightarrow{\sim} \mathfrak{U}_{\gamma\sigma}$ over ∞, induced by the isomorphisms $\gamma^*\Psi(\gamma\ell) \xrightarrow{\sim} \Psi(\ell)$. These isomorphisms give an action of U_F^2 on $\mathfrak{X}$, whence a local isomorphism $\mathfrak{X} \to \mathfrak{X}/U_F^2$.

5. Now, $\mathfrak{X}/U_F^2 \cong (X(\Gamma)^{\mathrm{tor}})^\wedge_\infty$.

6. Let $\mathscr{E}^{(n)}$ be the pullback of $\mathcal{E}(nD)$ to $\mathfrak{X}/U_F^2$, and $\mathfrak{N} = \widetilde{\mathfrak{N}}/U_F^2$. Let $\underline{\mathscr{H}}^d(\mathscr{M})$, for a quasicoherent sheaf $\mathscr{M}$ on $\mathfrak{X}/U_F^2$, be the constructible sheaf over $\mathfrak{N}$ which has stalks $H^d(\mathfrak{U}_\sigma, \mathscr{M}|_{\mathfrak{U}_\sigma})$ over σ. There is a *nerve spectral sequence*

$$H^i(\mathfrak{N}, \underline{\mathscr{H}}^j(\mathscr{E}^{(n)})) \Rightarrow H^{i+j}(\mathfrak{X}/U_F^2, \mathscr{E}^{(n)}).$$

Each term of spectral sequence also admits a Hochschild–Serre spectral sequence

$$H^a(U_F^2, H^b(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^c(\mathscr{E}^{(n)}))) \Rightarrow H^{a+b}(\mathfrak{N}, \underline{\mathscr{H}}^c(\mathscr{E}^{(n)})).$$

7. There exists an $\mathbb{R}_{>0}$-valued *polarization function* pol on the positive cone $\mathbb{P}$ which is piecewise linear on each cone $\sigma \in \Sigma$.

8. For $\mathcal{E} = \omega^k$, we define $\mathrm{FJ}^{0,(\ell)}(\mathscr{E}) = H^0(\infty, \Psi(\ell))$, which coincides with [Lan2, Corollary 5.9], as $\mathscr{E}_0 = \mathcal{O}_\infty$ for $\mathcal{E} = \omega^k$.

Aruging similarly to [Lan2], we compute $\varinjlim_{n\geq 0} H^{d-1}(\mathfrak{X}/U_F^2, \mathscr{E}^{(n)})$.

Proposition 6.2.2. *Let $n \geq 0$.*

1. *The sheaf $\underline{\mathscr{H}}^c(\mathscr{E}^{(n)}) = 0$ for $c > 0$.*

2. *The natural map*

$$H^{d-1}(\mathfrak{X}/U_F^2, \mathscr{E}^{(n)}) \twoheadrightarrow H^{d-1}(U_F^2, H^0(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)}))),$$

becomes an isomorphism after taking the limit,

$$\varinjlim_{n\geq 0} H^{d-1}(\mathfrak{X}/U_F^2, \mathscr{E}^{(n)}) \xrightarrow{\sim} \varinjlim_{n\geq 0} H^{d-1}(U_F^2, H^0(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)}))).$$

3. *If $\mathcal{E} = \omega^k$, we have*

$$\varinjlim_{n\geq 0} H^{d-1}(U_F^2, H^0(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)})) \cong H^{d-1}\left(U_F^2, \bigoplus_{\ell\in -\mathbb{P}^{\vee,+}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E})\right).$$

Proof. (1) follows from the fact that $\mathfrak{U}_\sigma$'s are affine in our case. To prove (2), we need to argue similarly as [Lan2, Proposition 7.8]. That the natural map is surjective comes from the spectral sequence. To prove the full statement, it is sufficient to prove that $\varinjlim_{n\geq 0} H^i(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)})) = 0$ for $i > 0$. We will prove that the direct system $\{H^i(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)}))\}$ is eventually zero. Now the cohomology of simplicial complex $H^i(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)}))$ is identified with the local cohomology with support at $\widetilde{\mathfrak{N}} - \widetilde{\mathfrak{N}}^{\ell,(n)}$. As $\varinjlim_{n\geq 0} \widetilde{\mathfrak{N}}^{\ell,(n)} = \widetilde{\mathfrak{N}}$, any cohomology class over $\widetilde{\mathfrak{N}}$ would have vanishing image when restricted to local cohomology with support in $\widetilde{\mathfrak{N}} - \widetilde{\mathfrak{N}}^{\ell,(n)}$ for $n \gg 0$. To prove (3), first we note that $U_F^2 \cong \mathbb{Z}^{d-1}$, and one can take a basis $\epsilon_1, \cdots, \epsilon_{d-1}$ such that $\sigma_i(\epsilon_i) > 1$ and $\sigma_j(\epsilon_i) < 1$ for $j \neq i$. If $\ell_i > 0$, then $H^1(\epsilon_i^{\mathbb{Z}}, \prod_{\ell'\in\epsilon_i^{\mathbb{Z}}\cdot\ell} \mathrm{FJ}^{0,(\ell')}(\mathscr{E})) = 0$ as in the proof of [Lan2, Proposition 7.16] by Shapiro's lemma. Now

$$\varinjlim_{n\geq 0} H^{d-1}(U_F^2, H^0(\widetilde{\mathfrak{N}}, \underline{\mathscr{H}}^0(\mathscr{E}^{(n)}))) = H^{d-1}\left(U_F^2, \widehat{\bigoplus_{\ell\in\mathbb{S}}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E})\right),$$

where the completion is taken with respect to $\bigoplus_{\ell\in\mathbb{P}^{\vee,+}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E})$. Thus, the sub

$$H^{d-1}\left(U_F^2, \widehat{\bigoplus_{\ell\in\mathbb{S}-(-\mathbb{P}^{\vee,+})}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E})\right) \subset H^{d-1}\left(U_F^2, \widehat{\bigoplus_{\ell\in\mathbb{S}}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E})\right),$$

has consituents of the form $H^{d-1}(U_F^2, \widehat{\bigoplus_{\ell\in U_F^2\cdot\ell_0}} \mathrm{FJ}^{0,(\ell)}(\mathscr{E}))$ for $\ell_0 \notin -\mathbb{P}^{\vee,+}$, and this is

isomorphic to $\prod_{i=1}^{d-1} H^1(\epsilon_i^{\mathbb{Z}}, \widehat{\bigoplus_{\ell\in\epsilon_i^{\mathbb{Z}}\cdot\ell_0}} \mathrm{FJ}^{0,(\ell')}(\mathscr{E}^{\circ}))$. As one of the multiplicands is zero, this is zero. Thus, the result follows. $\square$

As algebraic and analytic Fourier–Jacobi coefficients coincide [Lan1], we see the following.

Proposition 6.2.3. *The image of $H^{d-1}(X(\Gamma)^{\mathrm{tor}}, \omega^k) \subset H^{d-1}(Y(\Gamma), \omega^k)$ under $q\exp^{d-1}$ is identified with the image of $\mathbb{C}$ in $(\mathbb{C}\langle q^{\nu}\rangle_{\nu\in\mathcal{O}_F^{\vee}})_{U_F^2}$. The space of principal parts of harmonic Hilbert Maass forms, $\frac{H_k^+}{M_k}$, is identified with the totally negative Laurent polynomials, up to U_F^2-coinvariance.*

Example 6.2.4 (Harmonic Maass forms)**.** As always, we would like to see how this is analogous to an existing feature of harmonic Maass forms. These correspond to the q-expansions of the principal parts of harmonic Maass forms. Recall that a harmonic Maass form f has a decomposition $f = f^+ + f^-$, where f^+ is the holomorphic part (*mock modular form*), and f^- is the non-holomorphic part. Furthermore, the datum of the shadow $\xi_k(f)$ is equivalent to the datum of f^-.

Now the key is that f^- is determined by f^+, or even better, the principal parts of f^+, by [DL, Proposition 2.6]. Namely, the principal parts define a linear functional on S_{2-k}, and this by Serre duality pins down the (complex conjugate of the) shadow.

Example 6.2.5. In the case of Hilbert modular surfaces, Proposition 6.2.3 implies that, given $\nu < 0$, there exists a nonholomorphic function $f_{\gamma,\nu}$, for each $\gamma \in U_F^2$, so that $\gamma \cdot f_\gamma - f_\gamma = q^\nu$. One would hope that $f_\gamma = \sum_{n=1}^{\infty} \gamma^{-n} \cdot q^\nu$ would formally work, but this is actually a divergent series! This is one of the reasons why the naïve generalization of harmonic Maass forms does not work.

Chapter 7

Borcherds lift of harmonic Hilbert Maass forms

In [Br2], in a view towards constructing Borcherds lift for Hilbert modular forms, a way to avoid Koecher's principle for Hilbert modular forms was suggested. This starts by realizing that the usual Borcherds lift is constructed for each non-holomorphic Poincaré series in the sense of [Br1], as the space of harmonic Maass forms is spanned by such forms. In the regularization process for singular theta integrals, one realizes that it is rather a Whittaker function that is used in the integral, which is the analogue of exponential function for the usual Poincaré series. Now, [Br2] uses the fact that the analogue of Whittaker functions has no Koecher's principle, and one can think such Whittaker functions (named *harmonic Whittaker forms*) as alternatives for then non-existent harmonic Hilbert Maass forms.

Using our new definition of harmonic Hilbert Maass forms and their q-expansions, we can now identify harmonic Whittaker forms with principal parts of harmonic Hilbert Maass forms. Namely,

$$\left\{\begin{matrix}\text{Principal parts of harmonic}\\ \text{Hilbert Maass forms}\end{matrix}\right\} = \left\{\begin{matrix}\text{Laurent polynomials, with totally}\\ \text{negative exponents}\end{matrix}\right\} / U_F^2.$$

Definition 7.0.1 (Scalar-valued version of [Br2, Definition 4.1])**.** A *Whittaker form* of weight k is a linear combination of functions of form

$$f_m(\tau, s) = \frac{(4\pi m_2)^{k-1} \cdots (4\pi m_d)^{k-1}}{\Gamma(s+1)\Gamma(k-1)^{d-1}} |4\pi m_1 v_1|^{-k/2} M_{\mathrm{sgn}(-m_1)k/2, s/2}(4\pi |m_1| v_1) e^{2\pi m_1 v_1} e(\mathrm{tr}(-m\overline{\tau})),$$

where

- $\tau = (\tau_1, \cdots, \tau_d) \in \mathbb{H}^d$, $\tau_j = u_j + iv_j$,
- $m \in \mathcal{O}_F^\vee$ acts on $(z_1, \cdots, z_d) \in \mathbb{H}^d$ via $m(z_1, \cdots, z_d) = (\sigma_1(m)z_1, \cdots, \sigma_d(m)z_d)$, where σ_i is the i-th real embedding of F,
- $\mathrm{tr}(z_1, \cdots, z_d) = z_1 + \cdots + z_d$,
- and M is the "bad Whittaker function" (it is "bad" because it blows up exponentially as $z \to \infty$, unlike the "good Whittaker function" that only has moderate growth towards infinity, which is usually used in the theory of automorphic forms); namely, $M_{\nu,\mu}(z)$ is a specific solution to the Whittaker differential equation

  $$M'' + \left(-\frac{1}{4} + \frac{\nu}{z} + \frac{\frac{1}{4} - \mu^2}{z^2}\right) M = 0,$$

  that is expressed in terms of the confluent hypergeometric function of the second kind,

  $$M_{\nu,\mu}(z) = e^{-z/2} z^{1/2+\mu} M\left(\frac{1}{2} + \mu - \nu, 1 + 2\mu, z\right),$$

  where

  $$M(a, b, z) = \sum_{n=0}^{\infty} \frac{\Gamma(a+n)/\Gamma(a)}{\Gamma(b+n)/\Gamma(b)} \frac{z^n}{n!}.$$

A *harmonic Whittaker form* of weight k is a linear combination of functions of form

$$f_m(\tau) := f_m(\tau, 1-k) = \frac{(4\pi m_2)^{k-1} \cdots (4\pi m_d)^{k-1}}{\Gamma(1-k)\Gamma(k-1)^{d-1}} (\Gamma(1-k) - \Gamma(1-k, 4\pi m_1 v_1)) e^{4\pi m_1 v_1} e(\mathrm{tr}(-m\overline{\tau})),$$

where $m \in \mathcal{O}_F^\vee$, $m \gg 0$, $\tau = (\tau_1, \cdots, \tau_d)$, $\tau_a = u_a + iv_a$, and $\Gamma(s, v)$ is the incomplete Gamma function.

It is computed that

$$v_1^{k-2}\overline{L_k^{(1)} f_m(\tau)} = \frac{(4\pi m_1)^{1-k}(4\pi m_2)^{k-1}\cdots(4\pi m_d)^{k-1}}{\Gamma(1-k)\Gamma(k-1)^{d-1}} e(\mathrm{tr}(m\tau)),$$

where $L_k^{(1)}$ is the lowering operator in variable τ_1.

Using this, we define the non-holomorphic Hilbert Poincaré series.

Definition 7.0.2 (Non-holomorphic Hilbert Poincaré series)**.** We define a *non-holomorphic Hilbert Poincaré series* F_m, for $m \in \mathcal{O}_F^\vee$ with $m \gg 0$, to be a harmonic Hilbert Maass form whose q-expansion is q^{-m}.

Note that this only defines a class in H_k^+/M_k, but this is enough as an input for Borcherds lift.

Definition 7.0.3 (Borcherds lift of harmonic Hilbert Maass forms)**.** Let (V, Q) be a quadratic space over F of dimension $n+2$ and signature $((n,2),(n+2,0),\cdots,(n+2,0))$, and let $L \subset V$ be a unimodular lattice. Let $H = \mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{GSpin}(V)$, $K \subset H(\mathbb{A})$ be an open compact subgroup and X_K be the Shimura variety associated with H and level K,

$$X_K = H(\mathbb{Q})\backslash(X \times H(\mathbb{A}_f))/K.$$

Here, the Hermitian symmetric space X for H is the Grassmannian of oriented negative-definite 2-dimensional subspaces of V_{σ_1}. For $f \in \frac{H_k^+}{M_k}$, the Borcherds lift $\Phi(f)$ is defined to be

$$\Phi(f) = \sum_{m \in \mathcal{O}_F^\vee, m \ll 0} c_m \Phi_{-m}(z, h),$$

where the q-expansion of f (see Proposition 6.2.3) is

$$q\exp^{d-1}(f) = \sum_{m\in\mathcal{O}_F^\vee, m\ll 0} c_m q^m,$$

and $\Phi_{-m}(z,h)$ is, as in [Br2, (5.2)], the "regularized theta lift" of the harmonic Whittaker form $f_{-m}(\tau)$:

$$\begin{aligned}\Phi_{-m}(z,h) &= \frac{1}{\sqrt{D}}\int_{\Gamma_\infty\backslash\mathbb{H}^d}^{\text{reg}} f_{-m}(\tau)\Theta_{Siegel}(\tau,z,h)(v_2\cdots v_d)^{(n+2)/2}d\mu(\tau)\\ &= \frac{1}{\sqrt{D}}\int_{v\in(\mathbb{R}_{>0})^d}\left(\int_{u\in\mathbb{R}^d/\mathcal{O}_F} f_{-m}(\tau)\Theta_{Siegel}(\tau,z,h)du\right)(v_2\cdots v_d)^{(n+2)/2}\frac{dv}{(v_1\cdots v_d)^2},\end{aligned}$$

where

- $D = \operatorname{disc} F$, $h\in H(\mathbb{A}_f)$, $z\in X$,
- and $\Theta_{Siegel}(\tau,z,h)$ is the Siegel theta function

$$\Theta_{Siegel}(\tau,z,h) = v_1\sum_{\lambda\in V(F)}\mathbf{1}_{L\otimes\widehat{\mathcal{O}_F}}(h^{-1}\lambda)e\left(\operatorname{tr}Q(\lambda_{z^\perp})\tau + \operatorname{tr}Q(\lambda_z)\overline{\tau}\right),$$

  where $\mathbf{1}_{L\otimes\widehat{\mathcal{O}_F}}$ is the characteristic function for $L\otimes\widehat{\mathcal{O}_F}\subset V\otimes\mathbb{A}_{F,f}$, and λ_z (resp. $\lambda_{z^\perp}$) for $\lambda\in V(F)\subset V(\mathbb{R})$ is the orthogonal projection of λ to z (resp. $z^\perp$), where $z\in X$ is regarded as a 2-dimensional real subspace of $V(\mathbb{R})$.

The functions $\Phi_{-m}(z,h)$ are functions over $X\times H(\mathbb{A}_f)$ which are naturally left-invariant under $H(\mathbb{Q})$ and right-invariant under K, so thus define functions over X_K. These are called *automorphic Green functions.*

Proposition 7.0.4. *Our definition of non-holomorphic Poincaré series coincides with the non-holomorphic Poincaré series in the case of elliptic modular forms. In particular, our Borcherds lift of harmonic Hilbert Maass forms coincides with the usual Borcherds lift in the case of harmonic Maass forms.*

Proof. This follows from the computation of Fourier expansion of non-holomorphic Poincaré series in the context of harmonic Maass forms. Indeed, [Br1, Proposition 1.0] computes that the principal part of holomorphic part of F_m is e^{-m}. Furthermore, using the formula of *loc. cit.*, the shadow of F_m is the holomorphic Poincaré series associated to $\frac{(4\pi m)^{1-k}}{\Gamma(1-k)}e(m\tau)$, which coincides with the normalization of Definition 7.0.2. $\square$

This viewpoint, that *the harmonic Hilbert Maass forms in our sense are the source of the Borcherds lift*, demystifies several oddities of [Br2].

- The most apparent asymmetry is that [Br2] chooses a real embedding and works only with that coordinate; for example, in the course of defining harmonic Whittaker forms, holomorphicity is relaxed only for one coordinate, and the other coordinates seemed to stay the same. This can be explained by that a harmonic Maass form is a collection of d functions where the i-th function is holomorphic in all variables but the i-th variable. Namely, this comes from the downside of trying to use only one function, which is the precise reason why the classical approach could not overcome Koecher's principle.

- In general, the Borcherds lift, even though it is defined for all harmonic Maass forms, is zero when it is restricted to the space of cusp forms. This is also in accordance with the fact that the natural object to consider is the space of harmonic Maass forms modulo the space of modular forms.

Chapter 8

The arithmetic of harmonic Maass forms

8.1 Mod p and p-adic harmonic Maass forms

From our reinterpretation, we can define the notion of *p-adic* and *mod p* harmonic Maass forms.

Definition 8.1.1. Let $(N, p) = 1$, and let $X_1(Np^r)$ ($Y_1(Np^r)$, respectively) be the Katz–Mazur integral model of the compact (open, respectively) modular curve with moduli problem ([bal. $\Gamma_1(p^r)]^{\zeta-\mathrm{can}}$; $[\mu_N]$) (see [KM]) over $\mathbb{Z}_p[\mu_{p^r}]$. Let $D = X_1(Np^r) - Y_1(Np^r)$ with its reduced scheme structure. We define the space of principal parts of *p-adic* (*mod p*, respectively) *harmonic Maass forms* of weight k as

$$\frac{H^+_{k,\mathbb{Q}_p}}{M_{k,\mathbb{Q}_p}} := H^1_{D_{\mathbb{Q}_p(\mu_{p^r})}}(X_1(Np^r)_{\mathbb{Q}_p(\mu_{p^r})}, \omega^k) \qquad (\frac{H^+_{k,\mathbb{F}_p}}{M_{k,\mathbb{F}_p}} := H^1_{D_{\mathbb{F}_p}}(X_1(Np^r)_{\mathbb{F}_p}, \omega^k), \text{ respectively}).$$

The above local cohomology groups are algebraic local cohomology groups.

Note that the p-adic algebraic local cohomology group can be computed using the adic analytification via rigid GAGA; the foundation on the local cohomology of

adic spaces was developed in [BoPi, §2]. In both the p-adic and the mod p cases, we use excision so that the spaces are the same as the algebraic local cohomology of the *ordinary locus* with support at the cusps. In particular, the singularity of the integral model is irrelevant, and they can be explicitly considered using q-expansions; for example, the space of principal part of p-integral harmonic Maass forms are those with p-integral Fourier coefficients of principal parts. Since there is an exact sequence

$$0 \to H^0(Y^{\mathrm{ord}}_{\mathbb{F}_p}, \omega^k) \to H^0(Y^{\mathrm{ord}}_{\mathbb{F}_p} - D_{\mathbb{F}_p}, \omega^k) \to H^1_{D_{\mathbb{F}_p}}(Y^{\mathrm{ord}}_{\mathbb{F}_p}, \omega^k) \to 0,$$

we see that

$$\frac{H^+_{k,\mathbb{F}_p}}{M_{k,\mathbb{F}_p}} = \frac{H^0(Y^{\mathrm{ord}}_{\mathbb{F}_p} - D_{\mathbb{F}_p}, \omega^k)}{H^0(Y^{\mathrm{ord}}_{\mathbb{F}_p}, \omega^k)}.$$

As the geometry $Y^{\mathrm{ord}}_{\mathbb{F}_p}$ is extremely well-known, one can for example prove the following

Theorem 8.1.2. *There is a natural isomorphism*

$$\frac{H^+_0(Np, \mathbb{F}_p)}{M_0(Np, \mathbb{F}_p)} \cong \frac{q\exp\left(\bigoplus_k M^!_k(N, \mathbb{F}_p)\right)}{q\exp\left(\bigoplus_k M_k(N, \mathbb{F}_p)\right)} \oplus v\frac{q\exp\left(\bigoplus_k M^!_k(N, \mathbb{F}_p)\right)}{q\exp\left(\bigoplus_k M_k(N, \mathbb{F}_p)\right)},$$

where v is the automorphism of $X_1(Np)_{\mathbb{F}_p}$ as in [Gr1, (6.5)], $M^!_k(N, \mathbb{F}_p) = H^0(Y_1(N)_{\mathbb{F}_p}, \omega^k)$, $M_k(N, \mathbb{F}_p) = H^0(X_1(N)_{\mathbb{F}_p}, \omega^k)$, and $q\exp$ is the map of q-expansions $q\exp : M^!_k(N, \mathbb{F}_p) \to \mathbb{F}_p((q))$.

Proof. By [Gr1, §7], $Y_1(Np)^{\mathrm{ord}}_{\mathbb{F}_p} = I_1(N)^{\mathrm{ord}} \coprod vI_1(N)^{\mathrm{ord}}$, where $I_1(N)^{\mathrm{ord}}$ is the Igusa curve, so it is sufficent to prove

$$\frac{\mathbb{F}_p(I_1(N)^{\mathrm{ord}} - \{\infty\})}{\mathbb{F}_p(I_1(N)^{\mathrm{ord}})} \cong \frac{q\exp\left(\bigoplus_k M^!_k(N, \mathbb{F}_p)\right)}{q\exp\left(\bigoplus_k M_k(N, \mathbb{F}_p)\right)},$$

where $\mathbb{F}_p(I_1(N)^{\mathrm{ord}})$ is the affine coordinate ring of $I_1(N)^{\mathrm{ord}}$ and similarly for $I_1(N)^{\mathrm{ord}} - \{\infty\}$. As in [Gr1, Proposition 5.5], every function on $I_1(N)^{\mathrm{ord}} - \{\infty\}$ can be multiplied by a suitable power of Hasse invariant a such that it defines a regular section of ω^k

on $I_1(N) - \{\infty\}$ which descends down to a section of ω^k on $Y_1(N)$. $\square$

8.2 Rational structures on harmonic Maass forms

There is a very obvious notion of rational structure on $\frac{H_k^+}{M_k}$, coming from the $\mathbb{Q}$-structure on the modular curve. It should also be true that the exact sequence

$$0 \to M_k \to M_k^! \to \frac{H_k^+}{M_k} \to H^1(X, \omega^k) \to 0,$$

is compatible with the $\mathbb{Q}$-structures of each space. We interpret this compatibility using the result of [DL].

- In our convention, for a normalized Hecke eigen-cuspform f of weight $2-k$ defined over $\overline{\mathbb{Q}}$, a scalar multiple cf defines a $\overline{\mathbb{Q}}$-class in $H^1(X, \omega^k) = S_{2-k}^\vee$ if and only if $c\langle f, f\rangle \in \overline{\mathbb{Q}}$, where $\langle , \rangle$ is the Petersson inner product. This is immediate from the Serre duality.

- From [DL, Proposition 2.6], there is a harmonic Maass form $\widetilde{f} \in H_k^+$ whose shadow is cf and whose principal parts are in $\overline{\mathbb{Q}}$. Any two such harmonic Maass forms differ by a weakly holomorphic modular forms of weight k with principal parts in $\overline{\mathbb{Q}}$, so this is the $\overline{\mathbb{Q}}$-structure of $\frac{M_k^!}{M_k}$.

In general, we believe that the natural $\overline{\mathbb{Q}}$-structure to be imposed on H_k^+ is not that there is a natural underlying $\overline{\mathbb{Q}}$-vector space of maximal dimension but that it has a *two-step $\overline{\mathbb{Q}}$-structure*:

Definition 8.2.1. For a subfield $K \subset L$ and an L-vector space V, V has a *K-structure* if there is a sub-K-vector space $V_K \subset V$ such that $V_K \otimes_K L \xrightarrow{\sim} V$. On the other hand, V has an *n-step K-structure* if there is an n-step filtration $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ such that $\mathrm{gr}^i V = V_{i+1}/V_i$ has a K-structure for each $i = 0, 1, \cdots, n-1$.

Thus, the two-step $\overline{\mathbb{Q}}$-structure on H_k^+ is imposed by the filtration $0 \subset M_k \subset H_k^+$ and the natural $\overline{\mathbb{Q}}$-structures on each of the graded pieces.

Part II

The action of motivic cohomology on algebraic automorphic forms of irregular weight

Chapter 9

Introduction

The motivic action conjecture of Venkatesh posits that, roughly speaking, for a Hecke eigensystem h, there is a natural action of the motivic cohomology of the adjoint motive associated to h on the h-isotypic part of the rational cohomology of locally symmetric spaces. This has many incarnations which shed new light on various parts of the Langlands program. However, the conjectures have been mostly restricted to the case of "$\delta \neq 0$," namely when the reductive group in concern G has no compact Cartan subgroup. This in particular excludes the case when the locally symmetric space is a *Shimura variety*. As a locally symmetric space that is not a Shimura variety so far has not been related to algebraic geometry (although see [GoTa]), the motivic action conjectures seemed to be extremely difficult to approach.

On the other hand, there have been expectations that a similar conjecture would exist for automorphic forms over Shimura varieties with *irregular weight*. The easiest instance is the case of weight one modular forms; they appear in both H^0 and H^1 of the modular curve of the same line bundle. The main purpose of the paper is to give a formulation of such conjecture for general Shimura varieties, and provide somewhat intricate evidences using well-known results and conjectures regarding periods of automorphic forms. The following is a generalization of the archimedean motivic action

conjecture to general Shimura varieties.

Conjecture 1 (Archimedean motivic action conjecture for Shimura varieties)**.** *Let λ be a nondegenerate singular analytically integral character, and let Π satisfy Assumption 10.1.5, with $\Pi_\infty \in \mathfrak{P}_\lambda$. Let $\mathcal{M} = H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$ and $\mathcal{H}^i = H^i(X_G(\Gamma)_{\overline{\mathbb{Q}}}, [V])[\Pi_f]$, where both are regarded as $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian bilinear forms, induced from a fixed admissible bilinear form on $\mathfrak{g}_{\mathbb{C}}$ (see §10.2). Then, there is an isometry between graded $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian metrics,*

$$\wedge^* \mathcal{M}^* \otimes \mathcal{H}^{i_{\min}} \cong \bigoplus_{i=i_{\min}}^{i_{\max}} \mathcal{H}^i,$$

where V is the automorphic vector bundle coming from Levi (see Notation) such that Π_f appears in multiple degrees of its cohomology, $i_{\min} = \min\{i \mid H^i(X_G(\Gamma), [V])[\Pi_f] \neq 0\}$, and $i_{\max}$ is defined analogously.

This is the analogue of [PV, Conjecture 1.2.1], although new subtleties arise in the irregular weight case as we will see. Under some standard conjectures on periods, we show that the conjecture indeed holds in a few low-dimensional cases.

Theorem 2. *Under certain mild conditions and several standard conjectures on periods (see below), for $G = \mathrm{Sp}_4$ and $\mathrm{SU}(2,1)$, Conjecture 1 is true. In other words, the action of adjoint motivic cohomology group on the coherent cohomology groups of Shimura variety respects $\overline{\mathbb{Q}}^\times$-structure.*

The "mild conditions" are that, informally speaking, the finite part is globally generic, and that there are newforms; see Assumption 10.1.5. These conditions exist to have a clean statement of the conjectures. More importantly, the "standard conjectures on periods" are summarized in Assumption 10.4.5, which includes the Lapid–Mao conjecture and the Beilinson conjectures. The proof of Theorem 10.4.1 uses the same idea of [PV], but requires more machinery, as the whole setup is about comparing periods of *different automorphic representations.*

The cases where Theorem 2 is proved are when the Hecke eigensystem appears in two consecutive degrees of cohomology. In this case, or more generally, when we restrict the statement of Conjecture 1 into the relation between the top and bottom degrees, the conjecture can be made into a statement that does not refer to motivic cohomology.

More precisely, let $\Pi = \Pi_f \otimes \Pi_\infty$ be a cuspidal automorphic representation satisfying Assumption 10.1.5, and let λ be the infinitesimal character of Π_∞, which is singular and nondegenerate (see Notation). Let V be the automorphic vector bundle coming from the Levi (see Notation), such that $H^i(\mathfrak{p}, K; V \otimes \Pi_\infty) \neq 0$ for some i, and $\Pi_{\min}, \Pi_{\max}$ be the members of the archimedean L-packet of Π_∞ such that the degree that $\Pi_{\min}$ ($\Pi_{\max}$, respectively) has nontrivial $(\mathfrak{p}, K)$-cohomology with coefficient in V is the minimum (maximum, respectively) in the L-packet. Let $i_{\min}$ ($i_{\max}$, respectively) be the degree, and let $f_{\min} \in \Pi_f^{\mathrm{new}} \otimes \Pi_{\min}^{\mathrm{new}}$ and $f_{\max} \in \Pi_f^{\mathrm{new}} \otimes \Pi_{\max}^{\mathrm{new}}$ (see Definition 10.1.6) such that $[f_{\min}], [f_{\max}] \in H^*(X_G(\Gamma), V)$ (the harmonic Dolbeault forms corresponding to the automorphic forms; see Definition 10.1.8 for a precise definition) are defined over $\overline{\mathbb{Q}}$. Then, under the Beilinson conjectures, the information on the top and bottom degrees in Conjecture 1 is precisely that

$$\frac{\langle f_{\min}, f_{\min}\rangle}{\langle f_{\max}, f_{\max}\rangle} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left| \pi^{i_{\min} - i_{\max}} \frac{L_\infty(1, \Pi, \mathrm{Ad})}{L_\infty(0, \Pi, \mathrm{Ad})} \cdot \frac{L(1, \Pi, \mathrm{Ad})}{\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi)} \right|^2,$$

where $\langle \cdot, \cdot \rangle$ is the Petersson inner product, and the volume is computed with respect to the metric induced by any weak polarization (see [PV, §2.2.3])[1]. In particular, this statement is *equivalent to Conjecture 1 if there are only two degrees that* Π_f *appear in the cohomology*, assuming Beilinson conjectures.

At first sight, the information on top and bottom degrees might seem to less interesting, as one might guess there is an extra duality that one can compare the top and bottom degrees. Indeed, in [PV], the top and bottom degrees were complemen-

[1]The volume is independent of the choice of weak polarization, see [PV, Lemma 2.2.2].

tary, so that they are related via duality. On the other hand, in the irregular setting, there are numerous cases where the top and bottom degrees are not complementary. Rather, they are determined by the position of the infinitesimal character inside the Weyl chambers. Indeed, we work with two examples, $G = \mathrm{Sp}_4$ and $\mathrm{SU}(2,1)$, and both cases we work with the choice of λ where the minimum and maximum degree of appearances are H^0 and H^1, respectively.

9.1 The lack of rigidity for L-packets

We now explain the new subtleties of the irregular weight case. In view of automorphic cohomology, the phenomenon of a weight one modular form appearing in H^0 and H^1 actually involves *two different representations*. If we denote ω to be the weight one line bundle or the corresponding representation of $\mathrm{SO}(2)$, then for a weight one modular newform f,

$$H^0(X,\omega)[f] = H^0(\mathfrak{p}, \mathrm{SO}(2); D_0^+ \otimes \omega),$$

$$H^1(X,\omega)[f] = H^1(\mathfrak{p}, \mathrm{SO}(2); D_0^- \otimes \omega),$$

where $\mathfrak{p} = \mathfrak{p}_- \oplus \mathfrak{so}(2)$, $\mathfrak{p}_-$ is the anti-holomorphic tangent space and D_0^+ and D_0^- are the holomorphic and the antiholomorphic discrete series, respectively. Indeed, it is $\overline{f}d\overline{z}$ that appears in $H^1(X,\omega)$ as a class in Dolbeult cohomology, and $\overline{f}$ is an *antiholomorphic* modular form of weight -1.

In general, if the same Hecke eigensystem appears in multiple degrees of cohomology of the same automorphic vector bundle of a Shimura variety, then each such instance is actually represented by the so-called $(\mathfrak{p}, K)$-cohomology of different automorphic representations. More precisely, the finite part ($G(\mathbb{A}_f)$-representation) remains the same, while the infinity type varies inside an archimedean L-packet. Such phenomenon happens precisely when the infinitesimal character of the infinity type

lies on the walls of Weyl chambers.

The action of $\wedge^* \mathfrak{a}_G^*$ in [PV] is most naturally thought as the self-Ext-algebra of a representation. On the other hand, in our case, the representations corresponding to the target and the source of the action is different, so the action cannot be thought as an Ext-*algebra*, but merely as an Ext-group. Furthermore, the translation of the action into the automorphic cohomology context also depends on the choice of *automorphic realization maps* $\Pi_f \otimes \Pi_\infty \hookrightarrow \mathcal{A}(G)$ for each Π_∞; namely, each realization map can be always scaled by a scalar, but we want to compare them as a whole. This problem did not arise in *op. cit.*, as a choice of a single automorphic realization map would rigidify the situation. In turn, we had no choice but to formulate a slightly weaker Conjecture 1 that asserts the motivic action conjecture on the level of metric spaces.

9.2 Comparison of periods of different automorphic representations

We briefly explain the strategy of the proof of Theorem 2. As done in [PV], we most notably assume *Beilinson's conjectures (for Chow motives)*. The main new feature in this paper is that, because we need to compare periods of two different representations, we need two different conjectures on periods, one for each representation. In both evidences, we will compare periods of a *holomorphic automorphic form* and a *generic automorphic form*. For the holomorphic form, we will need the *refined Gan–Gross–Prasad conjecture* (referred as Ichino–Ikeda conjecture in *op. cit.*), although in some instances this requirement can be avoided by using the doubling construction of standard L-functions. For the generic form, we will need the *Lapid–Mao conjecture*, which relates the value of a Whittaker function to a certain L-value.

One also needs a way to detect rationality of classes for both types of forms. For

the holomorphic forms, we can use Fourier expansion, but for those appearing in higher coherent cohomology, we need a new machinery. We will develop the so-called *cohomological period integrals* for higher coherent cohomology, which realizes integral representations of L-functions as cup product pairings in coherent cohomology. The cohomological interpretation of such integral representations (or, at least, their appearance in the literature) is relatively new ([LPSZ], or Part III of this thesis).

9.3 Generalized complex conjugations

The archimedean motivic action conjecture as stated in [PV] gives a recipe of rational cohomology classes, whereas our Conjecture 1 is a statement on metrics. To formulate a similar conjecture in the setting of coherent cohomology of Shimura varieties, we need a way to rigidify between different automorphic representations in an archimedean L-packet. Indeed, in retrospect, even in the easy case of modular forms, one needs complex conjugation to go between holomorphic and antiholomorphic limits of discrete series. Unfortunately, beyond the case of modular forms, there is no known general operation that can move between different infinity types. We will tentatively name such an operation a *generalized complex conjugation*, which should send an automorphic form of a certain infinity type to an automorphic form of another infinity type in the same L-packet. It seems inevitable to come up with such an operation to formulate the full conjecture on rational cohomology classes.

The generalized complex conjugations should be naturally understood in the context of a "derived" local-global compatiblity in some sense, and their existence is also suggested by the existence of similar operations in the analogous settings over the p-adic fields (Kottwitz's conjectures, e.g. [FaMa]) and over the function fields (excursion operators, e.g. [Laf]). Following the suggestions of Joseph Wolf, we will investigate the nature of generalized complex conjugations using the theory of Penrose

transforms.

On the other hand, if the associated Hermitian symmetric space is a product of copies of the upper half planes, one can come up with an operation that changes one infinity type to another by taking complex conjugation at certain variables. This is a *partial complex conjugation*, studied in [Har4]. Using partial complex conjugations, we can formulate a conjecture on rationality of cohomology classes in the case of Hilbert modular forms of partial weight one. There exists a prior work of [Ho] on the motivic action conjecture for Hilbert modular forms of parallel weight one, which similarly uses partial complex conjugations. We compare our conjecture in the Hilbert modular form case with the conjecture of [Ho], and explain the evidences given in *op. cit.* are also consistent with our conjecture.

9.4 Summary

In §10, we take an efficient route to the statement of the Archimedean motivic action conjecture (Conjecture 10.3.1) and its more accessible variant, the Period conjecture (Conjecture 10.3.3). The objective of the section is to set the conjecture in a context. We in particular defer the abstract discussion of how to derive the conjectures, parallel to those of [PV, §2-§5], to later sections, as it requires more advanced theory on representation theory of real groups.

In §11 and §12, we provide our main evidence for the Period conjecture (Conjecture 10.3.3). Similarly to [PV, §7], we prove that, for $G = \mathrm{Sp}_4$ and $\mathrm{SU}(2,1)$, the Period conjecture is compatible with several well-accepted conjectures on periods of automorphic forms, such as the Beilinson's conjectures, the Lapid–Mao conjecture and the refined Gan–Gross–Prasad conjectures. To use these, we review how certain period integrals can be interpreted as cup product pairings of (higher) coherent cohomology classes on Shimura varieties.

In §13, we discuss the issue on formulating a motivic action conjecture on rationality of coherent cohomology classes. Most notably, we suggest the notion of *generalized complex conjugations*, which move between different members of a single archimedean L-packet. In §13.1, we formulate a precise conjecture in the case of Hilbert modular forms using partial complex conjugations, and compare our conjecture with the conjecture of [Ho]. In §13.2, we spell out conditions that the generalized complex conjugations should satisfy, and formulate the full conjecture assuming their existence. In §14, we review the formulation of Beilinson's conjecture for motives over a general number field, as many references state the conjecture for only $\mathbb{Q}$-motives. Finally, we develop a representation theory background in §15, parallel to [PV, §2-§4]. Although §15 is independent of the development of the rest of the paper, the section is suggestive of a correct foundation in which the motivic action conjectures need to be developed.

9.5 Problems and questions

There are several interesting questions that arise in this work.

1. Place the generalized complex conjugations in the context of some form of "derived" local-global compatibility, motivated by the strong form of Arthur conjectures as realized in the function field case via excursion operators as in [Laf]. A correct formulation should be in accordance with the existing statements of derived local-global compatibility as in [Fe] and [Zhu].

2. The compatibility between the Period Conjecture, Conjecture 10.3.3, and the existing period conjectures relies on the yet-to-be-calculated archimedean zeta integrals. These can be conducted using explicit integral formulae of (generalized) Whittaker functions, e.g. [KO], [Od1].

3. As we deal with motives over a more general number field and Shimura varieties over a number field other than $\mathbb{Q}$, in every aspect of our discussion, the choice of a complex embedding is always implicit. In particular, there must be a relation between the conjectures in this paper for the *conjugates of Shimura varieties* (e.g. [Va]).

4. It seems extremely hard to detect rationality of coherent cohomology classes if the Hecke operators can only cut a space that is of dimension larger than one. For example, if X is a Hilbert modular surface, and if ω is the parallel weight one line bundle, then it seems extremely difficult to determine whether a class in $H^1(X, \omega)$ is defined over $\overline{\mathbb{Q}}$, or even to produce a class in it.

5. It is expected that the motivic action conjecture will involve L-packets even in the case of $\delta > 0$. It may be possible to formulate the conjecture for the same eigensystem appearing in cohomology with *different coefficients*.

9.6 Notation

Let G be a connected reductive algebraic group over $\mathbb{Q}$. For simplicity, let us assume that G is quasisplit, $G(\mathbb{R})$ is connected, and that the center of G does not have a nontrivial $\mathbb{R}$-split torus. Also, we assume that there exists a *twisting element* in the sense of [BG, Definition 5.2.1][2]. Let $\mathfrak{g}_{\mathbb{Q}}$ be the $\mathbb{Q}$-Lie algebra of G, and let $\mathfrak{g}_{\mathbb{R}}, \mathfrak{g}_{\mathbb{C}}$ be its base change to $\mathbb{R}$ and $\mathbb{C}$, respectively. We occasionally drop the subscript for $\mathbb{C}$. Let W_G be the Weyl group of G. We endow an invariant, θ-invariant, $\mathbb{R}$-invariant bilinear form B on $\mathfrak{g}_{\mathbb{R}}$, such that $B(X, \theta(X))$ is negative definite, where θ is the Cartan involution. We will use this to talk about inner product on weight space, Riemannian metric on the Hermitian symmetric domain, etc. In specific examples, we may and will choose B to induce a preferred Riemannian metric on the Hermitian symmetric

[2]This is to avoid the subtlety of difference between C-algebraicity and L-algebraicity.

space (for example, one may want the Riemannian metric to be $dxdy$ on the upper half plane $\mathbb{H} = \{x + iy \mid y > 0\}$). We also use B to any other bilinear form induced from B.

Suppose further that G gives rise to a *Shimura variety*, which means that there is a symmetric space X for $G(\mathbb{R})$ which can be endowed with a structure of Hermitian symmetric domain (which we will fix). Fix a point $h \in X$, which gives rise to a *Hodge cocharacter* $h : \mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{R}}$ which in turn induces a real Hodge structure of weight 0 on $\mathfrak{g}_{\mathbb{R}}$, $\mathfrak{g} = \mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{0,0} \oplus \mathfrak{g}^{1,-1}$. Given an open compact subgroup $\Gamma \subset G(\mathbb{A}_f)$, there exists a quasi-projective variety $Y_G(\Gamma)$, a Shimura variety, defined over a number field E, whose complex points[3] have an analytification isomorphic to the double quotient $G(\mathbb{Q})\backslash(X \times G(\mathbb{A}_f)/\Gamma)$.

Let $K \subset G(\mathbb{R})$ be the stabilizer of h. Let T be the Cartan subgroup of K. Then, $X \cong G(\mathbb{R})/K$ and $\mathfrak{g}^{0,0} = \mathfrak{k} := \mathrm{Lie}(K)_{\mathbb{C}}$. We denote $\mathfrak{p}_+ = \mathfrak{g}^{-1,1}$, $\mathfrak{p}_- = \mathfrak{g}^{1,-1}$ and $\mathfrak{p} = \mathfrak{k} \oplus \mathfrak{p}_-$. Then, $\mathfrak{p}$ is a parabolic subalgebra of $\mathfrak{g}$, giving rise to a parabolic subgroup $P \subset G_{\mathbb{C}}$ with $\mathrm{Lie}\, P = \mathfrak{p}$. We also fix once and for all a positive system of roots for $\mathfrak{k}$. The holomorphic tangent space of X at h is identified with $\mathfrak{p}_+$, so there is a $G(\mathbb{R})$-equivariant embedding of complex manifolds $X \to \check{D} := G(\mathbb{C})/P(\mathbb{C})$, sending $h \mapsto P(\mathbb{C})$. In this regard, $K = G(\mathbb{R}) \cap P(\mathbb{C})$, and $K(\mathbb{C})$ is the Levi subgroup of $P(\mathbb{C})$. Also, any finite-dimensional holomorphic representation V of $P(\mathbb{C})$ gives rise to an algebraic vector bundle over $Y_G(\Gamma)$, an *automorphic vector bundle*, denoted $[V]$, which is an algebraization of the pullback of the holomorphic vector bundle on X which in turn is the restriction of the vector bundle $G(\mathbb{C}) \times^{P(\mathbb{C})} V \to \check{D}$ on $\check{D}$. If V factors through $K(\mathbb{C})$, namely if it is induced from a representation of $K(\mathbb{C})$, we will call $[V]$ an automorphic vector bundle *coming from the Levi*. If not, we will call V

[3]Note that the choice of a point in a Hermitian symmetric domain gives the reflex field as a subfield of $\mathbb{C}$, so there is a *preferred complex embedding*; e.g. [Va, Notation 4.6]. In particular, one can expect that the statement of the conjecture depends a priori on the choice of a Hermitian symmetric domain. It could be interesting to check if our conjecture is consistent with conjugation of Shimura varieties.

nearly, following [LPSZ].

To save space, we may abbreviate some words with repeated appearances: discrete series into DS, limit of discrete series into LDS, and nondegenerate limit of discrete series (see the paragraph before Theorem 10.1.4 for its definition) into NLDS. All real group representations are thought as $(\mathfrak{g}, K)$-modules.

For automorphic representations, their L-functions are normalized so that $\frac{1}{2}$ is the center of symmetry. For pure motives, theire L-functions are normalized so that, if w is its weight, $\frac{w+1}{2}$ is the center of symmetry. For both kinds of L-functions, w is called the *motivic weight* of the L-function. For an automorphic representation $\Pi = \Pi_f \otimes \Pi_\infty$ of $G(\mathbb{A})$, the field of rationality F_Π is the fixed field of the isomorphism class of Π_f as a $G(\mathbb{A}_f)$-representation ([Cl, §3.1]). An inner product on the space of automorphic forms, denoted $\mathcal{A}(G)$, can be given as either the L^2-norm on $G(\mathbb{Q})/G(\mathbb{A})$ with respect to the Tamagawa measure or the measure coming from the Riemannian metric of the symmetric space, as the norms are all scaled by the same scalar factor. The second norm is the same as the usual *Petersson norm*, which we will denote as $\langle , \rangle_P$.

For the integral representations, we fix a nontrivial additive character $\psi = \prod_p \psi_p$ of $\mathbb{A}/\mathbb{Q}$. For a cuspidal automorphic form $\varphi = \prod_p \varphi_p$ of cuspidal automorphic representation $\pi = \bigotimes_p' \pi_v$ of $G(\mathbb{A})$, its Whittaker transform $W_\varphi(g)$ is defined as $W_\varphi(g) = \int_{N_G(\mathbb{Q})\backslash N_G(\mathbb{A})} \varphi(ng)\psi(n^{-1})dn$, for some choice of N_G that needs to be specified when talking about Whittaker model. Locally, a generic $G(\mathbb{Q}_p)$-representation π_p is isomorphic to the space of functions $\mathcal{W}_p := \{W_p : G(\mathbb{Q}_p) \to \mathbb{C} \mid W_p(ng) = \psi_p(n)W_p(g) \text{ for } n \in N_G(\mathbb{Q}_p)\}$; one can choose isomorphisms (a local *Whittaker model*) $\pi_p \xrightarrow{\sim} \mathcal{W}_p$, $f_p \mapsto W_{f_p}$ that are compatible with the global Whittaker model; namely, $W_\varphi(g) = \prod_p W_{\varphi_p}(g_p)$.

We use notational convention for motives as in [PV, §2], except that we will deal with motives over a more general number field. In that case, we put the complex

embedding in the subscript, such as M_σ, $\mathrm{comp}_{B,\mathrm{dR},\sigma}$, etc.

Chapter 10

Archimedean L-packets and the motivic action conjecture

In this section, we take the shortest path to the statement of the Archimedean motivic action conjecture for Shimura varieties, Conjecture 10.3.1. More abstract justification of the formulation of the Conjecture, including a parallelism between [PV] and our conjecture, is discussed in §15 and §13.2.

10.1 $(\mathfrak{p}, K)$-cohomology and automorphic forms

Firstly, we quickly review how coherent cohomology of Shimura varieties is related to automorphic forms via the theory of $(\mathfrak{p}, K)$-cohomology. Recall that, as the singular cohomology of locally symmetric spaces can be calculated in terms of $(\mathfrak{g}, K)$-cohomology, the coherent cohomology of Shimura varieties can be calculated in terms of the so-called $(\mathfrak{p}, K)$-cohomology. By reinterpreting what the Dolbeault cohomology calculates in the setting of Shimura varieties, one gets the following

Proposition 10.1.1 (See [Su, (2.12)])**.** *We have*

$$H^i(Y_G(\Gamma), [V]) \cong H^i(\mathfrak{p}, K; C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma)^{K-\text{finite}} \otimes V),$$

where the left hand side is analytic cohomology, and V is understood as a $(\mathfrak{p}, K)$-module with trivial $\mathfrak{p}$-action.

Furthermore, there is an analogue of Franke's theorem for coherent cohomology of Shimura varieties.

Theorem 10.1.2 (Su, [Su, Theorem 6.7])**.** *For any sufficiently refined polyhedral cone decomposition Σ, there is a natural Hecke-equivariant isomorphism*

$$H^i(X_G^\Sigma(\Gamma), [V]^{\mathrm{can}}) \cong H^i(\mathfrak{p}, K; \mathcal{A}(G)^\Gamma \otimes V),$$

where $X_G^\Sigma(\Gamma)$ is the corresponding toroidal compactification, $[V]^{\mathrm{can}}$ is the canonical extension of $[V]$, and $\mathcal{A}(G)$ is the space of automorphic forms, namely the space of right K-finite, $Z(\mathfrak{g})$-finite smooth functions on $G(\mathbb{Q})\backslash G(\mathbb{A})$ of moderate growth.

Remark 10.1.3. We would be only interested in a part of coherent cohomology localized at a cuspidal Hecke eigensystem, so it is unlikely that the full power of Su's theorem is required.

The calculation of coherent cohomology is therefore about $(\mathfrak{p}, K)$-cohomology of automorphic representations. As far as the coherent cohomology is concerned, the choice of Σ is ineffective, so we may occasionally drop the superscript Σ if there is no confusion.

We will be interested in the situation where a Hecke eigensystem appears in multiple degrees of coherent cohomology. Namely, for an admissible $G(\mathbb{A}_f)$-representation Π_f and an automorphic vector bundle $\mathcal{E}$ of $Y_G(\Gamma)$, we are interested in

$$H^*(X_G(\Gamma), \mathcal{E}^{\mathrm{can}})[\Pi_f].$$

By Theorem 10.1.2, we have a canonical isomorphism

$$H^*(X_G(\Gamma), \mathcal{E}^{\mathrm{can}}) \cong \left(\bigoplus_{\Pi_f \otimes \Pi_\infty \subset \mathcal{A}(G)} H^*(\mathfrak{p}, K; \Pi_\infty \otimes E) \right) \otimes_{\mathbb{C}} \Pi_f^\Gamma,$$

where the sum runs over all automorphic representations with the finite part being Π_f, and E is the algebraic $P(\mathbb{C})$-representation such that $[E] = \mathcal{E}$. Thus, it is possible that *several different* Π_∞*'s can appear in the decomposition*, if $H^*(\mathfrak{p}, K; \Pi_\infty \otimes E) \neq 0$ for several different Π_∞'s. Indeed, this can be the case if, for example, some of Π_∞ is a nondegenerate limit of discrete series (NLDS) and not a discrete series (DS); recall that a *nondegenerate limit of discrete series* is a limit of discrete series whose infinitesimal character is not orthogonal to any compact root. Unlike the case of "$\delta > 0$" as in [PV], the appearance of single Hecke eigensystem in multiple cohomological degrees in our setting necessarily implies that, by the following Theorem, there are *many different archimedean representations involved*:

Theorem-Definition 10.1.4 (See [VZ], [Sc2])**.** Let Π_∞ be the $(\mathfrak{g}, K)$-module associated to a DS or a NLDS representation of $G(\mathbb{R})$. Then, there is a unique $0 \leq i \leq \dim X$ and a finite-dimensional irreducible K-representation V such that $H^i(\mathfrak{p}, K; \Pi_\infty \otimes V) \neq 0$. Furthermore, $\dim_{\mathbb{C}} H^i(\mathfrak{p}, K; \Pi_\infty \otimes V) = 1$. We will denote i_{Π_∞} and V_{Π_∞} for the i and V corresponding to Π_∞.

Indeed, the above Theorem says that a single archimedean representation can only contribute to a single degree. We will see that the *raison d'être* of appearance of a Hecke eigensystem in multiple degrees is that the infinity type Π_∞ can change in an archimedean L-packet without changing the finite part. We will see in detail in §15 how an archimedean L-packet (rather than a single $G(\mathbb{R})$-representation) appears in the context of motivic action. For now, we move on to the formulation of the "metric" conjecture which does not involve abstract real group representation theory nor Lie algebra cohomology. From now on, for the sake of simplicity, we assume the following

Assumption 10.1.5. Let $\Pi = \Pi_f \otimes \Pi_\infty$, with $\Pi_f = \prod_{p<\infty} \Pi_p$, be a cuspidal automorphic automorphic representation with Π_∞ an NLDS (see Notation)[1]. We hereafter assume the following:

$$\Pi_f \text{ is globally generic.} \tag{GEN}$$

For each $p < \infty$, there exists a compact open subgroup $\Gamma_p \leq G(\mathbb{Z}_p)$ such that $\dim_{\mathbb{C}} \Pi_p^{\Gamma_p} = 1$.

(NEW)

If $G = \mathrm{Sp}_4$, a holomorphic Siegel modular newform f_Π in $\Pi_f \otimes \Pi_\infty^{\mathrm{hol}}$, where $\Pi_\infty^{\mathrm{hol}}$ is a holomorphic (limit of) discrete series, has a nontrivial special Bessel period $B(f_\Pi, F) \neq 0$ for some imaginary quadratic field F. Here,

$B(f_\Pi, F) = \sum_{A=\left(\begin{smallmatrix} a & b/2 \\ b/2 & c \end{smallmatrix}\right), 4ac-b^2=D_F, A\sim M^T AM \text{ for } M\in \mathrm{SL}_2(\mathbb{Z})} a_A$, where $f_\Pi(Z) = \sum_S a_S e^{2\pi i \operatorname{tr}(SZ)}$ is the q-expansion of f_Π, and $-D_F$ is the discriminant of F.

(Sp$_4$)

If $G = \mathrm{SU}(2,1)$, defined using an imaginary quadratic field K, (1) there exists a split place v such that Π_v is supercuspidal, and (2) if p is not split in K where Π_p is also not unramified, either Π_p is supercuspidal or the stabilizer of an anisotropic vector $\mathrm{SU}(1,1)(\mathbb{Q}_p) \subset \mathrm{SU}(2,1)(\mathbb{Q}_p)$ is compact.

(SU(2, 1))

The purpose of Assumption 10.1.5 is to isolate the effect of archimedean L-packet phenomenon amongst others and to use existing results on the refined Gan–Gross–Prasad conjectures. We believe that, for example, it would be not difficult to formulate the Conjectures without (NEW).

Definition 10.1.6. Under Assumption 10.1.5, let $\Gamma(\Pi_f) = \Gamma := \prod_p \Gamma_p$. Also, when we say a vector $f \in \Pi$ is a *newform*, it means $f = \prod_p f_p \otimes f_\infty$ such that, not only $f_p \in \Pi_p^{\Gamma_p}$, but also f_∞ is a highest weight vector of the minimal K-type of Π_∞ (which exists as we will be only concerned about either DS or LDS). We define

[1] This would ensure that F_Π, the field of definition, is a number field, by our assumption in Notation that there exists a twisting element in the sense of [BG, Definition 5.2.1]. Note that the twisting element indeed exists for $G = \mathrm{Sp}_4$ (as it is split and has simply-connected derived subgroup) and $\mathrm{SU}(2,1)$ (as the half-sum of positive roots is integral).

$\Pi_f^{\text{new}} = \prod_p \Pi_p^{\Gamma_p}$, and also Π_∞^{new} to be the one-dimensional $\mathbb{C}$-vector subspace of Π_∞ generated by highest weight vectors of the minimal K-type. In particular, f being a newform means that $f \in \Pi_f^{\text{new}} \otimes \Pi_\infty^{\text{new}}$.

In view of Theorem-Definition 10.1.4, the above "newform" appears in

$$H^{i_{\Pi_\infty}}(X_G(\Gamma), [V_{\Pi_\infty}]^{\text{can}})[\Pi_f],$$

which we denote by $H^{i_{\Pi_\infty}}(X)[\Pi_f]$.

Note that a choice of a highest weight vector of V_{Π_∞} induces a natural isomorphism

$$H^{i_{\Pi_\infty}}(\mathfrak{p}, K; \Pi_\infty \otimes V_{\Pi_\infty}) \cong \Pi_\infty^{\text{new}},$$

defined as follows. Given $v \in \Pi_\infty^{\text{new}}$, define a K-homomorphism

$$f : \left(\wedge^{i_{\Pi_\infty}}(\mathfrak{p}/\mathfrak{k})\right) \otimes V_{\Pi_\infty}^* \to \Pi_\infty,$$

by sending the highest weight vector (induced from the choice of a highest weight vector of V_{Π_∞} and the roots of $\mathfrak{g}$) of the highest K-type of the source to v and sending all other K-types to zero. This defines a class in $\text{Hom}_K(\wedge^{i_{\Pi_\infty}}(\mathfrak{p}/\mathfrak{k}), \Pi_\infty \otimes V_{\Pi_\infty})$ which is closed in the corresponding Chevalley–Eilenberg complex for the $(\mathfrak{p}, K)$-cohomology, thus a class in $H^{i_{\Pi_\infty}}(\mathfrak{p}, K; \Pi_\infty \otimes V_{\Pi_\infty})$.

Remark 10.1.7. In this paper, each conjecture will consider a fixed finite type Π_f and a coefficient vector bundle V, so in that context we firstly fix a choice of a highest weight vector of V before anything else.

Definition 10.1.8. For $f = \prod_{p \leq \infty} f_p \in \Pi_f^{\text{new}} \otimes \Pi_\infty^{\text{new}}$, we define $[f] \in H^{i_{\Pi_\infty}}(X)[\Pi_f]$

to be the class corresponding to

$$[f_\infty] \otimes \prod_{p<\infty} f_p \in H^{i_\Pi\infty}(\mathfrak{p}, K; \Pi_\infty \otimes V_{\Pi_\infty}) \otimes \Pi_f^{\mathrm{new}} \subset H^{i_\Pi\infty}(X)[\Pi_f],$$

where $[f_\infty] \in H^{i_\Pi\infty}(\mathfrak{p}, K; \Pi_\infty \otimes V_{\Pi_\infty})$ corresponds to $f_\infty \in \Pi_\infty^{\mathrm{new}}$ via the above natural isomorphism $H^{i_\Pi\infty}(\mathfrak{p}, K; \Pi_\infty \otimes V_{\Pi_\infty}) \cong \Pi_\infty^{\mathrm{new}}$.

10.2 Metrics on cohomology

To formulate the main Conjecture 10.3.1, we need to define metrics on the coherent cohomology of Shimura varieties as well as motivic cohomology. To define metrics on both motivic cohomology and Dolbeault cohomology, we fix an *admissible bilinear form* on $\mathfrak{g}_\mathbb{C}$ in the following sense.

Definition 10.2.1 (Admissible bilinear form)**.** An *admissible bilinear form* B on $\mathfrak{g}_\mathbb{C}$ is an invariant, θ-invariant Hermitian bilinear form on $\mathfrak{g}_\mathbb{C}$ such that the following conditions are satisfied.

1. It is a natural extension of a $\mathbb{R}$-valued bilinear form on $\mathfrak{g}_\mathbb{R}$ such that $B(X, \theta(X))$ is negative definite, where θ is the Cartan involution that fixes $\mathfrak{k}_\mathbb{R}$.

2. It is $\overline{\mathbb{Q}}$-valued on $\mathfrak{g}_{\overline{\mathbb{Q}}}$.

For example, if $G(\mathbb{R})$ is semisimple, the extention of the Killing form as a Hermitian bilinear form on $\mathfrak{g}_\mathbb{C}$ is an admissible bilinear form.

Firstly, the coherent cohomology of toroidal compactifications of Shimura varieties

$$H^*(X_G(\Gamma), [V]^{\mathrm{can}}),$$

can be given a Hermitian metric[2], induced from a Hermitian metric on the Hermitian

[2]This coincides with the metric on the Lie algebra cohomology (see e.g. [BW, §II.2]), and this will be reviewed later in §15.

symmetric domain and the automorphic vector bundle (which is always possible by the compactness of K, see e.g. [BKK, §5]), which is in turn induced from our choice of admissible bilinear form on $\mathfrak{g}_{\mathbb{C}}$. The choice of Hermitian metric induces Hermitian metrics on the entries of Dolbeault complex $\mathscr{A}^{0,*}([V]^{\mathrm{can}})$. By taking the formal adjoint $\overline{\partial}^*$ of $\overline{\partial}$, one can define the Laplacian $\Delta = \overline{\partial}\overline{\partial}^* + \overline{\partial}^*\overline{\partial}$ on each entry of Dolbeault complex. By Hodge theory, the Dolbeault cohomology $H^i([V]^{\mathrm{can}})$ is identified with the space of *harmonic* $(0,i)$*-forms* $\mathscr{H}^i([V]^{\mathrm{can}})$, which is just the kernel of the Laplacian Δ. The restriction of Hermitian metric on $\mathscr{A}^{0,i}([V]^{\mathrm{can}})$ to the space of harmonic $(0,i)$-forms gives rise to a Hermitian metric on the Dolbeault cohomology.

On the other hand, the motivic cohomology that would have to appear in the motivic action conjectures is that of the adjoint motive. As in [PV, §4.2], we assume a conjecture on the existence of adjoint motive.

Conjecture 10.2.2. *For* $\Pi = \Pi_f \otimes \Pi_\infty$ *satisfying Assumption 10.1.5, there exists an adjoint motive* $\mathrm{Ad}\,\Pi$*, in the sense of [PV, Definition 4.2.1], over the reflex field* E *with coefficients in* F_Π*.*

To endow a Hermitian metric on the adjoint motivic cohomology $H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$, consider the Beilinson regulator for $\mathrm{Ad}\,\Pi$. Recall that the Beilinson regulator is a map from motivic cohomology to Deligne cohomology,

$$H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \mathbb{Q}(1)) \to H^1_{\mathscr{D}}((\mathrm{Ad}\,\Pi)_{\mathbb{R}}, \mathbb{R}(1)),$$

where the Deligne cohomology group of a motive M over a number field k is defined as (see [Ra, (6.1.22)])

$$H^i_{\mathscr{D}}(M_{\mathbb{R}}, A) = \prod_{w \text{ complex places of } k} H^i_{\mathscr{D}}(M \times_{k,w} \mathbb{C}, A) \times \prod_{w \text{ real places of } k} H^i_{\mathscr{D}}(M \times_{k,w} \mathbb{R}, A).$$

Under the Beilinson's conjectures, the Beilinson regulator gives rise to an isomorphism

$$H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \mathbb{Q}(1)) \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} H^1_{\mathscr{D}}((\mathrm{Ad}\,\Pi)_{\mathbb{R}}, \mathbb{R}(1)).$$

Remark 10.2.3. Note that the Betti realization, on which the Beilinson regulator depends, *depends on the complex embedding* $E \hookrightarrow \mathbb{C}$. In our discussion, we use the preferred embedding that came with the datum of reflex field[3]. In the following discussions, we always use this complex embedding.

From §15.3, we know that the target of the Beilinson regulator is identified with $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})}$, where $\varphi : W_{\mathbb{C}/\mathbb{R}} \to {}^L G$ is the corresponding Lanvlands parameter, and that this is identified as a Lie subalgebra of $\widehat{\mathfrak{t}}$. We define a Hermitian bilinear form on $\widehat{\mathfrak{t}}$ as the dual Hermitian bilinear form of the one we chose for $\mathfrak{t}$, and this restricts to a Hermitian bilinear form on the Deligne cohomology.

Definition 10.2.4. Let X be a finite-dimensional $\overline{\mathbb{Q}}$-vector space, together with an embedding $\iota : \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$. A *Hermitian bilinear form* on X is a Hermitian metric on $X \otimes_{\overline{\mathbb{Q}},\iota} \mathbb{C}$.

By [PV, Lemma 2.2.2], the volume of $H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$ is in fact independent of choice of the admissible Hermitian bilinear form.

10.3 Archimedean motivic action conjecture for Shimura varieties

Now, we are able to state the Archimedean motivic action conjecture for Shimura varieties as follows.

Conjecture 10.3.1 (Archimedean motivic action conjecture for Shimura varieties, metric version)**.** *Let* $\Pi = \Pi_f \otimes \Pi_\infty$ *satisfy Assumption 10.1.5, with* Π_∞ *being a NLDS*

but not being a DS. Let $\mathcal{M} = H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$ *and* $\mathcal{H}^i = H^i(X)[\Pi_f]$*, where both are regarded as* $\overline{\mathbb{Q}}$*-vector spaces equipped with a Hermitian bilinear form (see Definition 10.2.4), induced from a fixed admissible Hermitian bilinear form on* $\mathfrak{g}_{\mathbb{C}}$*. Then, there is an isomorphism of graded* $\overline{\mathbb{Q}}$*-vector spaces equipped with Hermitian metrics,*

$$\wedge^* \mathcal{M}^* \otimes \mathcal{H}^{i_{\min}} \cong \bigoplus_{i=i_{\min}}^{i_{\max}} \mathcal{H}^i,$$

where $i_{\min}$ *and* $i_{\max}$ *are the bottom and top degrees, respectively, of appearance of* Π_f *in the cohomology* $H^*(X)[\Pi_f]$*.*

Remark 10.3.2. It seems that, to descend the coefficient field from $\overline{\mathbb{Q}}$ to a number field, one may have to take a field larger than EF_Π (the compositum of the reflex field and the field of definition of Π_f) even in the case of $\mathrm{SL}_2(\mathbb{Q})$; see [Ho, Corollary 4.6].

The relationship between the above Conjecture and the philosophy of motivic action conjectures will be fully discussed in §15.

A special subset of the Main conjecture (Conjecture 10.3.1) concerning the norms of *top and bottom degrees* can be formulated without reference to motivic cohomology, assuming Beilinson's conjectures for Chow motives [PV, Conjecture 2.1.1].

Conjecture 10.3.3 (Comparison of top and bottom in Conjecture 10.3.1)**.** *Let* Π *be as in Conjceture 10.3.1. Let* V *be the automorphic vector bundle coming from the Levi, such that* $H^i(\mathfrak{p}, K; V \otimes \Pi_\infty) \neq 0$ *for some* i*, and* $\Pi_{\min}, \Pi_{\max}$ *be the members of the archimedean* L*-packet of* Π_∞ *such that the degree that* $\Pi_{\min}$ *(*$\Pi_{\max}$*, respectively) has nontrivial* $(\mathfrak{p}, K)$*-cohomology with coefficient in* V *is the minimum (maximum, respectively) cohomological degree, denoted* $i_{\min}$ *(*$i_{\max}$*, respectively) in the* L*-packet. Let* $f_{\min} \in \Pi_f^{\mathrm{new}} \otimes \Pi_{\min}^{\mathrm{new}}$ *and* $f_{\max} \in \Pi_f^{\mathrm{new}} \otimes \Pi_{\max}^{\mathrm{new}}$ *such that* $[f_{\min}], [f_{\max}] \in H^*(X)[\Pi_f]$ *are defined over* $\overline{\mathbb{Q}}$*. Then,*

$$\frac{\langle f_{\min}, f_{\min}\rangle_P}{\langle f_{\max}, f_{\max}\rangle_P} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left| \pi^{i_{\min} - i_{\max}} \frac{L_\infty(1, \Pi, \mathrm{Ad})}{L_\infty(0, \Pi, \mathrm{Ad})} \cdot \frac{L(1, \Pi, \mathrm{Ad})}{\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi)} \right|^2,$$

where the volume is computed with respect to the metric induced by any weak polarization (see [PV, §2.2.3])[3].

Proposition 10.3.4. *Assuming the Beilinson conjecture, Conjecture 10.3.3 is equivalent to the isometry statement in Conjecture 10.3.1 for top and bottom degrees,*

$$\wedge^{\mathrm{top}}\mathcal{M}^* \otimes \mathcal{H}^{i_{\min}} \cong \mathcal{H}^{i_{\max}}.$$

In particular, if $i_{\max} = i_{\min} + 1$, Conjecture 10.3.1 and Conjecture 10.3.3 are equivalent.

Proof. The key is to compare the top and the bottom degrees of the graded vector spaces and relate them with the volumes, which is independent of the choice of metric. Namely, we need to prove the analogue of [PV, Lemma 2.2.2],

$$\mathrm{vol}_S\, H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1)) \sim_{\overline{\mathbb{Q}}^\times} \frac{L^*(0, \mathrm{Ad}\,\Pi)}{\mathrm{vol}_S\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi)},$$

for S the Hermitian inner product induced by a weak polarization, and the statement will follow after applying functional equation. This now follows from the Beilinson's conjecture over a general number fields, as in [Ra, §6]. Namely, there is a fundamental exact sequence, [Ra, (6.4.2)],

$$0 \to F^1 H_{\mathrm{dR}}((\mathrm{Ad}\,\Pi)_{\mathbb{C}}) \to H^0_B((\mathrm{Ad}\,\Pi)_{\mathbb{R}}, \mathbb{C}) \to H^1_{\mathscr{D}}((\mathrm{Ad}\,\Pi)_{\mathbb{R}}, \mathbb{C}(1)) \to 0, \qquad (\star)$$

where

$$H^0_B((\mathrm{Ad}\,\Pi)_{\mathbb{R}}, \mathbb{C}) = \prod_{w \text{ complex places of } E} H^0_B(M \times_{E,w} \mathbb{C}, \mathbb{C}) \times \prod_{w \text{ real places of } E} H^0_B(M \times_{E,w} \mathbb{R}, \mathbb{C}),$$

is similarly defined as the "real Deligne cohomology". The Beilinson's conjecture over

[3] The volume is independent of the choice of weak polarization, see [PV, Lemma 2.2.2].

a general number field says that the determinant of the fundamental exact sequence ($\star$) has incompatible $\overline{\mathbb{Q}}$-rational structures, and are off precisely by $L^*(0, \operatorname{Ad}\Pi)$ (see [Ra, §6.4]):

$$\det(H^0_B((\operatorname{Ad}\Pi)_{\mathbb{R}}, \overline{\mathbb{Q}}))L^*(0, \operatorname{Ad}\Pi) \sim_{\overline{\mathbb{Q}}^\times} \det F^1 H_{\mathrm{dR}}((\operatorname{Ad}\Pi)_{\overline{\mathbb{Q}}})\cdot\det(H^1_M((\operatorname{Ad}\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))).$$

Regarding this as an equality inside the determinant of the fundamental exact sequence ($\star$), computing the volumes would give the desired statement. $\square$

We will later prove Conjecture 10.3.1 in special cases where Conjecture 10.3.3 is equivalent to Conjecture 10.3.1 (namely, the appearances of Hecke eigensystem span over two degrees).

Example 10.3.5 (Sanity check: SL_2)**.** The simplest example is the case of $G = \mathrm{SL}_{2,\mathbb{Q}}$, where the conjecture is about *weight one elliptic modular forms.* Let $f \in S_1(\Gamma)$ be a weight one cuspidal new eigenform with Fourier coefficients in $\overline{\mathbb{Q}}$, generating an automorphic representation Π. Then, the complex conjugate $\overline{f} \in \overline{\Pi}$, which also satisfies $\langle \overline{f}, \overline{f}\rangle_P = \langle f, f\rangle_P$. Thus, we are led to the following question: for what $c \in \mathbb{C}^\times$ does $c\overline{f}$ define a $\overline{\mathbb{Q}}$-coherent cohomology class in $H^1(X(\Gamma), \omega)$? After calculating the archimedean L-factors (which is elementary), Conjecture 10.3.3 says that

$$|c|^{-2} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left(\pi^{-2}\frac{L(1, \Pi, \operatorname{Ad})}{\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad}\Pi)}\right)^2.$$

Note that Π is a pure weight 0 motive, and so is $\operatorname{Ad}\Pi$; therefore, $F^1 H_{\mathrm{dR}}(\operatorname{Ad}\Pi) = 0$, which means that

$$|c| \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \frac{\pi^2}{L(1, \Pi, \operatorname{Ad})}.$$

It is well-known that $L(1, \Pi, \operatorname{Ad})\pi^{-2} \sim_{\overline{\mathbb{Q}}^\times} \langle f, f\rangle_P$. Thus, the conjecture says

$$|c| \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \frac{\pi}{\langle f, f\rangle_P}.$$

Let $c\overline{f}d\overline{z}$ define a Dolbeault cohomology class $f^\vee \in H^1(X(\Gamma), \omega)$, which is, by definition, defined over $\overline{\mathbb{Q}}$. By Serre duality, $\langle f, f^\vee \rangle_S \in \overline{\mathbb{Q}}$, where $\langle -, - \rangle_S$ denotes the Serre duality.

On the other hand, the Serre duality in this case coincides with Petersson inner product scaled by the factor of $\frac{1}{2\pi i}$ (e.g. [DMOS, p. 22]), namely $2\pi i \langle f, f^\vee \rangle_S = \langle f, \overline{c} f \rangle_P = \overline{c} \langle f, f \rangle_P$. Thus, means that $c \sim_{\overline{\mathbb{Q}}^\times} \frac{1}{\langle f, f \rangle_P}$,[4] which is consistent with our conjecture. That these facts can be realized as an instance of motivic action conjecture was realized in [HV] and was explicitly spelled out in [Ho].

10.4 An approach towards the motivic action conjecture

Unlike [PV] or the case of SL_2, the top and bottom degrees are most of the time not complementary (namely, $i_{\Pi_{\min}} + i_{\Pi_{\max}} \neq \dim_{\mathbb{C}} X$), so the conjecture has nothing to do with any form of duality. We will nevertheless prove a form of the conjecture by relating this with *cohomological period integrals*. These are integral representations of certain L-functions that

- apply to automorphic forms appearing in higher coherent cohomology,
- and admit an interpretation as cup product pairing in coherent cohomology.

This is useful in verifying our conjectures as the coherent cohomological cup product can *detect rationality of higher coherent cohomology classes*.

We will show that a strategy similar to [PV, §7] can also show that our period conjecture, Conjecture 10.3.3, is true in certain cases.

Theorem 10.4.1. *Let G be either Sp_4 or $\mathrm{SU}(2,1)$, and let Π be a globally generic cuspidal automorphic representation of $G(\mathbb{A}_{\mathbb{Q}})$ satisfying Assumption 10.1.5. Assume*

[4] Indeed, $\langle f, f \rangle_P$ is a real number.

the working hypothesis on periods, Assumption 10.4.5. Then, Conjecture 10.3.3 is true, up to the factor of an archimedean zeta integral (see Remarks 11.3.1, 12.3.2).

In the two cases, we will compare periods of *holomorphic LDS* appearing in H^0 and *generic LDS* appearing H^1. The tools that we will use are summarized in the following table.

LDS type	Detecting rationality	Relation with Petersson norm
Holomorphic (H^0)	Rational Fourier coefficient	Doubling method, Refined GGP conjectures
Generic ($H^{>0}$)	Cohomological period integrals	Lapid–Mao conjecture

A slightly more detailed outline is as follows. We would need to know how $\overline{\mathbb{Q}}$-algebraicity of coherent cohomology classes in H^0 and H^1 is related to Petersson norms. For H^0, the classes are represented by holomorphic automorphic forms, where their algebraicity is detectable by Fourier coefficients. In the language of periods, these are related to Bessel or Fourier–Jacobi periods, which are related to Petersson norms via the refined Gan–Gross–Prasad conjectures. For H^1, the classes are represented by generic automorphic forms[5]. The Petersson norms of corresponding automorphic forms are related to the *Whittaker periods* via the *Lapid–Mao conjecture* [LM]. Due to its simple statement, we recall the conjecture here:

Conjecture 10.4.2 ([LM])**.** *Let Π be a globally generic representation, satisfying (NEW) of Assumption 10.1.5. Let $f = \otimes_p f_p \in \Pi$ be a newform. Then,*

$$\frac{\langle f, f\rangle}{|W(1)|^2} \sim_{\mathbb{Q}^\times} \frac{L(1, \Pi, \mathrm{Ad})}{\Delta_G(1)|W_\infty(1)|^2},$$

where $\langle , \rangle$ is the L^2-norm (as opposed to the Petersson norm), $\Delta_G(s)$ is the L-function

[5]In both cases of our concern, the infinity type corresponding to H^1 belongs to a generic (L)DS, which is a numerical coincidence that only happens in certain special examples.

of the dual to the Artin motive attached to G as defined in [Gr2], and W (W_∞, respectively) is the Whittaker function of f (f_∞, respectively).

The Whittaker periods are then related to $\overline{\mathbb{Q}}$-algebraicity of coherent cohomology classes via cohomological period integrals. On one hand, the period integral has automorphic interpretation, which connects to Whittaker periods. On the other hand, the period integral has cohomological interpretation, so that in particular it descends to $\overline{\mathbb{Q}}$, hence detects $\overline{\mathbb{Q}}$-algebraicity.

Finally, as the conjectures are formulated using invariants coming from the motivic formalism, we would be working with the corresponding motives and compute motivic invariants (dubbed "Hodge-linear algebra"). We will work with the adjoint motive, as in Conjecture 10.2.2, as well as the motive of the given automorphic representation[6]:

Conjecture 10.4.3. *For Π as in Assumption 10.1.5, there exists a motive M_Π that is uniquely characterized by [Cl, §4.3.3].*

Remark 10.4.4. The adjoint motive of Conjecture 10.2.2 is the motive associated to $L(s, \Pi, \mathrm{Ad})$ in the above sense.

Thus, the "working hypothesis" is as follows.

Assumption 10.4.5 (Working hypothesis)**.** We assume the following conjectures[7]. There are numerous instances of these conjectures being verified, and we do not attempt to list them here.

1. *Beilinson's conjecture for Chow motives*, Conjecture 2.1.1 of [PV].

2. *Lapid–Mao conjecture*, Conjecture 10.4.2.

3. *Existence of motives*, Conjecture 10.2.2 and Conjecture 10.4.3.

[6]The construction of those motives is a subtle matter, as we work with limits of discrete series. Indeed, the corresponding Galois representations have been constructed, but only by using congruences.

[7]Even though the refined Gan–Gross–Prasad conjectures are being mentioned throughout the paper, they are not required as an assumption, because we have an unconditional alternative result using the doubling method (and they are equivalent if one assumes Beilinson's conjectures anyways).

Chapter 11

Evidence I: The case of Sp_4

We first provide the evidence for the motivic action conjecture (Conjecture 10.3.1) for the case of certain irregular automorphic forms on $G(\mathbb{A}) = \mathrm{Sp}_4(\mathbb{A})$. In this case, we will use an integral reprsentation of the spinor L-function by Novodvorsky ([No]), whose coherent cohomological interpretation was given by [LPSZ].

Let Π be a globally generic cuspidal automorphic representation of $G = \mathrm{GSp}_4(\mathbb{A}_{\mathbb{Q}})$, namely that the Whittaker transform (see §9.6, Notation) defines a realization of Π as a space of functions on $G(\mathbb{Q})\backslash G(\mathbb{A})$ satisfying a transformation property under the $N(\mathbb{A})$-action. We choose unramified vectors $\varphi_v^0 \in \Pi_v$ for v finite with Π_v unramified such that, if ψ_v is unramified, $W_{\varphi_v^0}(1) = 1$. Suppose that Π satisfies Assumption 10.1.5.

Definition 11.0.1. Let $M \supset F_\Pi$ be a number field. A ψ-Whittaker function W on G is called to be *defined over* M if it takes values in $M(\mu_\infty)$ and satisfies

$$\sigma(W(g)) = W(w(\kappa(\sigma))g),$$

for all $g \in G(\mathbb{A}_f)$ and $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/M)$, where $w(x) = \mathrm{diag}(x^3, x^2, x, 1)$ and $\kappa : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \widehat{\mathbb{Z}}^\times$ is the cyclotomic character.

Remark 11.0.2.

1. The convention is made so that, if ψ is unramified, the unramified ψ-Whittaker function W with $W(1) = 1$ is defined over F_Π. As $L(s, \Pi, \mathrm{Ad})$ is regular at $s = 1$, $W(1)$ is nonzero by [LM, §3.1]. Thus, W is defined over M if and only if $W(1) \in M^\times$.

2. Indeed, our conditions on G and Π imply that F_Π is a number field (see Notation). Thus, the space of ψ-Whittaker functions, which naturally has an action of $\mathrm{Aut}(\mathbb{C}/\mathbb{Q})$ as in [GHL, §4.1], is fixed by $\mathrm{Aut}(\mathbb{C}/F_\Pi)$. Thus, there exists a nontrivial ψ-Whittaker function defined over M.

11.1 Whittaker periods via cohomological period integrals

The purpose of this subsection is to prove the following

Theorem 11.1.1. *Let $\varphi \in \Pi^{\mathrm{new}} = \Pi_f^{\mathrm{new}} \otimes \Pi_\infty^{\mathrm{new}}$. Let F be an imaginary quadratic field such that the special Bessel period of Π for F is not identically zero (see (Sp_4) of Assumption 10.1.5). If $[\varphi]$ (see Definition 10.1.8) is defined over a number field $F' \supset FF_\Pi$, then $\Lambda(1/2, \Pi)\Lambda(1/2, \Pi \otimes \chi_F)W_\varphi$ is a nontrivial ψ-Whittaker function defined over F', where χ_F is the quadratic character associated to F.*

Proof. Let $C \in \mathbb{C}^\times$ be such that $\frac{W_\varphi}{C}$ is defined over F', which exists by Remark 11.0.2(2). We would like to show that

$$C\Lambda(1/2, \Pi)\Lambda(1/2, \Pi \otimes \chi_F) \in F'^\times.$$

We first prove that the above quantity is nonzero. This is because, by [FuMo, (1.26)], the nonvanishing of $L(1/2, \Pi)L(1/2, \Pi \otimes \chi_F)$ is equivalent to nonvanishing of the

special Bessel period $B(f_\Pi, F)$ in (Sp_4) of Assumption 10.1.5; that the special Bessel period defined in this paper is the Bessel period (up to explicit nonzero scalar) in [FuMo] follows from the calculation of [DPSS, Proposition 3.5].

We now consider Novodvorsky's integral representation of spinor L-function. This is, roughly speaking, the period integral of φ times an Eisenstein series over an embedded product of two modular curves. More precisely, there is an embedding of $H = \mathrm{GL}_2 \times_{\mathrm{GL}_1} \mathrm{GL}_2$ into G, which is most naturally thought as $\mathrm{SO}(2,2) \hookrightarrow \mathrm{SO}(3,2)$. Let $B \subset H$ be the upper triangular Borel. For $\Phi_1, \Phi_2 : \mathbb{A}^2 \to \mathbb{C}$ and χ_1, χ_2 a unitary Grössencharcter, we define an Eisenstein series on H with respect to B, $E(h, \chi_i, \Phi_i, s_i)$. The following are well-known (e.g. [LPSZ, Proposition 7.3]).

Proposition 11.1.2. *Suppose we are taking the "weight k-section" for $\Phi_{i,\infty}$.*

1. *For $-\frac{k}{2} + 1 \leq s_1, s_2 \leq \frac{k}{2}$ half-integral, equivalent modulo 1 to $\frac{k}{2}$, this defines a nearly holomorphic form on H, which can be thought as an H^0 class of some automorphic vector bundle over a Shimura variety for H.*

2. *If χ_i's and Φ_i's are valued in a number field F, then the H^0 class is defined over F.*

Let (λ_1, λ_2) be the Harish–Chandra character of the (necessarily generic) (L)DS Π_∞. Let $\Gamma_H = H \cap \Gamma$, and let $i : X_H(\Gamma_H) \hookrightarrow X_G(\Gamma)$ be the closed embedding of certain toroidal compactifications of the corresponding Shimura varieties[1]. Then, Novodvorsky's integral representation can be understood via a cup product pairing

$$H^2(X_G, V) \otimes_{\mathbb{C}} H^0(X_H, W) \xrightarrow{(\mathrm{id}, i_*)} H^2(X_G, V) \otimes_{\mathbb{C}} H^1(X_G, W') \xrightarrow{\cup} H^3(X_G, V \otimes W') \xrightarrow{S} H^0(X_G, \mathcal{O}) = \mathbb{C},$$

where $\cup$ is the cohomological cup product and S is the Serre duality pairing, induced

[1] That the closed immertion of open Shimura varieties extends to a closed immersion of toroidal compactifications with respect to certain refinements of polyhedral cone decomposition is achieved by [Lan4], but the situation is simplier in this case, because X_H, being a productof modular curves, is unique.

from a morphism of algebraic $\overline{\mathbb{Q}}$-representations of K_∞, $V \otimes W' \to \mathfrak{g}^{-1,1}$ (namely, the pairing is normalized such that, on the level of representations of K_∞, the $\overline{\mathbb{Q}}$-structures are compatible). Here, W and W' are certain automorphic vector bundles over X_H and X_G, respectively, corresponding to $[V_H(\lambda_1 - \lambda_2 - 1, 0)] \otimes \omega_H(1,1)$ and $[\widetilde{L}_1]$, if we use the notation of [LPSZ, §6]. The reinterpretation, done in [LPSZ, §7.4], of Novodvosky's integral asserts that, given an imaginary quadratic field F, there is a coherent cohomology class $[E] \in H^0(X_H(\Gamma_H), W)$ defined over FF_{Π_f}, which corresponds to a nearly-holomorphic Eisenstein series E under the Hodge splitting of [LPSZ, §6.3], such that

$$\langle [\varphi], [E] \rangle = C\Lambda\left(1/2, \Pi\right) \Lambda\left(1/2, \Pi \otimes \chi_F\right),$$

where χ_F is the Hecke character corresponding to F. As $\langle , \rangle$ is defined over $\mathbb{Q}$, the left hand side is in F', which is what we are looking for. $\square$

11.2 Hodge-linear algebra

We now put the relevant Hodge-linear algebra that proves the case of Sp_4. For the sake of simplicity, we assume we work with a parallel weight $(2,2)$ Siegel modular form, or Harish–Chandra parameter $(1,0)$, although the Hodge-linear algebra calculation stays the same for general weights. Our goal is to convert, using elementary linear algebra, $\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ into an expression that involves Deligne's periods $c^+(M), c^-(M), \delta(M)$ whose definitions will be recalled later. We will then be able to express $\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ with L-values, using Deligne's conjectures. We will prove

Proposition 11.2.1. *For a motive M associated to Π (in the sense of Conjecture 10.4.3), we have*

$$\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\sqrt{c^+(M)c^-(M)}^3}{\delta(M)^{3/2}}.$$

Proof. The motive M of Π should be of rank 4 and weight 1, with the Hodge decomposition

$$H_B(M) \otimes_{\mathbb{Q}} \mathbb{C} = H^{1,0}(M) \oplus H^{0,1}(M), \quad \dim_{\mathbb{C}} H^{1,0}(M) = \dim_{\mathbb{C}} H^{0,1}(M) = 2,$$

which is of the type of the Hodge structure defined by the corresponding archimedean Langlands parameter. In this case,

$$\delta(M) \ \left(c^{\pm}(M), \text{ resp.}\right) \ \in \mathbb{C}^{\times}/\mathbb{Q}^{\times},$$

is the determinant of the comparison map

$$H_B(M) \otimes \mathbb{C} \xrightarrow{\sim} H_{\mathrm{dR}}(M) \otimes \mathbb{C} \ \left(H_B(M)^+ \otimes \mathbb{C} \to H_B(M) \otimes \mathbb{C} \xrightarrow{\sim} H_{\mathrm{dR}}(M) \otimes \mathbb{C} \twoheadrightarrow (H_{\mathrm{dR}}(M)/F^1 H_{\mathrm{dR}}(M)) \otimes \mathbb{C}, \text{ resp.}\right),$$

with respect to the bases coming from the underlying $\mathbb{Q}$-structures on both sides.

Let e_1^+, e_2^+ be a $\mathbb{Q}$-basis of $H_B(M)^+$ and e_1^-, e_2^- be a $\mathbb{Q}$-basis of $H_B(M)^-$. Let $f_1, f_2 \in F^1 H_{\mathrm{dR}}(M)$ be a $\mathbb{Q}$-basis, and $g_1, g_2 \in H_{\mathrm{dR}}(M)/F^1 H_{\mathrm{dR}}(M)$ be a $\mathbb{Q}$-basis, and $\widetilde{g_1}, \widetilde{g_2} \in H_{\mathrm{dR}}(M)$ be lifts of g_1, g_2. Given two $\mathbb{C}$-bases of $H_B(M) \otimes \mathbb{C} \cong H_{\mathrm{dR}}(M) \otimes \mathbb{C}$, we can write an expression

$$\begin{pmatrix} e_1^+ & e_2^+ & e_1^- & e_2^- \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & \widetilde{g_1} & \widetilde{g_2} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

for $A, B, C, D \in M_2(\mathbb{C})$. Note that by definition $\delta(M) = \det \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.

We have canonical isomorphisms

$$H^{1,0}(M) \cong F^1 H_{\mathrm{dR}}(M) \otimes \mathbb{C}, \quad H^{0,1}(M) \cong \frac{H_{\mathrm{dR}}(M)}{F^1 H_{\mathrm{dR}}(M)} \otimes \mathbb{C}.$$

Let $f_{1,B}, f_{2,B} \in H^{1,0}(M)$ and $g_{1,B}, g_{2,B} \in H^{0,1}(M)$ be the images of f_1, f_2, g_1, g_2 under the above canonical isomorphisms. Then $f_{i,B}$ and f_i coincide as elements of $H_B(M) \otimes \mathbb{C} \cong H_{\mathrm{dR}}(M) \otimes \mathbb{C}$, whereas $\widetilde{g_i} - g_{i,B} \in H^{1,0}(M)$. So,

$$\begin{pmatrix} f_1 & f_2 & \widetilde{g_1} & \widetilde{g_2} \end{pmatrix} = \begin{pmatrix} f_{1,B} & f_{2,B} & g_{1,B} & g_{2,B} \end{pmatrix} \begin{pmatrix} 1_2 & M \\ 0_2 & 1_2 \end{pmatrix}.$$

In particular, if we write

$$\begin{pmatrix} e_1^+ & e_2^+ & e_1^- & e_2^- \end{pmatrix} = \begin{pmatrix} f_{1,B} & f_{2,B} & g_{1,B} & g_{2,B} \end{pmatrix} \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix},$$

then $\det \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) = \det \left(\begin{smallmatrix} A' & B' \\ C' & D' \end{smallmatrix}\right)$.

Also, there must be relations

$$c_B(f_{1,B}) = a g_{1,B} + b g_{2,B},$$

$$c_B(f_{2,B}) = c g_{1,B} + d g_{2,B}.$$

Using that $F_\infty e_i^+ = e_i^+$, $F_\infty e_i^- = -e_i^-$ and that $F_\infty(f_{i,B}) = c_B(f_{i,B})$ and $F_\infty(g_{i,B}) = c_B(g_{i,B})$ (as in [PV, Lemma, §8.2.1]), one has

$$\begin{aligned} c_B(A'_{1i} f_{1,B} + A'_{2i} f_{2,B}) &= C'_{1i} g_{1,B} + C'_{2i} g_{2,B}, \\ c_B(B'_{1i} f_{1,B} + B'_{2i} f_{2,B}) &= -(D'_{1i} g_{1,B} + D'_{2i} g_{2,B}), \end{aligned}$$

for $i = 1, 2$. This can be packaged into $C' = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) A'$, $D' = -\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B'$. So

$$\delta(M) = \det \begin{pmatrix} A' & B' \\ \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) A' & -\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B' \end{pmatrix} = \det \begin{pmatrix} A' & B' \\ 0_2 & -2\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B' \end{pmatrix} = 4 \det \begin{pmatrix} a & c \\ b & d \end{pmatrix} \det A' \det B'.$$

Note on the other hand that $c^+(M) = \det C$, $c^-(M) = \det D$. As $C' = C$, $D' = D$, we see that

$$c^+(M)c^-(M) \sim_{\overline{\mathbb{Q}}^\times} \delta(M) \det \begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

Note that, as M is self-dual, $M^\vee \cong M(1)$, which implies $\operatorname{Ad} M = (\operatorname{Ad} M)^* = (\operatorname{Sym}^2 M)(1)$. We are also led to calculate $\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ (we know it does not depend on the choice of weak polarization up to $\overline{\mathbb{Q}}^\times$-ambiguity). We know that $(\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M))^2 \sim_{\overline{\mathbb{Q}}^\times} \lambda$, where $\varphi(c_B(v^+)) = \lambda v^-$. Here, v^+, v^- are $\mathbb{Q}$-bases vectors for $\det F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ and $\det\left(\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}\right)$, respectively, and $\varphi : \wedge^{\dim F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)}(H_{\mathrm{dR}}(M) \otimes \mathbb{C}) \to \det\left(\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}\right)$ is the natural projection. As $\operatorname{Ad} M = (\operatorname{Sym}^2 M)(1)$, we can take $f_1^2, f_1 f_2, f_2^2$ and $g_1^2, g_1 g_2, g_2^2$ as $\mathbb{Q}$-bases of $F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ and $\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}$, respectively. Now from the known relations,

$$\begin{aligned} c_B(f_1^2) &\equiv a^2 g_1^2 + 2ab g_1 g_2 + b^2 g_2^2, \\ c_B(f_1 f_2) &\equiv ac g_1^2 + (ad + bc) g_1 g_2 + bd g_2^2, \\ c_B(g_1 g_2) &\equiv c^2 g_1^2 + 2cd g_1 g_2 + d^2 g_2^2, \end{aligned}$$

where $\equiv$ is mod $F^0 H_{\mathrm{dR}}(\mathrm{Ad}\, M) \otimes \mathbb{C}$. So

$$\lambda = \det \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad+bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix} = (ad-bc)^3.$$

Therefore, we obtain

$$\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\, M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\sqrt{c^+(M)c^-(M)}^3}{\delta(M)^{3/2}}.$$

□

11.3 Completion of the proof

Proof of Theorem 10.4.1 for Sp_4. Now we apply the relevant period conjectures for this case. Let f_{hol}, f_{gen} be newforms (see Assumption 10.1.5) in $\Pi_f \otimes \Pi_{\mathrm{hol}}$, $\Pi_f \otimes \Pi_{\mathrm{gen}}$, respectively, where Π_{hol} is the corresponding holomorphic NLDS in the L-packet of Π_{gen}. Let us assume that $[f_{\mathrm{hol}}]$ and $[f_{\mathrm{gen}}]$ are defined over $\overline{\mathbb{Q}}$. Let F be the imaginary quadratic field as in Theorem 11.1.1. Then, a theorem of Furusawa–Morimoto[2] [3] implies that

$$\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}} \rangle}{|B(f_{\mathrm{hol}}, F)|^2} \sim_{\overline{\mathbb{Q}}^\times} \pi^{-6} \frac{L(1, \Pi, \mathrm{Ad})}{L(1/2, \Pi) L(1/2, \Pi \otimes \chi_F)}. \tag{A}$$

Since $[f_{\mathrm{hol}}]$(see Definition 10.1.8) is defined over $\overline{\mathbb{Q}}$, we know that $B(f_{\mathrm{hol}}, F)$, a $\mathbb{Q}$-linear combination of Fourier coefficients of f_{hol}, is in $\overline{\mathbb{Q}}$. Thus, in the setting of Theorem 10.4.1, $\langle f_{\mathrm{hol}}, f_{\mathrm{hol}} \rangle \sim_{\overline{\mathbb{Q}}^\times} \pi^{-6} \frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_F)}$.

[2] See [FuMo, Theorem 1] for the case of discrete series; the case of limit of discrete series is also an upcoming work of them.

[3] This is a specific case of the refined Gan–Gross–Prasad conjecture for Bessel periods, [Li, Conjecture 2.5]. This special case is also sometimes called *Böcherer's conjecture*.

We now relate the $\overline{\mathbb{Q}}$-rationality of f_{gen} with $\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle$. Note that Theorem 11.1.1 is about the relationship between rationality of Whittaker functions and that of coherent cohomology classes, for the anti-generic LDS (namely, those appearing in H^2 of coherent cohomology). Thus, as $[f_{\mathrm{gen}}]$ is defined over $\overline{\mathbb{Q}}$, by Serre duality,

$$\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P = (2\pi i)^3 \langle [f_{\mathrm{gen}}], [\overline{f}_{\mathrm{gen}}]\rangle_{\mathrm{coh}}, \tag{B}$$

where $\langle -, -\rangle_{\mathrm{coh}}$ is the cohomological cup product defined over $\overline{\mathbb{Q}}$ normalized so that it is induced from the $\overline{\mathbb{Q}}$-morphism of algebraic K_∞-representations $W \otimes \mathrm{Hom}(W, \mathfrak{g}^{-1,1}) \to \mathfrak{g}^{-1,1}$. If we let $C \in \mathbb{C}^\times$ be such that $C[\overline{f}_{\mathrm{gen}}]$ is defined over $\overline{\mathbb{Q}}$, we see that the RHS is $\sim_{\overline{\mathbb{Q}}^\times} \pi^3 C^{-1}$. On the other hand, by Theorem 11.1.1,

$$\Lambda(1/2, \Pi)\Lambda(1/2, \Pi \otimes \chi_F) W_{C\overline{f}_{\mathrm{gen}}} = C\Lambda(1/2, \Pi)\Lambda(1/2, \Pi \otimes \chi_F) W_{\overline{f}_{\mathrm{gen}}},$$

is defined over $\overline{\mathbb{Q}}$. We now invoke the Lapid–Mao conjecture, Conjecture 10.4.2:

$$\frac{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle}{|W(1)|^2} \sim_{\overline{\mathbb{Q}}^\times} \pi^9 \cdot \frac{L(1, \Pi, \mathrm{Ad})}{|W_\infty(1)|^2}.$$

By Remark 11.0.2(1), $W_{\overline{f}_{\mathrm{gen}}}(1) \neq 0$, so that

$$C\Lambda(1/2, \Pi)\Lambda(1/2, \Pi \otimes \chi_F) W_{\overline{f}_{\mathrm{gen}}}(1) \in \overline{\mathbb{Q}}^\times. \tag{C}$$

Since $W_{\overline{f}_{\mathrm{gen}}}(1) = \overline{W_{f_{\mathrm{gen}}}(1)}$, we have

$$\pi^9 \frac{L(1, \Pi, \mathrm{Ad})}{|W_\infty(1)|^2} |W_{f_{\mathrm{gen}}}(1)|^2 \overset{\text{Lapid–Mao}}{\sim_{\overline{\mathbb{Q}}^\times}} \langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle \overset{\text{(B)}}{\sim_{\overline{\mathbb{Q}}^\times}} \pi^3 C^{-1} \overset{\text{(C)}}{\sim_{\overline{\mathbb{Q}}^\times}} \pi^3 \overline{W_{f_{\mathrm{gen}}}(1)} \Lambda(\frac{1}{2}, \Pi)\Lambda(\frac{1}{2}, \Pi \otimes \chi_F),$$

or

$$W_{f_{\mathrm{gen}}}(1) \sim_{\overline{\mathbb{Q}}^\times} \pi^{-6}\frac{|W_\infty(1)|^2\Lambda(\frac{1}{2},\Pi)\Lambda(\frac{1}{2},\Pi\otimes\chi_F)}{L(1,\Pi,\mathrm{Ad})} \sim_{\overline{\mathbb{Q}}^\times} \pi^{-10}\frac{|W_\infty(1)|^2 L(\frac{1}{2},\Pi)L(\frac{1}{2},\Pi\otimes\chi_F)}{L(1,\Pi,\mathrm{Ad})}. \tag{D}$$

We need to compute $\frac{\langle f_{\mathrm{hol}},f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}},f_{\mathrm{gen}}\rangle_P}$, which can be now seen as follows.

$$\begin{aligned}
\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P} &\overset{\text{Lapid–Mao + (A)}}{\sim_{\overline{\mathbb{Q}}^\times}} \frac{\pi^{-6}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_F)}}{\pi^9|W_{f_{\mathrm{gen}}}(1)|^2\frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2}} \\
&= \pi^{-15}\frac{|W_\infty(1)|^2}{|W_{f_{\mathrm{gen}}}(1)|^2 L(1/2,\Pi)L(1/2,\Pi\otimes\chi_F)} \\
&\overset{\text{(D)}}{\sim_{\overline{\mathbb{Q}}^\times}} \pi^{-15}\frac{|W_\infty(1)|^2}{\pi^{-20}\frac{|W_\infty(1)|^4 L(1/2,\Pi)^3 L(1/2,\Pi\otimes\chi_F)^3}{L(1,\Pi,\mathrm{Ad})^2}} \\
&\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^9}{|W_\infty(1)|^2}\left(\frac{L_\infty(1,\Pi,\mathrm{Ad})}{L_\infty(0,\Pi,\mathrm{Ad})}\cdot\frac{L(1,\Pi,\mathrm{Ad})}{\frac{\sqrt{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}^3}{\pi^3}}\right)^2,
\end{aligned} \tag{E}$$

which involves elementary calculation of archimedean L-factors.

On the other hand, by Deligne's conjectures, Proposition 11.2.1 implies that

$$\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\, M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\sqrt{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}^3}{\pi^3}.$$

Thus, the parenthized term in (E) is precisely the RHS of Conjecture 10.3.3, which finishes the proof. □

Remark 11.3.1. Unfortunately, for now, the author has been unable to calculate $W_\infty(1)$. It is however very believable that the archimedean zeta integral against a preferred, nice test vector is equal to an archimedean local L-factor, which is $\sim_{\overline{\mathbb{Q}}^\times}$ half-integral powers of π. In our case, there is even an explicit integral expression of $W_\infty(1)$, as written in [CI]:

$$W_\infty(1) = 16e^{-2\pi}\pi^{\frac{7}{2}}\int_{c-i\infty}^{c+i\infty}\pi^{-2s}\Gamma(s+\frac{1}{2})^2 U(s+\frac{1}{2},1,4\pi)\Gamma(s)\frac{ds}{2\pi i},$$

where $U(a,b,z) = \frac{1}{\Gamma(a)} \int_0^\infty e^{-zt} t^{a-1} (1+t)^{b-a-1} dt$ is the confluent hypergeometric function of the second kind.

Chapter 12

Evidence II: The case of $\mathrm{SU}(2,1)$

In this section, we provide the evidence for the Period conjecture (Conjecture 10.3.3) for the case of certain irregular automorphic forms on $G(\mathbb{A}) = \mathrm{SU}(2,1)(\mathbb{A})$. In this case, we will use an integral reprsentation of the base-change L-function by Gelbart and Piatetski-Shapiro ([GPS], completed by [KO]), whose coherent cohomological interpretation was given by the Part III of this thesis.

Let Π be a globally generic cuspidal automorphic representation of $G = \mathrm{SU}(2,1)(\mathbb{A}_{\mathbb{Q}})$ as in the previous subsection that also satisfies Assumption 10.1.5.

Definition 12.0.1. Let $M \supset F_{\Pi}$ be a number field. A ψ-Whittaker function W on G is called to be *defined over* M if it takes values in $M(\mu_\infty)$ and satisfies

$$\sigma(W(g)) = W(w(\kappa(\sigma))g),$$

for all $g \in G(\mathbb{A}_f)$ and $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/M)$, where $w(x) = \mathrm{diag}(x, 1, x^{-1})$ and $\kappa : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \widehat{\mathbb{Z}}^\times$ is the cyclotomic character.

Again, by Remark 11.0.2(1), W is defined over M if and only if $W(1) \in M^\times$.

12.1 Whittaker periods via cohomological period integrals

The purpose of this section is to prove the following

Theorem 12.1.1. *Let $\varphi \in \Pi^{\mathrm{new}} = \Pi_f^{\mathrm{new}} \otimes \Pi_\infty^{\mathrm{new}}$. If $[\varphi]$ (see Definition 10.1.8) is defined over a number field $F' \supset F_\Pi$, then $\Lambda(1/2, BC(\Pi))\, W_\varphi$ is a nontrivial ψ-Whittaker function defined over F'.*

Proof. Let $C \in \mathbb{C}^\times$ be such that $\frac{W_\varphi}{C}$ is defined over F', which exists by Remark 11.0.2(2). We would like to show that

$$C\Lambda(1/2, BC(\Pi)) \in F'^\times.$$

First of all, this is indeed nonzero as $BC(\Pi)$ is cuspidal and tempered.

We consider Gelbart–Piatetski-Shapiro integral representation of base change L-function. This is, roughly speaking, the period integral of φ times an Eisenstein over an embedded modular curve. More precisely, there is an embedding of $H = \mathrm{U}(1,1)$ into G. Let $B \subset H$ be the upper-triangular Borel. For $\Phi : \mathbb{A}^2 \to \mathbb{C}$ and χ a unitary Grössencharacter of the imaginary quadratic field F used for the definition of unitary groups, we define an Eisenstein series on H with respect to B, $E(h, \chi, \Phi, s)$. The same arithmeticity condition as Proposition 11.1.2 applies, as the objects involved are elliptic modular forms.

Using the notation of Example 15.1.3(3), let $(m, n) = (a-b, b-c)$ be the Harish–Chandra character[1] of Π_∞. Let $\Gamma_H = H \cap \Gamma$, and let $i : X_H(\Gamma_H) \hookrightarrow X_G(\Gamma)$ be the closed embedding of closed Shimura varieties, as before. Then, Gelbart–Piatetski-

[1]Note that G is now $\mathrm{SU}(2,1)$, so only the differences matter.

Shapiro's integral representation can be understood via a cup product pairing

$$H^1(X_G,V)\otimes_{\mathbb{C}}H^0(X_H,W)\xrightarrow{(\mathrm{id},i_*)}H^1(X_G,V)\otimes_{\mathbb{C}}H^1(X_G,W')\xrightarrow{\cup}H^2(X_G,V\otimes W')\xrightarrow{S}H^0(X_G,\mathcal{O})=\mathbb{C},$$

where $\cup$ is the cohomological cup product, S is a Serre duality pairing, normalized as in the proof of Theorem 11.1.1 (namely, the pairing induced from a morphism of K_∞-representations defined over $\overline{\mathbb{Q}}$). Here, W and W' are certain automorphic vector bundles over X_H and X_G, respectively, corresponding to $[V_H(|m-n|-1)]\otimes\omega_H(1,0)$ and $[\widetilde{L}_1]$, if we use the notation analogous to [LPSZ, §6]. The reinterpretation, done in Part III of this thesis, of Gelbart–Piatetski-Shapiro's integral asserts that there is a coherent cohomology class $[E]\in H^0(X_H(\Gamma_H),W)$ defined over FF_{Π_f}, which corresponds to a nearly-holomorphic Eisenstein series E under the Hodge splitting as in [LPSZ, §6.3], such that

$$\langle[\varphi],[E]\rangle = C\Lambda\left(1/2, BC(\Pi)\right),$$

where BC means base-change. As $\langle,\rangle$ is defined over $\mathbb{Q}$, the left hand side is in F'. On the other hand, the right hand side is nonzero as observed above. Thus, the desired statement follows. □

12.2 Hodge-linear algebra

We now conduct relevant calculations in Hodge-linear algebra to prove Theorem 10.4.1 for $G=\mathrm{SU}(2,1)$. For the sake of simplicity, we assume that we work with the case of Harish-Chandra character $(1,1,0)$; the same calculation yields the proof for general weights. Our goal is, as in §11.2, to convert $\mathrm{vol}\, F^1H_{\mathrm{dR}}(\mathrm{Ad}\, M)$ into an expression that involves Deligne's periods. We will prove

Proposition 12.2.1. *For a motive M associated with Π in the sense of Conjecture*

10.4.3, we have

$$\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^3 c^+(BC(M))^{3/2}}{\delta(M_{\overline{\sigma}})^{1/2}}.$$

Proof. In this case, the motive M is a motive over F with coefficients in $\mathbb{Q}$. If we denote $\sigma : F \to \mathbb{C}$ by the preferred complex embedding, then as the minimal K-type is just $(1,1,1)$, by the recipe in [HLSu, 2.3],

$$H_B(M_\sigma) \otimes_{\mathbb{Q}} \mathbb{C} = \underbrace{\mathbb{C}v_1 \oplus \mathbb{C}v_2}_{H^{1,0}} \oplus \underbrace{\mathbb{C}v_3}_{H^{0,1}}, \quad H_B(M_{\overline{\sigma}}) \otimes_{\mathbb{Q}} \mathbb{C} = \underbrace{\mathbb{C}\overline{v}_1 \oplus \mathbb{C}\overline{v}_2}_{H^{-1,0}} \oplus \underbrace{\mathbb{C}\overline{v}_3}_{H^{0,-1}}.$$

We can choose v_i and $\overline{v}_i$ so that $F_\infty(v_i) = \overline{v}_i$. Then, $BC(M) := \operatorname{Res}_{F/\mathbb{Q}} M_F = M_\sigma \oplus M_{\overline{\sigma}}(-1)$ and $\operatorname{Ad} M = M_\sigma \otimes M_{\overline{\sigma}}$. The Deligne period δ of a motive, as before, is the determinant of the Betti-to-de Rham comparison map with respect to natural underlying $\mathbb{Q}$-structures. Also, $c^+(BC(M))$ in this case would be the determinant of the map

$$H_B(BC(M))^+ \otimes \mathbb{C} \hookrightarrow H_B(BC(M)) \otimes \mathbb{C} \xrightarrow{\sim} H_{\mathrm{dR}}(BC(M)) \otimes \mathbb{C} \twoheadrightarrow (H_{\mathrm{dR}}(BC(M))/F^1 H_{\mathrm{dR}}(BC(M))) \otimes \mathbb{C},$$

with respect to the natural underlying $\mathbb{Q}$-structures.

Let e_1^+, e_2^+, e_3^+ be a $\mathbb{Q}$-basis of $H_B(BC(M))^+$, e_1^-, e_2^-, e_3^- be a $\mathbb{Q}$-basis of $H_B(BC(M))^-$, $f_1 f_2, f_3 \in F^1 H_{\mathrm{dR}}(BC(M))$ be an F-basis, $g_1, g_2, g_3 \in H_{\mathrm{dR}}(BC(M))/F^1 H_{\mathrm{dR}}(BC(M))$ be an F-basis, and $\widetilde{g}_1, \widetilde{g}_2, \widetilde{g}_3 \in H_{\mathrm{dR}}(BC(M))$ be lifts of g_1, g_2, g_3. We can further assume that $f_1, f_2 \in F^1 H_{\mathrm{dR}}(M_\sigma)$, $f_3 \in F^1 H_{\mathrm{dR}}(M_{\overline{\sigma}})$, $\widetilde{g}_1, \widetilde{g}_2 \in H_{\mathrm{dR}}(M_{\overline{\sigma}})$, $\widetilde{g}_3 \in H_{\mathrm{dR}}(M_\sigma)$. Given two $\mathbb{C}$-bases of $H_B(M) \otimes \mathbb{C} \cong H_{\mathrm{dR}}(M) \otimes \mathbb{C}$, we can express the map into a

matrix,

$$\begin{pmatrix} e_1^+ & e_2^+ & e_3^+ & e_1^- & e_2^- & e_3^- \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & f_3 & \widetilde{g}_1 & \widetilde{g}_2 & \widetilde{g}_3 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

for $A, B, C, D \in M_3(\mathbb{C})$.

Under the canonical isomorphisms

$$H^{1,0}(BC(M)) \cong F^1 H_{\mathrm{dR}}(BC(M)) \otimes \mathbb{C}, \quad H^{0,1}(BC(M)) \cong \frac{H_{\mathrm{dR}}(BC(M))}{F^1 H_{\mathrm{dR}}(BC(M))} \otimes \mathbb{C},$$

let

$$f_{1,B}, f_{2,B}, f_{3,B} \in H^{1,0}(BC(M)), \qquad g_{1,B}, g_{2,B}, g_{3,B} \in H^{0,1}(BC(M)),$$

be the images of $f_1, f_2, f_3, g_1, g_2, g_3$ under the above isomorphisms. Then, by the same argument as in §11.2,

$$\begin{pmatrix} e_1^+ & e_2^+ & e_3^+ & e_1^- & e_2^- & e_3^- \end{pmatrix} = \begin{pmatrix} f_{1,B} & f_{2,B} & f_{3,B} & g_{1,B} & g_{2,B} & g_{3,B} \end{pmatrix} \begin{pmatrix} A' & B' \\ C & D \end{pmatrix}.$$

As $c_B(f_{1,B}), c_B(f_{2,B}), c_B(f_{3,B})$ and $g_{1,B}, g_{2,B}, g_{3,B}$ are two $\mathbb{C}$-bases of $H^{0,1}(BC(M))$, there is a system of linear relations

$$\begin{pmatrix} c_B(f_{1,B}) & c_B(f_{2,B}) & c_B(f_{3,B}) \end{pmatrix} = \begin{pmatrix} g_{1,B} & g_{2,B} & g_{3,B} \end{pmatrix} X,$$

for some $X \in \mathrm{GL}_3(\mathbb{C})$. Using the same technique as [PV, Lemma, §8.2.1], we see that $C = XA', D = XB'$, which similarly implies that

$$c^+(BC(M))^2 \sim_{\overline{\mathbb{Q}}^\times} \delta(BC(M)) \det X,$$

where we used $c^+(BC(M)) \sim_{\overline{\mathbb{Q}}} c^-(BC(M))$ due to the decomposition $BC(M) = M_\sigma \oplus M_{\overline{\sigma}}(-1)$ as in [HL, (8)]. Furthermore, as $g_{1,B}, g_{2,B} \in H^{0,1}(M_\sigma)$ and $g_{3,B} \in H^{0,1}(M_{\overline{\sigma}}(-1))$, it turns out that X is of form

$$\begin{pmatrix} Y & \\ & \mu \end{pmatrix}, \quad Y \in \mathrm{GL}_2(\mathbb{C}), \mu \neq 0.$$

We know that $(\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\, M))^2 \sim_{\overline{\mathbb{Q}}^\times} \lambda$, where $\varphi(c_B(v^+)) = \lambda v^-$, $v^+ = (f_1 \otimes f_3) \wedge (f_2 \otimes f_3)$ and $v^- = (g_1 \otimes g_3) \wedge (g_2 \otimes g_3)$. This implies that $\lambda = \mu^2 \det Y = \mu \det X$.

Now, as in [Har6, §1.2], consider the determinant motive $\det(M)$. An F-rational basis vector of $H_{\mathrm{dR}}(\det(M_\sigma))$ can be taken as $v_\sigma := f_1 \wedge f_2 \wedge \widetilde{g}_3$, and similarly $v_{\overline{\sigma}} := \widetilde{g}_1 \wedge \widetilde{g}_2 \wedge f_3$ can be taken as an F-rational basis vector of $H_{\mathrm{dR}}(\det(M_{\overline{\sigma}}))$. On the other hand, if we take e_σ to be a $\mathbb{Q}$-rational basis vector of $H_B(\det(M_\sigma))$, then $F_\infty(e_\sigma) =: e_{\overline{\sigma}}$ is a $\mathbb{Q}$-rational basis vector of $H_B(\det(M_{\overline{\sigma}}))$. Then $e_\sigma = \delta(M_\sigma) v_\sigma$ and $e_{\overline{\sigma}} = \delta(M_{\overline{\sigma}}) v_{\overline{\sigma}}$, so

$$c_B(v_\sigma) = \delta(M_\sigma)^{-1} e_{\overline{\sigma}}.$$

On the other hand,

$$c_B(v_\sigma) = F_\infty(v_\sigma) = (\det Y \cdot \mu^{-1}) v_{\overline{\sigma}},$$

so

$$\delta(M_{\overline{\sigma}}) = \mu^{-1} \delta(M_\sigma) \det Y.$$

On the other hand, due to the polarization, we have [Har6, (1.2.5)],

$$\delta(M_{\overline{\sigma}}) = \delta(M_\sigma)^{-1} (2\pi i)^{-6}.$$

Thus,

$$\pi^6 \delta(M_\sigma)^2 \det Y \sim_{\overline{\mathbb{Q}}^\times} \mu,$$

so

$$\mu^2 \sim_{\overline{\mathbb{Q}}^\times} \pi^6 \delta(M_\sigma)^2 \det X \sim_{\overline{\mathbb{Q}}^\times} c^+(BC(M))^2 \pi^6 \frac{\delta(M_\sigma)}{\delta(M_{\overline{\sigma}})} \sim_{\overline{\mathbb{Q}}^\times} c^+(BC(M))^2 \pi^{12} \delta(M_\sigma)^2,$$

or

$$\mu \sim_{\overline{\mathbb{Q}}^\times} \pi^6 \delta(M_\sigma) c^+(BC(M)),$$

which implies that

$$(\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M))^2 \sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^6}{\delta(M_{\overline{\sigma}})} c^+(BC(M))^3,$$

as desired. □

12.3 Completion of the proof

Proof of Theorem 10.4.1 for $\mathrm{SU}(2,1)$. Let $f_{\mathrm{hol}}, f_{\mathrm{gen}}$ be newforms (see Assumption 10.1.5) in $\Pi_f \otimes \Pi_{\mathrm{hol}}$ and $\Pi_f \otimes \Pi_{\mathrm{gen}}$, respectively, such that $[f_{\mathrm{hol}}], [f_{\mathrm{gen}}]$ are defined over $\overline{\mathbb{Q}}$. Then, under the assumptions of $(\mathrm{SU}(2,1))$ of Assumption 10.1.5[2], [Zha, Theorem 1.2][3] implies that

$$\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}} \rangle_P}{|FJ(f_{\mathrm{hol}})|^2} \sim_{\overline{\mathbb{Q}}^\times} \pi^{-2} \frac{L(1, \Pi, \mathrm{Ad})}{L(1/2, BC(\Pi))}, \tag{F}$$

where $FJ(f_{\mathrm{hol}})$ is the special Fourier–Jacobi period of f_{hol}, defined by

$$FJ(f_{\mathrm{hol}}) = \int_{\mathrm{SU}(1,1)(\mathbb{Q}) \backslash \mathrm{SU}(1,1)(\mathbb{A})} f_{\mathrm{hol}}(h) dh,$$

[2]There is an extra condition on large residue characterstic in *loc. cit.*, but this restriction is recently removed by [BoPi, Theorem 1].

[3]This is a special case of the refined Gan–Gross–Prasad conjecture for Fourier–Jacobi periods (see e.g. [Xu, §1.1]), which is also often referred as the Ichino–Ikeda conjecture.

integrated against the Tamagawa measure. This period is in turn expressed as an inner product of (algebraic) theta functions,

$$FJ(f_{\mathrm{hol}}) \sim_{\overline{\mathbb{Q}}^\times} \langle a(0, f_{\mathrm{hol}})(v), 1\rangle,$$

where $a(0, f_{\mathrm{hol}})$ is the zero-th Fourier–Jacobi coefficient of f_{hol} and 1 is the constant function (regarded as a trivial theta function). Note that the nonvanishing of $FJ(f_{\mathrm{hol}})$ (and thus $a(0, f_{\mathrm{hol}})$) is also a part of the content of [Zha, §1.1].

By [Lan1], $a(0, f_{\mathrm{hol}})$ is identified with the algebraic Fourier–Jacobi coefficient, and in particular is in $\overline{\mathbb{Q}}^\times$, as $[f_{\mathrm{hol}}]$ is defined over $\overline{\mathbb{Q}}$. Thus, we have $\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P \sim_{\overline{\mathbb{Q}}^\times} \pi^{-2}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,BC(\Pi))}$.

On the other hand, we exploit the fact that H^1 is the middle degree of the Shimura variety. Note that the infinity type of $\overline{f}_{\mathrm{gen}}$ is a generic LDS as well (with different infinitesimal character from f_{gen}). Suppose $C \in \mathbb{C}^\times$ is a constant where $C[\overline{f}_{\mathrm{gen}}]$ is defined over $\overline{\mathbb{Q}}$. Then,

$$\begin{aligned}\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P &= (2\pi i)^2\langle [f_{\mathrm{gen}}], [\overline{f}_{\mathrm{gen}}]\rangle_{\mathrm{coh}} \\ &\sim_{\overline{\mathbb{Q}}^\times} \pi^2 C^{-1},\end{aligned} \tag{G}$$

where the cohomological cup product pairing $\langle -, -\rangle_{\mathrm{coh}}$ is the Serre duality pairing induced from a $\overline{\mathbb{Q}}$-morphism of algebraic K_∞-representations $V \otimes \mathrm{Hom}(V, \mathfrak{g}^{-1,1}) \to \mathfrak{g}^{-1,1}$.

By Theorem 12.1.1,

$$\Lambda(1/2, BC(\Pi))W_{C\overline{f}_{\mathrm{gen}}} = C\Lambda(1/2, BC(\Pi))W_{\overline{f}_{\mathrm{gen}}},$$

is defined over $\overline{\mathbb{Q}}$. We now invoke the Lapid–Mao conjecture (Conjecture 10.4.2),

which says

$$\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P \sim_{\mathbb{Q}^\times} |W_{f_{\mathrm{gen}}}(1)|^2 \frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2}.$$

By Remark 11.0.2(1), $W_{\overline{f}_{\mathrm{gen}}}(1) \neq 0$, and $C\Lambda(1/2, BC(\Pi))W_{\overline{f}_{\mathrm{gen}}}(1) \in \overline{\mathbb{Q}}^\times$ as observed in the beginning of the section. Thus, we have

$$|W_{f_{\mathrm{gen}}}(1)|^2 \frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2} \overset{\text{Lapid–Mao}}{\sim_{\overline{\mathbb{Q}}^\times}} \langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P \overset{\text{(F)}}{\sim_{\overline{\mathbb{Q}}^\times}} \pi^2 \Lambda(1/2, BC(\Pi))\overline{W_{f_{\mathrm{gen}}}(1)}.$$

By using $W_{\overline{f}_{\mathrm{gen}}}(1) = \overline{W_{f_{\mathrm{gen}}}(1)}$, we have

$$W_{f_{\mathrm{gen}}}(1) \sim_{\overline{\mathbb{Q}}^\times} \pi^2 \frac{|W_\infty(1)|^2 \Lambda(1/2, BC(\Pi))}{L(1,\Pi,\mathrm{Ad})}. \tag{H}$$

On the other hand, Theorem 12.1.1 says

$$W_{f_{\mathrm{gen}}}(1) \sim_{\overline{\mathbb{Q}}^\times} \frac{1}{\Lambda(1/2, BC(\Pi))},$$

which gives an extra relationship that we can utilize; namely,

$$\Lambda(1,\Pi,\mathrm{Ad}) \sim_{\overline{\mathbb{Q}}^\times} \Lambda(1/2, BC(\Pi))^2 |W_\infty(1)|^2 \pi^2.$$

For example, by combining (F) and (G), we have

$$\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P \sim_{\overline{\mathbb{Q}}^\times} \pi^4 \frac{|W_\infty(1)|^2 \Lambda(1/2, BC(\Pi))^2}{L(1,\Pi,\mathrm{Ad})}. \tag{I}$$

Remark 12.3.1. Note that this is already observed in [Har6, Corollary 1.3.5], under the assumption of Deligne's conjectures. In particular, assuming Deligne's conjectures, we deduce that $|W_\infty(1)| \sim_{\overline{\mathbb{Q}}^\times \pi^{\mathbb{Z}}} 1$.

Combining these, we get

$$\begin{aligned}
\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P} &\overset{\text{Lapid–Mao + (H)}}{\sim_{\overline{\mathbb{Q}}^\times}} \frac{\pi^{-2}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,BC(\Pi))}}{\pi^4\frac{|W_\infty(1)|^2\Lambda(1/2,BC(\Pi))^2}{L(1,\Pi,\mathrm{Ad})}} \\
&\sim_{\overline{\mathbb{Q}}^\times} \frac{1}{|W_\infty(1)|^2}\frac{L(1,\Pi,\mathrm{Ad})^2}{L(1/2,BC(\Pi))^3} \\
&\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^{18}}{|W_\infty(1)|^2}\left(\frac{L_\infty(1,\Pi,\mathrm{Ad})}{L_\infty(0,\Pi,\mathrm{Ad})}\cdot\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,BC(\Pi))^{3/2}}\right)^2, \\
&\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^{21}}{|W_\infty(1)|^2}\left(\frac{L_\infty(1,\Pi,\mathrm{Ad})}{L_\infty(0,\Pi,\mathrm{Ad})}\cdot\frac{L(1,\Pi,\mathrm{Ad})}{\mathrm{vol}\,F^1H_{\mathrm{dR}}(\mathrm{Ad}\,M)}\right)^2,
\end{aligned}$$

by Deligne's conjecture applied to Proposition 12.2.1. We are done, as the paranthesized term is the RHS of Conjecture 10.3.3. □

Remark 12.3.2. Similarly to Remark 11.3.1, $W_\infty(1)$ is given by the inverse Mellin transform of the formulae given in [KO, Theorem 5.5], which the author at the moment is unable to compute.

Chapter 13

Towards the refined motivic action conjecture

The original motivic action conjecture of [PV] involves the rational structure of singular cohomology. To derive a similar conjecture, we would have to come up with a way to normalize all the (choice of newforms of) automorphic representations at once. Recall that, in the case of modular forms, this is done by using complex conjugation. Unfortunately, so far there is no general construction of an operation that can move between different infinity types. We will tentatively name such an operation a *generalized complex conjugation*. Approaching the generalized complex conjugations using the theory of cycle spaces and Penrose transform will be the subject of the author's forthcoming work. For now, we will have to content ourselves with a preliminary analysis on what a generalized complex conjugation should be.

On the other hand, as the name suggests, the usual complex conjugation can be used when the symmetric space is the upper half plane. More generally, if the symmetric space is a product of upper half planes (e.g. in the case of Hilbert modular varieties), then the complex conjugations with respect to each variable would be a good candidate for generalized complex conjugation; these are called *partial complex*

conjugations [Har4] in the literature. In this case, we can deduce a precise conjecture on the $\overline{\mathbb{Q}}$-rational structure of coherent cohomology, from the generalities of §15. There is an existing work of Horawa exactly on this problem [Ho], and we will compare our conjecture with that of *op. cit.* In particular, we observe that the numerical evidences given in *op. cit.* are compatible with both conjectures.

13.1 The case of Hilbert modular forms: comparison with [Ho]

The work [Ho] states a similar conjecture on what archimedean motivic action should be for Hilbert modular forms of partial weight one. It uses the partial complex conjugation, which utilizes the fact that every (limit of) discrete series for $\mathrm{SL}_2(\mathbb{R})^d$ is holomorphic or antiholomorphic in each variable.

Definition 13.1.1 (Partial complex conjugations)**.** Let F be a totally real field of degree d and φ be a holomorphic automorphic form for $G = \mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_{2,F}$, seen as a holomorphic function on the symmetric space for $G(\mathbb{R})$, $(\mathbb{C}-\mathbb{R})^d$, of weight $(k_1, \cdots, k_d; r)$ where $k_i \geq 1$, $k_i \equiv r (\mathrm{mod}\, 2)$ for $i = 1, \cdots, d$. For $I \subset \{1, \cdots, d\}$, φ^I is the automorphic form for $G = \mathrm{Res}_{F/\mathbb{Q}}\mathrm{SL}_{2,F}$, defined by $\varphi^I(g) = \varphi(gJ^I)$ for $J^I = (J^I_1, \cdots, J^I_d)$ given by $J^I_j = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ if $j \in I$ and $J^I_j = \mathrm{id}_2$ if $j \notin I$. This is called the *partial complex conjugation.*

The main objective of op. cit. is to describe the rationality of cohomology classes in terms of partial complex conjugations. In particular, it implies that the decomposition $H^i(X)[\Pi_f] = H^i(\mathfrak{p}, K; \omega \otimes I(\lambda)) \otimes \Pi_f^\Gamma = \left(\bigoplus_{\pi \in \mathfrak{P}_\lambda} H^i(\mathfrak{p}, K; \omega \otimes \pi)\right) \otimes \Pi_f^\Gamma$ descends to $\overline{\mathbb{Q}}$.

We have not been successful in approaching the conjecture. On the other hand, based on the materials developed in §15, we suggest a slightly different conjecture. We use the language of [Ho, §3] below.

Conjecture 13.1.2 (Motivic action conjecture for Hilbert modular varieties; compare with [Ho, Conjecture 3.21])**.** *Let f be a parallel weight one form. For each $u \in U_f^\vee$, let $u_i \in (\mathrm{Ad}^0 M \otimes_\iota \mathbb{C})^{\sigma_i c_0 \sigma_i^{-1}} \cong \mathbb{C}$ be the σ_i-component of $U_f^\vee \otimes_\iota \mathbb{C}$ as in [Ho, Proposition 3.2], where the isomorphism is given by the natural $\overline{\mathbb{Q}}$-structure on $\mathfrak{sl}_2^d$. Then, for every $u \in U_f^\vee$,*

$$2\pi i \sum_{i=1}^{d} u_i \omega_f^{\{i\}} \in H^1(X(\Gamma), \omega),$$

defines a cohomology class over $\overline{\mathbb{Q}}$.

In other words, Conjecture 13.1.2 asserts that the following two $\overline{\mathbb{Q}}$-lattices coincide:

$$\begin{tikzcd}
H^1(X(\Gamma)_{\overline{\mathbb{Q}}}, \omega)[f] \arrow[r, hook] \arrow[d, equal, "\text{Conj. 13.1.2}"'] & H^1(X(\Gamma), \omega)[f] \arrow[r, equal] & \bigoplus_{i=1}^{d} \mathbb{C} \cdot 2\pi i \omega_f^{\{i\}} \arrow[d, equal] \\
U_f^\vee \otimes \overline{\mathbb{Q}} \arrow[r, hook, "\text{Regulator}"'] & U_f^\vee \otimes \mathbb{C} \arrow[r, equal] & \bigoplus_{i=1}^{d} \mathbb{C} \cdot u_i
\end{tikzcd}$$

Indeed, this coincides with the main conjecture of an updated version of [Ho].

An interesting statement that is contained in Conjecture 13.1.2 is the following:

Some cohomological automorphic representation may have no $\overline{\mathbb{Q}}$-rational automorphic form.

Namely, it may happen that, for some automorphic representation Π with nonvanishing cohomology, the part of $H^*(X)$ corresponding to Π may not have any $\overline{\mathbb{Q}}$-rational class. This cannot happen for Π with sufficiently regular weight, as proven in [Har3], so this is a phenomenon of *irregular weight automorphic forms*. Recall that Π with sufficiently regular weight defines a $\overline{\mathbb{Q}}$-rational class as the Hecke operators, which are $\overline{\mathbb{Q}}$-rational, can distinguish Π from other automorphic representations in cohomology; namely, the information on finite part is enough. On the other hand, if Π has a singular weight, there are multiple automorphic representations in the same cohomology space with the same finite part but with different infinity part. Unlike the finite part,

there is no $\overline{\mathbb{Q}}$-rational geometric correspondence that distinguishes various different infinity types. Thus, it is fair to say that the $\overline{\mathbb{Q}}$-rationality breaks down precisely because of the issue of archimedean L-packets.

In [Ho, §5], a numerical evidence in favor of the conjecture of [Ho] is given for base change forms in the case of Hilbert modular forms for real quadratic fields. On the other hand, the base change forms are very special in our context, as there is a significant numerical coincidence happening only for base-change forms.

Proposition 13.1.3. *Let f be a Hilbert modular eigen-cuspform of parallel weight one for a real quadratic field F. If f is a base change form, then Conjecture 13.1.2 implies that, for $i = 1, 2,$,*

$$\mathbb{C} \cdot 2\pi i \omega_f^{\{i\}} \subset H^1(X(\Gamma), \omega)[f],$$

descends to a $\overline{\mathbb{Q}}$-subspace of $H^1(X(\Gamma)_{\overline{\mathbb{Q}}}, \omega)[f]$.

In other words, for base-change forms, the Hodge-theoretic splitting of $H^1(X(\Gamma), \omega)[f]$ is $\overline{\mathbb{Q}}$-rational. This is a very rare phenomenon as this means the logs of units have linear independence relations.

Proof. Let σ_1, σ_2 be the two real embeddings of F, and suppose f is a base change form of f_0. Indeed, it is shown in [Ho, Corollary 5.2] that the space of $\overline{\mathbb{Q}}$-Stark units for f, $U_f \otimes \overline{\mathbb{Q}}$, is naturally isomorphic to $(U_{f_0} \otimes \overline{\mathbb{Q}})^{\oplus 2}$, and the decomposition is compatible with the Beilinson regulator. In particular, the $\overline{\mathbb{Q}}$-vector space spanned by the log of Stark units of f is exactly the $\overline{\mathbb{Q}}$-vector space spanned by the log of Stark units of f_0. In particular, Conjecture 13.1.2 implies that

$$\overline{\mathbb{Q}}\omega_f^{\{1\}} \oplus \overline{\mathbb{Q}}\omega_f^{\{2\}} = \frac{\log(u_{f_0})}{2\pi i} H^1(X(\Gamma)_{\overline{\mathbb{Q}}}, \omega),$$

where $u_{f_0} \in U_{f_0}$. $\square$

This can be more conceptually explained by the following. Indeed, a base change form satisfies an extra symmetry with respect to the "change of two upper half planes", namely

$$\mathbb{H}^2 \xrightarrow{(x,y)\mapsto(y,x)} \mathbb{H}^2,$$

and this extra symmetry guarantees that the $\overline{\mathbb{Q}}$-splitting of our form is compatible with the $\overline{\mathbb{Q}}$-splitting of the form in [Ho, Conjecture 3.21]. Thus, in this case there is extra geometric correspondence that can further distinguish automorphic representations of coinciding finite types.

13.2 Desiderata for generalized complex conjugations

To have a normalized choice of newforms simultaneously, we would like a certain way to relate different newforms. Example 10.3.5 suggests that the complex conjugation should play a role in the tentative statement of the full motivic action conjecture regarding rationality of cohomology classes. On the other hand, the complex conjugation can go back and forth between only two types of LDS's. For example, it sends a vector in the "holomorphic LDS" to that in the "anti-holomorphic LDS." Since a general motivic action conjecture involves many more LDS's, we suggest that there are *generalized complex conjugations* that can go between any of $\pi \in \mathfrak{P}_\lambda$.

We denote a generalized complex conjugation, sending an automorphic form $v \in \Pi \otimes A_C(\lambda)$ to another automorphic form in $\Pi \otimes A_{C'}(\lambda)$, by $c_{C,C'}$. There are several desired properties:

- $c_{C,C'}$ is $\mathbb{C}$-linear,
- If $C_{\text{hol}}, C_{\text{antihol}}$ are in $\mathfrak{C}_\lambda$, $c_{C_{\text{hol}},C_{\text{antihol}}}(f) = \overline{f}$,
- $\langle f, f\rangle_P = \langle c_{C,C'}(f), c_{C,C'}(f)\rangle_P$ (Condition (B)),

- $c_{C',C''} \circ c_{C,C'} = c_{C,C''}$ and $c_{C,C} = \mathrm{id}$,

- $c_{C_1' \times C_2', C_1 \times C_2} = (c_{C_1',C_1}, c_{C_2',C_2})$, for $G(\mathbb{R}) = G_1(\mathbb{R}) \times G_2(\mathbb{R})$.

It is still unclear how to formulate a set of conditions which will uniquely characterize $c_{C,C'}$'s. Although the nature of generalized complex conjugations still remains mysterious, using the ideas of Penrose transform and its related geometry, it could be possible to construct the purported generalized complex conjugations, following the suggestion by Joseph Wolf. This is the subject of the author's forthcoming work.

Remark 13.2.1. The last bullet point suggests that the *partial complex conjugation* (e.g. [Har4], [Sh1]) serves the role of generalized complex conjugations in the case of Hilbert modular forms. Unfortunately, for C_{hol} and C_{anithol} to be both in $\mathfrak{C}_\lambda$, λ has to be orthogonal to all compact roots, and this is allowed only if there is no compact root (as we exclude degenerate limit of discrete series from our discussion). Thus, we cannot use the usual complex conjugation besides when the associated symmetric space is a product of several upper half planes.

Remark 13.2.2. It is a relatively well-accepted technique in the case of unitary groups to use theta correspondence to move between different infinity types, as suggested by the recipe of [Pr]. Indeed, for a unitary group, [HLSu] proves that the theta correspondences and character twists act transitively upon the full Vogan L-packet (see [Vo2]). However, due to the idiosyncrasies of the recipe for the theta correspondence, it is still unclear whether the theta correspondence should be the generalized complex conjugation in this case.

For example, if $\Pi = \Pi_f \otimes \Pi_\infty$ is a globally generic automorphic representation of $\mathrm{U}(2,1)$ with Π_∞ being the generic limit of discrete series with Harish–Chandra paraemter $(a, b; b)$ with $a > 0 > b$, then the theta lift of Π into the same group $\mathrm{U}(2,1)$ should be $\Pi_f \otimes \Pi_{\mathrm{hol}}$ where Π_{hol} is the holomorphic limit of discrete series with the same Harish–Chandra character.

Whether this is actually the desired generalized complex conjugation is arguable. However, this is a certain operation that interacts with the complex conjugation in a very nontrivial way. First, note that, in this case, the holomorphic chamber and the antiholomorphic chamber are never adjacent to each other, because this will mean that the Harish–Chandra character is zero, making the LDS's solely consisted of degenerate LDS. Rather, the complex conjugation sends Π to $\Pi_f \otimes \overline{\Pi}_\infty$, where $\overline{\Pi}_\infty$ is the generic LDS with Harish–Chandra parameter $(-b, -a; -b)$. This lies on the wall between the generic chamber and the anti-holomorphic chamber. Theta correspondence now sends this nontrivially to $\mathrm{U}(1,2)$ (the same group at the finite places, and the signature is flipped only at the archimedean place) to $\Pi_f \otimes \Pi'_\infty$, where Π'_∞ is the holomorphic LDS with Harish–Chandra parameter $(-b; -b, -a)$. Regarding this with the unique isomorphism $\mathrm{U}(1,2) \cong \mathrm{U}(2,1)$, its complex conjugation is the antiholomorphic LDS with Harish–Chandra parameter $(a, b; b)$. In short,

$$\begin{tikzcd}
\boxed{\text{Generic, } \lambda = (a,b;b)} \arrow[r, "\Theta"] \arrow[d, "c"'] & \boxed{\text{Holomorphic, } \lambda = (a,b;b)} \arrow[dr, "c"] & \\
\boxed{\text{Generic, } \lambda = (-b,-a;-b)} \arrow[dr, "\Theta"'] & & \boxed{\text{Antiholomorphic, } \lambda = (-b,-a;-b)} \arrow[dl, dashed, "L\text{-packet}"] \\
& \boxed{\text{Holomorphic, } \lambda = (-b,-a;-b)} &
\end{tikzcd}$$

If one thinks of theta correspondence as a geometric correspondence that should interact well with generalized complex conjugation, another "geometric correspondence", then one can speculate that the dotted arrow should be an instance of generalized complex conjugation.

A caveat here, however, is that the holomorphic LDS with Harish–Chandra parameter $(-b, -a; -b)$ is a *degenerate* LDS (thus is not an element in $\mathfrak{P}_\lambda$).

Remark 13.2.3. We also speculate that this is the archimedean version of *excursion operators* (first appeared in the work of [Laf] on the global Langlands correspon-

dence over function fields, and extended to the context of mixed characteristic local Langlands via Kottwitz's conjecture, e.g. [FaMa], [RV]). Indeed, the isotypic decomposition with respect to the excursion algebra canonically decomposes the automorphic spectrum into L-indistinguishable pieces, which would mean that the excursion operators can go around different members of an L-packet.

Under the hypothesis on existence of generalized complex conjugations $c_{C,C'}$, we can formulate the motivic action conjecture in the Shimura variety context in its full form.

Conjecture 13.2.4. *Let λ, Π as in Conjecture 10.3.1. Let $f_h \in \Pi_f^{\mathrm{new}} \otimes \pi_h^{\mathrm{new}}$ be a newform such that $[f_h]$ (see Definition 10.1.8) is defined over $\overline{\mathbb{Q}}$. Let $\mathcal{E}_\lambda \cong \mathrm{Ext}^1_{(\mathfrak{p},K)}(I(\lambda), \pi_h)$ be defined such that $\mathbb{C}\alpha_i \cong \mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_{\{1,\cdots,n_\lambda\}-\{i\}}, \pi_h)$ sends $1\alpha_i$ to the homomorphism $c_{C_h, C_{\{1,\cdots,n_\lambda\}-\{i\}}}(f_h) \mapsto f_h$ (using the identification from Proposition 15.2.2). Then, for $v \in H^1_M((\mathrm{Ad}^* \Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1)) \subset H^1_{\mathscr{D}}((\mathrm{Ad}^* \Pi)_{\mathbb{R}}, \mathbb{C}(1))$, $a(v) \cdot f_l$ defines a coherent cohomology class $[a(v) \cdot f_l] \in H^{i_l+1}(X_G(\Gamma), [V_{A(\lambda)}])$ that is defined over $\overline{\mathbb{Q}}$.*

One can easily state a similar conjecture for the action of $\bigwedge^* H^1_M((\mathrm{Ad}\,\Pi)_{\mathcal{O}_E}, \overline{\mathbb{Q}}(1))$, but it is no deeper than the conjecture stated above.

Chapter 14

Beilinson's conjecture over a general number field

In this section, we recall the statement of *Beilinson's conjecture* we will need in the paper. A usual formulation of the conjecture involves motives over $\mathbb{Q}$ with coefficients in $\mathbb{Q}$, but we would have to relax both to be arbitrary number fields. A standard reference of this matter is [Ra, §6].

14.1 Chow motives

We recall the definition of Chow motives over a number field k, $\mathscr{M}_{k,\mathrm{rat}}$ in [PV, §2.1.1]. defined by cohomological correspondences up to rational equivalence. If k is a number field, then for a Chow motive $M \in \mathscr{M}_{k,\mathrm{rat}}$, there are the following cohomology theories, motivated by the cohomology theories of smooth proper k-varieties.

- For each prime ℓ, there is ℓ-adic cohomology $H^i(M_{\overline{K}}, \mathbb{Q}_\ell(r))$, which is a finite-dimensional ℓ-adic representation of $\mathrm{Gal}(\overline{K}/K)$.

- For each embedding $\sigma : k \hookrightarrow \mathbb{C}$, there is Betti cohomology $H^i_B(M_\sigma, \mathbb{Q}(r))$, which is a pure $\mathbb{Q}$-Hodge structure. If σ is a real embedding, it is equipped with

the *infinite Frobenius* $\mathrm{Fr}_{\sigma,\infty}$. On $H^i_B(M_\sigma, \mathbb{Q}(r)) \otimes_{\mathbb{Q}} \mathbb{C}$, the involution $\mathrm{Fr}_{\sigma,\infty} \otimes c_B$ preserves the Hodge decomposition, where c_B is the complex conjugation on the second factor. If σ is a complex embedding, $\mathrm{Fr}_{\sigma,\infty}$ is rather an isomorphism of $\mathbb{Q}$-vector spaces

$$\mathrm{Fr}_{\sigma,\infty} : H^i_B(M_\sigma, \mathbb{Q}(r)) \xrightarrow{\sim} H^i_B(M_{\overline{\sigma}}, \mathbb{Q}(r)).$$

- There is de Rham cohomology $H^i_{\mathrm{dR}}(M)(r)$, which is a finite-dimensional k-vector space, equipped with a decreasing filtration $F^k H^i_{\mathrm{dR}}(M)(r)$.

We will not care much about ℓ-adic realization, as it plays no role in the paper. For each embedding $\sigma : k \hookrightarrow \mathbb{C}$, there is a comparison isomorphism

$$\mathrm{comp}_\sigma : H^i_B(M_\sigma, \mathbb{Q}(r)) \otimes_{\mathbb{Q}} \mathbb{C} \cong H^i_{\mathrm{dR}}(M)(j) \otimes_{k,\sigma} \mathbb{C},$$

such that $\oplus_{p \geq k} H^{p,q}$ corresponds to $(F^k H_{\mathrm{dR}}) \otimes \mathbb{C}$. If σ is a real embedding, $\mathrm{Fr}_{\sigma,\infty} \otimes c_B$ corresponds to $1 \otimes c_{\mathrm{dR}}$, where c_{dR} is the complex conjugation on the second factor of $H^i_{\mathrm{dR}}(M)(j) \otimes_{k,\sigma} \mathbb{C}$. If σ is a complex embedding, $\mathrm{Fr}_{\sigma,\infty} \otimes c_B$ corresponds to $1 \otimes c_{\mathrm{dR}}$ in the sense that there is a commutative diagram

$$\begin{tikzcd}
H^i_B(M_\sigma, \mathbb{Q}(r)) \otimes_{\mathbb{Q}} \mathbb{C} \arrow[r, "\mathrm{comp}_\sigma"] \arrow[d, "\mathrm{Fr}_{\sigma,\infty} \otimes c_B"'] & H^i_{\mathrm{dR}}(M)(j) \otimes_{k,\sigma} \mathbb{C} \arrow[d, "1 \otimes c_{\mathrm{dR}}"] \\
H^i_B(M_{\overline{\sigma}}, \mathbb{Q}(r)) \otimes_{\mathbb{Q}} \mathbb{C} \arrow[r, "\mathrm{comp}_{\overline{\sigma}}"'] & H^i_{\mathrm{dR}}(M)(j) \otimes_{k,\overline{\sigma}} \mathbb{C}
\end{tikzcd}$$

Another key player is the Deligne cohomology. For a complex smooth projective variety X and a subring $A \subset \mathbb{C}$ invariant under complex conjugation, the Deligne cohomology $H^i_{\mathscr{D}}(X, A(r))$ is defined as the hypercohomology of the complex

$$A(r)_{\mathscr{D}} : A(r) \to \mathcal{O}_X \xrightarrow{d} \Omega^1_X \xrightarrow{d} \cdots \xrightarrow{d} \Omega^{r-1}_X,$$

regarded as a complex of analytic sheaves. This admits a complex conjugation of coefficients, denoted $c_{\mathscr{D}}$, which is induced from the complex conjugation on $A(r)_{\mathscr{D}}$. Similarly, there is infinite Frobenius for Deligne cohomology.

In the formulation of Beilinson's conjecture over $\mathbb{Q}$, a central role is played by the cohomology of $M_{\mathbb{R}}$. It plays the same role in Beilinson's conjecture over general number fields, even though it may sound peculiar to consider the base-change of M to $\mathbb{R}$ even if k is not a real field.

Definition 14.1.1. Given a subfield $A \subset \mathbb{C}$ stable under complex conjugation, define

$$H^i_{\mathrm{dR}}(M_{\mathbb{R}})(r) = \left(\bigoplus_{\sigma : k \hookrightarrow \mathbb{C}} H^i_{\mathrm{dR}}(M)(r) \otimes_{k,\sigma} \mathbb{C} \right)^{1 \otimes c_{\mathrm{dR}}}$$

$$H^i_B(M_{\mathbb{R}}, A(r)) = \left(\bigoplus_{\sigma : k \hookrightarrow \mathbb{C}} H^i_B(M_\sigma, \mathbb{Q}(r)) \otimes_{\mathbb{Q}} A \right)^{\oplus_\sigma \mathrm{Fr}_{\sigma,\infty} \otimes c_B}$$

$$H^i_{\mathscr{D}}(M_{\mathbb{R}}, A(r)) = \left(\bigoplus_{\sigma : k \hookrightarrow \mathbb{C}} H^i_{\mathscr{D}}(M_\sigma, A(r)) \right)^{\oplus_\sigma \mathrm{Fr}_{\sigma,\infty} \otimes c_{\mathscr{D}}}.$$

Concretely, cohomology of $M_{\mathbb{R}}$ is the part fixed by $\mathrm{Fr}_\infty \otimes c_B = 1 \otimes c_{\mathrm{dR}}$ via Betti-de Rham comparison isomorphism. Furthermore, there is a Chern class map

$$r_{\mathscr{D}} : H^i_M(M, \mathbb{Q}(r)) \to H^i_{\mathscr{D}}(M_{\mathbb{R}}, \mathbb{R}(r)).$$

The source of Chern class map is too large, and we choose a subspace $H^i_M(M_{\mathcal{O}_k}, \mathbb{Q}(r)) \subset H^i_M(M, \mathbb{Q}(r))$ consisting of classes that "extend to a good proper model of M over $\mathcal{O}_k$". If $M = h(X)$ for a smooth proper k-variety X, which has a regular proper model $\mathfrak{X}$ over $\mathcal{O}_k$, then

$$H^i_M(M_{\mathcal{O}_k}, \mathbb{Q}(r)) = H^i_M(M, \mathbb{Q}(r)) \cap (\mathrm{im}(K_{2r-i}\mathfrak{X} \to K_{2r-i}X) \otimes \mathbb{Q}),$$

where the latter image of K-theory groups is the image via the Chern class characters,

and this definition is independent of choice of $\mathfrak{X}$. In general, using alterations, Scholl defined this subspace in [Sch, Theorem 1.1.6] and showed that this is a unique way to assign subspaces satisfying various natural properties. The restriction of Chern class charcater into the integral subspace,

$$r_{\mathscr{D}} : H^i_M(M_{\mathcal{O}_k}, \mathbb{Q}(r)) \to H^i_{\mathscr{D}}(M_{\mathbb{R}}, \mathbb{R}(r)),$$

is called the Beilinson regulator.

14.2 Beilinson's conjecture for Chow and Grothendieck motives

From now on, we assume that $r \geq \frac{i+1}{2}$[1]. Beilinson's conjecture is formulated using fundamental exact sequences, which we review. From the definition of Deligne cohomology, for a complex smooth projective variety X, there is a long exact sequence

$$\cdots \to H^{i-1}_B(X, A(r)) \to \frac{H^{i-1}_B(X, \mathbb{C})}{F^r H^{i-1}_{\mathrm{dR}}(X)} \to H^i_{\mathscr{D}}(X, A(r)) \to H^i_B(X, A(r)) \to \cdots .$$

As this long exact sequence intertwines the involutions that define the cohomology of $M_{\mathbb{R}}$, for a Chow motive M defined over a number field k,

$$H^i_{\mathscr{D}}(M_{\mathbb{R}}, \mathbb{R}(r)) = \frac{H^{i-1}_B(M_{\mathbb{R}}, \mathbb{C})}{F^r H^{i-1}_{\mathrm{dR}}(M_{\mathbb{R}}) + H^{i-1}_B(M_{\mathbb{R}}, \mathbb{R}(r))} = \frac{H^{i-1}_B(M_{\mathbb{R}}, \mathbb{R}(r-1))}{F^r H^{i-1}_{\mathrm{dR}}(M_{\mathbb{R}})},$$

which gives rise to two fundamental exact sequences,

$$0 \to F^r H^{i-1}_{\mathrm{dR}}(M_{\mathbb{R}}) \to H^{i-1}_B(M_{\mathbb{R}}, \mathbb{R}(r-1)) \to H^i_{\mathscr{D}}(M_{\mathbb{R}}, \mathbb{R}(r)) \to 0,$$

[1] This is equivalent to that the weight of M, $i - 2r$, is negative. This is not a restriction as one can always reduce to this case possibly after using functional equation.

$$0 \to H_B^{i-1}(M_{\mathbb{R}}, \mathbb{R}(r)) \to \frac{H_{\mathrm{dR}}^{i-1}(M_{\mathbb{R}})}{F^r H_{\mathrm{dR}}^{i-1}(M_{\mathbb{R}})} \to H_{\mathscr{D}}^i(M_{\mathbb{R}}, \mathbb{R}(r)) \to 0.$$

The first two entries of the two fundamental exact sequences as above have natural $\mathbb{Q}$-structures, yielding the $\mathbb{Q}$-structure on $\det H_{\mathscr{D}}^i(M_{\mathbb{R}}, \mathbb{R}(r))$. Let $\mathscr{R}$ be the one from the first sequence, and $\mathscr{D}\mathscr{R}$ be the one from the second exact sequence.

Conjecture 14.2.1 (Beilinson's conjecture)**.** *Suppose either $r > \frac{i}{2} + 1$, or $r = \frac{i}{2} + 1$ and there is no Tate cycle, namely $H_{\text{ét}}^i(M_{\overline{k}}, \mathbb{Q}_\ell(i/2))^{\mathrm{Gal}(\overline{k}/k)} = 0$. Then, the following hold.*

1. *The Beilinson regulator*

$$r_{\mathscr{D}} : H_M^{i+1}(M_{\mathcal{O}_k}, \mathbb{Q}(r)) \otimes \mathbb{R} \to H_{\mathscr{D}}^{i+1}(M_{\mathbb{R}}, \mathbb{R}(r)),$$

   is an isomorphism.

2. *Let $\mathscr{M}$ be the $\mathbb{Q}$-structure defined on $\det H_{\mathscr{D}}^{i+1}(M_{\mathbb{R}}, \mathbb{R}(r))$ via the Beilinson regulator and the $\mathbb{Q}$-structure of the motivic cohomology. Then,*

$$\mathscr{M} = L(h^{-i}(M^\vee), 1-r)^* \mathscr{R} = L(h^i(M), r)\mathscr{D}\mathscr{R}.$$

Remark 14.2.2. The motives we will be working with come from automorphic representations in the sense of [Cl, §4.3.3], and for that purpose, we would rather like to work with *Grothendieck motives*, where the equivalence relation used is numerical equivalence, which is equivalent to homological equivalence under the Standard Conjecture D, which we have to assume. Fortunately, [PV, §2.1.9] works with arbitrary base field, so a similar set of assumptions would naturally lead to Beilinson's conjecture for Grothendieck motives.

Chapter 15

Deligne cohomology and Lie algebra cohomology

In this section, we develop a representation theoretic background parallel to [PV, §2-§4]. This is to correctly guess the motivic action conjectures, namely Conjecture 10.3.1 and Conjecture 13.2.4.

15.1 Nondegenerate limit of discrete series as constituents of reducible principal series

We need to understand what kinds of infinity types can appear in the coherent cohomology of Shimura varieties, given the finite part and the coefficient. Fortunately, the $(\mathfrak{p}, K)$-cohomology of unitary $G(\mathbb{R})$-representations is computed by Vogan–Zuckerman. As far as $G(\mathbb{R})$-representations are concerned, we will be only interested in discrete series, or in general nondegenerate limits of discrete series.

For a (L)DS, its L-packet consists of all (L)DS with the same infinitesimal character. Thus each such L-packet is consisted of $|W_G|/|W_K|$ elements, and in particular, upon choosing a system of positive roots Δ_K^+ for K, can be indexed by Weyl chambers

which makes all roots in Δ_K^+ positive. Given an infinitesimal character λ and a Weyl chamber C, let $A_C(\lambda)$ be the corresponding (L)DS. This is in accordance with the notation of [VZ]. In particular, $A_C(\lambda) = A_{\mathfrak{q}}(\lambda_C)$ for $\mathfrak{q} = \mathfrak{k} \oplus \mathfrak{p}_C$, where $\mathfrak{p}_C$ is the subalgebra of noncompact roots in Δ_C^+, the system of positive roots determined by C, and λ_C is the Weyl conjugate of λ that is contained in C. By the relatively straightforward nature of K-multiplicities of such representations, we have the following formulae.

Proposition 15.1.1. *Given λ and C as above, we have the following,*

$$i_{A_C(\lambda)} = \dim_{\mathbb{C}}(\mathfrak{p}_- \cap \mathfrak{p}_C),$$

$$V_{A_C(\lambda)} = V(\lambda_C + 2\rho(\mathfrak{p}_+)),$$

the highest weight representation of K with highest weight $\lambda_C + 2\rho(\mathfrak{p}_+)$, where $2\rho(\mathfrak{p}_+)$ is the sum of all roots in $\mathfrak{p}_+$.

Therefore, if λ lies on some wall of the Weyl chambers, then it can happen that $V_{A_C(\lambda)} = V_{A_{C'}(\lambda)}$ for different Weyl chambers C, C'. This is the setting we will be interested in. In such cases, we drop the subscript C if there is no issue of confusion.

Theorem 15.1.2. *Let λ be a nondegenerate analytically integral character. Let $\mathfrak{C}_\lambda = \{C \text{ Weyl chamber} \mid \lambda \in C\}$, and $\mathfrak{P}_\lambda = \{A_C(\lambda) \mid C \in \mathfrak{C}_\lambda\}$. Then, there is a parabolic subgroup $Q \subset G(\mathbb{R})$ and a discrete series representation ρ of the Levi M_Q such that*

$$I(\lambda) := \operatorname{Ind}_Q^G \rho = \bigoplus_{\pi \in \mathfrak{P}_\lambda} \pi,$$

where $\operatorname{Ind}_Q^G \rho$ is the normalized induction. These satisfy the following properties.

1. *Q is a parabolic subgroup which is minimal with respect to the property that the Langlands parameter $\varphi : W_{\mathbb{C}/\mathbb{R}} \to {}^L G$ corresponding to the representations in $\mathfrak{P}_\lambda$ can be arranged so that $\varphi(W_{\mathbb{C}/\mathbb{R}}) \subset {}^L Q$. Furthermore, if we denote T_{M_Q} by*

a Cartan subgroup of M_Q, then one can further assume that $\varphi(\mathbb{C}^\times) \subset {}^L T_{M_Q}$ and that $\varphi(W_{\mathbb{C}/\mathbb{R}})$ normalizes ${}^L T_{M_Q}$.

2. *$|\mathfrak{C}_\lambda|$ is a power of 2, and $\lambda^\perp \subset \mathfrak{g}$ is spanned by a superorthogonal set of real roots.*

3. *The infinitesimal character of ρ is the restriction of λ.*

Proof. We freely use the terminology of [Kn]. The property (3) is clear. The content of [Kn, §14.15] implies that any NLDS appears as a direct summand of such principal series with multiplicity one; this is via repeated application of generalized Schmid identities [Kn, Theorem 14.68]. The relation between Q and the Langlands parameter follows from the discussion before [Lang, Lemma 1]. Since the parabolic subgroup appears as Cayley transform of the minimal parabolic in the sense of [KZ], and one chooses noncompact simple roots for the Cayley transform, which in fact form a superorthogonal set of roots by [Kn, Theorem 14.64], based only on the infinitesimal character and not the chamber, it follows that

$$\{\mathrm{Ind}_Q^G \widetilde{\rho} \mid \widetilde{\rho} \text{ DS, infinitesimal character} = \text{the restriction of } \lambda\},$$

sees every NLDS with infinitesimal character λ at least once as its constituent. The number of constituents is exactly $|W_G|/|W_K|$, so each such NLDS appears exactly once. Each $\mathrm{Ind}_Q^G \rho$ has constituents $\pi = A_C(\lambda)$ such the representative of λ (thought as a Weyl orbit) in C is a fixed character (namely, C's that appear are all adjacent to a single representative of the Weyl orbit of λ). This is an equivalence relation, so this implies $I(\lambda)$ is precisely consisted of $A_C(\lambda)$ where C contains λ (regarded as an actual character). As the R-group is a direct sum of bunch of $\mathbb{Z}/2\mathbb{Z}$ by [Kn, §14.15], we get (2). □

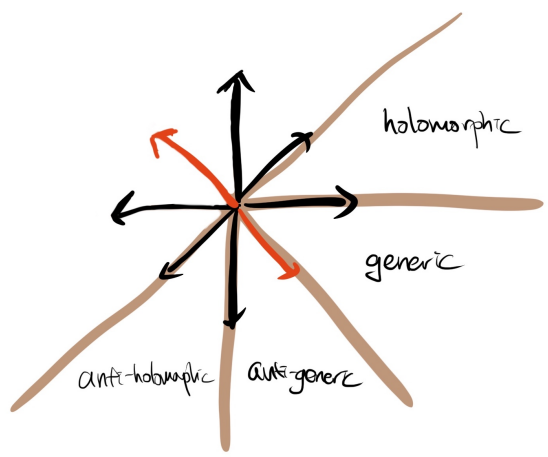
Example 15.1.3. We explain how Theorem 15.1.2 is realized in some examples. In

the figures, red arrows are the compact roots, so NLDS's are those lying on a wall not orthogonal to red arrows.

1. $\mathrm{SL}_2(\mathbb{R})$. Let $Q \subset \mathrm{SL}_2(\mathbb{R})$ be the upper triangular Borel, and let $\rho = \det / |\det| : Q \to \{\pm 1\}$. Then $I_Q^{\mathrm{SL}_2(\mathbb{R})}\rho = D_0^+ \oplus D_0^-$, the sum of the two NLDS, holomorphic ("weight 1") and anti-holomorphic ("weight -1").

2. $\mathrm{Sp}_4(\mathbb{R})$ (e.g. [Mu]). There are four types of LDS, two of them being holomorphic and anti-holomorphic, respectively, and the other two being large (i.e., maximal Gelfand–Kirillov dimension). We call the one adjacent to the holomorphic chamber *generic* and the other one adjacent to the anti-holomorphic chamber *anti-generic*. Using the notation of [Mu], a singular infinitesimal character is of one of the forms $(p, 0)$, $(0, -p)$ or $(p, -p)$ with $p \in \mathbb{N}$. The Langlands parameter φ_λ for infinitesimal character $\lambda = (a, b)$ is given by

$$\varphi_\lambda(re^{i\theta}) = \mathrm{diag}(e^{i(a+b)\theta}, e^{i(a-b)\theta}, e^{-i(a+b)\theta}, e^{-i(a-b)\theta}), \quad \varphi_\lambda(j) = \begin{pmatrix} 0 & (-1)^{a+b} I_2 \\ I_2 & 0 \end{pmatrix}.$$



- $\lambda = (p, 0)$ lies on the wall between the holomorphic chamber and the generic chamber. Then, Q is the so-called *Klingen parabolic*, whose Levi is

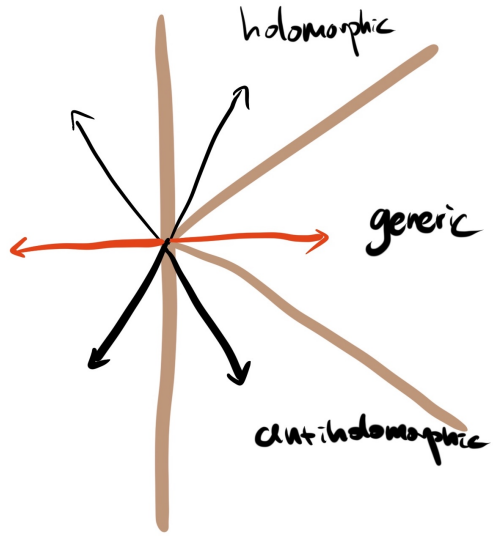
$\mathrm{GL}_1(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, and ρ is the (trivial extension of the) holomorphic DS D_p^+ of $\mathrm{SL}_2(\mathbb{R})$ of infinitesimal character p (or, equivalently, weight $p+1$).

- $\lambda = (p, -p)$ lies on the wall between the generic chamber and the anti-generic chamber. Then, Q is the so-called *Siegel parabolic*, whose Levi is $\mathrm{GL}_2(\mathbb{R})$, and ρ is the DS D_{2p} of $\mathrm{GL}_2(\mathbb{R})$ with central charcater $2p$ (which is as $\mathrm{SL}_2(\mathbb{R})$-representation the same as $D_{2p}^+ \oplus D_{2p}^-$).
- $\lambda = (0, -p)$ lies on the wall between the anti-generic chamber and the anti-holomorphic chamber. This situation is complex-conjugate to the situation of $\lambda = (p, 0)$. Thus, Q is again the Klingen parabolic, but ρ is the anti-holomorphic DS of the same infinitesimal character.

3. $\mathrm{U}(2,1)$ (e.g. [Wa], [Ro, §12]). There are three types of LDS, *holomorphic*, *generic* and *anti-holomorphic*. There are two typse of NLDS's , those lying on the wall between the holomorphic chamber and the generic chamber, and those lying on the wall betweent the generic chamber and the anti-holomorphic chamber. The Langlands parameters for (L)DS are of the form

$$\varphi(z) = \mathrm{diag}((z/\overline{z})^a, (z/\overline{z})^b, (z/\overline{z})^c), \quad \varphi(j) = \begin{pmatrix} & & 1 \\ & -1 & \\ 1 & & \end{pmatrix},$$

for $a, b, c \in \mathbb{Z}$, and the parameter only depends on the unordered set $\{a, b, c\}$. Ordering $a \geq b \geq c$, the two types of NLDS can occur for $a = b > c$ and $a > b = c$.



- $a = b > c$ lies on the wall between the holomorphic chamber and the generic chamber. Then, Q is the upper-triangular Borel (when $\mathrm{U}(2,1)$ is seen as the unitary group for the diagonal Hermitian matrix such as $\mathrm{diag}(1,1,-1)$), whose Levi is $\mathbb{C}^\times \times S^1$, and ρ is the character $\mathrm{diag}(\alpha, \beta, \overline{\alpha}^{-1}) \mapsto \alpha^a \beta^b \overline{\alpha}^{-c}$ for $\alpha \in \mathbb{C}^\times$, $\beta \in S^1$.
- $a > b = c$ lies on the wall between the generic chamber and the anti-holomorphic chamber. The situation is completely symmetric to the previous case.

15.2 Action of the Ext-space

We can now build an archimedean realization of motivic action from abstract nonsense. Recall that for a $(\mathfrak{p}, K)$-module M, $H^i(\mathfrak{p}, K; M)$ can be regarded as $\mathrm{Ext}^i_{(\mathfrak{p},K)}(\mathbf{1}, M)$, the i-th Ext group in the category of $(\mathfrak{p}, K)$-modules, where $\mathbf{1}$ is the trivial module. Thus, there is a natural action

$$\mathrm{Ext}^m_{(\mathfrak{p},K)}(M, N) \times \mathrm{Ext}^n_{(\mathfrak{p},K)}(\mathbf{1}, M) \to \mathrm{Ext}^{m+n}_{(\mathfrak{p},K)}(\mathbf{1}, N).$$

In particular, if λ is a nondegenerate singular character as above, with $\pi_1, \pi_2 \in \mathfrak{P}_\lambda$ with $i_{\pi_1} < i_{\pi_2}$, there is a natural action

$$\mathrm{Ext}^{i_{\pi_2}-i_{\pi_1}}_{(\mathfrak{p},K)}(\pi_1, \pi_2) \times H^{i_{\pi_1}}(\mathfrak{p}, K; \pi_1 \otimes V_{A(\lambda)}) \to H^{i_{\pi_2}}(\mathfrak{p}, K; \pi_2 \otimes V_{A_\lambda}),$$

where $V_{A(\lambda)}$ is the common coefficient for all $\pi \in \mathfrak{P}_\lambda$. Thanks to Theorem 15.1.2, the situation can be vastly simplified.

Theorem 15.2.1. *Let λ be as above.*

1. *There is a unique $\pi_l \in \mathfrak{P}_\lambda$ such that $i_{\pi_l} = i_l := \min\{i_\pi \mid \pi \in \mathfrak{P}_\lambda\}$.*
2. *For each $\pi \in \mathfrak{P}_\lambda$, $\mathrm{Ext}^{i_\pi - i_{\pi_l}}_{(\mathfrak{p},K)}(\pi_l, \pi)$ is one-dimensional.*
3. *Let $n_\lambda = \log_2 |\mathfrak{P}_\lambda|$, and $I_\lambda = \{\pi \in \mathfrak{P}_\lambda \mid i_\pi = i_{\pi_l} + 1\}$. Then, $n_\lambda = \#I$.*

Proof. As in the proof of Theorem 15.1.2, [Kn, §14.15] gives us a recipe of how λ is constructed in terms of superorthogonal roots $\{\alpha_1, \cdots, \alpha_{n_\lambda}\}$, it follows that the chambers in $\mathfrak{C}_\lambda$ are of the following form: there is one chamber C_l that is the "most holomorphic" chamber (namely, $\dim(C_l \cap \mathfrak{p}^+)$ is the largest) and a set of superorthogonal noncompact roots $\{\alpha_1, \cdots, \alpha_n\} \subset C_l$ such that all other chambers $C \in \mathfrak{C}_\lambda$ are of the form

$$C_{\{i_1, \cdots, i_k\}} := (C_l - \{\alpha_{i_1}, \cdots, \alpha_{i_k}\}) \cup \{-\alpha_{i_!}, \cdots, -\alpha_{i_k}\}.$$

These imply (1) and (3). To show (2), we use the Hochschild–Serre spectral sequence for Lie algebra cohomology, [Vo1, Proposition 6.1.29]. In our setting, we apply a spectral sequence

$$E_2^{p,q} = \mathrm{Ext}^p_{(\mathfrak{k},K)}(X, H^q(\mathfrak{p}_-, Y)) \Rightarrow \mathrm{Ext}^{p+q}_{(\mathfrak{p},K)}(X, Y),$$

for $X = \pi_l$, $Y = \pi$. Since the category of $(\mathfrak{k}, K)$-modules is semisimple, there is no higher Ext and the spectral sequence trivially degnerates at the E_2-page:

$$\mathrm{Ext}^m_{(\mathfrak{p},K)}(\pi_l, \pi) = \mathrm{Hom}_{(\mathfrak{k},K)}(\pi_l, H^m(\mathfrak{p}_-, \pi)).$$

On the other hand, the main result of [Wi] says that, if we choose the maximal nilpotent subalgebra $\mathfrak{b}_- \subset \mathfrak{k}$ such that $\mathfrak{n} := \mathfrak{b}_- \oplus \mathfrak{p}_-$ is a maximal nilpotent subalgebra of $\mathfrak{g}$, then $H^*(\mathfrak{n}, \pi)$ can only have weights $W_K\lambda + \rho$, where ρ is the half sum of positive roots in $\mathfrak{g}$, where the positivity is defined so that $\mathfrak{n}$ is spanned by the negative roots. Furthermore, each such weight occurs in exactly one cohomological degree. To relate this result to our setting, we use another Hochschild–Serre spectral sequence,

$$E_2^{p,q} = H^p(\mathfrak{b}_-, H^q(\mathfrak{p}_-, \pi)) \Rightarrow H^{p+q}(\mathfrak{n}, \pi).$$

Suppose V_τ be a K-type that appears in $H^m(\mathfrak{p}_-, \pi)$, with highest weight τ. Then, by Kostant's theorem [Kos, Theorem 5.14], $H^p(\mathfrak{b}_-, V_\tau)$ has a nonzero μ-isotypic part if and only if $\mu = w(\tau + \rho_{\mathfrak{k}}) + \rho_{\mathfrak{k}}$ for some $w \in W_K$ of length p, where $\rho_{\mathfrak{k}}$ is the half sum of positive roots in $\mathfrak{k}$, where the positive system is defined so that $\mathfrak{b}_-$ is spanned by the negative roots in $\mathfrak{k}$. In particular, μ determines p and τ, which implies that this spectral sequence also degenerates at the E_2-page. This implies that the K-types of $H^*(\mathfrak{p}_-, \pi)$ are of highest weight $\lambda + \rho_{w\mathfrak{p}_-} - \rho_{\mathfrak{k}}$ for $w \in W_G/W_K$, and these are precisely the minimal K-types of $\pi' \in \mathfrak{P}_\lambda$.

Let τ_l be the highest weight of the minimal K-type of π_l. Then, all other minimal K-types of $\pi' \in \mathfrak{P}_\lambda$ are of highest weight of the form $\tau_l - \sum_i^{n_\lambda} \epsilon_i \alpha_i$ for $\epsilon_i \in \{0, 1\}$. By Blattner's formula for π_l [Sc2], all K-types of π_l are of highest weight $\tau_l + \delta$, where δ is a positive linear combination of noncompact positive simple roots, where positivity is defined by C_l. In particular, this is a cone lying in the direction towards $\alpha_1, \cdots, \alpha_{n_\lambda}$, so the only K-type that appears both in π_l and $H^m(\mathfrak{p}_-, \pi)$ are τ_l, whence

$\dim \mathrm{Ext}^m_{(\mathfrak{p},K)}(\pi_l, \pi) \leq 1$. We have an exact formula for when nonzero Ext group occurs, which turns out to be $i_\pi - i_{\pi_l}$. Note that the same argument can be used to deduce the following: if $I \subset J \subset \{1, 2, \cdots, n_\lambda\}$, then $\mathrm{Ext}^{\#J-\#I}_{(\mathfrak{p},K)}(\pi_I, \pi_J)$ is one-dimensional. $\square$

Proposition 15.2.2. *For $I \subsetneq \{1, \cdots, n_\lambda\}$ with $i \notin I$, there is a natural identification*

$$\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{i\}}) \cong \mathrm{Hom}_{\mathbb{C}}(\pi_I^{\mathrm{new}}, \pi_{I\cup\{i\}}^{\mathrm{new}}).$$

Proof. We think the Ext group as the group of actual extensions. In particular, $\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{\alpha_i\}})$ is naturally identified with the one-dimensional vector space of the short exact squences of $(\mathfrak{p}, K)$-modules,

$$0 \to \pi_{I\cup\{\alpha_i\}} \to V \to \pi_I \to 0.$$

Such extension is uniquely identified by a scalar $\lambda \in \mathbb{C}$ in a way as follows. The sequence splits as a short exact sequence of K-modules, so the minimal K-type τ_I of π_I has a canonical lift in V, also denoted as τ_I. Let $v \in \tau_I$ be the chosen highest weight vector. As $\alpha_i \notin I$, v as a vector in π_I satisfies $\alpha_i \cdot v = 0$, where α_i is seen as an element of $U(\mathfrak{p})$. However, if v is seen as a vector in V, $\alpha_i \cdot v$ is nonzero precisely when the extension is nonsplit, and is sent to the line of highest weight vector of the minimal K-type of $\pi_{I\cup\alpha_i\}}$. Upon fixing such a highest weight vector v', we have $\alpha_i \cdot v = \lambda v'$ for some $\lambda \in \mathbb{C}$, and this gives an isomorphism $\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{\alpha_i\}}) \cong \mathbb{C}$. $\square$

Remark 15.2.3 (An important new difficulty in the $\delta = 0$ case)**.** It is very important to notice that, in our setting, the motivic action depends on the choice of *highest weight vectors of minimal K-types* of NLDS's, which was not necessary in the $\delta \neq 0$ case of [PV]. This is because our setting involves many different representations, whereas in the setting of [PV] one deals with a single representation. This will give rise to the conjecture on "generalized complex conjugations" in Section 13.2.

In the same vein, what naturally acts on the cohomology is not an Ext-algebra, but merely the Ext-group $\mathrm{Ext}^1(I(\lambda), \pi_h)$. However, one can identify this with an exterior algebra as follows. There is a natural identification of $(\mathfrak{p}, K)$-cohomology of π_I for $I \subset \{1, \cdots, n_\lambda\}$,

$$\begin{aligned} H^*(\mathfrak{p}, K; \pi_I \otimes V_{A(\lambda)}) &= \mathrm{Hom}_K(\wedge^* \mathfrak{p}_-, \pi_I \otimes V_{A(\lambda)}) \\ &= \mathrm{Hom}_K(\wedge^* \mathfrak{p}_- \otimes \sigma_\emptyset, \sigma_I) \\ &= (\wedge^* \mathfrak{p}_+)^{\alpha_I}, \end{aligned}$$

where σ_J is the minimal K-type of π_J, and $\alpha_I = \sum_{i \in I} \alpha_i$. In particular, this is identified with $(\wedge^* \mathcal{E}_\lambda)^{\alpha_I}$, where $\mathcal{E}_\lambda \subset \mathfrak{p}_+$ is the span of $\{\alpha_1, \cdots, \alpha_{n_\lambda}\}$, because these form a superorthogonal set of roots. Thus, one can think of the exterior algebra $\bigwedge^* \mathcal{E}_\lambda$ as acting on the $(\mathfrak{p}, K)$-cohomologies of π's in $\mathfrak{P}_\lambda$. Note that, however, the choice of highest weight vectors of minimal K-types underlies everything.

We will also occasionally use another notation, $\lambda^\perp$, for $\mathcal{E}_\lambda$, to be more indicative of its definition.

The action can be connected back to coherent cohomology of Shimura varieties. Consider the Π_f-isotypic part of the coherent cohomology of $X_G(\Gamma)$ with coefficient $[V_{A(\lambda)}]$; by Theorem 10.1.2, we have a natural isomorphism

$$H^{i_{\Pi_\infty}}(X)[\Pi_f] \cong H^i(\mathfrak{p}, K; \Pi_\infty \otimes V_{A(\lambda)}) \otimes_{\mathbb{C}} \Pi_f^\Gamma,$$

for each $\Pi_\infty \in \mathfrak{P}_\lambda$. Thus, using the notation of Theorem 15.2.1, we obtain an action

$$\wedge^* \mathcal{E}_\lambda \otimes_{\mathbb{C}} H^{i_l}(X)[\Pi_f] \to H^{i_l + *}(X)[\Pi_f].$$

15.3 Deligne cohomology as an Ext-space

We now relate the Ext-space with motivic cohomology. By [PV, (2.2.8)] and the definition of adjoint motive in Conjecture 10.2.2, there is a natural isomorphism $H^1_{\mathscr{D}}((\operatorname{Ad}\Pi)_{\mathbb{C}}, \mathbb{R}(1)) \cong H_B((\operatorname{Ad}\Pi)_{\mathbb{C}}, \mathbb{R})^{W_{\mathbb{C}/\mathbb{R}}}$, and its complexification $H_B((\operatorname{Ad}\Pi)_{\mathbb{C}}, \mathbb{C})^{W_{\mathbb{C}/\mathbb{R}}}$ can be naturally identified with $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})}$, the centralizer of the image of $W_{\mathbb{C}/\mathbb{R}}$ under the Archimedean Langlands parameter $\varphi : W_{\mathbb{C}/\mathbb{R}} \to {}^L G$.

We arrange φ in a way that is done in Theorem 15.1.2. Then, the centralizer of $\varphi(\mathbb{C}^\times)$ is the Levi component $\widehat{M_Q A_Q}$, where $\operatorname{Lie}\widehat{M_Q A_Q} = \widehat{\mathfrak{t}} \oplus \widehat{\lambda^\perp_{\mathbb{C}}}$. Thus, $\operatorname{Lie}\widehat{M_Q}$ is a direct sum of n_λ-copies of $\mathfrak{sl}_2(\mathbb{C})$. Since the centralizer of $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ in $\mathfrak{sl}_2(\mathbb{C})$ is a one-dimensional torus, $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})} = \mathbb{C}^{\oplus n_\lambda}$ is an n_λ-dimensional abelian Lie algebra, one coming from each α_i. It is therefore noncanonically isomorphic to $\lambda^\perp$. This also shows that $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})}$ gives a Cartan subalgebra $\widehat{\mathfrak{h}} = \widehat{\mathfrak{t}} \cap \operatorname{Lie}\widehat{M_Q}$ of $\operatorname{Lie}\widehat{M_Q}$.

There is on the other hand a natural map $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})} \to \operatorname{Ext}^1_{(\mathfrak{g},K)}(I(\lambda), \pi_h)$ whose definition does not depend on any arbitrary choice. Namely, twisting σ by a character of $H \subset M_Q$ gives a family of deformations of $I(\lambda) = \operatorname{Ind}_Q^G(\sigma)$, and the same deformation can be applied to extensions: namely, taking Ind_Q^G of $\operatorname{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1})$ gives rise to a map

$$\operatorname{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) \to \operatorname{Ext}^*_{(\mathfrak{g},K)}(\operatorname{Ind}_Q^G(\sigma), \operatorname{Ind}_Q^G(\sigma)).$$

Note that $\operatorname{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) = \bigwedge^* \widehat{\mathfrak{h}}$, so we have a natural map

$$\wedge^* \widehat{\mathfrak{h}} = \operatorname{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) \to \operatorname{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), I(\lambda)) = \bigoplus_{\pi \in \mathfrak{P}_\lambda} \operatorname{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), \pi) \twoheadrightarrow \operatorname{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), \pi_h),$$

where the last map is the projection map. This isomorphism is natural, so this defines a *degree-descending* action of $\bigwedge^* H^1_{\mathscr{D}}(\operatorname{Ad}\Pi, \mathbb{C}(1))$ on $H^{i_h - *}(X)[\Pi_f]$. Dually, this defines a *degree-ascending* action of $\bigwedge^* H^1_{\mathscr{D}}(\operatorname{Ad}\Pi, \mathbb{C}(1))^*$ on $H^{i_l + *}(X)[\Pi_f]$.

Part III

Higher Hida theory and the p-adic L-functions

Chapter 16

Introduction

In view of Kato's explicit reciprocity law [Ka], there has been a constant need of a "de Rham" construction of p-adic L-functions. The study of p-adic variation of automorphic forms in the de Rham or coherent cohomology context has been extremely limited. In particular, most results can only deal with holomorphic automorphic forms, namely those whose infinity type belongs to holomorphic (limit of) discrete series.

On the other hand, a very recent development of Pilloni has come up with the notion of *higher Hida/Coleman theory*, which p-adically interpolates the higher coherent cohomology groups of Shimura varieties. This opened a way to an approach for a de Rham construction of p-adic L-functions for automorphic representations that are not holomorphic. Indeed, such strategy was realized in [LPSZ] for the spin L-function of GSp_4.

Our objective is to use a similar strategy to p-adically interpolate the base change L-function for generic forms of $\mathrm{U}(2,1)$. Our main theorem is as follows.

Theorem 1. *Suppose* Π *is a generic, stable, cuspidal automorphic representation of* $\mathrm{GU}(2,1)$ *of infinitesimal character* $(k_1, k_2, k_3; c)$ *(see §16.5). If* $k_1 - k_2 \geq 2$ *and* $k_2 + k_3 \geq 2$*, then there is an element* $\mathcal{L}(\Pi, \mathbf{j}) \in \Lambda_{\overline{\mathbb{Q}}_p}(\mathbb{Z}_p^\times)$*, the Iwasawa algebra, such*

that, for any Dirichlet character χ of p-power conductor and any critical value a of $L(\Pi, \chi, s)$,

$$\mathcal{L}(\Pi, a + \chi) = Z_\infty(\Pi, \chi, a) Z_p(\Pi, \chi, a) L(BC(\Pi) \otimes \chi^{-1}, a),$$

where Z_∞ and Z_p are certain local factors at ∞ and p, respectively.

As in [LPSZ], the two main ingredients are

- *cohomological period integrals* (period integrals relating values of L-functions to coherent cohomology of Shimura varieties), and
- the higher Hida theory.

In our case, the cohomological period integral in use is the one of Gelbart–Piatetski-Shapiro [GPS]. On the other hand, most of the higher Hida theory for $\mathrm{GU}(2,1)$ is developed in [Ng]. We expect the same strategy will work for the Gan–Gross–Prasad type period integral for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$ for cusp forms of infinity type (generic, holomorphic).

16.1 Higher Hida theory for $\mathrm{U}(2,1)$

To make use of the cohomological interpretation of a period integral, we need to interpolate the classes appearing in higher coherent cohomology, which is exactly what higher Hida theory does. Fortunately, the analogue of higher Hida theory was developed in a Lyon thesis of M. T. Nguyen in [Ng]. The situation for $\mathrm{GU}(2,1)$ is actually easier than that for GSp_4, as the Hecke correspondences T_p and U_p are represented by a miniscule diagonal matrix, so that there is no need to compose two different Hecke correspondences. In particular, there are strengthenings in several parts, most notably a proof of an integral control theorem; this is in contrast with [Pi], where

the integral control theorem is only proved only for a sufficiently regular weight (with inexplicit bounds on the regularity). As a result, [Ng] did not have to invoke automorphic representations in the course of the proof of control theorem. We complement this with several calculations and results relating automorphic representations and the ordinary coherent cohomology.

16.2 Generalized Whittaker models: Unfolding of period integrals

The period integral we need for the purpose of this paper is already developed in [GPS]. On the other hand, as in the case of Novodvorsky's integral used in [LPSZ], this period integral is a certain degenerate case of a partly conjectural formula of Ichino–Ikeda type (also called the refined Gan–Gross–Prasad conjecture). Although the Ichino–Ikeda formula is now established by an amazing recent development by W. Zhang and his collaborators (cf. [Zha], [BPC+], [BPL+]), the proof of the formula is extremely roundabout. Furthermore, the Ichini–Ikeda formula is about the *norm* of the Gan–Gross–Prasad period integral, so the formula does not completely characterize the value of the period integral.

We show, on the other hand, that the period integral itself unfolds nicely into a certain local model, just as the classical Rankin–Selberg integral unfolds into the Whittaker model, so that the period integral can be manifestly seen as an Euler product of local factors. The local model we use is the so-called *generalized Whittaker model*, in the sense of the Japanese school of mathematicians (Shintani, Yamashita, Murase–Sugano, $\cdots$), who has systematically studied certain generalizations of Whittaker models still enjoying the finite multiplicity property. We use this name to distinguish the model from the *degenerate Whittaker models*, the spaces of functionals transforming under a smaller unipotent subgroup, which are typically infinite-dimensional as

soon as one takes the unipotent group smaller than that which appears on the wave-front set. Such infinite-dimensionality can be salvaged by also considering the action of a part of the Levi.

We also use freely the language of Fourier expansions with respect to closed subgroups. Typically the widely-known Fourier expansions of automorphic forms are Whittaker–Fourier expansion, q-expansion and Fourier–Jacobi expansion. The latter two only apply to holomorphic automorphic forms, whereas the former only applies to generic automorphic forms. The Whittaker–Fourier expansion is just the Fourier expansion associated to the maximal unipotent subgroup, q-expansion to the unipotent radical of the Siegel parabolic, and Fourier–Jacobi expansion to the unipotent radical of the "Jacobi parabolic." Making sense of more general Fourier expansions for more general types of automorphic forms requires overcoming some analytic issues coming from the continuous spectrum.

Using the above ingredients, we show that the generalized Whittaker model associated to the Jacobi parabolic of $\mathrm{U}(2,1)$ can be used to show that the GGP period integral for $\mathrm{U}(2,1) \times \mathrm{U}(1,1)$ is Eulerian; the local zeta integral is an integral of a local generalized Whittaker function (of $\mathrm{U}(2,1)$-form) times a Whittaker function (of $\mathrm{U}(1,1)$-form). We also explain how this line of ideas can explain the mystery of *Rankin–Selberg integrals with non-unique models*. This characterization of Eulerianity could be useful for the computation of local factors (especially at p and ∞).

Finally, it would be extremely interesting to understand how the generalized Whittaker model interacts with more general GGP period integrals. On one hand, the unfolding involves a branching of unipotent group restricted to a smaller unipotent group. On the other hand, the unfolding uses a Fourier expansion, which involves the wave-front of the involved representations. It is natural to expect that the group-theoretic nature of both ingredients would interact in an interesting way. For example, it would be interesting to discover an example of unfolding of a GGP period integral

into generalized Whittaker models for a pair of non-generic representations.

16.3 Prospects: The square-root p-adic L-function for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$

We expect that the strategy developed in this paper can be ultimately applied to the construction of the "square-root p-adic L-function" for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$, for automorphic forms of infinity type (generic, holomorphic), as follows.

- Unfolding of the GGP period integral into generalized Whittaker model, in turn, gives rise to integral formula for the local factors of the integral, which is a more refined information that what the Ichino–Ikeda formula conveys. Thus, the local factors can be explicitly computed given explicit formulae for generalized Whittaker functions. The archimedean generalized Whittaker functions for $\mathrm{U}(n,1)$ are studied in [Ts1, Ts2, Ts3], whereas the nonarchimedean generalized Whittaker functions are studied in [MS1]. Indeed, we have computed the unramified local factors for $\mathrm{U}(2,1) \times \mathrm{U}(1,1)$, and the full computation of all local factors will be contained in a forthcoming work of the author.

- The GGP period integral for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$ would unfold similarly to the unfolding process of that for $\mathrm{U}(2,1) \times \mathrm{U}(1,1)$ dealt in §19. The main analytic difficulty does not arise, as the extra reductive part for the generalized Whittaker function is compact (namely, just $\mathrm{U}(1)$).

- The higher Hida theory for $\mathrm{GU}(n,1)$ would enjoy many simplifications. Firstly, as in the case of $\mathrm{GU}(2,1)$, the Hecke operators T_p and U_p would be represented as a single cohomological correspondence given by the parahoric level Shimura vareity. Moreover, even though the $\mathrm{GU}(n,1)$-Shimura variety would have many more Ekedahl–Oort strata, the stratification has a nested structure. Namely,

there are $n+1$ strata, and the closure relation is a total order, so that there is unique closed stratum, which is contained in the closure of another stratum, which is contained in the closure of another stratum, etc., until you reach the unique open stratum, the ordinary locus. Thus, one can extend the argument used in the higher Hida theory for $\mathrm{GU}(2,1)$ to go through an induction over the depth of strata using generalized Hasse invariants (established in [Box]). Similarly, the embedding of Igusa towers for the $\mathrm{GU}(n-1,1)$-Shimura variety to that of the $\mathrm{GU}(n,1)$-Shimura variety can be given completely analogously.

16.4 Summary

In §16.5, we set the basic notations. Then, in §17, we review the higher Hida theory for $\mathrm{U}(2,1)$ developed in [Ng]. In the course of the review, we complement it with several missing ingredients necessary for our development in this paper. In §18, we define the p-adic pushforward map from the smaller Shimura variety to the bigger Shimura variety, by specifying the embedding of partial Igusa towers.

In §19, we develop another upshot of the paper, namely the use of *generalized Whittaker models* in the study of period integrals. We first review the integral representation of Gelbart–Piatetski-Shapiro for the base-change L-function for $\mathrm{U}(2,1)$. This can be regarded as a degenerate case of the Gan–Gross–Prasad period integral for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$, whose explicit relation to values of L-functions is given by the refined Gan–Gross–Prasad conjectures. We develop a way to directly unfold the period integral involving the generalized Whittaker models for the GGP period integral for $\mathrm{U}(2,1)\times\mathrm{U}(1,1)$. We also show that the same strategy can be used for other Rankin–Selberg-type period integrals that were not known to unfold to nice local models.

In §20, we consider how coherent cohomology of the $\mathrm{GU}(2,1)$-Shimura variety is

related to automorphic forms for $\mathrm{GU}(2,1)$. We describe the period integral as well as p-adic interpolation of it in terms of cup products in coherent cohomology. We also describe the theory of *nearly holomorphic and overconvergent modular forms* in terms of coherent cohomology of automorphic vector bundles.

In §23, we explicitly calculate the local factors of the Gelbart–Piatetski-Shapiro integral. Important are the calculations of the factors at p and at ∞. In §22, we gather all the ingredients we developed to construct the desired p-adic L-function.

16.5 Setup

16.5.1 Imaginary quadratic field

Once and for all we choose an embedding $\iota_\infty : \overline{\mathbb{Q}} \to \mathbb{C}$ and $\iota_p : \overline{\mathbb{Q}} \to \mathbb{C}_p$, connected via $\iota : \mathbb{C} \xrightarrow{\sim} \mathbb{C}_p$. Let K be an imaginary quadratic field and p be a rational prime that splits in K. By choosing a preferred embedding $\sigma : K \to \overline{\mathbb{Q}}$, one gets a positivity in an imaginary direction (via $\sigma_\infty = \iota_\infty \circ \sigma$) and a preferred $\pi \mid p$ (via $\sigma_p = \iota_p \circ \sigma$); thus we have a decomposition $p = \pi\overline{\pi}$.

Let D_K be the discriminant of K. We have $\mathcal{O}_K \otimes_{\mathbb{Z}} \mathcal{O}_K[1/2D_K] = \mathcal{O}_K[1/2D_K]e^+ \oplus \mathcal{O}_K[1/2D_K]e^-$, where e^+ and e^- are the idempotents corresponding to the identity $\mathrm{id} : K \to K$ and the complex conjugation $\bar{\cdot} : K \to K$, respectively.

16.5.2 Groups

Let $V = Y_K \oplus W \oplus X_K$ be a skew-Hermitian K-space of dimension 3 with the skew-Hermitian form $\theta_{2,1}$ defined by

$$\theta_{2,1} = \begin{pmatrix} & & -1 \\ & \theta & \\ 1 & & \end{pmatrix}.$$

Here, the bases are given by $X_K = Kx$, $Y_K = Ky$, $W = Kw$, respectively, and $\theta \in K^\times$ is totally imaginary with $i\sigma_\infty(\theta) > 0$, where i is the usual i (not $-i$) in $\mathbb{C}$. Without loss of generality, we assume that $\mathrm{Tr}_{K/\mathbb{Q}}(\theta) \in \mathbb{Z}$. Let $\langle , \rangle : V \times V \to \mathbb{Q}$ be the alternating pairing defined by $\langle v, v' \rangle = \mathrm{Tr}_{K/\mathbb{Q}}(\theta_{2,1}(v, v'))$.

Let $G = \mathrm{GU}(V)$ be the $\mathbb{Q}$-group of unitary similitudes of $(V, \theta_{2,1})$ (namely, g such that $g\theta_{2,1}\overline{g}^T = \nu(g)\theta_{2,1}$), and $G' = \mathrm{U}(V)$ be the unitary group of $(V, \theta_{2,1})$. Let $\nu : G \to \mathbb{G}_{m,\mathbb{Q}}$ be the similitude morphism. Let also $H \subset G$ ($H' \subset G'$, respectively) be the stabilizer of W. Then, H, H' act on $V_H = Y_K \otimes X_K$. We may suggestively call G, G', H, H' as $\mathrm{GU}(2,1), \mathrm{U}(2,1), \mathrm{GU}(1,1) \times_{\mathrm{GL}_1} \mathrm{GU}(1), \mathrm{U}(1,1) \times \mathrm{U}(1)$, respectively. We use the notation $[A] = A(\mathbb{Q})\backslash A(\mathbb{A})$ for a $\mathbb{Q}$-group A.

16.5.3 Lattices

Denote by $\mathfrak{d}_K$ the different of K, we define a lattice $M = Y \oplus L \oplus X^\vee := \mathcal{O}_K y \oplus \mathcal{O}_K w \oplus \mathfrak{d}_K^{-1} x$. Then, $\langle , \rangle_{2,1}|_M$ is $\mathbb{Z}$-valued, and $M \otimes_{\mathbb{Z}} \mathbb{Z}_p =: M_p$ is self-dual with respect to $\langle , \rangle_{2,1}$. Consider the $\mathbb{Z}_p$-group schemes $\mathcal{G} = \mathrm{GU}(M_p)$ and $\mathcal{G}' = \mathrm{U}(M_p)$. Let $M_\Sigma := e^+ M_p$ and $M_{\Sigma^c} := e^- M_p$ be $\mathbb{Z}_p$-modules. We have identifications

$$G(\mathbb{Q}_p) \xrightarrow{\sim, g \mapsto (g|_{e^+V}, \nu(g))} \mathrm{GL}_3(K_\pi) \times K_\pi^\times,$$

$$G'(\mathbb{Q}_p) \xrightarrow{\sim, g \mapsto g|_{e+V}} \mathrm{GL}_3(K_\pi),$$

and the same identification gives

$$\mathcal{G}(\mathbb{Z}_p) \xrightarrow{\sim, g \mapsto (g|_{M_\Sigma}, \nu(g))} \mathrm{GL}_3(\mathcal{O}_{K,\pi}) \times \mathcal{O}_{K,\pi}^\times,$$

$$\mathcal{G}'(\mathbb{Z}_p) \xrightarrow{\sim, g \mapsto g|_{M_\Sigma}} \mathrm{GL}_3(\mathcal{O}_{K,\pi}).$$

Let $\mathcal{P}' \subset \mathcal{G}'$ be the parahoric subgroup stabilizing the plane through y, w, and let $\mathcal{P} = \mathbb{G}_m(\mathbb{Z}_p) \times \mathcal{P}'$. We will colloquially refer it as a *mirahoric subgroup*. We also consider the upper triangular Borels $B'_G \subset G'$, $B_G \subset G$, $B'_H \subset H'$ and $B_H \subset H$, defined as the stabilizers of Y_K.

16.5.4 Shimura varieties

Let $\Gamma = \Gamma^p \Gamma_p \subset G(\mathbb{A}_f)$ be a neat open compact subgroup. We will only be considering the two cases $\Gamma_p = \mathcal{G}$ (hyperspecial) or $\Gamma_p = \mathcal{P}$ (mirabolic). Consider the moduli problem $Y_G(\Gamma) : \mathrm{Sch}_{\mathcal{O}_{K,\pi}} \to \mathrm{Sets}$ parametrizing the equivalence classes of quadruples $(A/S, i, \overline{\lambda}, \eta^p(, C))$ where

- A/S is an abelian scheme of relative dimension 3 for a $\mathcal{O}_{K,\pi}$-scheme S,
- $i : \mathcal{O}_K \hookrightarrow \mathrm{End}_S(A) \otimes_{\mathbb{Z}} \mathbb{Z}_{(p)}$ satisfying the Kottwitz determinant condition,

  $$\det(X - i(k) | \operatorname{Lie} A) = (X - \sigma(c(b)))^2 (X - \sigma(b)) \in \mathcal{O}_{K,\pi}[X], \quad \text{for all } k \in \mathcal{O}_K,$$

- λ is a prime-to-p polarization of A over $A^\vee$ and $\overline{\lambda}$ is the $\mathbb{Z}_{(p),+}$-orbit of λ, inducing a Rosati involution, sending $i(k)$ to $i(\overline{k})^\vee$ for all $k \in \mathcal{O}_K$,
- η^p is a Γ^p-level structure in the sense of [Kot],
- and if $\Gamma_p = \mathcal{P}$, then C is a totally isotropic order p^2 subgroup of $A[\pi]$.

We tend to omit Γ from the notation and rather write as Y (for hyperspecial level at p) and Y_{Kl} (for mirahoric level at p) if there is no confusion. Note that the Shimura variety we consider is the base-change of the usual Shimura variety to a p-adic base $\mathcal{O}_{K,\pi}$.

There exists a universal abelian scheme $\mathcal{A}$ over Y_G, and the Hodge bundle $\omega = e^*_{\mathcal{A}/Y_G}\Omega^1_{\mathcal{A}/Y_G}$, where $e_{\mathcal{A}/Y_G}$ is the unit section, is a rank 3 bundle with the action of $\mathcal{O}_K$. Thus, it decomposes into $\omega = \omega_\sigma \oplus \omega_{\overline{\sigma}}$, according to the CM types; ω_σ is of rank 2, whereas $\omega_{\overline{\sigma}}$ is of rank 1, by the signature condition.

It is known that the moduli problem is represented by a quasiprojective $\mathcal{O}_{K,\pi}$-scheme, denoted also as $Y_G(\Gamma)$, which is also smooth when the level at p is hyperspecial. Via $H \to G$, there is a natural embedding $\iota : Y_H(\Gamma_H) \to Y_G(\Gamma)$ for $\Gamma_H = \Gamma \cap H(\mathbb{A}_f)$ and $Y_H(\Gamma_H)$ is the scheme representing a similar moduli problem for H. This is on the other hand just a product of a modular curve and a definite unitary Shimura variety (thus a finite scheme); they are identified via taking the product of the two identical elliptic curves (from the modular curve) and a CM elliptic curve (from the definite unitary Shimura variety).

These Shimura varieties admit arithmetic toroidal compactifications, and the datum can be chosen so that the embedding extends to that of compactifications. Note that these are $\mathbb{Z}_p$-smooth if $\Gamma_p = \mathcal{G}$ but not so in general. We denote the toroidal compactification by $X_G(\Gamma)$, etc. We also denote the boundary divisor as D, if there is no confusion in notation. Note that we do not need to make any choices for H as Y_H is basically a copy of several modular curves. The choices will not matter because the cohomology does not depend on the choice. Also, we denote the minimal compactifications by $X_G(\Gamma)^*$, etc.

16.5.5 Complex uniformization: two different ways

We will talk about automorphic cohomology, so it is beneficial to write down the details of the Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ and complex structure $\mathfrak{p}_{\mathbb{C}} = \mathfrak{p}_+ \oplus \mathfrak{p}_-$. Note first that as the center gets modded out in the construction of symmetric spaces, it is usually the simplest to work with special unitary groups. Various parts of the section came from [Lan3], [He].

Generalities

Let us abstractly summarize what happens. We freely use standard notations that are not necessarily compatible with what are used in the other paragraphs. Given a semisimple algebraic group G with $K \subset G(\mathbb{R})$ a maximal compact, as we have $\operatorname{rank}_{\mathbb{R}} K = \operatorname{rank}_{\mathbb{R}} G(\mathbb{R})$, there is a compact Cartan $T_c \subset K \subset G(\mathbb{R})$. From the choice of a Cartan involution, one has a Cartan decomposition $\mathfrak{g}_{\mathbb{R}} = \mathfrak{k}_{\mathbb{R}} \oplus \mathfrak{p}_{\mathbb{R}}$ (we have put the subscripts to distinguish them from their complexifications). Furthermore, the choice of a Hodge cocharacter, or equivalently the choice of the complex structure on $G(\mathbb{R})/K$, gives a decomposition $\mathfrak{p}_{\mathbb{C}} = \mathfrak{p}_+ \oplus \mathfrak{p}_-$. Let $P_-(\mathbb{C}) \subset G(\mathbb{C})$ be the parabolic subgroup whose Lie algebra is $\mathfrak{k}_{\mathbb{C}} \oplus \mathfrak{p}_-$. Then the obvious inclusion $G(\mathbb{R})/K \hookrightarrow G(\mathbb{C})/P_-(\mathbb{C})$ is an open immersion of complex manifolds.

Using the $P_-(\mathbb{C})$-torsor $G(\mathbb{C}) \to G(\mathbb{C})/P_-(\mathbb{C})$, for any algebraic representation V of $P_-(\mathbb{C})$, one can form a holomorphic vector bundle $[V] = G(\mathbb{C}) \times^{P_-(\mathbb{C})} V \to G(\mathbb{C})/P_-(\mathbb{C})$, which restricts to a holomorphic vector bundle over $G(\mathbb{R})/K$ denoted also as $[V]$. It is known that these descend down to arithmetic quotients, and these admit arithmetic models. These are called *automorphic vector bundles*.

Here comes why we put the phrase "two different ways": when talking about automorphic vector bundles, or anything that comes out of using a compact Cartan, sometimes one works with a noncompact maximal torus, which is also the case for unitary groups. We claim that it does not really matter. Indeed, let $T \subset G$ be a maximal

torus that is not necessarily $\mathbb{R}$-anisotropic. Still, $T(\mathbb{C})$ and $T_c(\mathbb{C})$ are conjugate, say $T(\mathbb{C}) = g^{-1}T_c(\mathbb{C})g$. Suppose further that there is a parabolic subgroup $T \subset P \subset G$ such that $P(\mathbb{C}) = g^{-1}P_-(\mathbb{C})g$. Then $G(\mathbb{C})/P_-(\mathbb{C}) \cong G(\mathbb{C})/P(\mathbb{C})$, $h \mapsto g^{-1}hg$. Using this, one can construct automorphic vector bundles for algebraic representations of $P(\mathbb{C})$. Furthermore, these constructions yield the same vector bundles as those using $P_-(\mathbb{C})$ and K, as algebraic representations of $P_-(\mathbb{C})$ and those of $P(\mathbb{C})$ are identified via the same conjugation. Also, later we will be interested in automorphic cohomology, in particular Lie algebra cohomology of real group representations, but then again the automorphic vector bundles built for T are literally the same as automorphic vector bundles built for T_c, so the calculation of Lie algebra cohomology can be done purely in terms of $T_c \subset K \subset P_-$.

In the case of $\mathrm{U}(2,1)$

There are two prominent ways of dealing with unitary groups. One can work with the antidiagonal matrix as we do, but one can also work with the diagonal matrix. In our case, it would be to use the Hermitian diagonal matrix $1_{2,1} := \begin{pmatrix} 1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$. Let $\mathrm{U}_{\mathrm{diag}}$ be the unitary group defined using $1_{2,1}$ (and similarly for other related groups). Then, $\mathrm{U}_{\mathrm{diag}}(\mathbb{R})$ is isomorphic to $\mathrm{U}(2,1)(\mathbb{R})$ via

$$\mathrm{U}_{\mathrm{diag}}(\mathbb{R}) \xrightarrow{g \mapsto cgc^{-1}} \mathrm{U}(2,1)(\mathbb{R}), \quad c = \begin{pmatrix} \frac{\sqrt{-k}}{2}(1+i) & & \frac{\sqrt{-k}}{2}(1+i) \\ & 1 & \\ \frac{\sqrt{-k}}{2}(-1+i) & & \frac{\sqrt{-k}}{2}(1-i) \end{pmatrix},$$

where $\theta = ik$. Namely,

$$c(\theta 1_{2,1})\overline{c}^T = \theta_{2,1}.$$

The unitary group in the form of $\mathrm{U}_{\mathrm{diag}}$ acts naturally on the complex ball $\mathcal{D} =$

$\{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^2 + |z_2|^2 < 1\}$ via $g \in \mathrm{U}_{\mathrm{diag}}$ acting on 1×3 matrix $\begin{pmatrix} z_1 & z_2 & 1 \end{pmatrix}$, namely

$$\begin{pmatrix} z & 1 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} zA + C & zB + D \end{pmatrix} \sim \begin{pmatrix} \frac{zA+C}{zB+D} & 1 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 & z_2 \end{pmatrix}.$$

A natural choice of the maximal compact subgroup is the stabilizer of $0 \in \mathcal{D}$, namely $K_{\infty,\mathrm{diag}} := \begin{pmatrix} \mathrm{U}_2 & \\ & \mathrm{U}_1 \end{pmatrix}$. From this, we see that $K_{\infty,\mathbb{C}}$ is conjugate to $M_P(\mathbb{C})$.

The unitary group in the form of $\mathrm{U}(2,1)$ acts naturally on the unbounded domain

$$X = \left\{ \tau = \begin{pmatrix} x \\ y \end{pmatrix} \in M_{2\times 1}(\mathbb{C}) \mid 2\sqrt{-k}\,\mathrm{Im}\, x > |y|^2 \right\},$$

via $g \in \mathrm{U}(2,1)$ acting on 3×1 matrix $\begin{pmatrix} \tau \\ 1 \end{pmatrix}$, namely

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \tau \\ 1 \end{pmatrix} = \begin{pmatrix} A\tau + B \\ C\tau + D \end{pmatrix}.$$

A natural choice of the maximal compact subgroup is the stabilizer of $\begin{pmatrix} i \\ 0 \end{pmatrix} \in X$, which corresponds to $K_{\infty,\mathrm{diag}}$ after the conjugation by c as denoted above.

The Hodge cocharacters correponding to the complex structures of the above realizations of symmetric spaces are

$$h_{\mathrm{diag}} : S^1 \to \mathrm{U}_{\mathrm{diag}}, \quad h_{\mathrm{diag}}(z) = \begin{pmatrix} z & & \\ & z & \\ & & \overline{z} \end{pmatrix},$$

$$h : S^1 \to \mathrm{U}(2,1), \quad h(x+yi) = \begin{pmatrix} x & & y \\ & x+yi & \\ -y & & x \end{pmatrix}.$$

The Lie algebra of $\mathrm{U}_{\mathrm{diag}}$ is given by

$$\mathfrak{g}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & b & c \\ -\overline{b} & d & e \\ -\overline{c} & \overline{e} & f \end{pmatrix} \mid a, d, f \in i\mathbb{R} \right\},$$

$$\mathfrak{p}_{\mathrm{diag}} = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & e \\ \overline{c} & \overline{e} & 0 \end{pmatrix} \right\}, \quad \mathfrak{k}_{\mathrm{diag}} = \left\{ \begin{pmatrix} a & b & 0 \\ -\overline{b} & d & 0 \\ 0 & 0 & f \end{pmatrix} \mid a, d, f \in i\mathbb{R} \right\}.$$

Under this, $\mathfrak{p}_{\mathrm{diag},\mathbb{C}} = \mathfrak{p}^+_{\mathrm{diag}} \oplus \mathfrak{p}^-_{\mathrm{diag}}$, where $\mathfrak{p}^-_{\mathrm{diag}}$ ($\mathfrak{p}^+_{\mathrm{diag}}$, respectively) is generated by $X^- = N^+ \otimes i - N^- \otimes 1$ and $Y^- = M^+ \otimes i - M^- \otimes i$ ($X^+ = N^+ \otimes i + N^- \otimes 1$,

$Y^+ = M^+ \otimes i + M^- \otimes 1$, respectively), where

$$N^- = \begin{pmatrix} & & & i \\ & & & \\ & & & \\ -i & & & \end{pmatrix}, N^+ = \begin{pmatrix} & & & 1 \\ & & & \\ & & & \\ 1 & & & \end{pmatrix}, M^- = \begin{pmatrix} & & & \\ & & & i \\ & & & \\ & -i & & \end{pmatrix}, M^+ = \begin{pmatrix} & & & \\ & & & 1 \\ & & & \\ & 1 & & \end{pmatrix}.$$

16.5.6 Automorphic vector bundles

Let $T \subset G$, $T' \subset G'$ be the maximal torus consisting of diagonal matrices. Note that T, T' can be identified with $K^\times \times K^\times$ and $K^\times \times K^1$, respectively, where $K^1 = \{z \in K^\times \mid z\overline{z} = 1\}$, via

$$T' = \left\{ \begin{pmatrix} a & & \\ & b & \\ & & \overline{a}^{-1} \end{pmatrix} : a \in K^\times, b \in K^1 \right\},$$

and similarly for T. Let Z be the center of $\mathrm{GU}(2,1)$, identified with $\mathbb{G}_m(\mathbb{Q})$. Then these fit into an exact sequence

$$1 \to \mu_2 \to Z \times T' \to T \to 1.$$

The character group $X^*(T)$ is identified with $\{(k_1, k_2, k_3; c) \in \mathbb{Z}^4 \mid k_1 + k_2 + k_3 \equiv c (\mathrm{mod}\, 2)\}$ via

$$\lambda(k_1, k_2, k_3; c) : \begin{pmatrix} za & & \\ & zb & \\ & & z\overline{a}^{-1} \end{pmatrix} \mapsto \sigma(a)^{k_1}\sigma(b)^{k_2}\sigma(\overline{a})^{k_3} z^c.$$

It is M_P-dominant if $k_1 \geq k_2$ and is G-dominant if $k_1 \geq k_2$ and $k_2 + k_3 \geq 0$. We write $W_G(k_1, k_2, k_3; c)$ and $V_G(k_1, k_2, k_3; c)$ for the representations of M_P and G with the corresponding highest weights, respectively. Note that T is also a maximal torus of H, and $M_{B_H} = T$; we write $W_H(k_1, k_2, k_3; c)$ for the corresponding character of $T = M_{B_H}$. It is H-dominant if $k_1 + k_3 \geq 0$; we write $V_H(k_1, k_2, k_3; c)$ for the corresopnding H-representation.

Recall that we have chosen a preferred embedding $\sigma : K \hookrightarrow \overline{\mathbb{Q}}$. The σ-part of the $\mathcal{O}_K$-action on the first de Rham cohomology sheaf $\mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)$ has a Hodge filtration

$$0 \to \omega_\sigma \to \mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)_\sigma \to \omega_{\overline{\sigma}}^\vee \to 0,$$

and the torsor of filtration-preserving trivializations of $\mathscr{H}^1_{\mathrm{dR}}(\mathcal{A}/Y_G)_\sigma$ gives a $P_{\mathcal{O}_{K,\pi}}$-torsor

$$\mathcal{T}_{G,\mathcal{O}_{K,\pi}}/Y_{G,\mathcal{O}_{K,\pi}}.$$

A similar construction works over an infinite place; $(\mathrm{Res}_{K/\mathbb{Q}}\mathcal{T})_\mathbb{R} \to (\mathrm{Res}_{K/\mathbb{Q}} Y)_\mathbb{R}$ can be thought as the restriction of the $P(\mathbb{C})$-torsor $G(\mathbb{C}) \to G(\mathbb{C})/P(\mathbb{C})$. Here, note that $M_P(\mathbb{R})$ is not a maximal compact (modulo center) subgroup of $G(\mathbb{R})$, so strictly speaking one cannot directly use $G(\mathbb{C})/P(\mathbb{C})$; on the other hand, as we will see, $M_P(\mathbb{C})$ is a $G(\mathbb{C})$-conjugate of $K_{\infty,\mathbb{C}}$, so after all we can work as if the diagonal

matrix we have is a compact Cartan (modulo center).

Given an algebraic representation V of P over $\mathbb{Z}_p$, one defines an automorphic vector bundle $[V] := V \times^{P_{\mathcal{O}_{K,\pi}}} \mathcal{T}_{\mathcal{O}_{K,\pi}}$ over $Y_{G,\mathcal{O}_{K,\pi}}$. In particular, an M_P-highest weight representation W_G can be made into an automorphic vector bundle $[W_G]$ which we also write as ω_G, and similarly for H as well. By considering what is center and what is not, we see that the underlying sheaf of ω_G (ω_H, respectively) depends only on $(k_1 + k_3, k_2 + k_3)$ ($k_1 + k_3$, respectively). Thus we may want to only parametrize these parameters, and write $\omega_G(m_1, m_2) = \omega_G(-m_2, -m_1 - m_2, m_1 + 2m_2; -6)$ and $\omega_H(m) = \omega_H(0, -m, m; -2)$. The whole construction works equally well over the toroidal compactification by using their canonical extensions (cf. [MP]).

Note that

$$\omega_G(k_1, k_2, k_3; k_1 + k_2 + k_3 - 6) \cong \mathrm{Sym}^{k_1 - k_2} \omega_\sigma \otimes (\det \omega_\sigma)^{k_2} \otimes \omega_{\overline{\sigma}}^{k_3}.$$

This reflects the fact that $K'_\infty = \mathrm{U}(2) \times \mathrm{U}(1)$. By our earlier observation, we see that $\det \omega_\sigma \cong \omega_{\overline{\sigma}}$. By Kodaira–Spencer, $\Omega^1_{X_G}(\log D_G)$ is the automorphic vector bundle corresponding to the adjoint representation on Lie N_P. Thus $\Omega^1_{X_G}(\log D_G) \cong \omega_G(1,1)$, $\Omega^2_{X_G}(\log D_G) \cong \omega_G(0,3)$, $\Omega^1_{X_H}(\log D_H) \cong \omega_H(1,1)$. The conormal bundle to $Y_H \to Y_G$ corresponds to the automorphic vector bundle associated with the character $\alpha_{G/H} = \omega_H(2,1,0)$ of M_{B_H}. As in [LPSZ], there are pullback/pushforward maps of coherent cohomology of automorphic vector bundles of H and G,

$$H^i(X_G, [V]) \xrightarrow{\iota^*} H^i(X_H, [V|_B]),$$

$$H^i(X_H, [V|_{B_H} \otimes \alpha_{G/H}^{-1}]) \xrightarrow{\iota_*} H^{i+1}(X_G, [V]),$$

and the corresponding maps for subcanonical extensions.

Chapter 17

Higher Hida theory for $\mathrm{GU}(2,1)$

As a preliminary, we need to develop the analogue of higher Hida theory, as in [Pi] and [LPSZ, §3], for $\mathrm{GU}(2,1)$. A Lyon thesis of M. T. Nguyen [Ng] has generalized the higher Hida theory to a certain extent, which we will review and complement appropriately.

17.1 Overview of [Ng]

In their thesis [Ng], Nguyen directly generalized the works of [Pi] to $\mathrm{U}(2,1)$. For this to be useful for our purpose, we would need a control theorem for the range of weights where coherent H^1 is supposed to be the single degree of contribution (i.e. infinity type being generic discrete series). In the notation of the previous section, it is the range $m_2 < 0$ and $m_1 + m_2 > 0$, or in the notation of [Ng], it is the range $k_2 + k_3 < 0$ and $k_1 + k_3 > 0$.

One would expect that the higher Hida theory for unitary groups with p split is easier to develop, as the spherical Hecke algebra is generated by Hecke operators which are miniscule (namely, the double cosets generated by the elements of the form $\mathrm{diag}(p, p, \cdots, p, 1, \cdots, 1)$). Thus, the Hecke correspondences T_p and U_p are defined using only one geometric correspondence, and the analysis of p-divisibility is easier.

This is indeed the case in [Ng]. Due to the simple nature of the correspondences, [Ng] proves *integral control theorems* for both H^0 and H^1 without much trouble. This contrasts with the situation of [Pi] (Theorem 1.1 of *op.cit.*), where the integral control theorem is only proven for weights satisfying a regularity hypothesis involving some unknown constant C, coming from Proposition 7.4.2.2 of *op.cit.*. To prove a characteristic zero control theorem for higher Hida theory, Pilloni uses the higher Coleman theory ([Pi, Theorem 14.8.1]). In particular, the classicality of ordinary classes in higher Coleman theory requires explicit calculations of Hecke eigenvalues of local representations; in *loc. cit.*, Proposition 5.1.5.2 is used in the proof of Lemma 14.8.3. Such discussion is unneccessary in [Ng]; on the other hand, such ingredients would be neccessary for our purpose, and we complement *op.cit.* with the necessary ingredients. We closely follow the structure of [Pi].

The T-operator

Definition 17.1.1. The naive Hecke operator T_p^{naive} is defined as a cohomological correspondence ([Pi, §3]) $p_2^*(-) \to p_1^*(-) \to p_1^!(-)$ of

$$\begin{tikzcd} & X_{\mathrm{Kl}} \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ X & & X \end{tikzcd}$$

where $p_1(A/S, i, \overline{\lambda}, \eta^p, C) = (A/S, i, \overline{\lambda}, \eta^p)$, and $p_2(A/S, i, \overline{\lambda}, \eta^p, C)$ is induced from $A' = A/(C + C^{\perp})$ and the polarization $\widetilde{\lambda} : A' \to (A')^t$ is descended from $p\lambda : A \to A^t$.

This corresponds to the double coset operator

$$\mathcal{G}(\mathcal{O}_v)\,\mathrm{diag}(p, p, 1)\mathcal{G}(\mathcal{O}_v).$$

Note that the cohomological correspondence of [Pi, §3] behaves well if the involved

schemes are all Cohen–Macaulay, which is indeed the case by computing the local model of X_{Kl} ([Ng, Proposition 10.2.1]). In particular, p_1, p_2 are finite flat (Lemma 10.2.1 of *op. cit.*).

Proposition 17.1.2 (Proposition 10.4.1 of *op. cit.*)**.** *If $k_1 + k_3 \geq 1$, T_p^{naive} factorizes through*

$$p^{k_2+1} p_1^! \omega_G(k_1, k_2, k_3; k_1 + k_2 - k_3 - 4).$$

Here, we follow the rule set in §16.5.6 for the weight of the central $\mathbb{G}_m$, so we will omit the $\mathbb{G}_m$-weight from now on. Accordingly, we define $T_p = \frac{1}{p^{k_2+1}} T_p^{naive}$.

Finiteness of the ordinary cohomology

Let X_n be the $\mathbb{Z}/p^n\mathbb{Z}$-reduction of X, and let $X_1^{\leq i}$ be the locus where the multiplicative rank of $\mathcal{A}[\pi]$, the universal p-torsion group, is $\leq i$. This gives rise to the Ekedahl–Oort stratification of X_1,

$$X_1^{\leq 0} \subset X_1^{\leq 1} \subset X_1^{\leq 2} = X_1.$$

These admit a generalized Hasse invariant in an optimal way. Namely, the generalized Hasse invariant as a section of weight $(p^2-1, p^2-1, 0)$ on $X_1^{=1}$ extends to a section of the same weight on $X_1^{\leq 1}$ and vanishes to order 1 over $X_1^{=0}$. The T_p-operator interacts with multiplication by (generalized) Hasse invariants in an expected way, and using that one can show the following

Proposition 17.1.3 (Proposition 10.6.3 and 11.2.3 of *op. cit.*)**.** *Let e_{Kl} be the ordinary projector for complexes associated with T_p. For $k_1 - k_2 \geq 1, k_1 + k_3 > 3$, T_p acts trivially on $R\Gamma(X_1^{=0}, \omega(k_1 + (p^2+p-2), k_2 + (p^2+p-2), k_3))$. Moreover,*

$$e_{\mathrm{Kl}} R\Gamma(X_1, \omega_G(k_1, k_2, k_3)) \xrightarrow{\sim} e_{\mathrm{Kl}} R\Gamma(X_1^{\geq 1}, \omega_G(k_1, k_2, k_3)).$$

Proof. We say a word about the first statement, which is analogous to [Pi, Proposition

7.4.2.2]. In [Pi], the analogous statement is proved for $k \geq C$ for an unknown yet universal constant $C > 0$. There, the T_p-operator involves three different levels, hyperspecial, Klingen and paramodular. In [Ng], only hyperspecial and Klingen levels appear, and we know much more about the correspondence.

- T_p is zero on $X^{=1,00}_{\mathrm{Kl},1}$, corresponding to $H^\perp$ being biconnected. Thus, it is sufficient to prove that the restriction of T_p to $X^{\leq 1,\text{ét}}_{\mathrm{Kl},1}$ vanishes on $X^{\leq 1,\text{ét}}_{\mathrm{Kl},1} \cap V(p_1^*\mathcal{I})$, where $X^{\leq 1,\text{ét}}_{\mathrm{Kl},1}$ is the closure of $X^{=1,\text{ét}}_{\mathrm{Kl},1}$ (corresponding to $H^\perp$ being étale) in $X^{\leq 1}_{\mathrm{Kl},1}$, and $\mathcal{I}$ is the ideal sheaf generated by Ha', defining $X_1^{=0}$ in $X_1^{\leq 1}$.

- The natural maps $p_1, p_2 : X^{\leq 1,\text{ét}}_{\mathrm{Kl},1} \to X_1^{\leq 1}$ are extremely easy. Firstly, p_2 is an isomorphism, as there is unique choice of étale subgroup $H^\perp \subset \mathcal{A}[\overline{\pi}]$ of order p. In this viewpoint, p_1 is Frob^2. Thus, $V(p_1^*(\mathcal{I})) = p^2 V(p_2^*(\mathcal{I})) = p^2 V(p_2^* \mathrm{Ha}')$.

- The compatibility with generalized Hasse invariant is a commutative diagram

$$\begin{tikzcd}
p_2^*\omega(k_1+(p-1), k_2+(p-1), k_3) \arrow[r, "T_p"] \arrow[d, "\times p_2^* \mathrm{Ha}'"'] & p_1^!\omega(k_1+(p-1), k_2+(p-1), k_3) \arrow[d, "\times p_1^* \mathrm{Ha}'"] \\
p_2^*\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3) \arrow[r, "T_p"] & p_1^!\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3)
\end{tikzcd}$$

over $X^{\leq 1}_{\mathrm{Kl},1}$. From the simple nature of $p_1, p_2 : X^{\leq 1,\text{ét}}_{\mathrm{Kl},1} \to X_1^{\leq 1}$, over $X^{\leq 1,\text{ét}}_{\mathrm{Kl},1}$, we know that $p_1^* \mathrm{Ha}'$ and $p_2^* \mathrm{Ha}'$ vanishes along $V(p_2^*(\mathcal{I}))$ to order p^2 and 1, respectively. Thus,

$$T_p : p_2^*\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3) \to p_1^!\omega(k_1+(p^2+p-2), k_2+(p^2+p-2), k_3),$$

vanishes along $V(p_2^*(\mathcal{I}))$ to order at least $p^2 - 1$. One can compensate for an extra order of vanishing by using this fact for $(k_1 - 1, k_2, k_3)$ and realizing that the universal isogeny for T_p for (k_1, k_2, k_3) is that for $(k_1 - 1, k_2, k_3)$ tensored with the universal isogeny $\omega_{\mathcal{A}'[\pi^\infty]} \to \omega_{\mathcal{A}[\pi^\infty]}$, which vanishes at $V(p_2^*\mathcal{I})$ as well.

$\square$

Igusa tower and families of sheaves

We now introduce the family of sheaves which will interpolate the classical ordinary cohomology.

Definition 17.1.4 ([Ng, §12])**.** Let $\mathfrak{X}^{\geq 1}$ be the completion of X along X_1. Let $\mathfrak{X}^{\geq 1}(p^m)$ be the formal moduli parametrizing $H_m \subset \mathcal{A}[\pi^\infty]$ étale-locally isomorphic to μ_{p^m}. Over it, we define a partial Igusa tower $\mathfrak{IG}^{\geq 1}(p^m) = \mathrm{Isom}_{\mathfrak{X}^{\geq 1}(p^m)}(\mu_{p^m}, H_m)$. Let $\mathfrak{X}^{\geq 1}(p^\infty)$ be the inverse limit of $\mathfrak{X}^{\geq 1}(p^m)$, and let $\mathfrak{IG}^{\geq 1}(p^\infty) = \mathrm{Isom}_{\mathfrak{X}^{\geq 1}(p^\infty)}(\mu_{p^\infty}, H_\infty)$. Let $\pi : \mathfrak{IG}^{\geq 1}(p^\infty) \to \mathfrak{X}^{\geq 1}(p)$ be the projection. Let $\Lambda = \mathbb{Z}_p[[\mathbb{Z}_p^\times]]$ be the Iwasawa algebra and $\kappa : \mathbb{Z}_p^\times \to \Lambda^\times$ be the universal character. We define the sheaf

$$\mathfrak{F}^\kappa = (\pi_* \mathcal{O}_{\mathfrak{IG}^{\geq 1}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} \Lambda)[\Gamma = \kappa - k_2] \otimes \omega(k_2, k_2, k_3),$$

where $\Gamma = \mathbb{Z}_p^\times$ acts diagonally on the tensor product, and $[\Gamma = \kappa - k_2]$ is the part where Γ acts as $\kappa - k_2$.. Given any character $\lambda : \mathbb{Z}_p^\times \to R^\times$, we define its specialization $\mathfrak{F}^\lambda := \mathfrak{F}^\kappa \widehat{\otimes}_{\Lambda,\lambda} R$.

One can also define the analogous sheaf using the partial Igusa tower $\mathfrak{IG}^{\geq 1}(p^m)$ instead of $\mathfrak{IG}^{\geq 1}(p^\infty)$ in a similar way.

The U-operator and its relation with the T-operator

Over $\mathfrak{X}_n^{\geq 1}(p^m)$, we can, in a somewhat ad hoc way, define the geometric correspondence that gives rise to the U_p-operator.

Definition 17.1.5 ([Ng, §12.3])**.** Let $\mathfrak{C}_{\mathfrak{Y}}(p^m)$ be the formal moduli over $\mathfrak{Y}^{\geq 1}(p^m)$ parametrizing $L \subset \mathcal{A}[\pi]$ of order p^2 such that $L \cap H_m = 0$. This lies in a correspon-

dence diagram

$$\begin{tikzcd} & \mathfrak{C}_{\mathfrak{Y}}(p^m) \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ \mathfrak{Y}^{\geq 1}(p^m) & & \mathfrak{Y}^{\geq 1}(p^m) \end{tikzcd}$$

where p_1 forgets L and p_2 sends (L, H_m, A) to $(\frac{H_m+L+L^\perp}{L+L^\perp}, \frac{A}{L+L^\perp})$. These can be compactified to

$$\begin{tikzcd} & \mathfrak{C}(p^m) \arrow[dl, "p_2"'] \arrow[dr, "p_1"] & \\ \mathfrak{X}^{\geq 1}(p^m) & & \mathfrak{X}^{\geq 1}(p^m) \end{tikzcd}$$

Lemma 17.1.6 ([Ng, Lemma 12.3.1])**.** *The formal scheme $\mathfrak{C}(p^m)$ is Cohen–Macaulay, and p_1 is finite flat.*

Using this, we can define the U_p-operator.

Definition 17.1.7. Let U_p^{naive} be the cohomological correspondence defined by the above correspondence diagram, and $U_p = \frac{1}{p^{k_2+1}} U_p^{naive}$.

As in Pilloni's work, the spherical ordinary cohomology and the mirahoric ordinary cohomology are geometrically related over the special fiber.

Theorem 17.1.8 ([Ng, §12-§13])**.**

1. *Over the mod p ordinary locus $X_1^{=2}$, $T_p = T_p^{1,1}$, which only accounts for the multiplicative rank 1 locus of $X_1^{=2}(p)$.*

2. *If $k_1 > k_2$, U_p acts locally finitely on $H^0(X_1^{=2}(p), \omega(k_1, k_2, k_3))$. Let f_{Kl} be the ordinary projector associated with U_p.*

3. *For $k_1 > k_2$ and $k_1 + k_3 > 1$, $\mathrm{tr} \circ U_p^n \circ \mathrm{res} = (T_p^{1,1})^n$ as elements of*

$$\mathrm{End}(e_{\mathrm{Kl}} H^0(\mathfrak{X}_1^{=2}, \omega(k_1, k_2, k_3))).$$

Consequently, the composition

$$e_{\mathrm{Kl}}H^0(X_1^{=2},\omega(k_1,k_2,k_3)) \xrightarrow{\mathrm{res}} e_{\mathrm{Kl}}H^0(X_1^{=2}(p),\omega(k_1,k_2,k_3)) \xrightarrow{U_p} f_{\mathrm{Kl}}H^0(X_1^{=2}(p),\omega(k_1,k_2,k_3)),$$

is a bijection.

4. *Over the mod p (multiplicative) rank 1 locus, $X_1^{=1} \cong X_1^{=1}(p)$ and $T_p = U_p$.*

5. *If $k_1 > k_2$ and $k_1 + k_3 > 3$, then each of the two morphisms*

$$e_{\mathrm{Kl}}R\Gamma(X_1,\omega(k_1,k_2,k_3)) \to e_{\mathrm{Kl}}R\Gamma(X_1^{\geq 1},\omega(k_1,k_2,k_3)) \to f_{\mathrm{Kl}}R\Gamma(X_1^{\geq 1}(p),\omega(k_1,k_2,k_3)),$$

are quasi-isomorphisms.

Integral control theorem

Now we come to the main upshot.

Theorem 17.1.9 (Comparison of integral ordinary cohomology; [Ng, Theorem 13.0.1])**.** *For $k_1 > k_2$ and $k_1 + k_3 > 3$, we have quasi-isomorphisms*

$$e_{\mathrm{Kl}}R\Gamma(\mathfrak{X},\omega(k_1,k_2,k_3)) \xrightarrow{\sim} e_{\mathrm{Kl}}R\Gamma(\mathfrak{X}^{\geq 1},\omega(k_1,k_2,k_3)) \xrightarrow{\sim} f_{\mathrm{Kl}}R\Gamma(\mathfrak{X}^{\geq 1}(p),\omega(k_1,k_2,k_3)).$$

As mentioned above, the integral control theorem comes right away as we know much more about the T_p-correspondence.

Definition 17.1.10. Let the Hida complex M_{κ,k_2,k_3} be the complex of Λ-modules defined as

$$M^{\bullet}_{\kappa,k_2,k_3} = f_{\mathrm{Kl}}R\Gamma(\mathfrak{X}^{\geq 1}(p),\mathfrak{F}^{\kappa}).$$

Theorem 17.1.11 (Integral control theorem; [Ng, Theorem B])**.** *The Hida complex $M^{\bullet}_{\kappa,k_2,k_3}$ is a perfect complex of Λ-modules concetnrated in degrees $[0,1]$. For any*

$k_1 > k_2$, $k_1 + k_3 > 3$, we have an integral control theorem

$$M^{\bullet}_{\kappa,k_2,k_3} \otimes^L_{\Lambda,k_1} \mathbb{Z}_p \cong e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)).$$

Remark 17.1.12. The $\mathbb{Q}_p$-control theorem would have more relaxed regularity condition, including limits of discrete series. This however requires invoking the higher Coleman theory and in particular the computation of Hecke eigenvalues.

17.2 Concentration in degree 1

We complement the higher Hida theory of [Ng] with a vanishing theorem.

Proposition 17.2.1. *If $k_1 + k_3 > 3$ and $k_2 + k_3 < 0$, the perfect complex*

$$M^{\bullet}_{\kappa,k_2,k_3} \otimes^L_{\Lambda,k_1} \mathbb{Z}_p \cong e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)),$$

is concentrated in degree 1.

Proof. We know the characteristic zero classes

$$e_{\mathrm{Kl}} R\Gamma(X, \omega(k_1, k_2, k_3)) \otimes^L_{\mathbb{Z}_p} \mathbb{Q}_p,$$

are concentrated in degree 1 by a Lie algebra cohomology computation. Thus, we only need to show the vanishing of

$$e_{\mathrm{Kl}} H^0(X_1, \omega(k_1, k_2, k_3)),$$

which by Theorem 17.1.8(5) is isomorphic to $f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega(k_1, k_2, k_3))$.

To show it, we closely follow the proof of [LPSZ, Proposition 3.10]. Let $C \subset \mathcal{A}[\pi]$

be the universal multiplicative subgroup of order p. Consider

$$\omega(k_1, k_2, k_3) = \mathrm{Sym}^{k_1-k_2}\,\omega_\sigma \otimes \omega_{\overline{\sigma}}^{k_2+k_3} \twoheadrightarrow \omega_C^{k_1-k_2} \otimes \omega_{\overline{\sigma}}^{k_2+k_3}.$$

As U_p is a multiple of p as an endomorphism of its kernel sheaf, we know that the map

$$f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega(k_1, k_2, k_3)) \to f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega_C^{k_1-k_2} \otimes \omega_{\overline{\sigma}}^{k_2+k_3}),$$

is injective. Thus, it is sufficient to show that $f_{\mathrm{Kl}} H^0(X_1^{\geq 1}(p), \omega_C^{k_1-k_2} \otimes \omega_{\overline{\sigma}}^{k_2+k_3}) = 0$.

Let E_0 be a CM ordinary elliptic curve defined over a field k of characteristic p which has a CM by $\mathcal{O}_K$. Choose a multiplicative subgroup $C \subset E_0$ and the appropriate level structure away from p. Then, there is a natural embedding

$$j : X_H(\mathcal{P}_H)_1 \times_{\mathbb{F}_p} k \to X_1^{\geq 1}(p),$$

(see §16.5.4), sending $(A', C') \mapsto (A' \times E_0, C' \times C)$; here the p-level $\mathcal{P}_H$ corresponds to parametrizing $C' \subset A'[\pi]$ of order p. As $j^*\omega_H = \mathcal{O}$, for any $a \in H^0(X_1^{\geq 1}(p), \omega_H^{k_1-k_2} \otimes \omega_{\overline{\sigma}}^{k_2+k_3})$, j^*a is a mod p holomorphic automorphic form for $\mathrm{U}(1,1)$ with negative weight, which must be zero. The same applies to any prime-to-p Hecke translate of the embedded modular curves. It is thus sufficient to prove that the set of prime-to-p Hecke translates of points of form $A' \times E_0$ forms a Zariski dense subset of X_1^{ord}, using the irreducibility of $X_1^{\mathrm{ord}}(p) \to X_1^{\mathrm{ord}}$. Suppose the contrary; the set then misses a Zariski open subset of X_1^{ord}. Thus, the set is contained in a finite union of curves. This finite union of curves is invariant under all prime-to-p Hecke correspondences, but this violates the fact that the ℓ-adic monodromy of a Hecke-invariant subvariety must contain the derived group of G (cf. [Ch]). $\square$

17.3 Representation theory of the ordinary part

Although many parts of higher Hida theory as in [Pi] is established in [Ng], the bridge between cohomology and automorphic representations is not explored. To analyze it, we need to compute the Hecke eigenvalues of certain local representations of $G(\mathbb{Q}_p)$. Fortunately, $G(\mathbb{Q}_p) \cong \mathrm{GL}_3(K_\pi) \times K_\pi^\times$. Using the Satake isomorphism [Gr3], we can compute the spherical Hecke algebra as follows.

Lemma 17.3.1. *The spherical Hecke algebra*

$$\mathcal{H}_p := C_c(G(\mathbb{Q}_p)//\mathcal{G}(\mathbb{Z}_p), \mathbb{Z}),$$

is isomorphic to $\mathbb{Z}[T_{p,0}, T_{p,0}^{-1}, T_{p,1}, T_{p,2}, T_{p,3}, T_{p,3}^{-1}]$, *where*

$$T_{p,0} = \mathcal{G}(\mathbb{Z}_p)(1,p)\mathcal{G}(\mathbb{Z}_p), \quad T_{p,1} = \mathcal{G}(\mathbb{Z}_p)(\mathrm{diag}(p,1,1),1)\mathcal{G}(\mathbb{Z}_p),$$

$$T_{p,2} = \mathcal{G}(\mathbb{Z}_p)(\mathrm{diag}(p,p,1),1)\mathcal{G}(\mathbb{Z}_p), \quad T_{p,3} = \mathcal{G}(\mathbb{Z}_p)(\mathrm{diag}(p,p,p),1)\mathcal{G}(\mathbb{Z}_p),$$

under the identification $G(\mathbb{Q}_p) \cong \mathrm{GL}_3(K_\pi) \times \mathbb{G}_m(K_\pi)$. *The Hecke polynomial is*

$$Q_p(X) = 1 - T_{p,1}X + p^3 T_{p,2}X^2 - p^6 T_{p,3}X^3.$$

Similarly, let $\mathcal{H}_\mathcal{P}^+$ be the subalgebra of the "mirahoric" Hecke algebra $C_c(G(\mathbb{Q}_p)//\mathcal{P}, \mathbb{Z})$ generated by

$$U_{p,0} = \mathcal{P}(1,p)\mathcal{P}, \quad U_{p,1} = \mathcal{P}(\mathrm{diag}(p,1,1),1)\mathcal{P},$$

$$U_{p,2} = \mathcal{P}(\mathrm{diag}(p,p,1),1)\mathcal{P}, \quad U_{p,3} = \mathcal{P}(\mathrm{diag}(p,p,p),1)\mathcal{P},$$

under the identification $G(\mathbb{Q}_p) \cong \mathrm{GL}_3(K_\pi) \times \mathbb{G}_m(K_\pi)$. This is identified with the center of the mirahoric Hecke algebra, and is naturally isomorphic to the spherical Hecke algebra.

Now assume that π_p is an irreducible complex smooth admissible represntation of $G(\mathbb{Q}_p)$ which is spherical, namely $\pi_p^{\mathcal{G}(\mathbb{Z}_p)} \neq 0$. Let $\theta_{\pi_p} : \mathcal{H}_p \to \mathbb{C}$ be the Hecke eigensystem, and let (α, β, γ) be the reciprocal of the roots of $\theta_{\pi_p}(Q_p(X))$, ordered in a way that $|\alpha|_p \geq |\beta|_p \geq |\gamma|_p$. By the very definition, we have the following

Lemma 17.3.2. *The eigenvalues for $T_{p,1}, T_{p,2}$ and $T_{p,3}$ acting on $\pi_p^{\mathcal{G}(\mathbb{Z}_p)}$ are, respectively,*

$$\alpha + \beta + \gamma, \quad p^{-3}(\alpha\beta + \beta\gamma + \gamma\alpha), \quad p^{-6}\alpha\beta\gamma.$$

Lemma 17.3.3. *If π_p is furthermore generic (namely, an irreducible spherical principal series), then $\pi_p^{\mathcal{P}}$ is three-dimensional, and the eigenvalues of $U_{p,1}, U_{p,2}, U_{p,3}$ on $\pi_p^{\mathcal{P}}$ are, respectively,*

$$\begin{array}{lccc} U_{p,1}: & \alpha & \beta & \gamma \\ U_{p,2}: & p^{-3}\beta\gamma & p^{-3}\gamma\alpha & p^{-3}\alpha\beta \\ U_{p,3}: & p^{-6}\alpha\beta\gamma & p^{-6}\alpha\beta\gamma & p^{-6}\alpha\beta\gamma. \end{array}$$

Proof. This follows from the computation of [GeTi, Cor. 3.2.2] adapted to the Satake transform of [Re, Theorem 2]. □

From this, the following is immediate.

Corollary 17.3.4. *Assume that the set $\{\alpha\beta, \beta\gamma, \gamma\alpha\}$ has 3 distinct elements. Then, the map*

$$\pi_p^{\mathcal{G}(\mathbb{Z}_p)} \hookrightarrow \pi_p^{\mathcal{P}} \twoheadrightarrow (\pi_p^{\mathcal{P}})_{\alpha\beta},$$

is an isomorphism, where $(\pi_p^{\mathcal{P}})_{\alpha\beta}$ is the $p^{-3}\alpha\beta$-eigenspace in $\pi_p^{\mathcal{P}}$ for $U_{p,2}$.

For example, such conditions hold for any Klingen-ordinary representation appearing in $\varinjlim_{K_p} H^1(X(K^pK_p)_{\overline{\mathbb{Q}}_p}, \omega(k_1, k_2, k_3)(-D))$ with $k_1 + k_3 > 3$, $k_2 + k_3 < -1$.

Chapter 18

The p-adic pushforward map

We construct the p-adic pushforward map between (higher) Hida theories of H and G, as done in [LPSZ, §4]. In this section, we fix a tame level $K^p \subset G(\mathbb{A}_f)$. To discuss the p-adic pushforward of relevant (higher) Hida theories, we would want the compatibility between different (partial) Igusa towers that are used for the construction of (higher) Hida theories. The following elementary group-theory computation is one of the main point that makes the construction work.

Proposition 18.0.1. *Let*

$$\gamma = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Then, the following holds.

1. *We have*

$$H(\mathbb{Q}_p) \cap \gamma \operatorname{Kl}(p^m)\gamma^{-1} = \Gamma_{0,\Delta}(p^m),$$

where $\Gamma_{0,\Delta}(p^m) \subset H(\mathbb{Z}_p)$ *is the preimage of* $\left\{ \left(\begin{pmatrix} a & & \\ & a & \\ b & & c \end{pmatrix}, * \right) \right\} \subset H(\mathbb{Z}/p^m\mathbb{Z})$ *via the reduction map* $H(\mathbb{Z}_p) \twoheadrightarrow H(\mathbb{Z}/p^m\mathbb{Z})$.

2. *The* $H(\mathbb{Q}_p)$*-orbit of* $[\gamma] \in G(\mathbb{Q}_p)/B(\mathbb{Q}_p)$ *is open, where* $B \subset G$ *is the standard upper-triangular Borel subgroup. In other words,* $H(\mathbb{Q}_p)\gamma B(\mathbb{Q}_p) \subset G(\mathbb{Q}_p)$ *is open.*

Proof. As p splits in K, the computation becomes a computation of 3×3-matrices. We thus use the notations G' and H', which are the unitary groups (as opposed to the unitary similitude group) as defined in the Notations. We have

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a & b & c \\ d & e & f \\ & & g \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} a+c & c & b \\ -a-c+g & g-c & -b \\ d+f & f & e \end{pmatrix},$$

so for the RHS to be in $H(\mathbb{Q}_p)$, we need $b = c = f = 0$, $a = g$, which would yield

$$H' \cap \gamma\mathcal{P}'\gamma^{-1} = H' \cap \gamma \begin{pmatrix} a & & \\ d & e & \\ & & a \end{pmatrix} \gamma^{-1} = \begin{pmatrix} a & & \\ & a & \\ d & & e \end{pmatrix}.$$

For 2, we only need to check that the dimension of the stabilizer is the correct dimension. As $h \in \mathrm{Stab}_{H(\mathbb{Q}_p)}([\gamma])$ iff $h\gamma = \gamma b$ for some $b \in B(\mathbb{Q}_p)$, $\mathrm{Stab}_{H(\mathbb{Q}_p)} = H(\mathbb{Q}_p) \cap \gamma B(\mathbb{Q}_p)\gamma^{-1}$. Since G'/B' is 3-dimensional and H' is 5-dimensional, we want

the stabilizer $\mathrm{Stab}_{H'(\mathbb{Q}_p)}([\gamma])$ to be 2-dimensional. By the above calculation, we further need $d = 0$, which would yield the stabilizer

$$H' \cap \gamma B'(\mathbb{Q}_p)\gamma^{-1} = \begin{pmatrix} a & & \\ & a & \\ & & e \end{pmatrix},$$

as desired. $\square$

Definition 18.0.2. We define several subgroups of $H'(\mathbb{Z}_p)$ by

$$\Gamma_0(p^m) = \left\{ \begin{pmatrix} a & & \\ & b & \\ c & & d \end{pmatrix} \pmod{p^m} \right\},$$

$$\Gamma_{0,\Delta}(p^m) = \left\{ \begin{pmatrix} a & & \\ & a & \\ c & & d \end{pmatrix} \pmod{p^m} \right\},$$

$$\Gamma_{1,\Delta}(p^m) = \left\{ \begin{pmatrix} 1 & & \\ & 1 & \\ c & & d \end{pmatrix} \pmod{p^m} \right\}.$$

We similarly define $\Gamma_0(p^m), \Gamma_{0,\Delta}(p^m), \Gamma_1(p^m) \subset H(\mathbb{Z}_p)$ as above, abusing the notation. This definition of $\Gamma_{0,\Delta}(p^m) \subset H(\mathbb{Z}_p)$ coincides with that of Proposition 18.0.1.

As far as the restriction is concerned, the corresponding tame level for H would be $K_H^p := H(\mathbb{A}_f) \cap K^p$, which we also omit from the notations along with K^p.

18.1 Partial Igusa tower and Hida theory for H

Now we define a partial Igusa tower for H to be embedded into $\mathfrak{IG}_G^{\geq 1}(p^m)$. Recall that $H = \mathrm{GU}(1,1) \times_{\mathrm{GL}_1} \mathrm{GU}(1)$, so it admits a projection map $p : H \to \mathrm{GU}(1,1)$ from which there are natural maps between the corresponding Shimura varieties. Let $X_{\mathrm{GU}(1,1)}$ be the[1] compactified Shimura variety for $\mathrm{GU}(1,1)$ with level $K^p_{\mathrm{GU}(1,1)}\mathrm{GU}(1,1)(\mathbb{Z}_p)$. Let $\mathfrak{X}_{\mathrm{GU}(1,1)}/\operatorname{Spf}\mathbb{Z}_p$ be the p-adic formal completion of $X_{\mathrm{GU}(1,1)}$, and $\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}$ be the ordinary locus of $\mathfrak{X}_{\mathrm{GU}(1,1)}$. Over $\mathfrak{X}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}$, there is an Igusa tower $\mathfrak{IG}^{\mathrm{ord}}_{\mathrm{GU}(1,1)}(p^m)$, which is a moduli space of embeddings $\alpha : \mu_{p^m} \hookrightarrow \mathcal{A}_{\mathrm{GU}(1,1)}[\pi]$, where $\mathcal{A}_{\mathrm{GU}(1,1)}$ is the universal semi-abelian scheme over $\mathfrak{X}_{\mathrm{GU}(1,1)}$. Similarly, consider $\mathfrak{X}_{\mathrm{GU}(1)}/\operatorname{Spf}\mathbb{Z}_p$ which is the p-adic formal completion of $X_{\mathrm{GU}(1)}$.

Definition 18.1.1. Over $\mathfrak{X}^{\mathrm{ord}}_H$, we define the partial Igusa tower $\mathfrak{IG}^{\mathrm{ord}}_H(p^m)$ to be the moduli space of embeddings $\alpha : \mu_{p^m} \hookrightarrow \mathcal{A}_{\mathrm{GU}(1,1)}[\pi]$ and $\beta : \mu_{p^m} \hookrightarrow \mathcal{A}_{\mathrm{GU}(1)}[\pi]$, where $\mathcal{A}_{\mathrm{GU}(1,1)}$ and $\mathcal{A}_{\mathrm{GU}(1)}$ are the universal semi-abelian schemes over $\mathfrak{X}_{\mathrm{GU}(1,1)}$ and $\mathfrak{X}_{\mathrm{GU}(1)}$, respectively. The partial Igusa tower is a $(\mathbb{Z}/p^m\mathbb{Z})^2$-torsor over $\mathfrak{X}^{\mathrm{ord}}_H$; we let

$$\mathfrak{X}^{\mathrm{ord}}_{H,\Delta} = \mathfrak{IG}^{\mathrm{ord}}_H(p^m)/\Delta(\mathbb{Z}/p^m\mathbb{Z}),$$

where $\Delta : \mathbb{Z}/p^m\mathbb{Z} \hookrightarrow (\mathbb{Z}/p^m\mathbb{Z})^2$ is the diagonal embedding. It is the compactification of the moduli space parametrizing the pair of CM abelian schemes A_1, A_2 of dimensions $2, 1$, respectively, and the level structure parametrizing an isomorphism $A_1[\pi]^\circ \cong A_2[\pi]^\circ$.

From this, we can define the space of p-adic modular forms for H.

[1] There is only one choice of compactification, as $Y_{\mathrm{GU}(1,1)}$ is a curve.

Definition 18.1.2. Let $\pi : \mathfrak{IG}_H^{\mathrm{ord}}(p^\infty) \to \mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p)$ be the projection map. For $k_3 \in \mathbb{Z}$, let

$$\mathfrak{F}_H^{k_3} = (\pi_* \mathcal{O}_{\mathfrak{IG}_H^{\mathrm{ord}}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} \Lambda) \otimes \omega_H(0, -k_3, k_3; -2).$$

This has a natural action of

$$\Gamma \dot{\times} \Gamma = \{a, b \in \Gamma : a \equiv b (\mathrm{mod}\, p)\}.$$

For any flat $\mathbb{Z}_p$-algebra R and characters $\kappa_1, \kappa_2 : \Gamma \to R$, let

$$\mathfrak{F}_H^{(\kappa_1,\kappa_2,k_3)} = (\pi_* \mathcal{O}_{\mathfrak{IG}_H^{\mathrm{ord}}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} \Lambda)[\Gamma \dot{\times} \Gamma = (\kappa_1, \kappa_2)] \otimes \omega_H(0, -k_3, k_3; -2),$$

be the sheaf on $\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p)$. Let the space of *$p$-adic modular forms for H* be defined as

$$\mathcal{M}_H^{(\kappa_1,\kappa_2,k_3)}(R) = H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \mathfrak{F}_H^{(\kappa_1,\kappa_2,k_3)})^{(\mathbb{Z}/p\mathbb{Z})^\times},$$

where $(\mathbb{Z}/p\mathbb{Z})^\times = \Gamma \times \Gamma / \Gamma \dot{\times} \Gamma$. Similarly, the space of p-adic cusp forms for H is defined as

$$\mathcal{S}_H^{(\kappa_1,\kappa_2,k_3)}(R) = H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \mathfrak{F}_H^{(\kappa_1,\kappa_2,k_3)}(-D))^{(\mathbb{Z}/p\mathbb{Z})^\times}.$$

We also define the spaces of classical modular forms for H as follows.

Definition 18.1.3. For $k_1, k_2, k_3 \in \mathbb{Z}$ and a compact open subgroup $\Gamma_p \subset H(\mathbb{Z}_p)$, let

$$M_H^{(k_1,k_2,k_3)}(\Gamma_p) := H^0(X_H(\Gamma_p), \omega_H(k_1, k_2, k_3)),$$

$$S_H^{(k_1,k_2,k_3)}(\Gamma_p) := H^0(X_H(\Gamma_p), \omega_H(k_1, k_2, k_3)(-D)).$$

Note that $\Gamma_0(p^m)/\Gamma_1(p^m) \cong (\mathbb{Z}/p^m\mathbb{Z})^\times \times (\mathbb{Z}/p^m\mathbb{Z})^\times$ acts naturally on $M_H(\Gamma_1(p^m))$ and $S_H(\Gamma_1(p^m))$. For $\chi_1, \chi_2 : (\mathbb{Z}/p^m\mathbb{Z})^\times \to L$ characters valued in L, a characteristic

zero field, let $(\chi_1, \chi_2) : \Gamma_0(p^m)/\Gamma_1(p^m) \to L$ be defined as

$$(\chi_1, \chi_2) \begin{pmatrix} a & & \\ & b & \\ c & & d \end{pmatrix} := \chi_1(a)\chi_2(b).$$

Define the space of modular forms with nebentypus (χ_1, χ_2) to be

$$M_H^{(k_1,k_2,k_3)}(p^m, \chi_1, \chi_2) := M_H^{(k_1,k_2,k_3)}(\Gamma_1(p^m))^{(\chi_1,\chi_2)},$$

$$S_H^{(k_1,k_2,k_3)}(p^m, \chi_1, \chi_2) := S_H^{(k_1,k_2,k_3)}(\Gamma_1(p^m))^{(\chi_1,\chi_2)}.$$

Since $\pi_H : \mathfrak{IG}_H^{\mathrm{ord}}(p^m) \to \mathfrak{X}_H^{\mathrm{ord}}(p)$ is a torsor, for $k_1, k_2, k_3 \in \mathbb{Z}$, there is a comparison isomorphism

$$\omega_H(k_1, k_2, k_3) \xrightarrow{\sim} \mathfrak{F}_H(k_1, k_2, k_3),$$

which implies that the classical modular forms can be viewed as p-adic modular forms,

$$M_H^{(k_1,k_2,k_3)}(p^m, \chi_1, \chi_2) \hookrightarrow \mathcal{M}_H^{(k_1+\chi_1,k_2+\chi_2,k_3)}(\mathcal{O}_L) \otimes_{\mathcal{O}_L} L.$$

This is Hecke-equivariant if $(\chi_1, \chi_2)|_{\Gamma_{0,\Delta}(p^m)} = 1$.

18.2 Maps between Igusa towers

Definition 18.2.1. We define the tower of schemes

$$\begin{tikzcd}
\mathfrak{IG}_H^{\mathrm{ord}}(p^m) \arrow[r, "\widetilde{\iota}_m"] \arrow[d] & \mathfrak{IG}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p^m) \arrow[r, "\iota_m"] \arrow[d] & \mathfrak{X}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_H \arrow[r, "\iota"] & \mathfrak{X}_G
\end{tikzcd}$$

as follows.

- ι is induced from the inclusion of Shimura data, induced from $H \hookrightarrow G$;
- ι_m sends $(A_1, A_2, A_1[\pi]^\circ \cong A_2[\pi]^\circ)$ to $(A_1 \times A_2, \Gamma)$, where A_1 and A_2 are CM abelian schemes of dimension 2 and 1, respeectively, and $\Gamma \subset A_1 \times A_2[\pi]$ is a multiplicative subgroup of order p corresponding to the graph of the isomorphism $A_1[\pi]^\circ \cong A_2[\pi]^\circ$.
- $\widetilde{\iota}_m$ sends $\alpha : \mu_{p^m} \subset A_1[\pi]$ and $\beta : \mu_{p^m} \subset A_2[\pi]$ to $\alpha + \beta : \mu_{p^m} \hookrightarrow (A_1 \times A_2)[\pi]$.

Proposition 18.2.2. *The horizontal morphisms ι_m and $\widetilde{\iota}_m$ are closed embeddings. Furthermore, the following two squares are both cartesian,*

$$\begin{tikzcd}
\mathfrak{IG}_H^{\mathrm{ord}}(p^m) \arrow[r, "\widetilde{\iota}_m"] \arrow[d] & \mathfrak{IG}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p^m) \arrow[r, "\iota_m"] \arrow[d] & \mathfrak{X}_G^{\geq 1}(p^m) \arrow[d] \\
\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p) \arrow[r, "\iota_1"] & \mathfrak{X}_G^{\geq 1}(p).
\end{tikzcd}$$

Proof. The same proof works as in [LPSZ, Proposition 4.6]. It is sufficient to prove that $\widetilde{\iota}_m$ is a closed embedding, as ι_m is obtained from $\widetilde{\iota}_m$ by taking the quotient

by $\mathbb{Z}/p^m\mathbb{Z}$ under the $(\mathbb{Z}/p^m\mathbb{Z})$-equivariance of $\widetilde{\iota}_m$. As in *loc. cit.*, $\widetilde{\iota}_m$ factors through $\mathfrak{X}_H \times_{\mathfrak{X}_G} \mathfrak{IG}_G^{\geq 1}(p^m)$, which parametrizes $\mu_{p^m} \hookrightarrow (A_1 \times A_2)[\pi]$. By the same reason as in *loc. cit.*, there is an open-and-closed subscheme $\mathfrak{Z}_m$ of $\mathfrak{X}_H \times_{\mathfrak{X}_G} \mathfrak{IG}_G^{\geq 1}(p^m)$ characterized by the additional conditions that the projected coordinates $\mu_{p^m} \to A_1[\pi]$ and $\mu_{p^m} \to A_2[\pi]$ are both embeddings. This subscheme is identified with $\mathfrak{IG}_H^{\mathrm{ord}}(p^m)$, so $\widetilde{\iota}_m$ is a closed embedding, as desired.

To show that the two squares are cartesian, it is sufficient to show that the formation of $\mathfrak{Z}_m$ is compatible with base-change, namely that $\mathfrak{Z}_m = \mathfrak{Z}_1 \times_{\mathfrak{IG}_G^{\geq 1}(p)} \mathfrak{IG}_G^{\geq 1}(p^m)$. It is then sufficient to show the following statement: for a homomorphism $f : \mu_{p^m} \to H$ of finite flat group schemes, f is injective if and only if $f|_{\mu_p}$ is injective. This is of course true, as the only subgroup schemes of μ_{p^m} are of form μ_{p^n}, $n \leq m$, so $\ker f$ has to be trivial if it does not contain μ_p. $\square$

Remark 18.2.3. The way the Igusa tower is embedded is slightly different from how they are embedded in the context of of p-adic L-function for holomorphic modular forms on unitary groups, as in [EHLS], [Har7], [HLSk]. The main objective of the prior works is to compare Hida theory (H^0) of the smaller group with the Hida theory (again H^0) of the bigger group, whereas our objective is to compare Hida theory of the smaller group with the *higher Hida theory* (H^1) of the bigger group. Accordingly, the partial Igusa tower over H should naturally embed into the partial Igusa tower $\mathfrak{IG}_G^{\geq 1}(p^m)$ over G. We are in a fortunate situation where the complementary part of GU(1) has no meaningful geometric structure, so that the situation is not too different from the case of prior works.

If we denote the projection $\mathfrak{IG}_G(p^\infty) \to \mathfrak{X}_G^{\geq 1}(p)$ ($\mathfrak{IG}_H(p^\infty) \to \mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p)$, respectively) as π_G (π_H, respectively), then the Cartesian diagram implies the following equality,

$$\iota_1^*(\pi_{G,*}\mathcal{O}_{\mathfrak{IG}_G(p^\infty)}) = \pi_{H,*}\mathcal{O}_{\mathfrak{IG}_H(p^\infty)},$$

which is furthermore Γ-equivariant, where the action of Γ on the right hand side is the restriction of the action of $\Gamma\dot{\times}\Gamma = \{(a,b) \in \Gamma \times \Gamma : a \equiv b (\mathrm{mod}\, p)\}$.

18.3 Pushforward and pullback of big sheaves

The cartesian diagram as above results in the following information about comparison of pullback for $\mathfrak{IG}_G$ to $\mathfrak{IG}_H$.

Proposition 18.3.1. *Let $\lambda_1, \lambda_2 : \Gamma \to R^\times$ be continuous characters, where R is a flat complete $\mathbb{Z}_p$-algebra. Let $k_2, k_3 \in \mathbb{Z}$. Let $\kappa = \lambda_1 + \lambda_2 - k_2$. Then, there is a natural inclusion*

$$(\pi_{H,*}\mathcal{O}_{\mathfrak{IG}_H^{\mathrm{ord}}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} R)[\Gamma\dot{\times}\Gamma = (\lambda_1 - k_2, \lambda_2 - k_2)] \hookrightarrow (\pi_{H,*}\mathcal{O}_{\mathfrak{IG}_H^{\mathrm{ord}}(p^\infty)} \widehat{\otimes}_{\mathbb{Z}_p} R)[\Delta\Gamma = \kappa - k_2],$$

where $\Delta\Gamma \subset \Gamma\dot{\times}\Gamma$ is the diagonal embedding of Γ, and this yields an inclusion of sheaves

$$\mathfrak{F}_H^{(\lambda_1,\lambda_2,k_3)} \otimes \omega_H(0,0,0;-4) \hookrightarrow \iota_1^* \mathfrak{F}_G^{(\kappa,k_2,k_3)}.$$

For any integers $\ell_1, \ell_2 \geq k_2$ with $\ell_1 + \ell_2 = k_1 + k_2$, we have the following diagram of sheaves on $\mathfrak{X}_H^{\mathrm{ord}}(p)$,

$$\begin{array}{ccc} \omega_H(\ell_1,\ell_2,k_3;-4) & \hookrightarrow & \iota_1^*\omega_G(k_1,k_2,k_3) \\ \downarrow & & \downarrow \\ \mathfrak{F}_H^{(\ell_1,\ell_2,k_3)} \otimes \omega_H(0,0,0;-4) & \hookrightarrow & \iota_1^*\mathfrak{F}_G^{(k_1,k_2,k_3)} \end{array}$$

Proof. This is an easy consequence of the weight decomposition of algebraic representations of M_P, as M_{B_H} is just a maximal torus. □

We now use the p-adic pushforward map

$$\iota_* : H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \iota_1^*\mathfrak{F}_G^{(\kappa,k_2,k_3)} \otimes \omega_H(-2,-1,0)(-D_H)) \to H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(\kappa,k_2,k_3)}(-D_G)).$$

We choose characters $\lambda_1, \lambda_2 : \Gamma \to R^\times$ with $\lambda_1 + \lambda_2 = \kappa + k_2 - 3$. Then, we have a natural morphism

$$\mathfrak{F}_H^{(\lambda_1+2,\lambda_2+1,k_3)} \otimes \omega_H(0,0,0;-4) \to \iota_1^* \mathfrak{F}_G^{(\kappa,k_2,k_3)},$$

which yields a functoriality between Hida theory of H and higher Hida theory of G:

$$\begin{aligned}
\iota_* : \mathcal{S}_H^{(\lambda_1,\lambda_2,k_3)}(R) &= H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \mathfrak{F}_H^{(\lambda_1,\lambda_2,k_3)}(-D))^{(\mathbb{Z}/p\mathbb{Z})^\times} \\
&= H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \mathfrak{F}_H^{(\lambda_1+2,\lambda_2+1,k_3)} \otimes \omega_H(-2,-1,0;-4)(-D_H))^{(\mathbb{Z}/p\mathbb{Z})^\times} \\
&\to H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \mathfrak{F}_H^{(\lambda_1+2,\lambda_2+1,k_3)} \otimes \omega_H(-2,-1,0;-4)(-D_H)) \\
&\to H^0(\mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p), \iota_1^* \mathfrak{F}_G^{(\kappa,k_2,k_3)} \otimes \omega_H(-2,-1,0)(-D_H)) \\
&\xrightarrow{\iota_*} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(\kappa,k_2,k_3)}(-D_G)).
\end{aligned}$$

18.4 Comparison with classical pushforward

The rigid analytification of

$$\iota_m : \mathfrak{X}_{H,\Delta}^{\mathrm{ord}}(p^m) \to \mathfrak{X}_G^{\geq 1}(p^m),$$

yields a morphism of rigid analytic varieties which is the restriction of the rigid analytification of the finite morphism between $\mathbb{Q}$-varieties

$$X_{H,\Delta}(p^m) \xrightarrow{H \hookrightarrow G} X_{G,\mathrm{Kl}}(p^m) \xrightarrow{\times\gamma} X_{G,\mathrm{Kl}}(p^m),$$

namely the composition of the natural map induced by the natural inclusion $H \hookrightarrow G$ with the right-translation by the element γ of Proposition 18.0.1. This yields a natural

commutative diagram

$$\begin{tikzcd}
S_H^{(\ell_1,\ell_2,k_3)}(p^m,\chi_1,\chi_2) \arrow[r, "\iota_{m,*}"] \arrow[d, hook] & H^1(X_{G,\mathrm{Kl}}(P^m)_{\mathbb{Q}},\omega_G(k_1,k_2,k_3)(-D_G))\otimes_{\mathbb{Q}} L \arrow[d, hook] \\
\mathcal{S}_H^{(\ell_1+\chi_1,\ell_2+\chi_2,k_3)}(\mathcal{O}_L)\otimes_{\mathcal{O}_L} L \arrow[r, "\iota_*"'] & H^1(\mathfrak{X}_{\overline{G}}^{\geq 1}(p),\mathfrak{F}_G^{(k_1,k_2,k_3)}(-D_G))\otimes L,
\end{tikzcd}$$

where L is a finite extension of $\mathbb{Q}_p$, $\chi_1,\chi_2:(\mathbb{Z}/p^m\mathbb{Z})^\times\to\mathcal{O}_L^\times$ are characters such that $\chi_1\chi_2=1$, and $\ell_1+\ell_2=k+k_2-3$ with $\ell_1+1,\ell_2>k_2$.

Chapter 19

Integral representations of L-functions

We use the integral representation appearing in [GPS] for the standard L-function of generic automorphic representation of $\mathrm{GU}(2,1)$. We extend the strategy to relate the GGP period integral for $\mathrm{GU}(1,1) \subset \mathrm{GU}(2,1)$ to the standard L-function for $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$, which we develop here. This "unfolding of integral" formula is independent of the recent fascinatin resolution of the Ichino–Ikeda conjecture ([Zha], [BPC+], [BPL+]) and sheds new light on the period integral. This uses the *generalized Whittaker model*[1] due to several mathematicians, most notably Shintani, Yamashita and Murase–Sugano.

19.1 Gelbart–Piatetski-Shapiro integral for $\mathrm{GU}(2,1)$

Motivated by [GPS], we are led to consider the zeta integral as follows.

[1]There are unfortunately no consensus on what the term "generalized Whittaker models" means across the literature. In particular, in some literature, generalized Whittaker models are special cases of degenerate Whittaker models in the sense of Moeglin–Waldspurger. Our meaning of generalized Whittaker model is different from this. Another name for the generalized Whittaker model in our sense that appears in the literature is the *"reduced generalized Gelfand–Graev representation (RGGGR)"*.

Definition 19.1.1. Let Π be a cuspidal automorphic representation of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$, and $f \in \Pi$. Let μ be an idele class character of $K^\times \backslash \mathbb{A}_K^\times$. Let Φ be a Schwartz–Bruhat function on $V_H(\mathbb{A}_K)$. Then,

$$Z(f, \mu, \Phi, s) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h) E^\Phi(h; \mu, s) dh,$$

where

$$E^\Phi(h; \mu, s) := \sum_{\gamma \in B_H(\mathbb{Q})\backslash H(\mathbb{Q})} f^\Phi(\gamma h; \mu, s),$$

$$f^\Phi(h; \mu, s) := \int_{\mathbb{A}_K^\times} (h \cdot \Phi)(ty)\mu(t)|t|_K^s d^\times t \qquad \textit{(“Mellin transform”).}$$

By unfolding, one gets a factorizable integral representation.

Proposition 19.1.2 ([GPS, Proposition 3.6])**.** *Suppose $\Phi = \prod_w \Phi_w$ is factorizable. Then*

$$\begin{aligned} Z(f, \mu, \Phi, s) &= \int_{Z_G(\mathbb{A})Z_{N_{B_G}}(\mathbb{A})\backslash H(\mathbb{A})} W_f^\psi(h) f^\Phi(h; \mu, s) dh \\ &= \prod_v \int_{Z_G(\mathbb{Q}_v)Z_{N_{B_G}}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_v(h) f_v^\Phi(h; \mu, s) dh \\ &=: \prod_v Z_v(f, \mu_v, \Phi_v, s), \end{aligned}$$

where the notation is as follows.

- *$Z_{N_{B_G}}$ is the center of the unipotent radical of B_G, so*

$$N_{B_G} = \left\{ \begin{pmatrix} 1 & -\overline{a}\theta & z \\ & 1 & a \\ & & 1 \end{pmatrix} : z - \overline{z} + \theta|a|^2 = 0 \right\},$$

$$Z_{N_{B_G}} = \left\{ \begin{pmatrix} 1 & & z \\ & 1 & \\ & & 1 \end{pmatrix} : z \in \mathbb{Q} \right\}.$$

- *ψ is a nontrivial additive character of $\mathbb{Q}\backslash\mathbb{A}_{\mathbb{Q}}$, and $\psi_{N_{B_G}}$ is the character of N_{B_G} defined by*

$$\begin{pmatrix} 1 & -\overline{a}\theta & z \\ & 1 & a \\ & & 1 \end{pmatrix} \mapsto \psi(\operatorname{Re} a),$$

where $\operatorname{Re} a = \frac{a+\overline{a}}{2}$.

- *A Whittaker coefficient $W_f^{\psi}(g)$ is defined as*

$$W_f^{\psi}(g) = \int_{N_{B_G}(\mathbb{Q})\backslash N_{B_G}(\mathbb{A}_{\mathbb{Q}})} f(ng)\overline{\psi_{N_{B_G}}(n)}dn,$$

which thus satisfies

$$W_f^{\psi}(ng) = \psi_{N_{B_G}}(n)W_f^{\psi}(g),$$

for $n \in N_{B_G}(\mathbb{A}_{\mathbb{Q}})$.

- *$W_v(h)$ is in the local Whittaker model*

$$W(\pi_v, \psi_v) = \{W_v : G(\mathbb{Q}_v) \to \mathbb{C} \mid W_v(ng) = \psi_v(n)W_v(g) \forall n \in N_{B_G}(\mathbb{Q}_v)\},$$

which is at most one-dimensional,

- *A Siegel section f_v^Φ is defined as*

$$f_v^\Phi(h;\mu,s) = \prod_{w|v} \int_{K_w^\times} (h\cdot\Phi_w)(ty)\mu_w(t)|t|_w^s d^\times t.$$

19.2 Generalized Whittaker model, after Shintani, Yamashita, Murase–Sugano

If we replace the Eisenstein series on $\mathrm{GU}(1,1)$ with a cusp form, then the unfolding picks up more degenerate Whittaker coefficients, because any character χ of B restricts to a trivial character on B_H which makes the integral corresponding to the Whittaker coefficient with respect to χ vanish by the cuspidality of the cusp form.

In this subsection, we introduce the ideas behind the so-called *generalized Whittaker model*, as developed by Shintani, Yamashita and Murase–Sugano. For the sake of introduction, the notations in this section are only used here and not related to those used anywhere else.

We first recall that the Fourier expansions of a modular form, or more generally the Whittaker–Fourier expansions of an automorphic form, are all obtained by seeing a form as a function over the unipotent radical N of a parabolic subgroup P. In this way, one observes that the chosen form is an N-intertwiner, from the (restriction to N of the) automorphic representation that the form generates, to the regular representation $L^2(N_\mathbb{Q}\backslash N_\mathbb{A})$. This is decomposed into the (maybe) continuous sum of unitary irreducible N-representations, and each Fourier coefficient is an intertwiner between the automorphic representation π and an irreducible N-constituent ρ, or by Frobenius reciprocity, $\mathrm{Hom}_G(\pi, \mathrm{Ind}_N^G \rho)$.

The Whittaker–Fourier coefficient corresponds to the case where N is the maximal unipotent subgroup and the irreducible N-representations are characters. However, there are many other choices of N, and also of a unitary irreducible representation

of N. The main reason why such general theory is ill-developed is because, over local field F, $\mathrm{Ind}_{N(F)}^{G(F)}\rho$ is not multiplicity-free, and in most cases not even finite-dimensional. Indeed, if N is smaller than those appearing in the "wave-front set" of a $G(F)$-representation σ, $\mathrm{Hom}_{G(F)}(\sigma, \mathrm{Ind}_N^G \rho)$ is infinite-dimensional (cf. [MW], [GGS])[2]. This prevents one from using this kind of model (at least naively) to unfold Rankin–Selberg integrals.

The idea of the generalized Whittaker model, as developed by Shintani, Yamashita, Murase–Sugano, is to extend the group N to a slightly bigger group $N \subset R \subset P$ and consider R-intertwiners. Over a local field, the group R is defined as follows. Let $P = L \ltimes N$ be the Levi decomposition. Let $S \subset L$ be the centralizer of $[\rho] \in \widehat{N}$, the class of ρ in the unitary dual of N. Then, one defines $R := S \ltimes N$. The N-representation ρ can be trivially extended to a S-representation, also denoted as ρ. Representations of R are of form $\rho \otimes \psi$, where ρ is the extension of an N-representation as above, and $\psi \in \widehat{S}$.

Definition 19.2.1. Let π be a G-representation. The *generalized Whittaker model* of π with respect to (N, ρ, ψ) is the intertwiner

$$\mathrm{Hom}_R(\pi|_R, \rho \otimes \psi) = \mathrm{Hom}_G(\pi, \mathrm{Ind}_R^G(\rho \otimes \psi)).$$

The induced representation $\pi_{\rho,\psi} := \mathrm{Ind}_R^G(\rho \otimes \psi)$ is called a *reduced generalized Gelfand–Graev representation (RGGGR).*

Surprisingly, the existing works suggest that these models have finite multiplicity property, and even multiplicity-free property in favorable situations.

Slogan. The RGGGR $\pi_{\rho,\psi}$ has the multiplicity one property, namely $\dim_G(\pi, \pi_{\rho,\psi})^\circ \leq 1$.

[2]We will call this often-infinite-dimensional model a *degenerate Whittaker model*, to distinguish it from the generalized Whittaker model we define in this section.

The superscript $\circ$ indicates moderate growth condition should be imposed in the archimedean case. Recall that, even in the classical theory of Jacquet, the space of Whittaker functionals is of dimension larger than one; but *there is exactly one-dimensional line spanned by Whittaker functionals of moderate growth.*

We exhibit how to use these models in some period integrals to show their Eulerianity, via the theory of Fourier expansions along closed subgroups (cf. [Od2]), which, formally speaking, is just the decomposition of an R-representation into η-isotypic parts with η running over $\eta \in \widehat{R}$.

Remark 19.2.2. One reason behind our interest towards this theory is for the construction of square-root p-adic L-function for $\mathrm{U}(n,1) \times \mathrm{U}(n-1,1)$. This more general case should use [MS2], [Ts1], [Ts2], [Ts4].

19.3 Example I: GGP period integral for $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$

We now consider an actual case of the period integral of Gan–Gross–Prasad type for $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$, where the automorphic form on $\mathrm{GU}(1,1)$ is now *cuspidal.*

Definition 19.3.1. Let Π, σ be cuspidal automorphic representations of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$ and $\mathrm{GU}(1,1)(\mathbb{A}_{\mathbb{Q}})$, respectively. Let μ be a character of $\mathrm{GU}(1)(\mathbb{A}_{\mathbb{Q}})$. Let $f \in \Pi$, $g \in \sigma$. Then,

$$Z(f,g,\mu,s) := \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} f(h)g(h)\mu(h)|\det h|^{s-\frac{1}{2}}dh,$$

where g and μ are regarded as forms over $H(\mathbb{A})$ by extending trivially.

We will show that this integral of "GGP form" can be unfolded by using the theory of generalized Whittaker models. To unfold, we need to recall some of the machineries used in [GPS]. There, the theory of Fourier–Jacobi expansions is developed (see also [GR]), which uses the unipotent radical $N_{B'_G}$, which is isomorphic to a Heisenberg

group. More precisely, for a local field F, the irreducible unitary representations of $N_{B'_G}(F)$ are precisely the characters (which factor through $N_{B'_G}/Z_{N_{B'_G}}$) and the Weil representations (characterized by the central character). Taking the Fourier expansion of f with respect to the compact abelian group $[Z_{N_{B'_G}}]$, we have an expansion analogous to the Fourier–Jacobi expansion[3],

$$f(h) = f_{00}(h) + \sum_{\psi \neq 1} f^{\psi}(h),$$

where the sum runs over nontrivial characters of $[Z_{N_{B'_G}}]$, $f_{00}(h) = \int_{[Z_{N_{B'_G}}]} f(nh)dn$, and

$$f^{\psi}(h) = \int_{[Z_{N_{B_G}}]} f(zh)\overline{\psi(z)}dz,$$

is the "Fourier–Jacobi coefficient." As noted above, f^{ψ} corresponds to the Weil representation with central character ψ, whereas f_{00} corresponds to all the representations with trivial central character. Thus, one can expect that this will further break down into a sum of the Fourier coefficients corresponding to the characters of $[N_{B'_G}]$, the "Whittaker–Fourier coefficients":

$$f_{00}(h) = \sum_{\chi \neq 1} W^f_{\chi}(h), \qquad W^f_{\chi}(h) = \int_{[N_{B_G}]} f(nh)\overline{\chi(n)}dn,$$

where χ runs over all (nontrivial[4]) characters of $[N_{B'_G}]$.

When g was an Eisenstein series, the integral unfolded so that the major term came from f_{00}. On the other hand, if we unfold a cuspidal g against f_{00}, since f_{00} is left-$Z_{N_{B_G}}(\mathbb{A})$-invariant, the integral vanishes because of the cuspidality of g; note

[3]This is indeed the classical Fourier–Jacobi expansion if f is a holomorphic automorphic form, because the holomorphic (limit of) discrete series does not admit a Whittaker mordel, so $f_{00}(g)$, the sum of Whittaker–Fourier coefficients, vanishes; on the other hand, $f^{\psi}(g)$ is a classical Fourier–Jacobi coefficient (cf. [Is]).

[4]The coefficient against the trivial character vanishes, because f is cuspidal. One can see that the (non)cuspidality also plays a role in this hierarchy of various depths of generalized Whittaker models.

$Z_{N_{B_G}} = N_{B_H}$. Thus, in the case of $\mathrm{GU}(2,1) \times \mathrm{GU}(1,1)$, we are led to consider the integral against the Fourier–Jacobi coefficients f^{ψ}.

The imminent problem is that the Fourier–Jacobi coefficient f^{ψ} does not have multiplicity free local model. Rather, we would need to use the generalized Whittaker model. If we let R to be the centralizer of $Z_{N_{B'_G}}$ in B'_G, then $R = S \ltimes N_{B_G}$, where $S = \mathrm{U}(1)$ is the reductive part. Thus, irreducible unitary representaitons of $R(F)$ are of form $\chi \otimes \omega_{\psi}$, where ω_{ψ} is the trivial extension of the Weil representation with central character ψ, and χ is a character of $\mathrm{U}(1)$.

In our case, the **Slogan** was already proven in the literature.

Proposition 19.3.2. *Let F be a local field.*

1. *If F is nonarchimedean,* $\mathrm{Ind}_{R(F)}^{G(F)}(\chi \otimes \omega_{\psi})$ *is multiplicity-free.*

2. *If F is archimedean,* $\mathrm{Ind}_{R(F)}^{G(F)}(\chi \otimes \omega_{\psi})^{\circ}$ *is multiplicity-free.*

Proof. For (1), see [GR]. For (2), see [Is]. □

This implies that the degenerate Whittaker model f^{ψ} decomposes into the sum of generalized Whittaker models,

$$f^{\psi}(g) = \sum_{\chi} f^{\psi,\chi}(g),$$

where χ runs over the unitary characters of $S(\mathbb{A}) = \mathrm{U}(1)(\mathbb{A})$, and $f^{\psi,\chi}(g)$ is a generalized Whittaker function, so that it transforms as $\chi \otimes \omega_{\psi}$ under the action of $R(\mathbb{A})$. By Proposition 19.3.2, $f^{\psi,\chi}$ is supported by a multiplicity one model. As

$\mathrm{U}(1,1)$-automorphic forms are basically elliptic modular forms, we thus have

$$\begin{aligned}
Z(f,g,\mu,1/2) &= \int_{[H]} \sum_{\psi\neq 1,\chi} f^{\psi,\chi}(h)g(h)\mu(h)dh \\
&= \int_{[\mathrm{GU}(1,1)]} \sum_{\psi\neq 1} f^{\psi,\mu^{-1}}(h)g(h)dh \\
&= \int_{N_{B'_H}(\mathbb{Q})\backslash \mathrm{U}(1,1)(\mathbb{A})} f^{\psi,\mu^{-1}}(h)g(h)dh \\
&= \int_{N_{B'_H}(\mathbb{A})\backslash \mathrm{U}(1,1)(\mathbb{A})} f^{\psi,\mu^{-1}}(h)\left(\int_{[N_{B'_H}]} g(nh)\psi(n)dn\right)dh \\
&= \int_{Z_H(\mathbb{A})N_{B_H}(\mathbb{A})\backslash H(\mathbb{A})} f^{\psi,\mu^{-1}}(h)W_g^{\psi}(h)dh,
\end{aligned}$$

where $W_g^{\psi}(h)$ is the Whittaker function for g with respect to ψ. Note that we ignored the power of $|\det|$ to unclog the formulae, but one can verbatim follow the same process with arbitrary choice of s. Now the last integral we yield is manifestly Eulerian due to the *multiplicity one properties of (generalized) Whittaker models*, and we have shown the following

Proposition 19.3.3. *We have*

$$\begin{aligned}
Z(f,g,\mu,s) &= \int_{Z_H(\mathbb{A})N_{B_H}(\mathbb{A})\backslash H(\mathbb{A})} f^{\psi,\mu^{-1}}(h)W_g^{\psi}(h)|\det h|^{s-\frac{1}{2}}dh \\
&= \prod_v \int_{Z_H(\mathbb{Q}_v)N_{B_H}(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_{f,v}^{\psi_v,\mu_v^{-1}}(h)W_{g,v}^{\psi_v}(h)|\det h|^{s-\frac{1}{2}}dh \\
&=: \prod_v Z_v(f,g,\mu_v,s),
\end{aligned}$$

where $W_{f,v}^{\psi_v,\mu_v^{-1}}(h)$ and $W_{g,v}^{\psi}(h)$ belong to the generalized Whittaker model for Π_v with respect to $\mu_v^{-1}\otimes\omega_{\psi_v}$ and the Whittaker model for σ_v with respect to ψ_v, respectively.

19.4 Example II: Kohnen–Skoruppa integral

We suggest that the generalized Whittaker model can also explain the so-called *Rankin–Selberg integrals with non-unique local models*. These are Rankin–Selberg integrals that are shown to represent an L-function not just by unfolding but also using some auxiliary argument (usually involving Hecke algebra). It is called to have a "non-unique" model because the naive approach unfolds the integral into a degenerate Whittaker model (see §19.2), which is infinite-dimensional as we exactly noted above. We show that, by considering the action of a part of the reductive part as in generalized Whittaker models, a formal manipulation can prove the Eulerianity of the Kohnen–Skoruppa integral[5], a Rankin–Selberg integral with non-unique local models for the spin L-function of Sp_4. The following material in this subsection is purely formal, and there are some analytic difficulties to be overcome.

The Kohnen–Skoruppa integral is recast in modern language by [PS]. We freely use the notations of *op. cit.*, although we would have to conjugate the embedding $\mathrm{GL}_{2,L}^* \hookrightarrow \mathrm{GSp}_4$ slightly. For simplicity, we use the split case, $H := \mathrm{GL}_2 \times_{\mathrm{GL}_1} \mathrm{GL}_2 \hookrightarrow G := \mathrm{GSp}_4$, given by

$$\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\right) \mapsto \begin{pmatrix} a & & b & \\ & a' & & b' \\ c & & d & \\ & c' & & d' \end{pmatrix}.$$

[5] A related suggestion was made in [Od2], although no further justifications were made. As suggested in *op. cit.*, the formal manipulation in this subsection, if made rigorous, shows that the Kohnen–Skoruppa integral would also give rise to an integral representation of the spin L-function for automorphic representations whose infinity type is a *generic discrete series*; so far, the integral representation property is only shown for those belonging to a *(anti-) holomorphic discrete series*.

The integral we consider is

$$J(\phi, s) = \int_{Z(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} E(g, s, \Phi)\phi(g)dg,$$

where ϕ is a cusp form of a cuspidal automorphic representation π on GSp_4, $E(g, s, \Phi)$ is the Klingen Eisenstein series, and Z is the center. This unfolds into

$$J(\phi, s) = \int_{B'(\mathbb{Q})Z(\mathbb{A})\backslash \mathrm{GL}^*_{2,L}(\mathbb{A})} f^{\Phi}(g, s)\phi(g)dg,$$

where $B'(\mathbb{Q}) = Q(\mathbb{Q}) \cap H(\mathbb{Q})$ and $Q(\mathbb{Q}) \subset \mathrm{GSp}_4(\mathbb{Q})$ is the Klingen parabolic,

$$Q = \begin{pmatrix} * & & * & * \\ * & * & * & * \\ * & & * & * \\ & & & * \end{pmatrix}.$$

Let $B' = LN$ be its Levi decomposition. Then, $L = \mathrm{GL}_2(\mathbb{Q}) \times T$, where

$$T = \begin{pmatrix} * & \\ & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Q}),$$

and N is abelian, just consisted of

$$N = \begin{pmatrix} 1 & & & \\ & 1 & & * \\ & & 1 & \\ & & & 1 \end{pmatrix}.$$

Since $[N]$ is a part of the region of integral, from the Fourier expansion $\phi(g) = \sum_{\psi \in \widehat{[N]}} \phi_\psi(g)$, only the coefficient for the trivial character $\mathbf{1}$ survives, and we get

$$J(\phi, s) = \int_{L(\mathbb{Q})N(\mathbb{A})Z(\mathbb{A})\backslash H(\mathbb{A})} f^\Phi(g, s)\phi_{\mathbf{1}}(g)dg.$$

Now consider U, the unipotent radical of Q, the Klingen parabolic. It is also a Heisenberg group, with center N. The irreducible representations of U with N acting trivially are precisely characters, so

$$\phi_{\mathbf{1}}(g) = \sum_{\chi \in \widehat{[U/N]}} \phi_\chi(g) = \sum_{\gamma \in T} \phi_{\chi_0}(\gamma g),$$

where subscript means the corresponding character for the Fourier coefficient and χ_0 is any choice of nontrivial character. We thus get

$$J(\phi, s) = \int_{\mathrm{GL}_2(\mathbb{Q})N(\mathbb{A})Z(\mathbb{A})\backslash H(\mathbb{A})} f^\Phi(g, s)\phi_{\chi_0}(g)dg,$$

where $\mathrm{GL}_2(\mathbb{Q})$ in the region is the "first factor." Now $\phi_{\chi_0}(g)$ is also an element of the degenerate Whittaker model with respect to Klingen parabolic, which is typically infinite-dimensional. We can try to use the generalized Whittaker model in this case,

noting that $S = \mathrm{GL}_2(\mathbb{Q})$ is the reductive part of the centralizer of N:

$$\phi_{\chi_0}(g) = \text{“} \sum_{\sigma \in [\widehat{\mathrm{GL}_{2,\mathbb{Q}}}]} \phi_{\sigma \otimes \chi_0} \text{”}. \tag{19.1}$$

One big problem is that the unitary dual of $\mathrm{GL}_2(F)$ is much more complicated, involving continuous spectrum. Let us pretend that such sum makes sense as we would want. Then, as the region of integral involves $[\mathrm{GL}_{2,\mathbb{Q}}]$, among $\{\phi_{\sigma \otimes \chi_0}\}_{\sigma \in [\widehat{\mathrm{GL}_{2,\mathbb{Q}}}]}$, only the one corresponding to the trivial representation of GL_2, $\phi_{\mathbf{1} \otimes \chi_0}$, survives:

$$J(\phi, s) = \text{“} \int_{R(\mathbb{A}) \backslash H(\mathbb{A})} f^{\Phi}(g, s) \phi_{\mathbf{1} \otimes \chi_0}(g) dg \text{”}.$$

Now we would win the Eulerianity of the integral if we have multiplicity one property of the generalized Whittaker models associated with $\phi_{\mathbf{1} \otimes \chi_0}(g)$. For non-archimedean local models, this is exactly achieved in Section 3 of [MS1]. For archimedean local models, this is achieved in [Hi], with the usual caveat that the multiplicty one holds when one imposes extra *moderate growth condition*. If one settles the question on how to make the sum (19.1) rigorous, then, as in Jacquet's classical theory, one should be able to show that, for cusp forms, only archimedean local generalized Whittaker functionals of moderate growth can appear.

Chapter 20

Automorphic cohomology

We start with a globally generic Π of $\mathrm{GU}(2,1)(\mathbb{A}_{\mathbb{Q}})$. Then, by Arthur's conjectures, which is verified in this case by Rogawski [Ro], Π appears[1] with multiplicity one in the automorphic spectrum. We furthermore assume that Π is unramified at p.

20.1 Generic discrete series and its K-types

There are three types of discrete series representation for $\mathrm{SU}(2,1)$: *holomorphic*, *generic* and *antiholomorphic*. For each type, one can parametrize them via the unique coefficient that gives rise to nonzero $(\mathfrak{sp}^-, SK_\infty)$-cohomology; here $\mathfrak{sp}^-$ is the corresponding $\mathfrak{p}^-$ for $\mathfrak{su}(2,1)$, and SK_∞ refers to a maximal compact subgroup of $\mathrm{SU}(2,1)$. The convention we are using for weights, etc., match with [KO], so we will freely borrow results from it.

We are mainly interested in the generic discrete series $\Pi_\infty^{\mathrm{gen}}(m_1, m_2)$, which means that

$$H^2(\mathfrak{su}(2,1)_{\mathbb{C}}, SK_\infty; \Pi_\infty^{\mathrm{gen}}(m_1,m_2) \otimes V_G(m_1,m_2)) \neq 0.$$

Recall that G-dominance means $m_1, m_2 \geq 0$. For this to hold, $\chi_{V_G(m_1,m_2)^\sim} = \chi_{\Pi_\infty^{\mathrm{gen}}(m_1,m_2)}$.

[1] In this case, as the half sum of positive roots is integral, there is no difference between C-algebraicity and L-algebraicity as in [BG].

Here, $V_G(m_1, m_2)^\sim$ is the contragredient of $V_G(m_1, m_2)$, which is $V_G(m_2, m_1)$. Note also that the Harish–Chandra parameter and the Blattner parameter coincide for generic discrete series in this case. Thus, it turns out that $\Pi_\infty^{\mathrm{gen}}(m_1, m_2)$ has minimal K-type $W_G(m_1 + m_2 + 2, -m_1 - 1)$. Furthermore, all K-types that appear in $\Pi_\infty^{\mathrm{gen}}(m_1, m_2)$ are of the form $W_G((m_1 + m_2 + 2, -m_1 - 1) + a(1, 1) + b(1, -2))$ for $a, b \geq 0$, and they all appear with multiplicity one (this follows from Blattner's formula; cf. [KO, Proposition 3.1]).

In terms of representations of $\mathrm{GU}(2, 1)(\mathbb{R})$, we denote the generic discrete series $\Pi_\infty^{\mathrm{gen}}(k_1, k_2, k_3)$ to be the infinity type that appears in the cohomology of $H^1(X_G, \omega_G(k_1, k_2, k_3))$. We assume that the globally generic Π has infinity type $\Pi_\infty = \Pi_\infty^{\mathrm{gen}}(k_1, k_2, k_3)$. Let E be the field of definition of Π_f, and let $\Pi'_f = \Pi_f \otimes \|\cdot\|^{-(k_1+k_2-k_3)}$. Let $H^1(\Pi_f)$ be the Π'_f-isotypical subspace of $H^1(X_{G,E}, [L_1](-D))$, where $L_1 = W_G(k_1, k_2, k_3)$.

Theorem 20.1.1. *The space $H^1(\Pi_f)$ satisfies*

$$\dim_E H^1(\Pi_f) = \dim_{\mathbb{C}}({\Pi'_f}^{(p)})^{K^p},$$

where ${\Pi'_f}^{(p)}$ is the prime-to-p part of the finite part Π'_f.

Proof. In general, by the $(\mathfrak{p}, K)$-cohomology computation,

$$\dim_E H^1(\Pi_f) = m_{\Pi'} \dim_{\mathbb{C}}(\Pi'_f)^{K^p G(\mathbb{Z}_p)},$$

where $m_{\Pi'}$ is the automorphic multiplicity of the automorphic representation Π'. As Π' is globally generic, the automorphic multiplicity is one, by Arthur's conjecture for $\mathrm{U}(2, 1)$, which is proved in [Ro]. Also, one could ignore the p-part as Π is assumed to be unramified at p. $\square$

20.2 Automorphic period pairing

On the hyperspecial level, there is a similar pushforward map of classical automorphic forms; namely, there is a morphism between algebraic B_H-representations

$$W_H(\ell_1+2,\ell_2+1,\kappa_3) \to W_G(\kappa_1,\kappa_2,\kappa_3)|_{B_H} \otimes \alpha_{G/H}^{-1},$$

for integers $\ell_1,\ell_2,\kappa_1,\kappa_2,\kappa_3$ satisfying $\ell_1+\ell_2=\kappa_1+\kappa_2-3$ and $\ell_1+1,\ell_2>\kappa_2$, so that there is the corresponding pushforward map

$$\iota_* : H^0(X_{H,\mathbb{Q}},\omega_H(\ell_1+2,\ell_1+1,\kappa_3)) \to H^1(X_{G,\mathbb{Q}},\omega_G(\kappa_1,\kappa_2,\kappa_3)).$$

Recall that X_H and X_G are Shimura varieties of hyperspecial level. After applying the Serre duality, we get

$$H^0(X_{H,\mathbb{Q}},\omega_H(\ell_1+2,\ell_1+1,\kappa_3)) \otimes_{\mathbb{Q}} H^1(X_{G,\mathbb{Q}},\omega_G(-\kappa_2,-\kappa_1,-\kappa_3+3)(-D)) \to \mathbb{Q},$$

from the expression of $\Omega^2_{X_G}$ as an automorphic vector bundle and the weight of the dual representation. Now let $-\kappa_2=k_1$, $-\kappa_1=k_2$, $-\kappa_3+3=k_3$, then we have $\kappa_3=3-k_3$, $\kappa_2=-k_1$, $\kappa_1=-k_2$. So, we have

$$H^0(X_{H,E},\omega_H(\ell_1,\ell_2,3-k_3)) \otimes_E H^1(\Pi_f) \to E\{k_1+k_2-k_3-4\},$$

where $\ell_1+\ell_2=-k_1-k_2-3$, $\ell_1+k_1+1,\ell_2+k_1>0$ and $\{m\}$ denotes the twist by a character $\ell \mapsto \ell^m$.

Theorem 20.2.1 (Harris, Su; see [LPSZ, Theorem 5.6])**.** *Let τ_1 be the minimal K-type of $\Pi_\infty^{\mathrm{gen}}(k_1,k_2,k_3)$, namely the K-type of highest weight* $(k_2+1,-k_1+k_2-k_1-$

$1, k_3)$. *Then, there is a canonical isomorphism of irreducible* $G(\mathbb{A}_f)$*-representations*

$$\mathrm{Hom}_K(\tau_1, \Pi)\left\{\frac{k_1+k_2-k_3}{2}\right\} \xrightarrow{\sim} H^1(\Pi_f)_{\mathbb{C}}.$$

Moreover, if $\eta \in H^1(\Pi_f)_{\mathbb{C}}$*, corresponding to some* K*-homomorphism* $F_\eta : \tau_1 \to \Pi$*, then for any* $\ell_1+\ell_2 = -k_1-k_2-3$*,* $\ell_1+k_1+1, \ell_2+k_1 > 0$*, and any* $f \in M_H^{(\ell_1,\ell_2,3-k_3)}$*, we have*

$$\langle \iota_*(f), \eta\rangle = \frac{1}{(2\pi i)^2}\int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} F_\eta(v_{\kappa_1,\kappa_2})f(h)dh,$$

where v_{κ_1,κ_2} *is the standard basis vector of* τ_2 *of weight* $(-\kappa_1+1, -\kappa_2-2)$*.*

Proof. As in [LPSZ, Theorem 5.6], it follows from [Su]. □

20.3 p-adic period pairing

Now we think of p-adic period pairing: let p be a prime such that $p = v\overline{v}$ is split in K and Π_p is unramified. We have seen in §17.3 that $\Pi_p^{\mathcal{P}}$ is 3-dimensional, and the $U_{p,2}$-operator

$$p^{5-k_2}[\mathcal{P}\,\mathrm{diag}(p,p,1)\mathcal{P}],$$

acts on $\Pi_p^{\mathcal{P}}$ with eigenvalues

$$\left\{\frac{\alpha\beta}{p^{k_2-2}}, \frac{\beta\gamma}{p^{k_2-2}}, \frac{\gamma\alpha}{p^{k_2-2}}\right\},$$

where $\{\alpha, \beta, \gamma\}$ is the Hecke-eigenvalues of Π_p. Note that the adjoint Hecke operator,

$$U'_{p,2} := p^{5-k_2}[\mathcal{P}\,\mathrm{diag}(1,p,p)\mathcal{P}],$$

also acts on $\Pi_p^{\mathcal{P}}$ with eigenvalues

$$\left\{\frac{\alpha\beta}{p^{k_2-2}}, \frac{\beta\gamma}{p^{k_2-2}}, \frac{\gamma\alpha}{p^{k_2-2}}\right\}.$$

By the standard "p-level descent argument" in Hida theory (e.g. [Ti, Lemma 2.21]), we have the following

Proposition 20.3.1. *For $m \geq 1$, we have a factorization of the diagram*

$$\begin{tikzcd}
\Pi_p^{\mathrm{Kl}(p^m)} \arrow[r, "U_{p,2}"] \arrow[d, hook] & \Pi_p^{\mathrm{Kl}(p^m)} \arrow[d, hook] \\
\Pi_p^{\mathrm{Kl}(p^{m+1})} \arrow[r, "U_{p,2}"'] \arrow[ur, dashed, "\xi_m"] & \Pi_p^{\mathrm{Kl}(p^{m+1})}
\end{tikzcd}$$

Using this, we define the following projector.

Definition 20.3.2. Let $\eta \in H^1(\Pi_f)^{\mathrm{Kl}(p)}$. Then, let $\eta_m \in H^1(\Pi_f)^{\mathrm{Kl}(p^m)}$ be characterized as follows.

1. For $v \in H^1(\Pi_f^\vee)^{\mathrm{Kl}(p^m), U_{p,2}\text{-nilpotent}}$, $\langle v, \eta_m \rangle = 0$.

2. For $v \in H^1(\Pi_f^\vee)^{\mathrm{Kl}(p)}$, $\langle v, \eta \rangle = \langle \xi_m(v), \eta_m \rangle$, where ξ_m is as defined in Proposition 20.3.1.

Proposition 20.3.3. *If η_1 lies in the $U_{p,2}$-eigenspace with eigenvalue λ, then*

$$\eta_m = \left(\frac{p^{1-k_2}}{\lambda}\right)^{m-1} \sum_{a \in \mathbb{Z}/p^{m-1}\mathbb{Z}} \begin{pmatrix} 1 & & p^m a \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} p & & \\ & p & \\ & & 1 \end{pmatrix}^{m-1} \eta.$$

Proof. The proof is the same as that of [LPSZ, Proposition 5.9]. From the character-

izing properties of η_m, we have the p-compatibility

$$\eta_m = \frac{1}{p^3} \sum_{g \in \mathrm{Kl}(p^m)/\mathrm{Kl}(p^{m+1})} g\eta_{m+1}.$$

Thus, it is sufficient to show the p-compatibility of the RHS of the expression. On the other hand, one can choose the coset representatives of the $U_{p,2}$-operator on $\mathrm{Kl}(p^m)$-level to be lower triangular and to commute with $\begin{pmatrix} 1 & & p^m a \\ & 1 & \\ & & 1 \end{pmatrix}$. Thus we can group the coset representatives to manifest the p-compatibility of the RHS. $\square$

Definition 20.3.4. For a place w of E above p, Π is said to be Klingen-ordinary at p with respect to w if $\Pi_p^{\mathrm{Kl}(p)}$ has an $U'_{p,2}$-eigenvalue λ which is a w-adic unit.

For such Π, we construct a p-adic interpolation of period pairing, and show that it has the desired interpolation property.

Definition 20.3.5 (Ordinary period pairing)**.** Let L be a completion of E at a place w above p, and assume that Π is Klingen-ordinary at p with respect to w, and $k_1, k_2, k_3 \in \mathbb{Z}$ satisfies $k_2 + k_3 \leq 0$ and $k_1 - k_2 \geq 3$. Choose $\eta \in (H^1(\Pi_f) \otimes_E L)^{U'_{p,2}-\mathrm{ord}}$. From Corollary 17.3.4, this corresponds to

$$\eta_1 \in H^1(X_G(\mathrm{Kl}(p))_L, \omega_G(k_1, k_2, k_3)(-D))^{\mathrm{Kl}(p), U'_{p,2}-\mathrm{ord}} \cong (H^1(\Pi_f) \otimes_E L)^{U'_{p,2}-\mathrm{ord}}.$$

Now we define

$$\langle -, \eta \rangle : f_{\mathrm{Kl}} H^1(X_G(\mathrm{Kl}(p))_L, \omega_G(-k_2, -k_1, 3 - k_3)) \to L,$$

to be the Serre duality pairing with respect to η_1.

Definition 20.3.6 (p-adic period pairing)**.** Retain the hypothesis as in Definition 20.3.5. Let R be a p-complete flat $\mathcal{O}_L$-algebra, and $\lambda_1, \lambda_2 : \Gamma \to R^\times$ be continuous

characters such that

$$\lambda_1 + \lambda_2 = -k_1 - k_2 - 3.$$

By the functoriality between the Hida theory for H and the higher Hida theory for G, we have a map

$$\begin{aligned}\iota_* : \mathcal{S}_H^{(\lambda_1,\lambda_2,3-k_3)}(R) &\to H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_{G,R}^{(-k_2,-k_1,3-k_3)}(-D_G)) \\ &= R\widehat{\otimes}_{\mathbb{Z}_p} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(-k_2,-k_1,3-k_3)}(-D_G)).\end{aligned}$$

Given $\mathcal{E} \in \mathcal{S}_H^{(\lambda_1,\lambda_2,3-k_3)}(R)$, we define $\langle \iota_*(\mathcal{E}), \eta \rangle \in R\left[\frac{1}{p}\right]$ to be the image of $\mathcal{E}$ under the following composition of morphisms:

$$\begin{aligned}\mathcal{S}_H^{(\lambda_1,\lambda_2,3-k_3)}(R) &\xrightarrow{\iota_*} R\widehat{\otimes}_{\mathbb{Z}_p} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(-k_2,-k_1,3-k_3)}(-D_G)) \\ &\to R[1/p]\widehat{\otimes}_{\mathbb{Z}_p} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(-k_2,-k_1,3-k_3)}(-D_G)) \\ &\xrightarrow{f_{\mathrm{Kl}}} R[1/p]\widehat{\otimes}_{\mathbb{Z}_p} f_{\mathrm{Kl}} H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}_G^{(-k_2,-k_1,3-k_3)}(-D_G)) \\ &\underset{\substack{\text{Integral control theorem} \\ \text{Theorem 17.1.11}}}{\cong} R[1/p] \otimes_{\mathbb{Q}_p} f_{\mathrm{Kl}} H^1(X_G(p)_{\mathbb{Q}_p}, \omega_G(-k_2,-k_1,3-k_3)(-D_G)) \\ &\to R[1/p] \otimes_{\mathbb{Q}_p} f_{\mathrm{Kl}} H^1(X_G(p)_{\mathbb{Q}_p}, \omega_G(-k_2,-k_1,3-k_3)) \\ &\xrightarrow{\langle -,\eta \rangle} R[1/p].\end{aligned}$$

20.4 Comparison of p-adic period pairing with automorphic period pairing

Proposition 20.4.1 (Comparison of p-adic pairing and automorphic pairing)**.** *Retain the assumptions of Definition 20.3.5 and Definition 20.3.6. Let $\phi : R \to L$ be a continuous ring homomorphism such that*

$$\phi \circ \lambda_1, \phi \circ \lambda_2 : \Gamma \to L^\times,$$

are algebraic characters

$$x \mapsto x^{\ell_1}, x^{\ell_2},$$

respectively, for integers ℓ_1, ℓ_2*. Suppose that* $\phi(\mathcal{E}) \in \mathcal{S}_H^{(\ell_1,\ell_2,3-k_3)}(L)$ *is identified with the restriction of a classical modular form* $E_\phi \in M_H^{(\ell_1,\ell_2,3-k_3)}(p, L)$ *to* $\mathcal{X}_H^{\mathrm{ord}}$*. Then, the p-adic period pairing of* $\mathcal{E}$ *and the automorphic period pairing of* E_ϕ *are compatible with each other in the following sense,*

$$\phi(\langle \iota_*(\mathcal{E}), \eta \rangle) = \langle \iota_{1,*}(E_\phi), \eta \rangle.$$

Proof. As in the proof of [LPSZ, Proposition 5.12], the only subtlety arises when E_ϕ is not cuspidal but nevertheless cuspidal at the ordinary locus. Such modular form might exist as our ordinary locus is a certain connected component of the usual ordinary locus of the singular fiber of $X_H(p)$.

We thus have to use the classicality of overconvergent cohomology, [Ng, Theorem 18.4.1, Theorem 18.4.2] for all classes of small slope (which, in particular, include ordinary classes). We have the analytic class

$$z^{\mathrm{an}} = \iota_*(\mathcal{E}) \in H^1(\mathcal{X}_G^{\geq 1}(p), \mathcal{F}_G^{(-k_2,-k_1,3-k_3)}(-D_G)).$$

By classicality of cuspidal classes in higher Coleman theory, there is a cuspidal *overconvergent* class

$$z^\dagger_{\mathrm{cusp}} \in H^1(\mathcal{X}_G^{\geq 1}(p)^\dagger, \omega_G(k_1, k_1 - k_2 - k_3, k_3 + 3)(-D)),$$

such that

$$(z^\dagger_{\mathrm{cusp}})|_{\mathcal{X}_G^{\geq 1}(p)} = f_{\mathrm{Kl}} z^{\mathrm{an}}.$$

By the obvious commutative diagram, if you let

$$z^\dagger \in H^1(\mathcal{X}_G^{\geq 1}(p)^\dagger, \omega_G(-k_2, -k_1, 3-k_3)),$$

be the image of $z^\dagger_{\mathrm{cusp}}$ by forgetting cuspidality, then

$$z^\dagger = (f_{\mathrm{Kl}} z^{\mathrm{alg}})^{\mathrm{an}}|_{\mathcal{X}_G^{\geq 1}(p)^\dagger},$$

where

$$z^{\mathrm{alg}} = \iota_{1,*}(E_\phi) \in H^1(X_G(p)_L, \omega_G(-k_2, -k_1, 3-k_3)),$$

is the algebraic pushforward of E_ϕ.

By classicality of *higher Hida theory*, there is a cuspidal *algebraic* class

$$z^{\mathrm{alg}}_{\mathrm{cusp}} \in H^1(X_G(p)_L, \omega_G(-k_2, -k_1, 3-k_3)(-D)),$$

such that

$$(z^{\mathrm{alg}}_{\mathrm{cusp}})^{\mathrm{an}}|_{\mathcal{X}_G^{\geq 1}(p)} = f_{\mathrm{Kl}} z^{\mathrm{an}}.$$

Moreover,

$$(z^{\mathrm{alg}}_{\mathrm{cusp}})^{\mathrm{an}}|_{\mathcal{X}_G^{\geq 1}(p)^\dagger} = z^\dagger_{\mathrm{cusp}}.$$

If we let

$$z^{\mathrm{alg}}_{\mathrm{cusp}\to\mathrm{mod}} \in H^1(X_G(p)_L, \omega_G(-k_2, -k_1, 3-k_3)),$$

be the class obtained from $z^{\mathrm{alg}}_{\mathrm{cusp}}$ by forgetting cuspidality, then

$$(z^{\mathrm{alg}}_{\mathrm{cusp}\to\mathrm{mod}})^{\mathrm{an}}|_{\mathcal{X}_G^{\geq 1}(p)^\dagger} = z^\dagger = (f_{\mathrm{Kl}} z^{\mathrm{alg}})^{\mathrm{an}}|_{\mathcal{X}_G^{\geq 1}(p)^\dagger}.$$

By classicality of *higher Coleman theory* of modular forms (without $(-D)$), we get

$z^{\mathrm{alg}}_{\mathrm{cusp}\to\mathrm{mod}} = f_{\mathrm{Kl}} z^{\mathrm{alg}}$. Thus,

$$\phi(\langle \iota_*(\mathcal{E}), \eta\rangle) = \langle z^{\mathrm{alg}}_{\mathrm{cusp}\to\mathrm{mod}}, \eta\rangle = \langle f_{\mathrm{Kl}} z^{\mathrm{alg}}, \eta\rangle = \langle \iota_{1,*}(E_\phi), \eta\rangle,$$

as desired. □

Proposition 20.4.2 (Comparison of p-adic pairing and automorphic pairing, higher level)**.** *Retain the assumptions of Definition 20.3.5 and Definition 20.3.6. Let $\phi : R \to L$ be a continuous ring homomorphism such that*

$$\phi \circ \lambda_1, \phi \circ \lambda_2 : \Gamma \to L^\times,$$

are locally algebraic characters

$$x \mapsto x^{\ell_1}\chi_1(x), x^{\ell_2}\chi_2(x),$$

respectively, for integers ℓ_1, ℓ_2 and finite order characters χ_1, χ_2 such that $\chi_1\chi_2 = 1$. Suppose that $\phi(\mathcal{E}) \in \mathcal{S}_H^{(\ell_1+\chi_1,\ell_2+\chi_2,3-k_3)}(L)$ is the p-adic modular form corresponding to a classical modular form $E_\phi \in M_H^{(\ell_1,\ell_2,3-k_3)}(p^m, \chi_1, \chi_2)_L$ for some $m \gg 0$. Then, the p-adic period pairing of $\mathcal{E}$ and the automorphic period pairing of E_ϕ are compatible with each other, in the following sense,

$$\phi(\langle \iota_*(\mathcal{E}), \eta\rangle) = \langle \iota_{m,*}(E_\phi), \eta_m\rangle.$$

Proof. The same proof as above, the proof of Proposition 20.4.1, works, as we have the compatibility of classical and p-adic modular forms for higher level specializations as well. □

Chapter 21

Nearly coherent cohomology

21.1 Nearly sheaves

We develop the theory of "nearly" sheaves in the sense of [LPSZ, §6], and connect it with nearly holomorphic modular forms and their incarnation as p-adic modular forms. It involves automorphic vector bundles that do not necessarily come from Levi. Namely, the "nearly-ness" comes from considering automorphic vector bundles $[V]$ where V is an algebraic representation of P_- which does not factor through K.

Definition 21.1.1. Let V be an algebraic representation of P_-. Then, let Fil^n be the sum of weight spaces of weight $\lambda(k_1, k_2, k_3; c)$ with $k_1+k_2+k_3+c \geq 2n$. This filtration is P_--stable, and the action of P_- on $\mathrm{gr}^n V = \mathrm{Fil}^n V / \mathrm{Fil}^{n+1} V$ factors through M.

We apply this to $V_G(k_1 - 1, -k_3, -k_2 - 1; k_1 + k_2 - k_3)$, where the grading spans for degrees $0 \leq n \leq k_1 - k_3 - 1$.

Definition 21.1.2. Let $\widetilde{L}_1$ be the P_--representation

$$\widetilde{L}_1 := \left(\mathrm{Fil}^{k_1+k_2-1} V_G(k_1 - 1, -k_3, -k_2 - 1; k_1 + k_2 - k_3)\right) \otimes W(1, -1, 0; 0),$$

and let $\widetilde{L}_1'$ be the top filtration quotient of $\widetilde{L}_1$. We similarly define the dual notion

$\widetilde{L^\vee}_1$ and $\widetilde{L^\vee}'_1$ for $L_1^\vee$.

There is a natural inclusion $L_1 \hookrightarrow \widetilde{L}'_1$.

Proposition 21.1.3. *From the natural morphisms $L_1 \hookrightarrow \widetilde{L}'_1 \twoheadleftarrow \widetilde{L}_1$,*

$$H^1(\Pi_f) \to H^1(X_{G,E}, [\widetilde{L}'_1](-D))[\Pi_f] \leftarrow H^1(X_{G,E}, [\widetilde{L}_1](-D))[\Pi_f],$$

are all isomorphisms.

Proof. This is because, again from Arthur's conjecture, Π_f can only contribute to $(\mathfrak{p}, K)$-cohomology via the minimal K-type of Π_{gen}, and such K-type does not occure in either $\widetilde{L}'_1/L_1$ or $\ker(\widetilde{L}_1 \twoheadrightarrow \widetilde{L}'_1)$. □

Definition 21.1.4. Given $\eta \in H^1(\Pi_f)$, let $\widetilde{\eta} \in H^1(X_{G,E}, [\widetilde{L}_1](-D))[\Pi_f]$ be the corresponding class under the above isomorphism. From Serre duality, we again obtain an automorphic period pairing

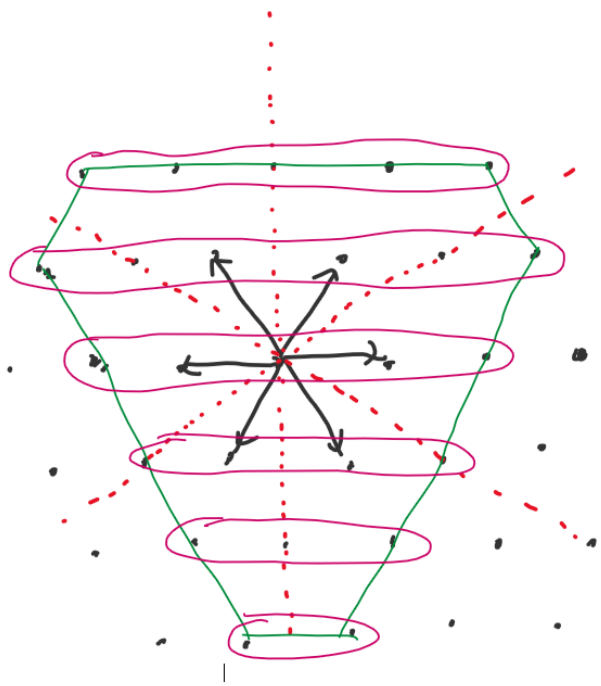
$$H^1(X_{G,E}, [\widetilde{L^\vee}_1]) \to E,$$

which factors through $\Pi_f^\vee$-isotypic part, and its restriction to $H^1(X_{G,E}, [\widetilde{L^\vee}'_1])$ factors through the natural map induced from the projection $\widetilde{L^\vee}'_1 \twoheadrightarrow L_1^\vee$. Recall that the natural morphisms are as follows,

$$L_1 \hookrightarrow \widetilde{L}'_1 \twoheadleftarrow \widetilde{L}_1,$$

$$L_1^\vee \twoheadleftarrow \widetilde{L^\vee}'_1 \hookrightarrow \widetilde{L^\vee}_1.$$

The branching of algebraic representations of GL_3 to GL_2 works pictorially as follows.



By branching law, $V_H := V_H(\kappa_1, \kappa_2)$ is an H-subrepresentation of $V_G(1-k_2, k_3, k_1 - 1; -k_1 - k_2 + k_3)$ if and only if

$$1 - k_2 \geq \kappa_1 \geq k_3 \geq \kappa_2 \geq 1 - k_1.$$

Note that $\widetilde{L^\vee}_1$ is

$$\widetilde{L^\vee}_1 = \frac{V_G(1-k_2, k_3, k_1 - 1; -k_1 - k_2 + k_3)}{\mathrm{Fil}^{-k_1-k_2} V_G(1-k_2, k_3, k_1 - 1; -k_1 - k_2 + k_3)} \otimes W(1, -1, 0; 0),$$

and $L_1^\vee$ is the bottom filtration piece of $\widetilde{L^\vee}'_1$.

Definition 21.1.5 (Pushforward of nearly sheaves)**.** From the above inclusion

$$V_H \hookrightarrow V_G(1-k_2, k_3, k_1 - 1; -k_1 - k_2 + k_3),$$

we obtain the pushforward of nearly sheaves

$$\iota_*^{\text{nearly}} : H^0(X_{H,\mathbb{Q}}, [V_H] \otimes \omega_H(1,-1,0;0)) \to H^1(X_{G,\mathbb{Q}}, [\widetilde{L^\vee}_1]).$$

Again, we will see that the source of ι_*^{nearly} are *nearly holomorphic modular forms* of weight (κ_1+1, κ_2+1).

21.2 Hodge splitting and nearly holomorphic modular forms

Using this, we have a cohomological interpretation of period integral for nearly holomorphic modular forms,

$$H^0(X_{H,E}, [V_H] \otimes \omega_H(1,-1,0;0)) \otimes H^1(\Pi_f) \to E,$$

$$(f, \eta) \mapsto \langle \iota_*^{\text{nearly}}(f), \widetilde{\eta} \rangle.$$

Definition 21.2.1. Let the *Hodge splitting* be the C^∞-splitting

$$[V_H] \otimes \omega_H(1,-1,0;0) \to \omega_H(\kappa_1+1, \kappa_2-1),$$

which exists as the category of finite dimensional representations of a compact Lie group is semisimple.

Comparing the K-types of Π_∞^{gen} and that of $\widetilde{L}_1$, one can compute the Π_f-isotypic part of $H^1(X_{G,\mathbb{C}}, [\widetilde{L}_1](-D))$ in terms of Hodge splitting.

Theorem 21.2.2. *Let $E \in H^0(X_{H,\mathbb{C}}, [V_H] \otimes \omega_H(1,-1,0;0))$ and let $E_\infty \in H^0(X_{H,\mathbb{C}}, \omega(\kappa_1+1, \kappa_2-1)_{C^\infty})$ be its image under the Hodge splitting. Let $\eta \in H^1(\Pi_f)_\mathbb{C}$ corresponding*

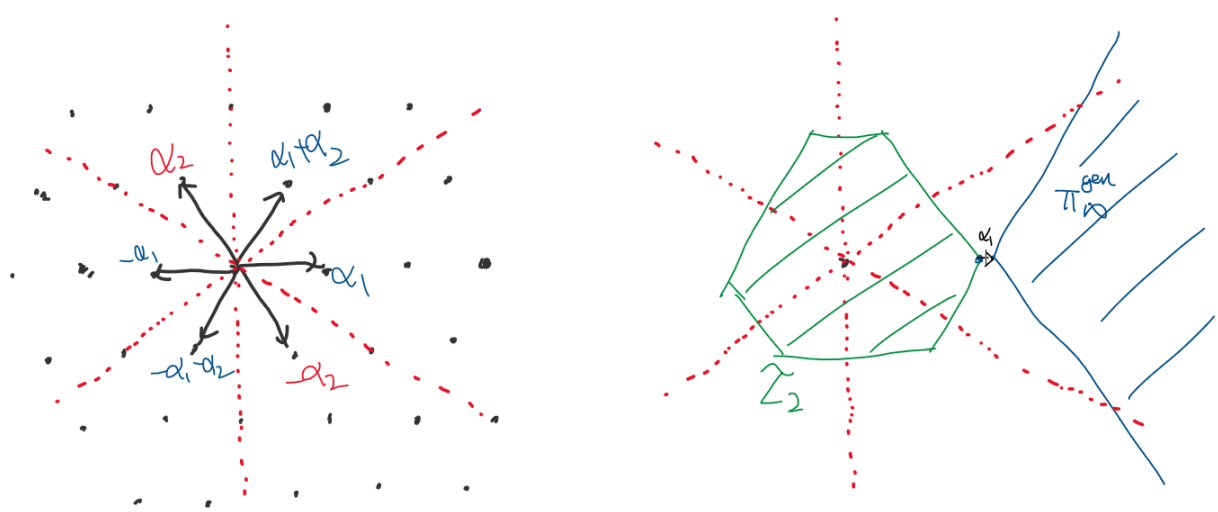
to a K_∞-homomorphism $F_\eta : \tau_1 \to \Pi$. Then,

$$\langle \iota_*^{\text{nearly}}(E), \widetilde{\eta} \rangle = \frac{1}{(2\pi i)^2} \int_{Z_G(\mathbb{A})H(\mathbb{Q})\backslash H(\mathbb{A})} F_\eta(v_{\kappa_1,\kappa_2}) E_\infty(h) dh.$$

Proof. The proof of this requires that the Chevalley–Eilenberg complex for $(\mathfrak{p}, K)$-cohomology is concentrated in degree 1 and is represented by the minimal K-type, which is the content of [LPSZ, Lemma 6.5]. In our setting, this is the following statement:

$$\dim_{\mathbb{C}} \operatorname{Hom}_{K_\infty}\left(\wedge^j(\mathfrak{p}^-) \otimes \left(\widetilde{L}_1\right)^\vee, \Pi_\infty^{\text{gen}}(k_1,k_2,k_3) \right) = \begin{cases} 0 & \text{if } j \neq 1, \\ 1 & \text{if } j = 1. \end{cases}$$

The proof of this can be done by comparing the K-types of the irreducible algebraic G-representation and that of the generic discrete series. With the aid of the following picture, we can prove the statement by noticing the following facts.



- In the above picture, $\alpha_2, -\alpha_2$ are the compact roots.
- The generic discrete series has K-types lying in the cone with a vertex on the minimal K-type and the rays to the directions of $\alpha_1 + \alpha_2$ and $-\alpha_2$.

- The minimal K-type of the generic discrete series is α_1 plus the rightmost K-type of $\widetilde{L}_1^\vee$. Thus, not only $\widetilde{L}_1^\vee$ has no intersection with generic discrete series, but also $\widetilde{L}_1^\vee \otimes \mathbb{C}\alpha_1$ contains the minimal K-type of generic discrete series with multiplicity one, and $\widetilde{L}_1^\vee \otimes \mathbb{C}(\alpha_1 + \alpha_2)$ has no K-type lying in the region of generic discrete series.

□

21.3 p-adic splitting of nearly sheaves

We would like to develop the analogous theory for p-adic modular forms. To do this, we need a p-adic analogue of Hodge splitting over $\mathfrak{X}_{\widetilde{G}}^{\geq 1}(p^m)$. In the context of classical Hida theory, the corresponding p-adic splitting is called the *unit root splitting*, coming from the decomposition of cohomology in terms of slopes with respect to the crystalline Frobenius. Namely, over $X_{G,\mathbb{Z}_p}$, there is a canonical sequence of vector bundles

$$0 \to \omega_\sigma \to \mathscr{H}^1_{\mathrm{dR}}(\overline{\mathcal{A}}/X_G)^{\mathrm{can}}_\sigma \to \omega^\vee_{\overline{\sigma}} \to 0,$$

which is the Hodge filtration of the σ-part of the $\mathcal{O}_K$-action on the canonical extension of the relative first de Rham cohomology sheaf $\mathscr{H}^1_{\mathrm{dR}}(\overline{A}/X_G)^{\mathrm{can}}$. Then, over $\mathfrak{X}_G^{\mathrm{ord}} \subset \mathfrak{X}_G$, the Hodge filtration has a canonical splitting

$$\mathscr{H}^1_{\mathrm{dR}}(\overline{A}/X_G)^{\mathrm{can}}_\sigma \cong \omega_\sigma \oplus \mathcal{U},$$

where $\mathcal{U}$ is the *unit root subsheaf* on which the crystalline Frobenius acts with p-adic unit eigenvalues. We will abuse the above notation to denote the unit root subsheaf over $\mathfrak{IG}_G^{\mathrm{ord}}(p^m)$ (where the whole picture is just pulled back from $\mathfrak{X}_G^{\mathrm{ord}}$). Over the ordinary locus, from the unit root splitting, we can reduce the $\mathcal{P}$-torsor $\mathcal{T}_G/Y_G$ (see Notations) to a $M_{\mathcal{P}}$-torsor where one can define splitting of $\mathcal{P}$-reprsentations into

$M_{\mathcal{P}}$-representaions.

This splitting does not extend over the ordinary locus, so in particular one cannot expect to have such a splitting over $\mathfrak{IG}_{\overline{G}}^{\geq 1}(p^\infty)$ (this is one of the major points of higher Hida theory). Thus, one has to use a different approach towards unit root splitting that works over other Ekedahl–Oort strata.

Proposition 21.3.1. *Let $[\alpha] \in H^0(\mathfrak{IG}_{\overline{G}}^{\geq 1}(p^\infty), \omega_{\overline{\sigma}}^\vee)$ be the natural universal morphism of p-divisible groups $\alpha : \mu_{p^\infty} \hookrightarrow A[\overline{\pi}^\infty]$. Then, there is a unique lifting*

$$[\alpha]^{\mathrm{nr}} \in H^0(\mathfrak{IG}_{\overline{G}}^{\geq 1}(p^\infty), \mathscr{H}^1_{\mathrm{dR}}(\overline{A}/X_G)_\sigma^{\mathrm{can}}),$$

such that its restriction to $\mathfrak{IG}_G^{\mathrm{ord}}(p^\infty)$ takes values in the unit root subsheaf $\mathcal{U} \subset \mathscr{H}^1_{\mathrm{dR}}(\overline{A}/X_G)_\sigma^{\mathrm{can}}$.

Proof. The uniqueness follows as $\mathfrak{IG}_G^{\mathrm{ord}}(p^\infty)$ is dense in $\mathfrak{IG}_G(p^\infty)$, where there is a unique such class due to the classical unit root splitting. Also, one can work over the open Shimura variety as the complement of the union of open Shimura variety and the ordinary locus is of codimension ≥ 2, so that one can extend the section by normality of schemes.

For an abelian scheme A/S, there is a *universal vectorial extension* of A, which is a short exact sequence

$$0 \to V \to I \to A \to 0,$$

of S-group schemes, where V is a vector group, such that it is universal among the short exact sequences of the above form. This also encodes the Hodge filtration of $H^1_{\mathrm{dR}}(A/S)$, as $\mathrm{Inv}(I/S)$, the space of invariant differentials of I over S, is identified with $H^1_{\mathrm{dR}}(A/S)$, and under this identification, $\mathrm{Inv}(A/S) \to \mathrm{Inv}(I/S)$ is identified with $\omega_{A/S} \hookrightarrow H^1_{\mathrm{dR}}(A/S)$.

As the universal vectorial extension is universal, any structure that is given on A can be extended to the whole exact sequence consisting the universal vectorial

extension. In particular, if A has a CM by $\mathcal{O}_K$, then the whole exact sequence has a CM by $\mathcal{O}_K$. In particular, the σ-part of the universal vectorial extension

$$0 \to V_\sigma \to I_\sigma \to A_\sigma \to 0,$$

satisfies

$$\mathrm{Inv}(I_\sigma/S) = H^1_{\mathrm{dR}}(A/S)_\sigma \supset \mathrm{Inv}(A_\sigma/S) = \omega_\sigma.$$

Replicating the argument of [LPSZ, Proposition 6.8], let us take the short exact sequence of corresponding formal groups,

$$0 \to \widehat{V}_\sigma \to \widehat{I}_\sigma \to \widehat{A}_\sigma \to 0.$$

For a morphism $\alpha : \widehat{\mathbb{G}}_m \hookrightarrow \widehat{A}_\sigma$, one can pullback the above short exact sequence to get another short exact sequence,

$$\begin{tikzcd}
0 \arrow[r] & \widehat{V}_\sigma \arrow[r] \arrow[d, equal] & \widehat{I}_{\sigma,\alpha} \arrow[r] \arrow[d, hook] & \widehat{\mathbb{G}}_m \arrow[r] \arrow[d, hook, "\alpha"] & 0 \\
0 \arrow[r] & \widehat{V}_\sigma \arrow[r] & \widehat{I}_\sigma \arrow[r] & \widehat{A}_\sigma \arrow[r] & 0
\end{tikzcd}$$

Since the top row is an extension of a multiplicative formal group by an additive formal group, it uniquely splits and yields the splitting $p : \widehat{\mathbb{G}}_m \to \widehat{I}_{\sigma,\alpha}$. This gives rise to a map

$$[\alpha] : \mathrm{Inv}(I_\sigma/S) = \mathrm{Inv}(\widehat{I}_\sigma/S) \xrightarrow{\alpha^*} \mathrm{Inv}(\widehat{I}_{\sigma,\alpha}/S) \xrightarrow{p^*} \mathrm{Inv}(\widehat{\mathbb{G}}_m/S) \cong \mathcal{O}_S,$$

or a class

$$[\alpha] \in H^0(\mathfrak{IG}_G^{\geq 1}(p^\infty)_{\mathfrak{Y}_G}, \mathscr{H}^1_{\mathrm{dR}}(A/Y_G)_\sigma),$$

where $\mathfrak{IG}_G^{\geq 1}(p^\infty)_{\mathfrak{Y}_G}$ is the open formal subscheme of $\mathfrak{IG}_G^{\geq 1}(p^\infty)$ that lies over $\mathfrak{Y}_G \subset$

$\mathfrak{X}_G$, the formal completion of $Y_G \subset X_G$.

Over $\mathfrak{IG}_G^{\mathrm{ord}}(p^\infty)_{\mathfrak{Y}_G}$, the universal abelian scheme is ordinary, so the bottom row

$$0 \to \widehat{V}_\sigma \to \widehat{I}_\sigma \to \widehat{A}_\sigma \to 0,$$

also always splits, as $\widehat{A}_\sigma$ itself is of multiplicative type. Therefore, there is a unique splitting $p_A : \widehat{A}_\sigma \to \widehat{I}_\sigma$ that lives over $\mathfrak{Y}_G^{\mathrm{ord}}$. By commutativity, the map $[\alpha] : \mathrm{Inv}(I_\sigma/S) \to \mathcal{O}_S$ is equal to the composition

$$\mathrm{Inv}(I_\sigma/S) = \mathrm{Inv}(\widehat{I}_\sigma/S) \xrightarrow{p_A^\sigma} \mathrm{Inv}(\widehat{A}_\sigma/S) \xrightarrow{\alpha^*} \mathrm{Inv}(\widehat{\mathbb{G}}_m/S) \cong \mathcal{O}_S.$$

As p_A is the splitting, the restriction of $[\alpha]$ to $\mathrm{Inv}(\widehat{A}_\sigma/S) \subset \mathrm{Inv}(\widehat{I}_\sigma/S)$ is just equal to

$$\alpha^* : \mathrm{Inv}(\widehat{A}_\sigma/S) \to \mathrm{Inv}(\widehat{\mathbb{G}}_m/S),$$

which is the class corresponding to the classical unit root splitting. Thus we have shown what we desired. □

Definition 21.3.2. Let $Q \subset P \cap \overline{B}$, where $\overline{B}$ is the lower-triangular Borel.

This is a parabolic subgroup of M_P, whose Levi is identified with T.

Definition 21.3.3. Let $\mathcal{T}^{\mathrm{Kl}}$ be the Q-torsor over $\mathfrak{X}_G^{\geq 1}(p^\infty)$ parametrizing bases f_1, f_2, f_3 of $\mathcal{H}^1_{\mathrm{dR}}(A)^{\mathrm{can}}_\sigma$ compatible with the filtration and the polarization, with the additional property that the pullback of f_1 to $\mathfrak{IG}_G^{\geq 1}(p^\infty)$ is a scalar multiple of $[\alpha]^{\mathrm{nr}}$.

This is a reduction of structure of the P-torsor $\mathcal{T}$. Using this torsor, one can define the unit root splitting over $\mathfrak{X}_G^{\geq 1}(p^\infty)$ as follows.

Proposition 21.3.4.

1. *Let $k_1 \geq -k_3$ be integers. Then, there is a unique-up-to-scalars Q-equivariant*

surjection

$$W_G(k_1, -k_3, -k_2; c) \to \lambda(k_1, k_2, k_3; c).$$

2. *Let $k_1 \geq -k_3 \geq k_2$ be integers. Then, there is a unique-up-to-scalars Q-equivariant surjection*

$$V_G(k_1, -k_3, -k_2; -k_1 - k_2 + k_3) \to \lambda(k_1, k_2, k_3; -k_1 - k_2 + k_3),$$

which factors through $V/\operatorname{Fil}^{k_3+1} V$. The restriction of this morphism to $\operatorname{gr}^{k_3} V$ is nontrivial, and this factors through the map of form (i).

3. *For any $k_1 \geq -k_3 \geq k_2$, one has the "unit root splitting",*

$$\begin{tikzcd}
H^1(\mathfrak{X}_G^{\geq 1}(p), [\widetilde{L^\vee}_1](-D)) \arrow[dr, dashed] & \\
H^1(\mathfrak{X}_G^{\geq 1}(p), [L_1^\vee](-D)) \arrow[u, hook] \arrow[r] & H^1(\mathfrak{X}_G^{\geq 1}(p), \mathfrak{F}^{(-k_2,-k_1,-k_3;-k_1-k_2+k_3)}(-D)).
\end{tikzcd}$$

Proof. The proof is entirely analogous to that of Lemma 6.11 and Corollary 6.15 of [LPSZ]. Namely, (i) is clear from the fact that T is the Levi of Q, (ii) is because the target weight is an extremal weight, and (iii) is because of the construction of (ii). □

From this, the compatibility of the unit root splitting and the Hodge splitting follows.

21.4 p-adic pushforward of nearly sheaves

We now need to extend the theory of p-adic period pairing of the previous section to nearly holomorphic modular forms.

Theorem 21.4.1. *Via the embedding $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty) \to \mathfrak{X}_G^{\geq 1}(p^\infty)$, the (classical) unit root splitting over $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty)$ and the unit root splitting over $\mathfrak{X}_G^{\geq 1}(p^\infty)$ of Proposition 21.3.4.3 are compatible with each other. Namely, for ℓ_1, ℓ_2 such that*

$$\ell_1 + \ell_2 = -k_1 - k_2 - 3,$$

$$\ell_1 + k_1 + 1 > 0,$$

$$\ell_2 + k_1 > 0,$$

the diagram

$$\begin{tikzcd}
H^0(\mathfrak{X}_H^{\mathrm{ord}}, [V_H(\ell_1 - 1, \ell_2 + 1, 3 - k_3))] \otimes \omega_H(1, -1, 0; 0)(-D_H)) \arrow[r] \arrow[d] & H^1(\mathfrak{X}_G^{\geq 1}, [\widetilde{L^\vee}_1](-D) \arrow[d] \\
H^0(\mathfrak{X}_H^{\mathrm{ord}}, \omega_H(\ell_1, \ell_2, 3 - k_3)) \arrow[r] & H^1(\mathfrak{X}_G^{\geq 1}, \mathfrak{F}^{(-k_2, -k_1, -k_3; -k_1 - k_2 + k_3)}(-D))
\end{tikzcd}$$

is commutative, where the left vertical map (the right vertical map, respectively) is the classical unit root splitting over $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty)$ (the unit root splitting of Proposition 21.3.4 over $\mathfrak{X}_G^{\geq 1}(p^\infty)$, respectively).

Proof. We check the commutativity over $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty)$ by pulling back. Then, the Theorem follows from the fact that the Q-torsor $\mathcal{T}^{\mathrm{Kl}}$ pulls back to a Q-torsor $\iota_\infty^*(\mathcal{T}^{\mathrm{Kl}})$ over $\mathfrak{X}_H^{\mathrm{ord}}(p^\infty)$, which has a reduction of structure to a T-torsor $\mathcal{T}^{\mathrm{ord}}$. □

Corollary 21.4.2. *Propositions 20.4.1, 20.4.2 can be relaxed so that $\phi(\mathcal{E})$ can be the image of a classical* nearly holomorphic *modular form. Namely, if $\phi(\mathcal{E})$ is the image of a classical nearly-holomorphic modular form E_ϕ^{nh}, then*

$$\phi(\langle \iota_*(\mathcal{E}), \eta \rangle) = \langle \iota_{m,*}(E_\phi^{\mathrm{nh}}), \eta_m \rangle.$$

Proof. By the diagram of Theorem 21.4.1, this amounts to the identification of the

two maps

$$H^0(\mathfrak{X}_H^{\mathrm{ord}}, [V_H(\ell_1{-}1, \ell_2{+}1, 3{-}k_3)]\otimes\omega_H(1,-1,0;0)(-D_H)) \to H^1(\mathfrak{X}_{\overline{G}}^{\geq 1}, \mathfrak{F}^{(-k_2,-k_1,-k_3;-k_1-k_2+k_3)}(-D)),$$

given by the composition going through the upper-right corner of the square and the composition going through the lower-left corner of the square. $\square$

Chapter 22

Construction of the p-adic L-function

We now gather all the ingredients to construct the desired p-adic L-function, modulo the local calculations that will be conducted in §23.

22.1 Eisenstein series over H

We now spell out the theory of classical and p-adic Eisenstein series over H fairly explicitly. As H is, up to center, the split unitary group of signature $(1,1)$, the Hermitian symmetric domain associated to $H(\mathbb{R})$ is the same as the usual upper half plane, and indeed an elliptic modular form can be regarded as a holomorphic modular form for H.

Definition 22.1.1. Given a holomorphic modular form f of weight $k = 2m$ and level N, we denote by ϕ_f the automorphic form on $H(\mathbb{A})$ defined by

$$\phi_f(\gamma g_\infty kz) = (ci + d)^{-m}(\overline{c}i + \overline{d})^{-m} f(gi),$$

where

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in H'(\mathbb{R}),$$

$z \in Z(H)(\mathbb{A})$, $\gamma \in H(\mathbb{Q})^+$, and $k \in \Gamma_1(N)(\mathbb{A}_f) \subset H(\mathbb{A}_f)$.

Even though, as above, holomorphic automorphic forms for $\mathrm{U}(1,1)$ are basically the same as the (elliptic) modular forms, the arithmetic changes slightly as we change the viewpoint to unitary Shimura varieties. We thus find it helpful to work directly with the formula for real analytic Eisenstein series (in Katz's terminology). Recall that $H \subset G = \mathrm{GU}(V)$ is the stabilizer of W, where $V_H = Y_K \oplus X_K$ is a skew-Hermitian K-space of dimension 2 with the skew-Hermitian form $\theta_{1,1}$ defined by

$$\theta_{1,1} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix},$$

with respect to the basis $X_K = Kx$, $Y_K = Ky$, and $V = Y_K \oplus W \oplus X_K$ with the skew-Hermitian form $\theta_{2,1}$ defined by

$$\theta_{2,1} = \begin{pmatrix} & & -1 \\ & \theta & \\ 1 & & \end{pmatrix}.$$

Consider the natural action of $H(\mathbb{R})^+$ acting on the upper half plane $\mathbb{H}$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+c}{bz+d}.$$

Recall the defininition of the Eisenstein series:

Definition 22.1.2 (Eisenstein series on H)**.** Let Φ be a Schwartz–Bruhat function on $V_H(\mathbb{A}_{\mathbb{Q}})$. Let μ be an idele class character of $K^\times \backslash \mathbb{A}_K^\times$. Then,

$$f^\Phi(h;\mu,s) := \int_{\mathbb{A}_K^\times} (h\cdot\Phi)(ty)\mu(t)|t|_K^s d^\times t,$$

be a Siegel section, so that

$$f^\Phi(bh;\mu,s) = \mu(b)\delta(b)^s f^\Phi(h;\mu,s),$$

for

$$b = \begin{pmatrix} \alpha & * \\ & \overline{\alpha}^{-1} \end{pmatrix} \in B'_H,$$

such that

$$\mu\begin{pmatrix} \alpha & * \\ & \overline{\alpha}^{-1} \end{pmatrix} = \mu(\alpha),$$

$$\delta\begin{pmatrix} \alpha & * \\ & \overline{\alpha}^{-1} \end{pmatrix} = |\alpha|_K^s.$$

The corresponding Eisenstein series is defined as

$$E^{\Phi}(h;\mu,s) := \sum_{\gamma\in B_H(\mathbb{Q})\backslash H(\mathbb{Q})} f^{\Phi}(\gamma h;\mu,s).$$

This implies that f^{Φ} is a section of an induced representation,

$$f^{\Phi}(h;\mu,s) \in \mathrm{Ind}_{B_H}^{H}(\mu\delta^{s-1/2}).$$

Whether $E^{\Phi}(h;\mu,s)$ is a (nearly) holomorphic modular form is completely determined by the data at infinity.

Definition 22.1.3. For $m_1, m_2, n_1, n_2 \geq 0$ integers, define $\Phi_{m_1,m_2,n_1,n_2,\infty} \in \mathcal{S}(V_H(\mathbb{R}))$ to be

$$\Phi_{m_1,m_2,n_1,n_2,\infty}(z_1,z_2) = z_1^{m_1}\overline{z}_1^{n_1} z_2^{m_2}\overline{z}_2^{n_2} e^{-\frac{|z_1|^2+|z_2|^2}{2}},$$

where $(z_1,z_2) \in V_H(\mathbb{R})$ is the coordinate with respect to $x \in X_K, y \in Y_K$.

Remark 22.1.4. The space spanned by $\Phi_{m_1,m_2,n_1,n_2,\infty}$'s is dense in $\mathcal{S}(V_H(\mathbb{R}))$.

The infinity type of μ is described by two numbers,

$$\mu_\infty(t) = |t|^{w_0}(t/|t|)^{w_1}, \qquad w_0 \in \mathbb{C}, w_1 \in \mathbb{Z}.$$

By elementary integral computation, one sees the following.

Proposition 22.1.5 ([KO, §6])**.** *The Siegel section*

$$f^{\Phi_f\otimes\Phi_{m_1,m_2,n_1,n_2,\infty}}(h;\mu,s),$$

vanishes unless $w_1 = n_1+n_2-m_1-m_2$. *Furthermore, if* $w_1 = n_1+n_2-m_1-m_2$, *then*

$$E^{\Phi_f\otimes\Phi_{m_1,m_2,n_1,n_2,\infty}}(h;\mu,s),$$

spans a K-type of weight

$$(2n_1 - 2m_1, n_1 + n_2 - m_1 - m_2),$$

and is nearly holomorphic for

$$-\frac{n_1 + n_2 - m_1 - m_2 + 2w_0}{2} \leq s \leq -w_0.$$

Finally, under the above conditions, if $\Phi_f \in \mathcal{S}(V_H(\mathbb{A}_f))$ is valued in a number field E, then the Eisenstein series

$$\frac{1}{\Gamma\left(s + \frac{w_0+n_1+n_2+m_1+m_2}{2}\right)} E^{\Phi_f \otimes \Phi_{m_1,m_2,n_1,n_2,\infty}}(h; \mu, s),$$

defines an E-rational nearly coherent cohomology class.

22.2 p-adic family of Eisenstein series

We fix the standard additive charcater $\psi : \mathbb{A}/\mathbb{Q} \to \mathbb{C}^\times$ such that $\psi(x_\infty) = \exp(2\pi i x_\infty)$ for $x_\infty \in \mathbb{R}$.

Definition 22.2.1 (Fourier coefficient for a nearly holomorphic Eisenstein series over H)**.** Assume that the Eisenstein series

$$E(h) := E^{\Phi_f \otimes \Phi_{m_1,m_2,n_1,n_2,\infty}}(h; \mu, s),$$

constructed in the previous section, is nearly holomorphic. Then, E has a Fourier expansion

$$E(h) = \sum_{\beta \in \mathbb{Q}} a_\beta(h, E),$$

$$a_\beta(h,E) = \int_{\mathbb{Q}\backslash\mathbb{A}} E\left(\begin{pmatrix} 1 & t \\ & 1 \end{pmatrix} h\right) \psi(2\beta t)dt.$$

From this computation in terms of Φ, we have the following theorem on the existence of p-adic interpolation of these Eisenstein series.

Theorem 22.2.2 ([Ei])**.** *We assume the following data.*

- *Let E be a number field, and let L be the completion of E at a prime above p.*
- *Let $\Phi^{(p)} \in \mathcal{S}(V_H(\mathbb{A}_f^{(p)}))$ be a Schwartz function with values in L.*
- *Let $\chi^{(p)}$ be a Dirichlet character of prime-to-p conductor such that*

$$\begin{pmatrix} a & \\ & a \end{pmatrix} \Phi^{(p)} = \widehat{\chi}(a)^{-1}\Phi^{(p)},$$

  for $a \in (\widehat{\mathbb{Z}}^{(p)})^\times$.

- *Let χ_1, χ_2 be Dirichlet characters of p-power conductor.*
- *Let χ be a Hecke character of K of type A_0 hat satisfies the following:*
  - *The prime-to-p part of χ factors through a finite quotient and is equal to $\chi^{(p)}$.*
  - *The p-part, χ_p, is identified with*

$$\chi_p = (\chi_1, \chi_2^{-1}),$$

  under the chosen identification

$$K_p = K_\pi \times K_{\overline{\pi}},$$

(see Notations).

– *The ∞-part, χ_∞, is identified with*

$$\chi_\infty(z) = |z|^{w_0}(z/|z|)^{w_1} = z^{\frac{w_0+w_1}{2}}\overline{z}^{\frac{w_0-w_1}{2}},$$

for $w_0 \equiv w_1 (\mathrm{mod}\, 2)$.

We let

$$\widetilde{\chi} : K^\times \backslash \mathbb{A}_K^{\infty,\times} \to \overline{\mathbb{Q}}_p^\times,$$

be the associated p-adic Hecke character, namely

$$\widetilde{\chi}_v = \chi_v \qquad v \nmid p,$$

$$\widetilde{\chi}_p = \chi_p \cdot \sigma(z)^{\frac{w_0+w_1}{2}}\overline{\sigma}(z)^{\frac{w_0-w_1}{2}}.$$

- *Let $R = \Lambda_L(\mathbb{Z}_p^\times \times \mathbb{Z}_p^\times)$ with its two canonical characters κ_1, κ_2.*

Then, there is an element

$$\mathcal{E}^{\Phi^{(p)}}(\kappa_1, \kappa_2; \chi^{(p)}) \in \mathcal{S}_{H'}^{(\kappa_1+1,\kappa_2;-\kappa_1-\kappa_2-1)}(R),$$

whose specialization at $(a+\mu, b+\nu)$ for $a, b \geq 0$ is the p-adic modular form associated to the nearly holomorphic Eisenstein series

$$\frac{1}{\Gamma(\frac{w_0-3a+1}{2})} E^{\Phi^{(p)}\otimes\Phi_{\mu,\nu,p}\otimes\Phi_{-a-1,-b,-\frac{a}{2}-1,-\frac{a}{2},\infty}}(h, \chi^{(p)}\chi_1\chi_2^{-1}, \frac{b-a+1}{2}) \in M_H^{(a+1,b,-a-b-1),\mathrm{nh}},$$

where $\Phi_{\mu,\nu,p} \in \mathcal{S}(V_H(\mathbb{Q}_p))$ is the Schwartz function such that the local Whittaker

transform of $f^{\Phi_p}(h;\mu_p,\frac{b-a+1}{2})$ *is*

$$\begin{cases} \mu(x)\nu(y) & \text{if } (x,y)\in\mathcal{O}_{K,p}^{\times} \\ 0 & \text{otherwise.} \end{cases}$$

Proof. This follows from the more general construction of Eisenstein measure for Siegel Eisenstein series in [Ei, Theorem 20]. □

Remark 22.2.3. This can be directly compared with the p-adic Eisenstein series for GL_2, constructed in [LPSZ, Theorem 7.6].

22.3 Main theorem

We retain the notations used in the previous sections.

Theorem 22.3.1 (Theorem 1 of Introduction)**.** *Let* $R = L[[\mathbb{Z}_p^{\times}]]$*. If* $k_1 - k_2 \geq 2$ *and* $k_2 + k_3 \geq 2$*, there is an element* $\mathcal{L}(\Pi,\mathbf{j}) \in R$ *such that, for any Dirichlet character* χ *of* p*-power conductor and any critical value* a *of* $L(\Pi,\chi,s)$*,*

$$\mathcal{L}(\Pi, a+\chi) = \frac{Z_p(\Pi,\chi,a)\Lambda(BC(\Pi)\otimes\chi^{-1},a)}{\Omega^W(\Pi)},$$

where

$$Z_p(\Pi,\chi,a) = \frac{1}{L_p(BC(\Pi^{\vee}),\chi^{-1},1-a)\epsilon(BC(\Pi_p),a)},$$

and $\Omega^W(\Pi)$ *is the* Whittaker period*, defined in a way that the Whittaker function* W_f *corresponding to an* E*-rational vector* $f \in H^1(\Pi_f)$ *is such that* $\Omega^W(\Pi)W_f$ *is an* E*-rational Whittaker function.*

Proof. We will plug the following data to the general machinery we have developed so far. Namely, we will specialize $\langle \iota_*(\mathcal{E}^{\Phi^{(p)}}(\kappa_1,\kappa_2;\chi_{\Pi}^{-1})),\eta\rangle$ at appropriate weights and relate the period integral to the value of $L(BC(\Pi)\otimes\chi^{-1},a)$.

More precisely, we will show in §23 that a choice of finite linear combination of E-rational Schwartz functions away from $\{p, \infty\}$ would give rise to the desired local L-factor. We have also shown how to choose the data at v and ∞, respectively. Finally, there must be a factor of Whittaker period coming from the archimedean zeta integral, as it involves the value of Whittaker function at 1. $\square$

Chapter 23

Local calculations

We now conduct the final task, namely calculation of the local zeta integrals for the Gelbart–Piatetski-Shapiro integral for $\mathrm{GU}(2,1)$.

23.1 Nonarchimedean L-factors

Let F be a nonarchimedean local field over $\mathbb{Q}_v$, and let E/F be a quadratic extension. Let π_v be an irreducible admissible representation of $G(F)$ which is also *generic*. Recall the definition of the local zeta integral,

$$Z_v(f, \mu_v, \Phi_v, s) = \int_{Z_G(\mathbb{Q}_v) Z_{N_{B_G}}(\mathbb{Q}_v) \backslash H(\mathbb{Q}_v)} W_v(h) f_v^{\Phi}(h; \mu, s) dh,$$

where

$$f_v^{\Phi}(h; \mu, s) = \prod_{w|v} \int_{K_w^\times} (h \cdot \Phi_w)(ty) \mu_w(t) |t|_w^s d^\times t,$$

and $W_{f,v}(h)$ corresponds to f in the local Whittaker model for π_v with respect to the character ψ_v. Here the choice of an isomorphism between the local Whittaker model and π_v is chosen and fixed implicitly. The following local calculation is performed in [GPS].

Theorem 23.1.1. *1. There is a positive real number $R > 0$ depending on π_v, μ_v such that the local zeta integral $Z_v(f, \mu_v, \Phi_v, s)$ converges absolutely for $\mathrm{Re}(s) > R$ for all choices of f, Φ_v. Furthermore, each $Z_v(f, \mu_v, \Phi_v, s)$ has a meromorphic continuation with respect to s, and is a rational function of q_v^s.*

2. The vector space of functions on $\mathbb{C}$ spanned by $Z_v(f, \mu_v, \Phi_v, s)$, as the data (f, Φ_v) vary, is a fractional ideal of $\mathbb{C}[q_v^{\pm s}]$ containing the constant functions. The fractional ideal is generated by $L(s, \pi_v, \mu_v)$, which is the local L-factor corresponding to the standard L-function for a $\mathrm{GL}_3(E)$-representation $BC(\pi_v) \otimes \mu_v$, the base-change lift of π_v to E tensored with μ_v.

3. If everything in the data is unramified, $f_v^0 \in \pi_v$ is the unique spherical vector such that $W_{f_v^0,v}(1) = 1$, and $\Phi_v^0 = \mathbf{1}_{\mathcal{O}_E^2}$, then

$$Z_v(f_v^0, \mu_v, \Phi_v^0, s) = L(s, \pi_v, \mu_v).$$

Proof. This is precisely the content of [GPS, §4.4]. □

Now we check the rationality of the local zeta integral.

Definition 23.1.2. Let M be a number field. An irreducible admissible representation π_v of $G(F)$ is said to be *defined over* M if, for each $\sigma \in \mathrm{Aut}(\mathbb{C}/M)$, there is an isomorphism

$$\pi_v \otimes_{\mathbb{C},\sigma} \mathbb{C} \cong \pi_v,$$

such that the collection satisfies the cocycle condition.

Definition 23.1.3. Let M be a number field. A Whittaker function W is said to be *defined over* M if it takes values in $M(\mu_\infty)$ and satisfies

$$\sigma(W(g)) = W(w(\kappa_v(\sigma))g),$$

for all $g \in G(F)$ and $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/M)$, where $w(x) = \mathrm{diag}(x, 1, x^{-1})$ and κ_v is the v-adic cyclotomic character.

Proposition 23.1.4. *Suppose that π_v is defined over a number field M. Then, the Whittaker model $W(\pi_v)$ is spanned by the Whittaker functions defined over M.*

We denote the M-vector space of Whittaker functions defined over M by $W(\pi_v)_M$.

Proposition 23.1.5. *Retain the premise of Theorem 23.1.1. Suppose that every datum is unramified and the character as well as the Whittaker function is valued in M. Then,*

$$Z_v(f, \mu_v, \Phi_v, s),$$

is an E-valued Laurent polynomial with variables in q^s.

23.2 Calculations at p

We will be concerned with the "Klingen-ordinary" representations, which are in particular principal series representations as we saw in §17.3. The local zeta integral is that of $\mathrm{GL}_3 \times \mathrm{GL}_2$ by the hypothesis that p splits in K. Let $\pi = I(\chi)$ be a spherical principal series of $\mathrm{GL}_3(\mathbb{Q}_p)$.

Definition 23.2.1. For $r \geq 1$, let $\phi_r \in \pi^{\mathrm{Kl}(p^r)}$ be the characteristic function with support $B(\mathbb{Q}_p)\,\mathrm{Kl}(p^r)$, scaled so that $\phi_r(1) = p^{2r}$.

A straightforward calculation gives the following

Lemma 23.2.2. *Let $r \geq 1$.*

1. *The functions ϕ_r are compatible with taking traces, namely*

$$\frac{1}{[\mathrm{Kl}(p^r) : \mathrm{Kl}(p^{r+1})]} \sum_{\gamma \in \mathrm{Kl}(p^r)/\,\mathrm{Kl}(p^{r+1})} \gamma \phi_{r+1} = \phi_r.$$

2. *The vector* $\phi_r \in \pi^{\mathrm{Kl}(p^r)}$ *is an eigenvector for* $U_{p,1}$ *with eigenvalue* $p\chi(p)$.

3. *The vector* ϕ_r *satisfies*

$$\phi_r = \chi(p)^{1-r} \sum_{a \in \mathbb{Z}/p^{r-1}\mathbb{Z}} \begin{pmatrix} 1 & & p^r a \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} p & & \\ & p & \\ & & 1 \end{pmatrix}^{r-1} \phi_1.$$

Theorem 23.2.3. *Let* $\Phi \in \mathcal{S}^2(\mathbb{Q}_p^2)$ *be such that the local Siegel section* $f^\Phi(-;\mu,s)$ *is supported in the big cell* $B_H w N_{B_H}$ *where*

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then, for $r \gg 0$, *the zeta integral*

$$Z_v(\gamma \cdot \phi_r, \mu, \Phi, s) = \int_{Z_H(\mathbb{Q}_v)N_H(\mathbb{Q}_v)\backslash H(\mathbb{Q}_v)} W_{\phi_r}(h\gamma) f^\Phi(h;\mu,s)dh, \quad \gamma = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix},$$

stabilizes to a value

$$\frac{L(BC(\pi),s)}{L(BC(\pi^\vee),1-s)\epsilon(BC(\pi),s)} \int_{\mathbb{Q}_p^\times} W_\xi(\left(\begin{smallmatrix} x & 0 \\ 0 & 1 \end{smallmatrix}\right)) W^\Phi(\left(\begin{smallmatrix} x & 0 \\ 0 & 1 \end{smallmatrix}\right), \mu, s) \frac{\theta(x)}{|x|} d^\times x.$$

Proof. As h is supported in the big cell, there is unique $a \in \mathbb{Q}_p$ such that

$$Z_H N_H h = Z_H N_H \begin{pmatrix} \det h & \\ & 1 \end{pmatrix} w \begin{pmatrix} 1 & a \\ & 1 \end{pmatrix}.$$

Thus, changing the variables the zeta integral is easily seen to be equal to

$$Z = \frac{p}{p+1} \int_{a \in \mathbb{Q}_p} f^{\Phi} \left(w \begin{pmatrix} 1 & a \\ & 1 \end{pmatrix} \right) \int_{b \in \mathbb{Q}_p^\times} W_{\phi_r} \left(\begin{pmatrix} b & & \\ & 1 & \\ & & 1 \end{pmatrix} \begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 & & a \\ & 1 & \\ & & 1 \end{pmatrix} \gamma \right) v^{s-1} dad^\times b.$$

Calculating, we simplify into

$$Z = \frac{p}{p+1} \int_{a \in \mathbb{Q}_p} f^{\Phi} \left(w \begin{pmatrix} 1 & a \\ & 1 \end{pmatrix} \right) \int_{b \in \mathbb{Q}_p^\times} W_{\phi_r} \begin{pmatrix} -b & 0 & b \\ a & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} v^{s-1} dad^\times b.$$

Since

$$\begin{pmatrix} -b & 0 & b \\ a & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -b \\ & 1 & a \\ & & 1 \end{pmatrix} \begin{pmatrix} & & b \\ & 1 & \\ 1 & & \end{pmatrix},$$

by using the left-N_{B_G}-transformation property of W_{ϕ_r}, we obtain

$$Z = \frac{p}{p+1} \int_{a \in \mathbb{Q}_p} f^{\Phi} \left(w \begin{pmatrix} 1 & a & \\ & & \\ & & 1 \end{pmatrix} \right) \int_{b \in \mathbb{Q}_p^\times} W_{\phi_r} \begin{pmatrix} & & b \\ & 1 & \\ 1 & & \end{pmatrix} \psi_p(a) v^{s-1} da d^\times b.$$

This integral is exactly the one that shows up for the Rankin–Selberg integral for $\mathrm{GL}_3 \times \mathrm{GL}_2$ whose stabilization property as well as its relation to the modified local p-factor is shown in [Ja, §4]. □

23.3 Calculations at ∞

This is one of the main purposes of [KO]. In *op. cit.*, one explicitly obtains the generalized Whittaker functions via solving differential equations arising from the Schmid operators. More precisely, we know that, as seen above, a dense subspace of the Schwartz space is spanned by

$$\Phi_{(m_1,n_1,m_2,n_2)}(z_1, z_2) = z_1^{m_1} \overline{z}_1^{n_1} z_2^{m_2} \overline{z}_2^{n_2} \exp\left(-\frac{1}{2} \left(|z_1|^2 + |z_2|^2 \right) \right), \quad m_1, n_1, m_2, n_2 \in \mathbb{Z}.$$

Theorem 23.3.1 ([KO, §6])**.** *Let W be the Whittaker function of the generic discrete series $\Pi_\infty^{\mathrm{gen}}(\lambda_1, \lambda_2)$ contained in a K-type subspace of weight $(\lambda_1 + \lambda_2 + 2, -\lambda_1 - 1) + e(1,1) + f(1,-2)$. Then, $Z_\infty(W, \mu_\infty, \Phi_{(m_1,n_1,m_2,n_2)}, s) \neq 0$ if and only if $w_1 = n_1 + n_2 - m_1 - m_2$, $n_1 - m_1 = \lambda_1 + e$, $n_2 - m_2 = -e + f$, where $\mu_\infty(t) = |t|^{w_0} \left(\frac{t}{|t|} \right)^{w_1}$. Furthermore, it is equal to $C \cdot L(BC(\Pi_\infty^{\mathrm{gen}}(\lambda_1, \lambda_2)), s)$, where C is a constant that lies in*

$$C \in \mathbb{Q}^\times \cdot \frac{1}{\Gamma(s + w_0 + c(e))},$$

where

$$c(e) = \frac{1}{2}\left(\left|e + \frac{-f + \lambda_1 + w_1}{2}\right| + \left|e + \frac{-f + \lambda_1 - w_1}{2}\right|\right).$$

Proof. This follows from the explicit formulae of Proposition 6.7 and Theorem 6.8 of [KO]. □

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